



**GREEN
CLIMATE
FUND**

Meeting of the Board

15 – 18 July 2024

Songdo, Incheon, Republic of Korea

Provisional agenda item 10

GCF/B.39/02/Add.14

24 June 2024

Consideration of funding proposals – Addendum XIV

Funding proposal package for FP239

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL)";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title: ***Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL)***

Country(ies): *Djibouti, Ethiopia, Kenya, Somalia, and South Sudan*

Accredited Entity: The African Development Bank (AfDB).

Date of first submission: *[2023/05/27]*

Date of current submission: *[2024/06/18]*

Version number: *[V.007]*



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

Abbreviation	Definition
AAAP	Africa Adaptation Accelerator Program
ACS	Agricultural Cooperative Society
AD	Agro-dealer
ADF	African Development Fund
AE	Accredited Entity
AFAWA	Affirmative Finance Action for Women in Africa
AfDB	African Development Bank (AfDB).
AFOLU	Agriculture, Forests, and Other Land Use
AML/CFT	Anti-Money Laundering and Countering Financing of Terrorism
ARA	Adaptation Result Area
ASALs	Arid and Semi-arid Lands
AVC	Agropastoral Value Chain
BDCW	BREFOL Direct Credit Window
BDICW	BREFOL Direct Insurance Credit Window
BREFOL	Building Climate Resilience for Food and Livelihoods in the Horn of Africa
CBs	Commercial Banks
CDD	Maximum Number of Consecutive Dry Days
CDI	Combined Drought Indicator
CFRA	Country Fiduciary Risk Assessment
CIEWS	Climate Information and Early Warning System
CLMCP	Community Land Management Committees and Platforms
CMIPs	Coupled Model Inter-comparison Programs
COVID-19	Coronavirus 2019
CPAR	Country Procurement Assessment Report
CRU	Climatic Research Unit
CRVA	Climate Risk and Vulnerability Assessment
CSA	Climate-Smart Agriculture
CSOs	Civil Society Organizations
CSP	Country Strategy Papers
CWD	Consecutive Wet Day
DACCS	African Union-Draft Africa Climate Change Strategy
DRM	Disaster Risk Management
DRs	Discount Rates
DRSLP	Drought Resilience and Sustainable Livelihoods Programme
EA	Economic Analysis
EDRMC	Ethiopian Disaster Risk Management Commission
EE	Executing Entities
EFA	Economic and Financial Analysis
EIRR	Economic Internal rate of Return
ENPV	Economic Net Present Value
ESIAs	Environmental and Social Impact Assessments
ESMFs	Environmental and Social Management Frameworks
ESMPs	Environmental and Social Management Plans
EWS	Early warning systems
FA	Financial Analysis
FAA	Funded Activity Agreement
FAO	Food and Agricultural Organization
FBAs	Farm-based Associated
FCMB	First City Monument Bank
FEWS NET	Famine Early Warning Systems Network
FIRR	Financial Internal Rate of Return
FMA	Financial Management Arrangement
FNPV	Financial Net Present Value

FSI	Fragile State Index
FSNWG	Food Security and Nutrition Working Group
GALS	Gender Action Learning System
GCA	Global Centre on Adaptation
GDP	Gross Domestic Product
GFFA	Global Forum on Food and Agriculture
GGW	Great Green Wall
GHA	Greater Horn of Africa
GHGs	Green House Gases
HCVI	Hunger and Climate Vulnerability Index
HoA	Horn of Africa
HWF	Heat Wave Frequency
ICEWARN	IGAD Conflict Early Warning and Response Mechanism
ICG	International Crisis Group
ICPAC	IGAD Climate Prediction and Applications Center
ICPALD	IGAD Centre for Pastoral and Livestock Development
IDDRSI	IGAD's Drought Disaster Resilience and Sustainability Initiative
IES	IGAD Executive Secretariat
IFAD	International Fund for Agricultural Development
IFI	International Financial Institutions
IFRAH	IGAD Food Security, Nutrition and Resilience Analysis Hub
IGAD	Intergovernmental Authority on Development
IGADD	Intergovernmental Authority on Drought and Development
IGREENFIN	Inclusive Green Financing Initiative
ILRI	the International Livestock Research Institute
IMF	International Monetary Fund
IPC	Integrated Food Security Phase Classification
IPCC	Intergovernmental Panel on Climate Change
IPCU	IDDRSI Platform Coordinating Unit
IPMT	Integrated Pest Management Technology
IPRs	Independent Procurement Reviews
IRAPP	IGAD Regional Aids Programme
ISFM	Integrated Soil Fertility Management
ISSP	IGAD Security Sector Programme
LoC	Line of Credit
LPFIs	Local Private Financial Institutions
LUCF	Land-Use Change and Forestry
MAD	Meteorological and Agricultural Drought
MDAs	Ministries, Departments and Agencies
MENA	Middle East and North Africa
MFI	Microfinance institution
MIRR	Modified Internal Rate of Return
MoAF	Ministry of Agriculture and Forestry
MoLFR	Ministry of Livestock, Forestry and Range
MoA	Ministry of Agriculture
MoAWFL	Ministry of Agriculture, Water, Fisheries and Livestock
MoAWFLC	<u>Ministry of Agriculture, Water, Fisheries and Livestock, and Cooperatives</u>
MoCEF	Ministry of Commerce Economics and Finance
MoFE	Ministry of Finance, Ethiopia
MoFP	<u>Ministry of Finance and Planning, South Sudan</u>
MoFK	Ministry of Finance, Kenya
MoFS	Ministry of Finance, Somalia
MoIL	Ministry of Irrigation and Lowlands
MoU	Memorandums of Understanding

MRA	Mitigation Results Area
MSME	Micro, Small & Medium Enterprises
MtCO ₂ e	Metric tons of carbon dioxide equivalent
NAPA	National Adaptation Programmes of Action
NAPs	National Adaptation Plans
NDAs	National Designated Authorities
NDCs	Nationally Determined Contributions
ND-GAIN	Notre Dame Global Adaptation Initiative
NDVI	Normalized Difference Vegetation Index
NGOs	Non-governmental Organizations
NMA	National Meteorological Agency
NMS	National Meteorological Services
NPCU	National Program Coordination Unit
NTF	Nigeria Trust Fund
NTEP	National Treasury and Economic Planning
OECD	Organization for Economic Cooperation and Development
PARs	Project Appraisal Reports
PCU	Platform Coordinating Unit
PDAP	Procurement Development Action Plan
PEAPA	Programme Executing Agency Procurement Assessment
PEPs	Politically Exposed Persons
PFIs	Public Financial Institutions
PIA	Priority Intervention Areas
PIUs	Project Implementing Units
PMUs	Project Management Units
PPP	Private-Public Partnerships
PPM	Participatory Pasture Management
PTCs	Pastoralist Training Centers
RCP	Representative Concentration Pathway
RETs	Renewable energy technologies
RMCs.	Regional Member Countries
RPSC	Regional Program Steering Committee
SACCOs	Savings and Credit Cooperative Organizations
SBDs	Standard Bidding Documents
SC	Sub-Component
SME	Small and Medium-sized Enterprise
SMI	Small and Medium-sized Industry
SPEI	Standardized Precipitation Evapotranspiration Index
SPI	Standardized Precipitation Index
TAAT	Technologies for African Agricultural Transformation Program
TADs	Transboundary Animal Diseases
TFRA	Task Force on Rural Africa
ToC	Theory of Change
TOT	Training-of-Trainers
UA	Units of Account
UBA	United Bank for Africa
UNDP	United Nations Development Programme
UNHCR	United Nations High Commissioner for Refugees
USSD	Unstructured Supplementary Service Data
VHI	Vegetation Health Index



WABEs	Women in Agribusiness Enterprises
WASH	Water Sanitation and Hygiene
WFP	World Food Programme
WHO	World Health Organization
WMO	World Meteorological Organization
WOP	Without Project
WP	With Project
WSDI	Warm Spell Duration
YABEs	Youths in Agribusiness Enterprises

A. PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Programme	A.2. Public or private sector	Public
A.3. Request for Proposals (RFP)	Not applicable		
A.4. Result area(s)	The applicable GCF result area(s) that the overall proposed project/programme targets below.		
		GCF contribution	Co-financers' contribution ¹
	Mitigation total	18%	11%
	☒ Energy generation and access	7%	5%
	☐ Low-emission transport		
	☐ Buildings, cities, industries and appliances		
	☒ Forestry and land use	11%	6%
	Adaptation total	82%	89%
	☒ Most vulnerable people and communities	34.4 %	25.4%
	☒ Health and well-being, and food and water security	32.0%	58.5%
	☐ Infrastructure and built environment		
☒ Ecosystems and ecosystem services	15.6%	5.1%	
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	14.1 MtCO ₂ eq	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	Direct and indirect beneficiaries ~4.68 million direct beneficiaries ² (ADF 1,859,451.0 + GCF 2,835,280.0 million) at least 50% women ~15.55 million indirect beneficiaries (ADF 10,198,282 + GCF 5,350,100 million) at least 50% women 1.9% of total population 6.4% of total population
A.7. Total financing (GCF + co-finance ³)	USD 335,30 million (GCF 151.00 million + ADF & Others USD 184,30 million)	A.9. Project size	Large (Over USD 250 million)
A.8. Total GCF funding requested	USD 151.0 million		
A.10. Financial instrument(s) requested for the GCF funding	☒ Grant USD 90.70 Million ☐ Equity Enter number ☒ Loan USD 60.30 Million ☐ Results-based payment Enter number ☐ Guarantee Enter number		
A.11. Implementation period	7(84) years(months)	A.12. Total lifespan	25 (300) years(months)
A.13. Expected date of AE internal approval	9/30/2024	A.14. ESS category	Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category. B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐

A.19. Complementarity and coherence	Does the project/programme complement other climate finance funding (e.g., GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1. Yes ☐ No ☒
A.20. Executing Entity information	• The Government of Djibouti, acting through its Ministry of Commerce, Economy and Finance (MoCEF) and its Ministry of Agriculture, Water, Fisheries and Livestock (MoAWFL); • The Government of Ethiopia, acting through the Ministry of Finance of Ethiopia (MoFE) and the Ministry of Agriculture (MoA); • The Government of Kenya, acting through the National Treasury and Economic Planning (NTEP) and the Ministry of Agriculture, Water, Fisheries, Livestock and Cooperatives (MoAWFLC); • The Government of Somalia, acting through its Ministry of Finance, Somalia (MOFS) and Ministry of Livestock, Forestry and Range (MoLFR); • The Government of South Sudan, through its Ministry of Finance and Planning, South Sudan (MoFP) and Ministry of Agriculture and Forestry (MoAF); • The Intergovernmental Authority on Development (IGAD)
A.21. Executive summary (max. 750 words, approximately 1.5 pages)	
The HoA region faces increased vulnerability due to highly variable rainfall patterns leading to extreme weather events such as floods and droughts, resulting in humanitarian emergencies, food insecurity and infrastructure damage. Therefore, the proposed GCF-AfDB programme entitled "Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL)" aims to address the pressing issue of climate change and variability in the Horn of Africa (HoA) region, specifically targeting Djibouti, Ethiopia, Kenya, Somalia and South Sudan in addition to the IGAD region. The specific objective of the programme is to reduce the vulnerability and increase the resilience of agro-pastoral and pastoral communities to climate-induced food insecurity and to reduce greenhouse gas (GHG) emissions in the region. This will be achieved by providing each beneficiary community with a package of resilience solutions, including i) infrastructure for water harvesting and agro-pastoral development, ii) support for sustainable land management development, iii) access to innovative climate-smart technologies and best practices, iv) access to renewable energy, v) capacity building for cooperatives and agribusiness Micro, Small and Medium Enterprises (MSMEs) on entrepreneurship, vi) access to credit, and vii) access to climate services, early warning and index insurance. AfDB financing will focus on providing i) agro-pastoral production infrastructure, ii) support for sustainable land management, iii) access to innovation, iv) capacity building for cooperatives, and v) infrastructure to generate climate information. GCF funding will complement the required resilience package by providing climate finance and promoting low-carbon technologies to enable these communities to be better equipped and prepared to cope with climate and other environmental hazards. With an estimated budget of USD 335.29 million, the programme aims to build resilience to the impacts of climate change by integrating climate change considerations into agricultural practices. Key constraints to climate adaptation that the programme seeks to overcome include limited access to concessional finance, political instability, weak governance, infrastructure gaps and limited access to climate information. The outputs of the programme are expected to lead to three main outcomes: Increased adoption of climate-resilient solutions; enhanced carbon sequestration and reduced greenhouse gas emissions of agropastoral value chain (AVC) and; strengthened adaptive capacity of vulnerable communities. Additionally, the program aims to bring about co-benefits such improved livelihoods opportunities and ecosystem services in line with regional strategies and national commitments. The programme will be implemented by host countries' Ministries of Agriculture, with regional coordination through the Intergovernmental Authority on Development (IGAD) and its subsidiary bodies.	

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e., GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² See Annex 23 for computations

³ Refer to the Policy of Co-financing of the GCF.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

1. The Horn of Africa (HoA) is one of the most food insecure regions in the world. The eight countries in the HoA - including Djibouti, Ethiopia, Eritrea, Kenya, Somalia, Sudan, South Sudan and Uganda (Figure B.1.1) - have a combined population of about 250 million people, of whom 160 million (or about 64 per cent) live in areas where food is extremely⁴.

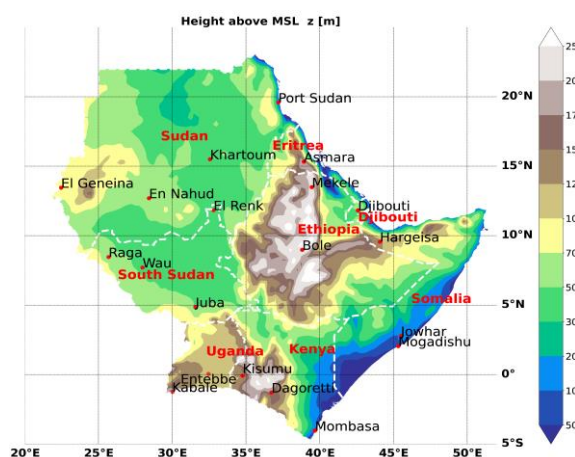


Figure B.1.1: Map of the Horn of Africa show station points used to plot the mean annual cycles of rainfall and temperature and elevation in meters above sea level.

2. In addition, about 80% of the region is classified as arid and semi-arid land (ASAL), with about 40% of the total land area classified as ecologically low productive land due to poor climatic conditions coupled with severe environmental degradation resulting from both natural conditions and human activities⁵. Since the 1970s, several of these countries have been threatened by famine at least once every decade (a frequency that has recently increased to once every three to five years). In Somalia, for example, the Famine Early Warning Systems Network (FEWS NET) predicts that the total number of people experiencing crisis-IPC 3⁶ or higher levels of acute food insecurity will increase significantly to more than 1.6 million. These projections are based on expected desert locust crop and pasture losses, the impact of the coronavirus (COVID-19) pandemic on domestic market supply chains and household food and income sources, increasing seasonal climate extremes, and the reduced ability of poor households to cope with shocks and agricultural losses. Reports from the Food Security and Nutrition Working Group (FSNWG) indicate that between 18.6 and 21.1 million people in Ethiopia, Kenya and Somalia are expected to experience crisis-IPC 3 or worsening levels of acute food insecurity due to drought. The number of people in need of emergency food assistance to meet their basic needs is expected to rise to as many as 26 million people in the three countries by February 2023, due to consecutive seasons of poor rainfall in 2022 and 2023⁷.

3. Similarly, in Djibouti, an estimated 60,000 Djiboutians and refugees in the country are at Crisis-IPC 3 or worse and in urgent need of food assistance. The situation is not much different in Ethiopia, where an estimated 8.1 million people are currently in Stressed or Crisis-IPC 2 and IPC 3 respectively. This has been attributed to localised flooding, conflict, insecurity, poor macroeconomic factors, desert locust infestation and the ongoing negative impact of past poor harvests. In South Sudan, an estimated 6 million people - about 51 percent of South Sudan's total population - could face crisis (IPC 3) or worse levels of acute food insecurity at the peak of the lean season from May to June, requiring urgent food

⁴ <http://www.unep.org/GEAS/>.

⁵ ICPAC (2018). Greater Horn of Africa Climate and Food Security Atlas. IGAD. <http://www.icpac.net/index.php/applications/research-development.html>.

⁶ Refers to the Integrated Food Security Phase Classification (IPC), which is a standardized tool that aims to classify the severity and magnitude of food insecurity. The IPC scale, which is comparable across countries, ranges from Minimal—IPC 1—to Famine—IPC 5.

⁷ USAID (2022): Horn of Africa – Complex Emergency. https://reliefweb.int/attachments/a062f05d-c9a6-49f4-aed0-a84de6508ddd/2022_08_19%20USG%20Horn%20of%20Africa%20Complex%20Emergency%20Fact%20Sheet%20%232.pdf.

assistance⁸. According to the latest IPC analysis, the main drivers of the high levels of acute food insecurity in the country include the negative impact of flooding and violence on livelihood activities, COVID-19, poor macroeconomic conditions, cereal production shortfalls, poor access to livestock products and increased food prices due to the COVID-19 pandemic.

4. The situation is exacerbated by the effects of climate change, population growth (estimated at 3% per year) and declining per capita agricultural production capacity. Climate change and climate extremes will exacerbate the fragility of food production systems and the natural resource base in environments prone to degradation, desertification and water stress, and in regions where poverty undermines the ability of rural people to take precautionary measures, thereby increasing their vulnerability. The HoA region is currently classified as one of the most disaster-prone regions in the world⁹. More than two-thirds of the total population is currently at greatest risk from seasonal droughts and recurrent flash floods, which are likely to increase livestock, crop and human diseases, land degradation, reduced crop production, increased crop failures, livestock diseases and deaths¹⁰. On the other hand, extreme demographic pressure (estimated at 3% per year) is a major concern in the region. Population growth continues to outpace growth in agricultural production, adding to the challenges of poverty and food insecurity in the region. This is particularly worrying for countries where the capacity to prepare and plan for impending disasters remains extremely low, as measured by their Notre Dame Global Adaptation Initiative (ND-GAIN) indices. The ND-GAIN indices¹¹ for these countries suggest a strong need for investment and innovation to improve preparedness and resilience to the impacts of climate change. With the exception of Djibouti, the HoA countries are among the 20 per cent most vulnerable to climate change in the world, and four (Eritrea, Ethiopia, Somalia and Sudan) are among the 10 per cent most vulnerable. Despite significant progress in reducing extreme poverty and food insecurity through several resilience-building programmes in the HoA, climatic shocks continue to pose a significant threat to food security, especially for the poor. Vulnerable people are trapped in a cycle of chronic hunger and poverty because they lack the capacity to cope with increasing, more frequent and intense risks and their impacts. Climate shocks are also increasing displacement and migration patterns. According to the World Meteorological Organization (WMO), by 2020, 12% of all new population displacements worldwide will occur in the East and Horn of Africa region, with more than 1.2 million new disaster-related displacements¹².

Climate Vulnerabilities, Impacts, and Adaptation Options

Climate Vulnerabilities:

5. The historical seasonal cycle of precipitation and temperature (mean, minimum and maximum) for selected stations in Ethiopia (Bole), Sudan (El Nahud), South Sudan (Wau), Kenya (Dagoretti), Somalia (Mogadishu), Djibouti (Djibouti), Uganda (Kabale) and Eritrea (Asmara) for the period 1981-2019¹³ is presented in Figure B.1.2 below. As observed (Figure B.1.2), stations across the HoA exhibit different characteristics depending on their geographical location. Some stations show unimodal precipitation seasons, while others show bimodal seasons. This corresponds to single temperature peaks at some stations and double peaks at others. For example, the observed rainfall at Bole is unimodal with a maximum monthly rainfall peak of about 264 mm/month in August. Double temperature peaks are also observed at the station between the seasons of March-May and September-November. Temperatures vary from about 7°C (min) to 26.3°C (max) at the station. En Nahud in the southern part of Sudan also shows a similar pattern of unimodal rainfall, but with distinct double temperature peaks in May (39°C) and October (37°C). The observed maximum mean monthly rainfall at this station is about 120 mm. Wau in Southern Sudan experiences an early onset of rainfall compared to the stations in Sudan. Although the temperature at Wau peaks in the same month as El Nahud, the magnitude is lower, and the mean monthly rainfall is higher (about 193 mm/month). Both Dagoretti in Kenya and

⁸ USAID (2020). Food Assistance Fact Sheet South Sudan

⁹ IPCC (2018). Summary for Policy Makers (Draft).

¹⁰ ICPAC (2018). Greater Horn of Africa Climate and Food Security Atlas. IGAD.

¹¹ University of Notre Dame (2020). Notre Dame Global Adaptation Initiative. URL: <https://gain.nd.edu/our-work/country-index/>

¹² WMO (2020) State of the Climate in Africa

https://reliefweb.int/sites/reliefweb.int/files/resources/1275_WMO_State_of_the_Climate_in_Africa_2020_en.pdf

¹³ The plotted [precipitation and temperature datasets](#) were retrieved from CHIRPSv2 and CRU, respectively

Mogadishu in Somalia experience bimodal rainy seasons with different first peaks in April and May respectively, but the second peak is observed in November. Both stations experience two temperature maxima. Djibouti experiences a year-round wet season with a single temperature maximum in July, although the magnitude of the mean monthly rainfall is lower (i.e. less than 35 mm/month). This can be explained by the influence of a precipitation area over the western equatorial Indian Ocean, which affects the distribution of rainfall along the HoA coast¹⁴. Perennial rainfall is also observed in Kabale, Uganda, with bimodal rainfall seasons in MAM (March-April-May) and SON (September-November), while Asmara, Eritrea, experiences unimodal rainfall peaking in July.

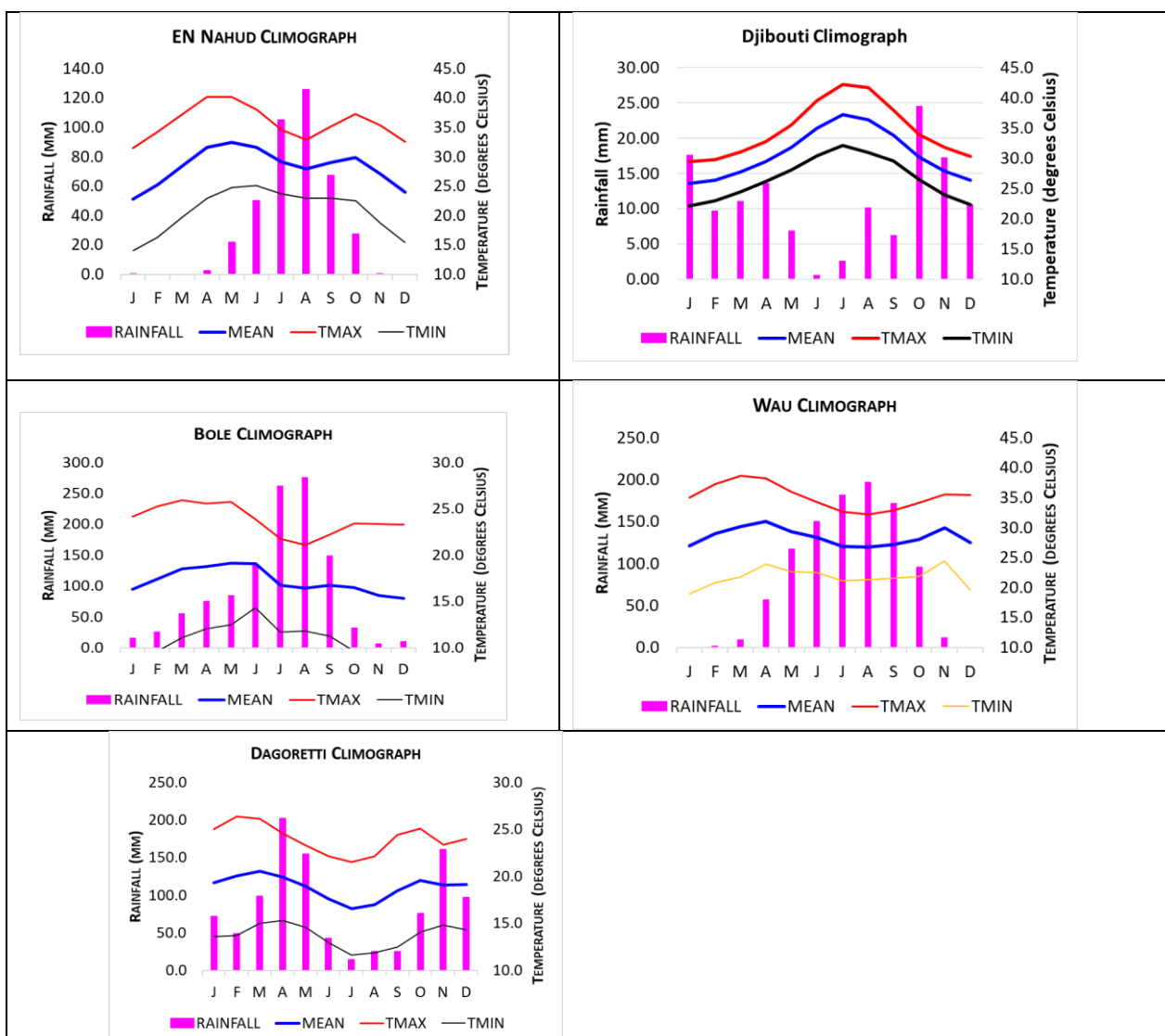


Figure B.1.2: Monthly Climatology of precipitation, mean, minimum and maximum temperature for El Nahud in Sudan, Bole in Ethiopia, Wau in South Sudan, Dagoretti in Kenya, Somalia, Djibouti city in Djibouti reference to 1981 -2019. The analysis is done based on observed station data.

6. Observed trends in precipitation, minimum and maximum temperature for the period 1976-2005 are shown in Figure B.1.3a. As observed, trends in precipitation range from -9 to 9 mm/month/decade over the entire HoA. Positive trends are observed over most of southern Sudan, southern Sudan, northwestern Uganda, northern Ethiopia, all of Djibouti and parts of Somalia. These trends are statistically significant only in the northeastern axis of Ethiopia. On the other hand, negative trends are

¹⁴ Greater Horn of Africa Climate Risk and Food Security Atlas

observed in most parts of Kenya, especially in the central and southern parts of the country. Declining trends are also observed in the southern parts of Ethiopia and the northern part of Eritrea. Some parts of Somalia and eastern Uganda have also experienced a decrease in rainfall. However, the observed negative trends are not statistically significant. With regard to temperature, a general nighttime warming trend is observed over the entire HoA, except for a few areas over the northeastern parts of Ethiopia, where a decreasing trend of about $-0.15^{\circ}\text{C}/\text{decade}$ was observed. The observed temperature trends are statistically significant over a larger spatial extent of the HoA. The observed warming trend in maximum temperature is relatively large in the central region of HoA and in the northwestern flank for minimum temperature. The observed precipitation deficits in some parts of the region and the large-scale increases in temperature are factors that contribute to increased evaporation, which in turn can lead to increased water demand and thus intensified drought impacts¹⁵.

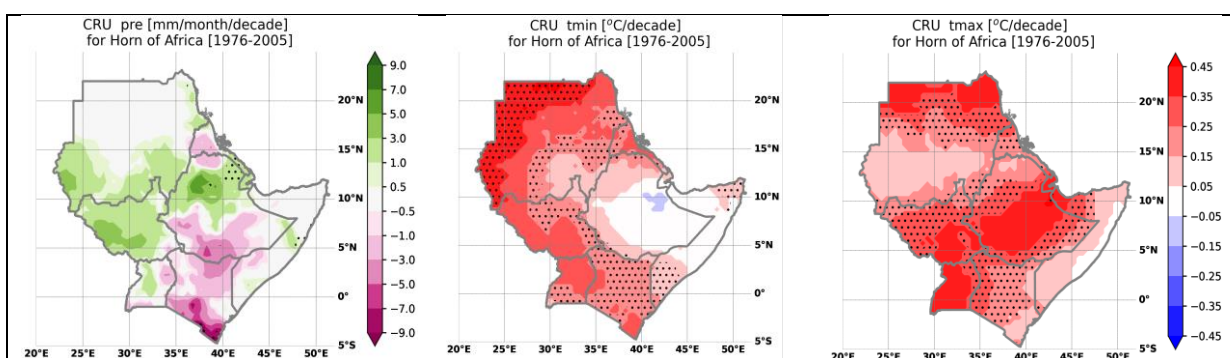
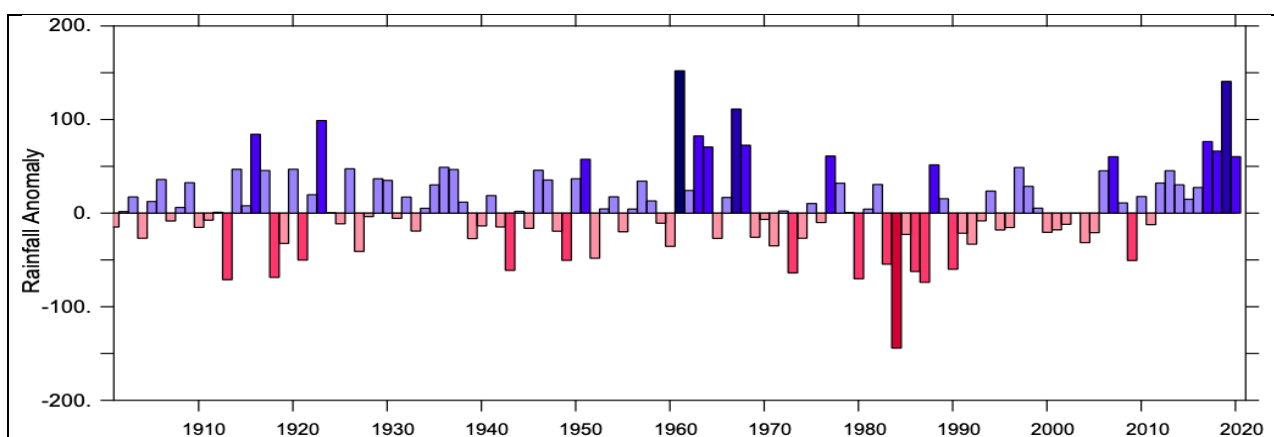


Figure B.1.3a: Observed trends in precipitation, minimum and maximum temperature for the period 1976-2005. Stippling denotes areas where trends are statistically significant at 0.05 level

7. 7. Analysis of area-averaged rainfall and temperature over the region based on CRU data for the period 1900-2020 is also shown in Figure B.1.3b. The results show that the region has experienced high inter-annual rainfall variability with no clear trend of increasing or decreasing rainfall. The period from the 1970s to the late 2000s was characterised by a decreasing rainfall trend, while the last decade (2011-2020) was characterised by a series of wet years. For temperature, however, there has been a consistent regional warming trend over the last forty years (1981 to 2020), and this is one of the key indicators of climate change in the region. Inter-annual variability in rainfall, intra-seasonal changes in rainfall and increases in temperature are threatening crop productivity, with farmers experiencing declining yields and some areas becoming less suitable for the production of major crops.



¹⁵ Kew, S. F., Philip, S. Y., Hauser, M., Hobbins, M., Wanders, N., van Oldenborgh, G. J., van der Wiel, K., Veldkamp, T. I. E., Kimutai, J., Funk, C., and Otto, F. E. L.: Impact of precipitation and increasing temperatures on drought trends in eastern Africa, *Earth Syst. Dynam.*, 12, 17–35, <https://doi.org/10.5194/esd-12-17-2021>, 2021.

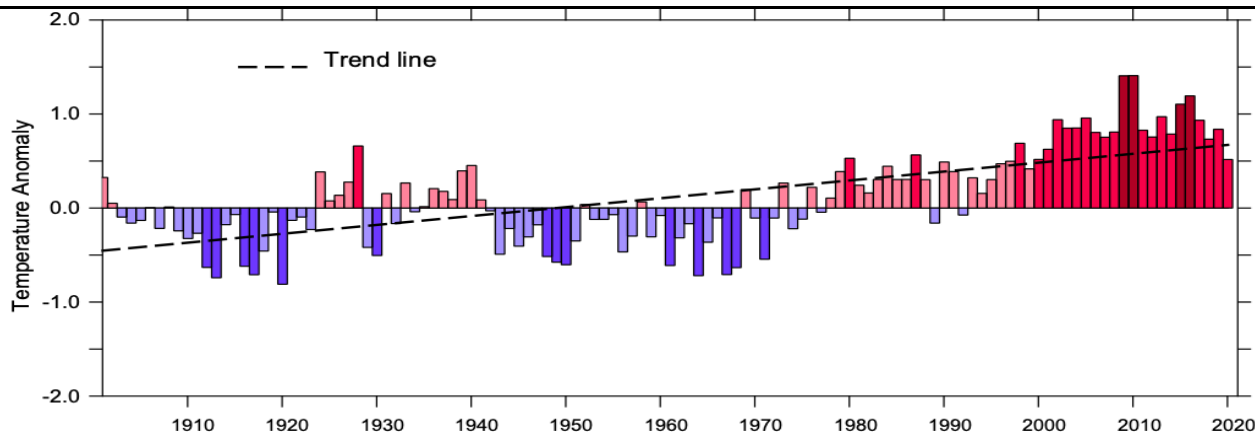


Figure B.1.3b: Anomalies of annual total rainfall (top) and temperature (bottom) (°C) for the IGAD region from CRU data for the period from 1900 to 2020 relative to 1961 to 1990 average. Blue bars indicate cooler-than-average years; red bars show warmer-than-average years.

8. Climate projections for the period 2021-2050¹⁶ show a general increase in annual mean temperatures under the RCP4.5 (medium-low) and RCP8.5 (high) emission scenarios. Under the medium-low emissions scenario, the annual mean near-surface temperature is projected to increase by more than **1°C** and **1.6°C**, respectively, in most countries of the HoA region (Figure B.1.4a). The difference between the projected period (2021-2050) and the baseline period (1976-2005) is statistically significant at the 0.05 level over some areas in the central and most of the southern parts of the region. Warming under RCP8.5 is projected to be high, reaching up to 2.4°C, as shown in Figure B.1.4. More intense warming is projected to occur during the March-April-May (MAM) season. The MAM season is important for most of the region, especially in sectors sensitive to rainfall variability. Therefore, increased temperature and precipitation variability during this season can negatively affect food security and key production sectors due to water scarcity and heat stress.

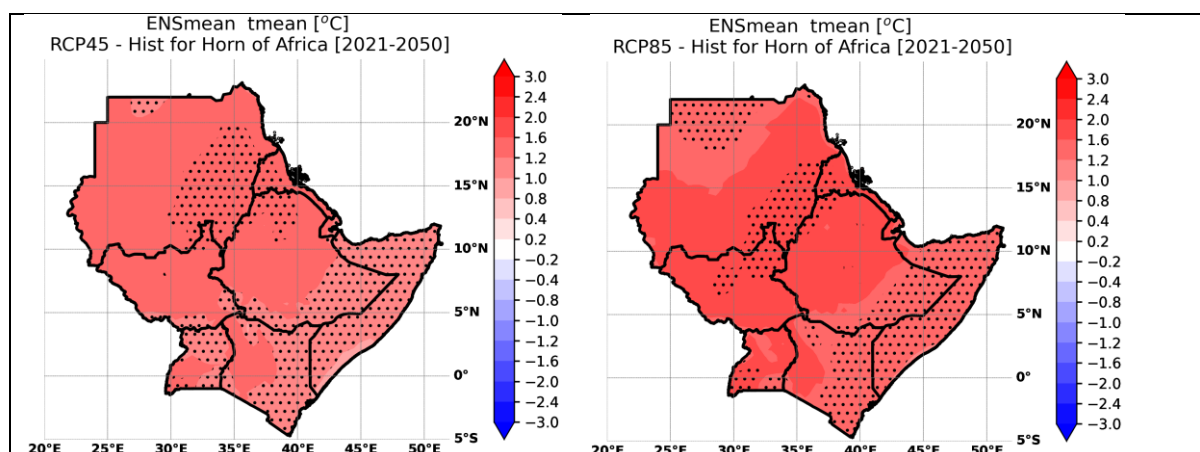


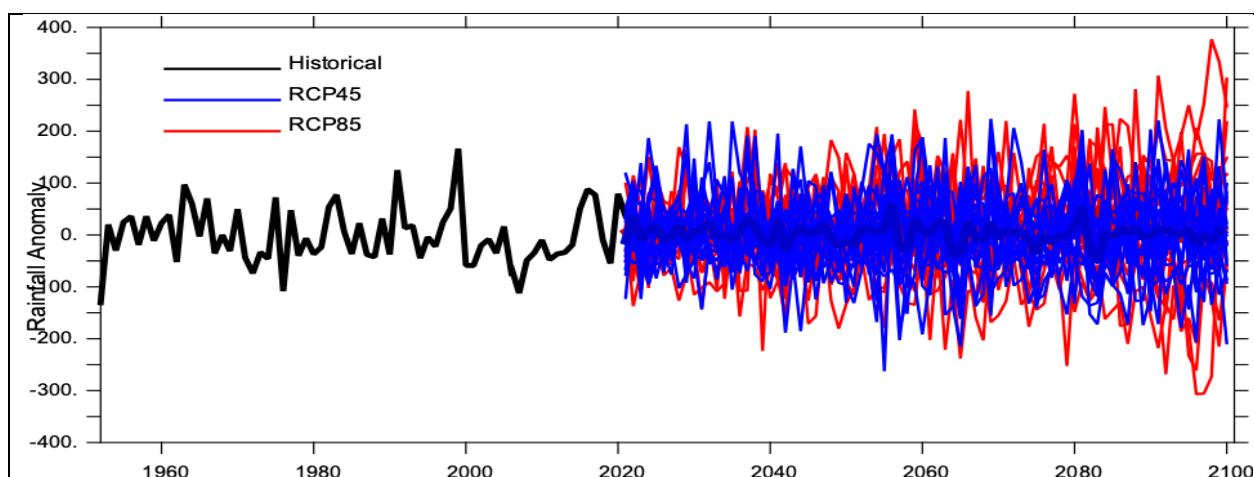
Figure B.1.4a: Projected change in mean temperature in the Horn of Africa for the period 2021-2050 under RCP 4.5 and RCP 8.5 emission scenarios. The change is computed relative to 1976-2005 base period. Stippling denotes areas where differences are statistically significant at 0.05 level.

9. Projected changes in the spatial distribution of precipitation under the RCP4.5 scenario indicate that widespread wet conditions are likely over the southern part of the region. On the other hand, precipitation deficits are projected under RCP4.5 over the whole of Sudan, Eritrea, central and northern

¹⁶ The assessment of future climate change in the GHA countries are based on projections produced by the Coupled Model Inter-comparison Programs (CMIPs), overseen by the World Climate Research Program. Data presented is [CMIP6](#), derived from the Sixth phase of the CMIPs. The CMIPs form the data foundation of the IPCC Assessment Reports. CMIP6 supports the [IPCC's Sixth Assessment Report](#). Projected data is presented at a 1.0° x 1.0° (100km x 100km) resolution.

parts of southern Sudan, northern and western Ethiopia and southern Uganda. Other parts, including all of Somalia, Djibouti, south-eastern Ethiopia, northern Uganda and Kenya, are projected to receive more precipitation under the same emissions scenario. The increase in precipitation is only significant in southern Kenya and a small part of Uganda. Under the RCP8.5 scenario, precipitation deficits are projected to increase in parts of the region. The spatial extent of the deficit extends further into the south of the region and is statistically significant mainly around the central area of the HoA. It is expected that regions with more precipitation are likely to experience an increase in the frequency of heavy rainfall and flood events, while those with projected deficits are likely to experience intense droughts in the HoA region. Flood events pose major risks to the populations living in these areas and often result in negative impacts such as human displacement, deaths, destruction and/or loss of property, damage to crops and arable land, as well as loss of productive assets such as livestock due to deaths¹⁷. The Greater Horn of Africa Climate Risk and Food Security Atlas shows that since the 1990s, the GHA region has experienced a number of flood events with varying impacts. The number of incidents varies across the region, with Ethiopia having the highest number of incidents (20), followed by Somalia, Sudan and Tanzania (18). Fewer people were affected in Djibouti than in the other countries. The highest number of people affected by a single flood event was in Somalia, followed by Sudan, Kenya and Uganda. The average number is highest in Sudan and South Sudan. These conditions are expected to worsen under wetter climate conditions, as expected under the RCP4.5 scenario. This variability calls for improved monitoring systems to enhance preparedness and planning.

10. In addition to projected changes in spatial distribution, time series analyses of annual precipitation and temperature anomalies averaged over the IGAD region based on multi-model ensemble simulations for the two scenarios (RCP4.5 and RCP8.5) have been presented to quantify uncertainties in future projections associated with climate models and scenarios (Figure B.1.4b). Under RCP4.5, models project a likelihood of small and consistent temperature increases in future periods. Under this scenario, the region is expected to experience a temperature increase of around 1.5-3.5°C. Under RCP8.5, higher temperature changes of up to 4-6°C are projected over the region. The temperature increase under this scenario is faster than under RCP4.5, especially between 2050 and 2100. For precipitation, the models show strong inter-annual variability in the future compared to the historical period. This variability is projected to be higher under RCP8.5 than under RCP4.5, with recurrent wet and dry periods.



¹⁷ Ibid.

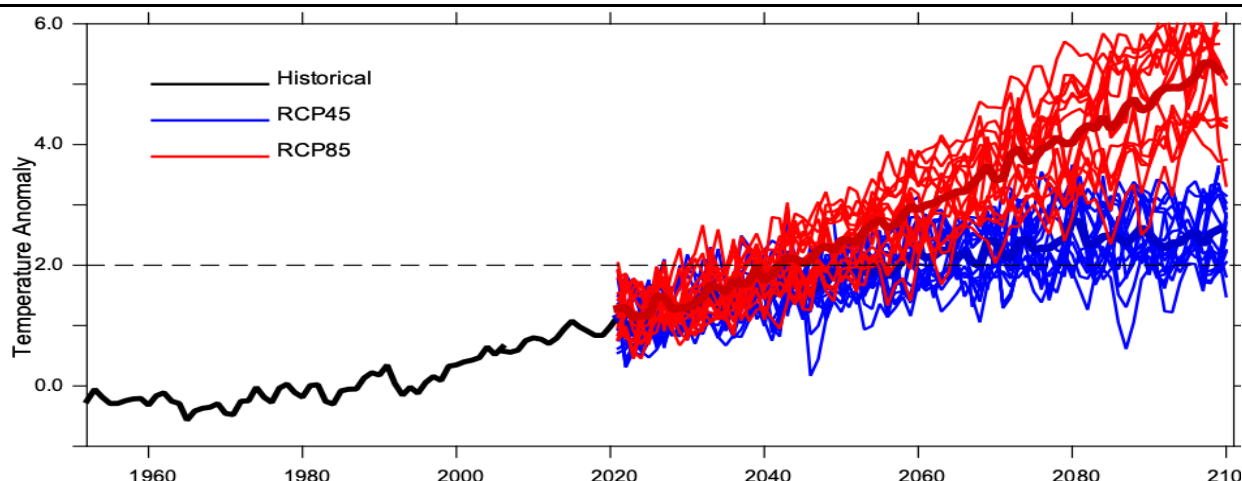


Figure B.1.4b: Time series of annual rainfall (top) and temperature (bottom) anomalies averaged over the IGAD region for the two scenarios (RCP4.5 and RCP8.5) for the period 1951-2100 with respect to 1971-2000 base period based on ensemble of simulations.

11. Other useful indicators for assessing climate change, impacts and associated costs include the World Meteorological Organization (WMO) Expert Team Core Sector Indicators such as Standardised Precipitation Index (SPI), Standardised Precipitation Evapotranspiration Index (SPEI), Warm Season Duration Index (WSDI), Heat Wave Frequency (HWF), Maximum Consecutive Dry Days (CDD), Maximum Consecutive Wet Days (CWD), Normalised Difference Vegetation Index (NDVI) and Meteorological and Agricultural Drought (MAD). The projected change in these indices was calculated for the period 2021-2050 (using scenario modelling) in the future and the reference period (1976-2005) under two representative concentration pathways (i.e. RCP4.5 and 8.5 emission scenarios). These extreme climate indicators are useful for designing sector-specific programmes, especially in regions that are extremely vulnerable to climate change and extreme climate events. **Appendix 1.1 (a&b)** shows the calculated SPI for the region. Under the RCP4.5 emissions scenario, moderate drought is projected to occur over most of Sudan, Eritrea, northern South Sudan and Ethiopia and southern Uganda, while moderate wetness is projected to occur over all of Somalia, northern Uganda, south-western and eastern Kenya and south-eastern Ethiopia. Under RCP8.5, moderate drought is expected to intensify across the HoA, with parts of Sudan, South Sudan, Ethiopia and Uganda likely to experience severe drought (i.e. SPI < -1.5). Similar but stronger signals are observed for SPEI **Appendix 1.1(c&d)**, where extreme drought (i.e. SPEI < -2) is expected to dominate the northern region of Sudan, although not statistically significant. In general, a statistically significant increase in warm spells is likely to occur across the HoA domain with WSDI, particularly under RCP4.5, where the hotspots (warm spells > 100 days) are in Ethiopia, Kenya and Uganda **Appendix 1.1(e&f)**. The change is expected to intensify under RCP8.5, but at a lower level of significance. The frequency of heatwaves is also projected to increase by 50 days over the entire HoA region under the RCP4.5 and 8.5 emission scenarios **Appendix (g&h)**. Consistent with the literature, days with extreme heat (DHI > 35°C) are projected to increase over most of the GHA region (Osima et al. 2018;¹⁸ Netherlands Ministry of Foreign Affairs, 2018;¹⁹ and Gudoshava et al. 2020²⁰). It is important to note that while global climate and weather monitoring systems are useful for providing an overview of the general situation, localised information systems are critical for providing information at the local level and are therefore useful for more detailed planning. As such, BREFOL will be more useful if it includes components aimed at generating and disseminating local information.
12. Similarly, **Appendix 1.2** shows that under RCP8.5, CDD (defined as a day without agriculturally significant rainfall with a threshold of 1 mm/ day)²¹ is projected to increase over most of the GHA region,

¹⁸ Sarah Osima et al. (2018). Environ. Res. Lett. 13 065004. <https://doi.org/10.1088/1748-9326/aaba1b>.

¹⁹ Netherlands Ministry of Foreign Affairs (2018). Climate Change Profile: Greater Horn of Africa. <http://www.government.nl/foreign-policy-evaluations>.

²⁰ Masilin Gudoshava et al 2020 Environ. Res. Lett. 15 034037. <https://doi.org/10.1088/1748-9326/ab6b33>.

²¹ <https://climateknowledgeportal.worldbank.org/>.

with more pronounced effects in north-eastern Ethiopia; the White Nile region of Sudan; the northern region of South Sudan; the coastal, north-eastern and Rift Valley regions of Kenya; central, western and northern parts of Somalia; and all of Djibouti and Eritrea. However, CDD is projected to decrease in the southern parts of the region under RCP4.5, although the change is not significant. The projected increase in CDD could affect crop growth and development, which in turn could affect food security in the affected regions. This is a clear indication of the need to formulate actionable policies aimed at adapting to drier days in the future. In **Appendix 1.3**, CWD (defined as a day with a rainfall threshold equal to or greater than 1 mm/day) is projected to decrease over most of the region, particularly in southern Sudan, western South Sudan, western Ethiopia, southern Eritrea, Uganda, and northwestern and southeastern Kenya. Some areas, including northern Ethiopia and central parts of southern Sudan, are expected to experience increases in CWD. Again, the projected increase in CWD in certain parts of the region calls for actionable strategies to adapt to or take advantage of localised impacts such as flooding in the region. Frequent flooding has had a strong impact on poverty reduction, food security, livelihoods and human capital of several communities in the GHA region, as shown in **Appendix 1.4**.

13. Appendices 1.5 and 1.6 present the Normalised Difference Vegetation Index (NDVI) and a proxy for MAD. While NDVI measures the presence or absence of live green vegetation and is very important for livestock management in terms of pasture availability in terms of abundance and scarcity, MAD is a proxy measure of meteorological and agricultural drought conditions on a seasonal time scale²². This new approach, developed in the Atlas, combines the traditional SPI and the Vegetation Health Index (VHI),²³ using a standardised method of layer combination described in detail in the Atlas, to produce a Combined Drought Indicator (CDI) for the region (**Appendix 1.5**). As can be seen from the NDVI maps, most of the pastoral areas, usually located in the eastern part of the region, tend to have lower vegetation performance compared to the non-pastoral areas of the region. This information is crucial for livestock management, especially in terms of pasture availability in terms of abundance and scarcity. Similarly, the CDIs (**Appendix 1.6**) show that the impact of drought varies across the region, affecting different locations at different times and to different degrees. In particular, the end of the 2008/2009 La Nina events, which led to the 2009 MAM drought conditions that affected almost the entire GHA region, is well represented in the combined map. The extent of the 2010/2011 drought in the GHA is also clearly visible on the CDI map.

14. The aggregated annual drought maps for the region show that drought is an inescapable feature of the GHA region, particularly in the semi-arid areas where an estimated 75 per cent of the labour force is engaged in smallholder rain-fed agriculture, and therefore its impact on crops and human populations can be acute, as illustrated in **Appendices 1.7 and 1.8**. For example, the severe drought of 2010/11 in Kenya, Somalia and south-eastern Ethiopia affected around ten million people, with more than 250,000 deaths in Somalia. Similarly, statistics from the EM DAT global database show that since 1990, Ethiopia, Kenya and Somalia have experienced more droughts than other countries in the region. The highest number (maximum) of people affected in any one episode was in Kenya, with over 20 million people affected, followed by Ethiopia and then Sudan. This led to animal deaths and a significant drop in their market prices, with milk production falling significantly for most affected households. Wildlife in these areas has also been severely affected. The food security situation in the region has also deteriorated. The increasing trend, especially in late 2015 and 2016, is linked to the El Niño drought that affected parts of the northern sector countries such as Ethiopia, Sudan and Djibouti. This led to poor production and livestock losses, affecting household food security in these countries. More than 9.5 million livestock - on which pastoralist families depend for food and livelihoods - have already died across the region, including 4 million in Ethiopia, 2.5 million in Kenya and over 3 million in Somalia, with many more at risk²⁴.

Sectoral impacts

15. Details of the sectoral impacts of climate change and extreme climate events on key sectors of the economy are presented in Table B.1.1 below, with key citations. These include impacts on: **1)**

²² ICPAC/WFP (2018). Greater Horn of Africa Climate Risk and Food Security Atlas.
<http://www.icpac.net/index.php/applications/research-development.html>.

²³ Both are proxies' measures used in assessing meteorological and agricultural drought conditions in the literature.

²⁴ <https://reliefweb.int/organization/ocha>.

Agriculture and livelihoods (i.e, changes in soil health/fertility, heat stress on crop and livestock production, scarcity of pasture and water for crop production and livestock rearing due to delayed onset, increased occurrence of pests/pathogens, desert locust infestation, occurrence of tick and skin diseases in camels, cattle, goats and sheep, increase in post-harvest losses and spoilage of animal products, shifts in planting/harvesting seasons with devastating impacts on productivity, etc.), **2) Water resources and energy** (large interannual and interdecadal rainfall variability resulting in increased frequency of floods and droughts affecting water availability for crop and livestock production, reduced surface water quality, increased run-off, erosion and subsequent sedimentation, Reduced quantity and quality of water in shallow wells and springs due to increased run-off and evaporation rates, increased damage to water infrastructure and transport networks, more variable hydrological regime leading to increased frequency of floods, which will complicate future dam management, especially during intense rainfall events, and unreliability of hydropower generation due to more variable river flows; **3) Forestry** (increased fire risk: drier conditions will increase fire risk, making forest fires more frequent and intense, leading to degradation, increase in invasive species such as *Prosopis juliflora* and *Acacia drepanolobium* in e.g. Afar and Somali regions where they cause significant socio-cultural damage, changes in forest-dependent livelihoods due to changes in forest extent, impacts on species with narrow ecological ranges (e. g. In Uganda, the disappearance of medicinal plant species has been consistently reported, while eucalyptus forests have been attacked by insect pests (tree lice), especially during the prolonged dry season, and increased incidence of pathogenic fungi has been observed to have moved into previously disease-free areas, causing widespread damage to forest plantations; and **4) Health** (incidence of vector-borne diseases, increased incidence of malaria in highland areas not currently suitable for transmission, and increased Rift Valley fever, more days of the year exceeding critical heat health thresholds (temperatures above 31°C are associated with increased mortality and risk of non-communicable diseases, which disproportionately affect children, the elderly, migrant workers, and the poor), (temperatures above 31°C are associated with increased mortality and risk of non-communicable diseases disproportionately affecting children, the elderly, migrant workers and those working outdoors), changes in the nutritional value of crops associated with lower child nutritional status, increased disease burden in the under-18 population, bacterial growth facilitated by contaminated water sources as a result of flooding, increasing the likely incidence of diarrhoeal and other water-related diseases.

Table B.1.1: Summary of Climate Hazards, Impacts and likely Adaptation Options for the HoA

Hazard/ Citations Consulted	Key	Likely Impacts	Potential Adaptation Options
Agriculture and Livelihoods			
• Rising temperatures especially seasonal mean temperature. • Prolonged dry spells • Rainfall variability • Increased extreme precipitation events		• Increased water and heat stress for crops and livestock, lowering the productivity of pastoral livelihoods and negatively impacting the production of important crops such as maize, wheat, cotton, and coffee. • Scarcity of pasture and water for crop production and livestock rearing because of delayed onset • Rangelands have become continuously degraded due to	• Promotion of agroforestry (mixing trees and agricultural crops): provides food security, erosion control (the trees keep the soil together, less will be lost), soil fertility (trees have a fertilizing effect on their environment) and moisture balance as well as an addition source of fuel (wood), all of which are important in climate change adaptation creating a buffer for the local environment when dealing with the changes • Investing in fodder banks, crop/livestock integration, improvements to grazing corridors, efficient watering technologies, vaccination, pest monitoring and control systems, intensification of cultivated pastures and resilient livestock breeds. • Promotion of robust policy and technological interventions that enhance the adoption of
Key Citations 1. Richardson, K., Calow, R., Pichon, F., New,			

S. and Osborne, R. (2022) <i>Climate risk report for the East Africa region</i>. Met Office, ODI, FCDO. https://reliefweb.int/attachments/5821f8e6-c172-4c3e-a6fa-cb91c95707bc/Climate%20risk%20report%20for%20the%20East%20Africa%20region.pdf. 2. World Bank Climate Change Knowledge Portal. http://sdwebx.worldbank.org/climateportal/countryprofile/home.cfm?page=country_profile&CCode=KEN&ThisTab=ClimateFuture... 3. Government of Netherlands (2019). <i>Climate Change Profile: Greater Horn of Africa</i>. https://reliefweb.int/attachments/1eea6572-e1bd-3eef-b9d6-09242b0c9316/Greater%20Horn%20of%20Africa.pdf. 4. IGAD Climate Prediction & Application Centre (ICPAC) / World Food Programme (WFP) Regional Bureau for East and Central Africa, Nairobi,	persistent droughts and overgrazing. • Flooding events from heavy precipitation resulting to livestock deaths and loss of agriculture. • Changes in soil health/fertility affecting crops yields and pasture growth for livestock productivity. • Land degradation and soil erosion will be exacerbated by more intense rainfall events, posing risks to the natural resource base, agricultural productivity and subsequently food security, particularly in already degraded areas. • Cash crops such as tea, coffee and cocoa are impacted as the ability to shift farming to higher (cooler) altitudes is limited • Increases in the occurrence of pests/pathogens – cassava mosaic, leaf rust, coffee rust, berry borer, potato taste defect, brown rot disease etc. • Post-harvest losses; spoilage of animal products because of the inability to effectively process agricultural products and safely store seeds, grains and other perishable products. • Shifts in timing of planting/harvesting as a result prolonged dry spell • Damage to road network caused by extreme weather events such as flooding, landslide, erosion etc., leading to decreasing access to finance, markets, and extension services.	climate-smart agriculture (CSA) practices such as precision farming and regenerative or conservation agriculture, adoption of climate adapted seed varieties, integrated soil fertility management (ISFM) programs, soil and water conservation techniques etc. • Diversification (crops and income), to deal with uncertain weather conditions: timing of the rains, temperatures, hours of sunshine / shade. • Use of improved seeds, crop varieties, and livestock breeds and other. • Sustainable agricultural practices and technologies such as integrated pest management, integrated soil management, and agricultural system diversification etc. • Improvements in supply chains logistics and distribution, appropriate packaging solutions, modern storage facilities, and cold chains including training in post-harvest management practices. • Promoting precision farming • Enhancing access to climate information early warning system (CIEWS) and agro-advisory services such as, planting time, good farm management practices, irrigation schedules, use of farm manure and choice of inorganic fertilizers, suitable crop types and varieties to be planted, weeding regimes, the available seed suppliers, crop pests and diseases prevention and control measures. • Introducing actions that include improved and climate-resilient crop varieties and livestock breeds; innovative adaptation practices and technologies; improving natural resource management (land, water, ecosystems, biodiversity) and use of other production inputs.). • Promoting access to innovative financing mechanisms such as crop/ livestock insurance programs, digital advisory services, and climate finance. • Developing entrepreneurship skills and increasing access to finance and markets for the most marginalized groups in the society. • Leveraging climate financing products from the private sector to invest in climate resilient livelihood option programs
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(2017). Greater Horn of Africa Climate Risk and Food Security Atlas. https://reliefweb.int/attachments/d82fb041-59a6-3b16-9421-f921049dd67e/WFP-0000098939.pdf.		
Water Resource & Energy		
• Rising temperatures • Increased frequency of intense precipitation events • Decreased dry season precipitation Key Citations 5. Oiro, S. et al. (2020). https://aura.abdn.ac.uk/bitstream/handle/2164/15773/Oiro_et_al_depletion_ground_water_hydrogeology_journal_vor.pdf?sequence=1&isAllowed=y. 6. Arcanjo, M. (2020). http://climate.org/wp-content/uploads/2020/05/Water-Security-in-the-Horn-of-Africa.pdf. 7. SIPRI (2020). https://www.sipri.org/sites/default/files/2020-03/sipripp54_0.pdf. 8. Hirwa, H. (2021). https://doi.org/10	• Reduced surface water quality • Increased sedimentation and erosion • Reduced water quantity and quality in shallow wells and springs due to increased rates of evaporation • Increased damage to water infrastructure and transportation networks • More variable hydrological regime leading to higher flood frequency that will complicate future dam management especially during intense precipitation. • Unreliability of hydroelectricity production due to more variable river flows	• Investments in safe water and sanitation (WASH) are, by their nature, risk-reducing, since they lessen people's dependence on more climate-vulnerable, poorer quality sources (rivers, streams, ponds etc.), and free up time – particularly for woman and girls – for more productive activities, as well as improving long-term health and nutrition • Promoting good watershed management practices that protect water sources from increased drought and water quality risk, while also protecting or rehabilitating landscapes that act as carbon sinks in the natural environment. • Investing in coastal protection, hydro-agro-meteorological early warning systems, improved water management, and urban planning to reduce the vulnerability of risks posed by flooding caused by intense precipitation. • Investing in early warning for early action food security Hub to support early detection and diagnosis of emerging food insecurity crises related to water scarcity. • Development of flood prediction system, avoiding development in flood prone areas and building flood preventing infrastructure (dykes and levees) were needed. • Investing in projects that increase energy access, including through renewable energy and improved energy efficiency through energy saving education plans. • Promoting technological improvements, including energy efficiency, battery storage, waste recycling to energy, green hydrogen, nuclear developments, carbon storage, and decarbonization techniques. Planning and implementing a wood-energy programme that includes legislation for forest management to minimize the cutting of natural forest but could also provide safe

<u>.3390/su13116483.</u>		wood stoves for those who use open fire or other forms of unsafe (self-made) equipment and education to use, or promotion of, other forms of energy.
Human Health		
• Rising temperatures especially daily maximum and minimum • Increased frequency of intense precipitation • Changing rainfall patterns and rising temperatures • Increasing temperature extremes • More flood events Key Citations 9. Muga et al., (2021). https://doi.org/10.1371/journal.pntd.0009166. 10. Nnko et al., (2021). https://doi.org/10.1371/journal.pntd.0009081. 11. Rother, (2020). Rother H.A. (2020) Controlling and preventing climate-sensitive noncommunicable diseases in urban sub-Saharan Africa. Sci Total Environ, 722:137772.	• More days of the year exceeding critical heat-health thresholds; temperatures above 31°C and associated with increased mortality and risks of non-communicable diseases which disproportionately affect children, the elderly, migrant workers, and those working outdoors. • Impacts on nutritional value of crops which is associated with lower nutritional status for children, increasing the disease burden as the under-18 population is projected to increase considerably. • More flood events that contaminate water sources and longer stretches of higher temperatures that facilitate bacterial growth increase the likely incidence of diarrheal and other water-related diseases. • Expanded ranges of disease vectors (e.g., mosquitos) and increased transmission of infectious diseases such as malaria and Monkeypox virus • Increased incidence of diarrheal diseases (particularly cholera)	• Priority adaptation measures concerning public health aim for the development of early warning systems, improved water basin management for potable water, urban planning, and environmental management. Additionally, improved training of health workers, better sanitation policies, expanded public education, and greater disease and epidemic monitoring will further reduce the risks to public health.
Climate Risk and Vulnerability Assessment (CRVA) 16. This subsection is largely based on the CRVA conducted in the Greater Horn of Africa Climate Risk and Food Security Atlas²⁵. <i>The Hunger and Climate Vulnerability Index (HCVI)</i> was developed in the		

²⁵ Greater Horn of Africa Climate and Food Security Atlas (2018), https://www.icpac.net/documents/3/17-00072_IGAD_Climate_Risk_and_Food_Security_Atlas.pdf

Atlas as a tool for understanding climate risks to food security at lower administrative levels in the HoA. It shows the relative vulnerability of the region's food-insecure population to climate risks at the district level. The HCVI analysis goes beyond the impact of climate on crop yields and provides a multi-dimensional analysis of vulnerability, while highlighting the critical role of adaptive capacity in determining vulnerability. The approach provides an important starting point for monitoring vulnerability, evaluating programme effectiveness and/or examining plausible impacts of climate change. It is based on the Intergovernmental Panel on Climate Change's (IPCC 2013)²⁶ definition of vulnerability to climate change. That is, the relationship between the degree of climate stress on populations (**exposure**), the degree of responsiveness to stress (**sensitivity**), and the ability of populations to adapt to climate change (**adaptive capacity**)²⁷. The composite index includes socio-economic and environmental indicators that are highly correlated and relevant to food insecurity in general. A total of six indicators were used in the assessment (Table B.1.2). Two for exposure (**frequency of floods and droughts**), two for sensitivity (**forest cover and rainfed agriculture**) and two for adaptive capacity (**population below poverty line and rural population**), as summarised in the table below.

17. Table B.1.2: Indicators considered in the Hunger and Climate Vulnerability Index (HCVI) for the HOA

Table B.11.2: Indicators considered in the Hunger and Climate Vulnerability Index (HCVI) for the 1		
Component	Profile	Indicator
Exposure	Climate hazards	· Frequency of droughts*
		· Frequency of floods*
		· Mean annual temperature*
		· Frequency of Storms
		· Long-term trend in temperature
		· Death toll
		· Mortality per 100,000 inhabitants
		· Economic losses (US \$)
		· Losses per GDP (%)
Sensitivity	Environment	· Forest cover* (% of total area)
	Agriculture	· Rain-fed agriculture* (% of total agriculture)
		· Crop yields (yield per ha – kg/ha)
		· Poverty index
Adaptive capacity	Infrastructure	· Market access*
		· Water access rural population (%)
		· Water access urban population (%)
		· Paved roads (% of all roads)
		· Governance
	Socioeconomic indicators	· Total population below poverty line* (\$ 2 per day PPP) (%)
		· Rural population* (%)
		· Decadal Population growth (%)
		· Vulnerable employment (%)
Hunger and Climate Vulnerability Index (HCVI) = Exposure X Sensitivity / Adaptive Capacity		
*Indicators used are those with good correlation (correlation coefficient ≥ 0.3) with Food Security Indicators		

18. Figures B.1.5a, b, c and d present the analysis of the HCVI for the HoA considering the three components (adaptive capacity, sensitivity and exposure). Starting with **exposure to climate stress**, as measured by the **frequency of floods and droughts (Figure B.1.5a)**, the analysis shows that the

²⁶ IPCC (Intergovernmental Panel on Climate Change). 2012. Special report on managing the risks of extreme events and disasters to advance climate change adaptation. Cambridge, UK: Cambridge University Press.

²⁷ https://www.icpac.net/documents/3/17-00072_IGAD_Climate_Risk_and_Food_Security_Atlas.pdf.

five countries are in the range of high to very high exposure to climate stress. In southern Sudan, where high to very high exposure is more pronounced in the western parts of the country, eastern Ethiopia, northern Somalia and high exposure in the south, including Kenya, where the eastern parts of the country suffer from high exposure to frequent floods and droughts. Very high exposure to frequent floods and droughts in Kenya is most pronounced in the south-eastern parts of the country, as shown in Figure B.1.5a.

19. In terms of **sensitivity** - measured in terms of **forest cover and rainfed agriculture (Figure B.1.5b)** - countries with high to very high sensitivity are South Sudan (the entire coastline from north to south), Ethiopia (north-central), northern Somalia and certain parts of western Kenya. In terms of **adaptive capacity** - measured by the **total population below the poverty line (\$2 per day PPP) and the percentage of the population living in rural areas** - South Sudan and Ethiopia have high to very high adaptive capacity, as shown in Figure B.1.5c. In contrast, most parts of Kenya and Somalia have the lowest adaptive capacity.
20. The results of the combined climate and food insecurity analysis (HCVI), which includes the components of exposure (a), sensitivity (b) and adaptive capacity (c), are presented in **Figure B1.5d** (HCVI). The results for Ethiopia show that part of the western highlands is highly vulnerable compared to the rest of the country. This is due to its heavy reliance on rain-fed production, and therefore rainfall variability has a major impact on production and food security. Other areas of high vulnerability include the northeastern region (parts of Afar and Tigray) and the southeastern pastoral areas. For Djibouti, taking into account the three components (adaptive capacity, sensitivity and exposure), parts of Obock, Tadjourah, Dikhill and Djibouti city are among the areas with the highest vulnerability to climate shocks and food insecurity. In Kenya, Figure B.1.5d shows that pastoral areas are more vulnerable compared to the rest of the country, while in Somalia, most parts of the country are highly vulnerable due to the fact that climatic shocks are prominent in most areas. The north is more vulnerable to climatic shocks than the rest of the country and the population has low adaptive capacity. The results are similar for Southern Sudan, where the HCVI identified many areas of the country as vulnerable due to high levels of food insecurity, underdeveloped agriculture and infrastructure, among other factors. Sensitivity to climate-related risks is high in agricultural production areas in the west and south.

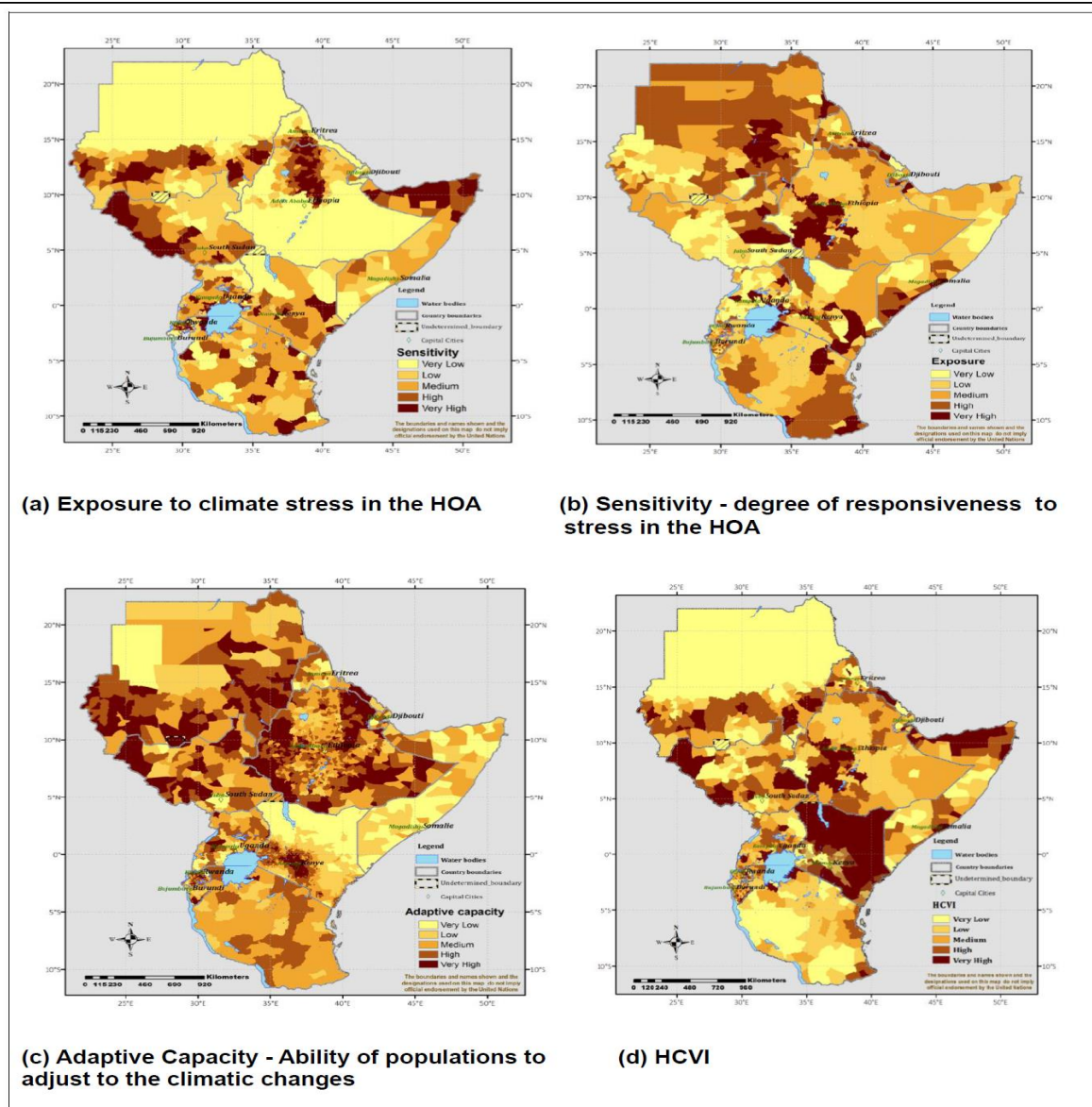


Figure B.1.5. Assessment of climate risk and vulnerability in HoA. Source: ICPAC (2018).

21. Green House Gases (GHGs) Emissions Profiles

Figure B.1 .6 below illustrates the GHG emissions profile for the five countries in the HoA, showing Ethiopia's emissions at 204.7 MtCO₂e, with agriculture as the main contributor (112. 5MtCO₂e), followed by land-use change and forestry (32.4MtCO₂e), energy (51.01MtCO₂e), waste (4.9MtCO₂e), industrial processes (3.9MtCO₂e) and bunker fuels (2.32MtCO₂e). Sudan recorded emissions of 130.64MtCO₂e, led by agriculture (75.6MtCO₂e), followed by energy (27.8MtCO₂e), Land-Use Change and Forestry (LUCF) (21.0MtCO₂e), waste (3.75MtCO₂e), industrial processes (2.52MtCO₂e), and bunker fuels (0.74MtCO₂e). Similarly, South Sudan's emissions in 2018 totalled over 68.5MtCO₂e, mainly from Agriculture (49.36MtCO₂e), LUCF (9.8MtCO₂e), Energy (6.7MtCO₂e), Waste (2.5MtCO₂e), Bunker Fuels (0.19MtCO₂e), and Industrial Processes (0.09MtCO₂e). In Somalia, emissions in 2018 were mainly from agriculture and LUCF (39.51MtCO₂e), followed by waste and energy (5.53MtCO₂e), with smaller contributions from industrial processes (0.28MtCO₂e) and bunker fuels (0.24MtCO₂e). In Kenya, emissions from different sectors were 47.7 MtCO₂e, -7.6 MtCO₂e, 28.5 MtCO₂e, 4.9 MtCO₂e, 3.9 MtCO₂e and 2.3 MtCO₂e, while in Djibouti the contributions were 0. 71MtCO₂e, 0.0MtCO₂e, 0.38MtCO₂e, 0.20MtCO₂e, 0.19MtCO₂e and 0.34MtCO₂e respectively, totalling 1.4MtCO₂e for 2019.

22. At current emission levels, these countries plan to keep their respective emissions close to the world average of 7.58 MtCO₂e by 2035 or neutral by 2050. This will be achieved by reducing their respective carbon footprints and by pursuing green growth pathways in all key economic sectors, particularly agriculture, forestry and other land use (AFOLU), including energy. By supporting activities to increase tree cover, BREFOL will help to improve GHG sequestration and thereby reduce emissions. BREFOL will support agro-forestry initiatives that lead to improved soil health, increased agricultural production while increasing tree cover, and increased household income from wood and non-wood farm products, thereby increasing household resilience.

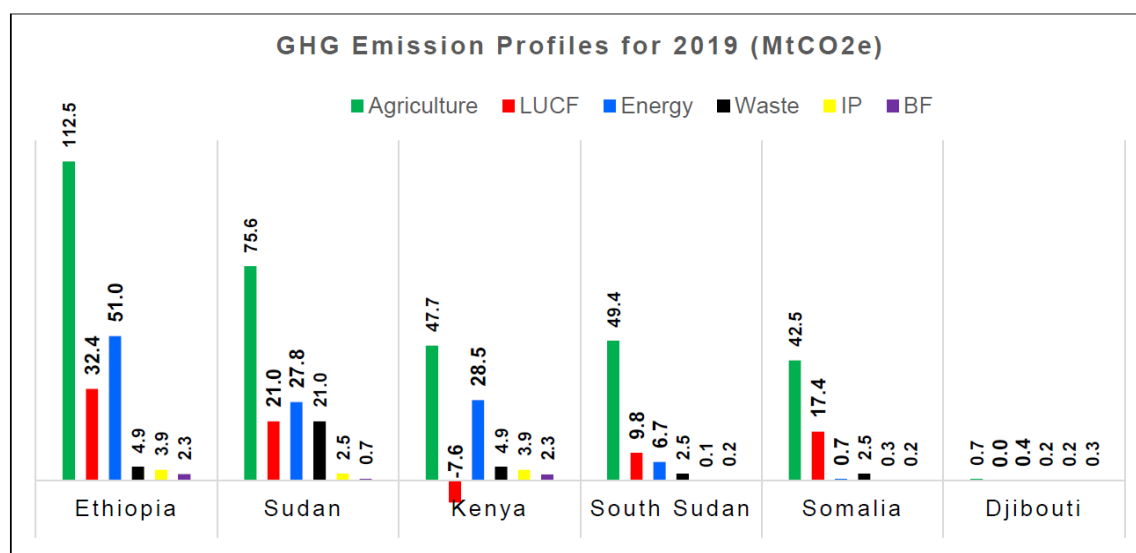


Figure B.1.6: GHGs emissions profiles for these five target countries and Sudan in the HoA as of 2019, for which the latest historical GHG emissions statistics are available at the NDC Partnership.

23. **Adaptation/mitigation needs:** The proposed programme, ***Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL)***, is highly consistent with the countries' legal frameworks or action plans on climate change (Table B.1.3). In particular, the focus on natural resource and environmental management; market access, trade and finance; improved production and livelihood diversification; disaster risk management; research, knowledge management and technology transfer; coordination, institutional strengthening and partnerships; and human capital, gender and social development is consistent with the ***IGAD Drought Disaster Resilience and Sustainability Initiative (IDDRSI) strategy*** and the ***Horn of Africa Regional Resilience Framework 3.0 (Table B.1.4)***. In addition, the programme is part of the AfDB and the Global Centre on Adaptation's Africa Adaptation Acceleration Programme, which aims to catalyse climate-resilient actions through a triple-win approach to address the impacts of COVID-19, climate change and the economy. In addition, the programme is aligned with the ***AfDB's Ten-Year Strategy (2013-2022)*** and supports its twin objectives (inclusive growth and transition to green growth) and its three special focus areas (agriculture and food security, fragile states, and gender). The programme will also contribute to achieving three of the AfDB's High 5 development priorities: Feeding Africa, Integrating Africa and Improving the Quality of Life for African People.

24. Table B.1.3: Country's Regulatory Framework/Action Plans on Climate Change

Country	Regulatory Framework or Action Plan	Adaptation/Mitigation Commitments	Targeted Sectors	Means of Implementation
Djibouti	- Green Economy Strategy - National Strategy on Climate Change - First NDC (2016)	- Reduce vulnerability to droughts - Protect against rising average sea levels; - Develop access to water; - Protect biodiversity; - Strengthen the resilience of rural populations. - Low energy lightbulbs - Administrative buildings thermal rehabilitation - Energy audit for buildings - Installation of 60 MW onshore wind turbines in Goubet by 2025 - Three (250MW) solar power plants by 2025 1000 ha reforestation - Efficient cook stoves - Efficient appliances for households operated using renewable energy resources and products - Energy from waste– Biodigester/Biogas Systems	Energy & Efficiency	Mitigation & Adaptation = \$6.413B
Ethiopia	- Climate Resilient Green Economy Strategy, - Programme of Adaptation to Climate Change, - National Strategy for CC adaptation and mitigation, - First NDC (2017) and Updated NDC 2021. - Ethiopia's Long-Term Low Emission and Climate Resilient Development Strategy (2020-2050)	- Sustainable agricultural land management practices - Reduction in preharvest losses - Carbon sequestration in grasslands - Lowlands Livelihood Resilience Project - Fuel switch and biomass efficiency (improved cookstoves) - Reforestation - Landscape restoration - Enhancement of livestock productivity - Agricultural mechanization - Increase in the share of poultry - Oilseed feeding to reduce emissions from enteric fermentation - Energy efficiency - Transport electrification - Public transport expansion - Industry fuel switch - Use of organic fertilizer and crop residues - Clinker substitution in cement - Reduction of waste per capita - Waste separation and composting - Wastewater management - Reduce preharvest losses in crop production - Increase climate resilient agricultural practices - Increase irrigation schemes	- AFOLU & Livestock. - Energy & Efficiency. - IPPU; - Infrastructure. - Managed Soils - Waste Management	Mitigation = \$281.3B; Adaptation = \$40.5B

Kenya	- Kenya Vision 2030, - Kenya National Adaptation Plan (NAP) 2015-2030 National Climate Change Response Strategy - National Policy on Climate Finance - First NDC and Updated NDC (2020): - Presidential directive to increase tree/forest cover to 15% by planting 15 billion trees in the next 5 years.	- Enhancing the adaptative capacity and climate resilience in all sectors and at the national and county levels, - Innovative livelihood strategies to enhance local communities' resilience, - Enhancing the adaptative capacity and climate resilience in all sectors and at the national and county levels - Build the capacity build of the private sector (formal and informal) so as to enhance the resilience of their investments e.g., through identification of new products and services that are more resilient to climate change impacts. - Develop fiscal incentive measures to encourage businesses to undertake investment in adaptation and resilience building measures. - Implement long term private sector investment in adaptation and resilience building measures. - Develop and implement a pilot project on climate resilient fish species and the related value chain. - Upscale sustainable aquaculture initiatives - Strengthen land use management systems including rangeland management, fodder banks and strategic reserves. - Develop new feeds. - Promote livelihood diversification and market access (camels, indigenous poultry, beekeeping, rabbits, emerging livestock - quails, guinea fowls, ostriches etc.). - Establish price stabilization schemes and strategic livestock-based food reserves. • Restore degraded grazing lands. - Innovative livelihood strategies to enhance local communities' resilience. - Renewable energy expansion - Enhancement of energy and resource efficiency across all sectors - Tree cover of at least 10% of the land area - Land degradation neutrality - Scaling nature-based solutions for mitigation - Clean cookstoves - Low-carbon and efficient transport - Climate Smart Agriculture - Sustainable waste management - Coastal zone management - Uptake of climate risk management tools, early warning system	- AFOLU, - Renewable Energy & Energy Efficiency, - Transport, - Waste management - Blue economy	Mitigation \$17.7B Adaptation \$43.9B	
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		• Update of adaptation technology • Strengthening tools for monitoring, evaluation and learning tools • Enhancement of blue and ocean economy		
Somalia	• National Adaptation Programme of Action (NAPA) • First NDC & Updated NDC (2021)	• Solar power lighting, • Efficient stoves & kilns for charcoal making, • Large scale water capture and storage facilities, • Biogas production from biodegradable feedstock, • Charcoal production from invasive Prosopis plant & replacement with crop production, • Action for E savings (turn off) and recycling	• Energy & Efficiency • AFOLU	N/A
South Sudan	National Adaptation Programme of Action (NAPA) and First NDC and Updated NDC (2021)	• Increase the use of clean and carbon-neutral energy • Declare 20% of natural forests as reserve forests to protect it from deforestation. • Prioritize the enhancement of climate resilience in the agricultural sector • Promote alternative sources of energy to reduce deforestation and the consequent loss of livelihood options. • Introduce fire management plans to prevent the spread of wildfires during periods of drought. • Establish emissions standards for vehicles	• Energy generation and energy end use • Transport • Land Use and Land Use Change	Mitigation & Adaptation = \$50B

Source: Adopted from Africa NDC Hub. Available at: <https://africandchub.org/>.

25. Table B.1.4: Alignment of BREFOL to IDDRSI Strategy and the HoRN Regional Resilience Framework 3.0

IDDRSI Priority Intervention Areas (PIAs)	HoA Resilience 3.0 Framework Intermediate Results (IRs)	BREFOL Intervention Areas-Sub-components (SC)
A. Natural Resources and Environment Management 1.1. Water resources development and management 1.2. Rangeland management and pasture development 1.3. Securing equitable access to natural resources 1.4. Environmental management including biodiversity 1.5. Development of renewable energy	IR 2.1. Strengthened collaboration, learning and adaptation of state, non-state actors, and communities IR 2.3. Strengthened capacity and systems of local, national, and regional institutions	SC 1-1: Developing climate-resilient infrastructure (Water resources, energy, markets, and ICTs etc.) SC 1-2: Supporting the sustainable management of agropastoral land.
B. Market Access, Trade and Financial Resources 2.1. Transport, market, and infrastructure development	IR 1.1. Diversified, sustainable and resilient economic opportunities IR 1.3. Maintaining livelihood opportunities and access to financial services	SC 2-1: Increasing access to digital advisory services and markets

2.2. Security livestock mobility 2.3. Access to secure and affordable financial services 2.4. Strengthening regional and cross-border trade 2.5. Development and harmonization of financial services in the IGAD region	IR 1.4. Increased and secure access to market infrastructure IR 1.5. Prioritizing private sector-led trade and investment approaches IR 4.2. Improved and sustained health, nutrition, and hygiene practices IR 4.3. Increased and sustained availability of, access to, and utilization of WASH services.	SC 2-2: Developing entrepreneurship skills and increasing access to finance SC 2-3: Increasing access to innovative financing for farmers organizations and MSMEs
C. Enhanced Production and Livelihood Diversification 3.1. Livestock production, health, and nutrition 3.2. Crop production and productivity 3.3. Fisheries development 3.4. Income diversification 3.5. Productive safety net (chronically food insecure) 3.6. Productive safety net 2 (graduating from safety net) 3.7. Transboundary Disease and SPS Measures and Standard	IR 1.2. Intensified, sustainable and resilient agricultural and livestock market systems IR 1.1. Diversified, sustainable and resilient economic opportunities IR 1.2. Intensified, sustainable and resilient agricultural and livestock market systems	SC 1-3: Improving access to climate-smart innovations, and technologies
D. Disaster Risk Management 4.1. Early warning systems and Response 4.2. Contingency planning 4.3. Disaster risk reduction and climate change adaptation 4.4. Reduction of vulnerability to climate related hazards and natural disasters	IR 2.1. Strengthened collaboration, learning and adaptation of state, non-state actors, and Communities	SC 3-1: Enhancing climate services infrastructure
E. Research, Knowledge Management, and Technology Transfer 5.1. Support to applied adaptive research 5.2. Advisory and extension services 5.3. Knowledge management and communication 5.4. Promote the network of national and regional dryland collaborative, adaptive, and applied research centres 5.5. Alignment of applied and adaptive research with development priorities of the resilience agenda	I.R 2.2. Strengthened knowledge management and communication of innovative research and practices. IR 2.1. Strengthened collaboration, learning and adaptation of state, non-state actors, and communities	SC 3-2: Strengthening operational capacity for resilience
F. Coordination, institutional strengthening, and partnerships 7.1. Coordination and platform management 7.2. Institutional strengthening and capacity building 7.3. Enhancing partnerships 7.4. Resource mobilization	IR 2.1. Strengthened collaboration, learning and adaptation of state, non-state actors, and communities IR 2.3. Strengthened capacity and systems of local, national, and regional institutions	SC 3-3: Strengthening operational capacity for resilience

7.5. Monitoring, Evaluation, and Learning		
G. Human Capital, Gender, and Social Development 7.1. Access to health and nutrition 7.2. Access to education and training 7.3. Promote gender equality, women's empowerment.	IR 3.1 Improved educational attainment IR 3.2 Improved vocational, entrepreneurship and employment skills IR 3.3 Financial capabilities IR 3.4 Aspirations, self-efficacy, and confidence to adapt IR 4.1. Increased and sustained availability of access to, and utilization of health services IR 4.2. Improved and sustained health, nutrition, and hygiene practices	SC 1-1: Developing climate-resilient infrastructure (Water resources, energy, markets, and ICTs etc.) SC 2-2: Developing entrepreneurship skills and increasing access to finance for women and youths SC 3-2: Strengthening operational capacity for resilience

26. Root causes to climate change adaptation and mitigation

Table B.1.5 below summarises the drivers, barriers and risks to climate change adaptation and mitigation in the HoA region.

27. Root cause No. 1: Extreme poverty and food insecurity: Djibouti, Ethiopia, Kenya, Somalia and South Sudan have a combined population of approximately 250 million, which is projected to increase by 5% by 2050. Within this population, an estimated 160 million people (or nearly 64%) live in rural areas that are highly vulnerable to environmental degradation. Alarming, about 40% of these rural areas are classified as having low environmental production potential, rendering them economically unproductive due to adverse climatic conditions and severe land and environmental degradation resulting from unsustainable land use practices or human-induced activities. As a result, the rapid growth of the rural population exacerbates the challenge of increasing agricultural productivity under uncertain climatic conditions. This challenge is particularly acute given that more than 40% of the region's population lives below the \$1 a day poverty line, adding to the urgency of addressing extreme poverty and food insecurity.

28. Root Cause No. 2: Climate change and environmental degradation – This factor has exacerbated the challenges facing the Horn of Africa (HoA) and its rural poor. Over the past two decades, severe droughts have become more frequent, including in 2002-2003 in Ethiopia, 2005-2006 in equatorial East Africa, and 2010-2011 across the Horn of Africa. In recent years, below-average rainfall has led to prolonged droughts in Somalia, southern and south-eastern Ethiopia, and northern and eastern Kenya, with devastating impacts on agriculture and livelihoods. At the same time, severe floods, such as those in South Sudan in 2020 and 2021, have caused extensive damage to infrastructure and loss of life. These climate-related disasters have had a profound economic impact, particularly on rural communities, leading to crop failures, famine and increased migration to urban areas. According to the UN Office for the Coordination of Humanitarian Affairs, **more than 36.1 million people across the Horn of Africa have been affected by drought since October 2020, with millions of livestock lost.**

29. Root cause No. 3: Fragility - According to the OECD (2022)²⁸, fragility is the combination of exposure to risks and insufficient coping capacity of the state, system and/or communities to manage, absorb or mitigate those risks. In the latest 2022 Fragile State Index (FSI)²⁹, Ethiopia, South Sudan, Kenya, Somalia and Djibouti rank 11th, 8th, 4th, 32nd, 2nd and 49th respectively, with four out of the five countries (Somalia, South Sudan and Ethiopia) classified as vulnerable to collapse³⁰. This is largely due to political violence; protests sparked by economic grievances, for example in Somalia; successive failed rains leading to drought conditions that have left over 36.1 million people severely food insecure and hundreds of thousands displaced from their homes; armed conflict; pastoralist and communal resource conflicts; increasing poverty; and humanitarian disasters. All of these have affected many vulnerable communities and poor rural households. Unfortunately, these poor rural communities and households do not have the resources to absorb or mitigate such risks. Fragility also creates an

²⁸ OECD (2022), *States of Fragility 2022*, OECD Publishing, Paris, <https://doi.org/10.1787/c7fedf5e-en>.

²⁹ The Fund for Peace (2022). <https://fragilestatesindex.org/wp-content/uploads/2022/07/22-FSI-Report-Final.pdf>.

³⁰ <https://fragilestatesindex.org/>.

unhealthy investment environment for private sector growth and investors. Given the fragile nature of the region, the **Leveraging Strategic Private Sector Investment for Fragile States Project**³¹ has become a key initiative through which the AfDB seeks to promote private sector development in the region.

30. Root cause No. 4: Conflicts - Conflict in the HoA plays a major role in food insecurity in the region. In South Sudan, for example, conflict-induced famines in 2017 and 2021 left more than 2.5 million people on the brink of starvation and death. These conflicts have affected the country's main 'breadbasket', where mixed cereal-root crop livelihoods include maize, sorghum, millet and livestock. As reported in (Eloff, 2021)³², in times of conflict, agriculture is disrupted, and the mobility of pastoralists is restricted. Rural populations are forced to flee, prioritising their physical security over food security. The same dynamic is playing out in other conflict-affected countries in the region. For example, violence is currently causing severe food insecurity in the Tigray region of Ethiopia, and in Uganda, conflict has been linked to food insecurity.

Table B.1.5: Summary matrix of the root causes, barriers and risks to climate change adaptation and mitigation in the HoA region

S/N	Root Causes	Barriers to be addressed	Risks/Problem
1	Extreme poverty and food insecurity: (1) Political instability and conflict (World Bank, 2021 ³³), (2) Weak governance and institutional capacity (UNDP, 2022 ³⁴), (3) Economic challenges and inequalities (IMF, 2021 ³⁵), (4) Environmental degradation and climate change (FAO, 2018 ³⁶), (5) Limited access to education and healthcare (WHO, 2023 ³⁷), (6) Food price volatility and market access (FAO, 2020 ³⁸), and (7) Displacement and	To address these root causes would comprehensive strategies that prioritize peacebuilding, good governance, economic development, environmental sustainability, social protection, and inclusive growth (United Nations, 2021 ⁴⁰).	The associated risks include: • Malnutrition and health risks: (WFP, 2020⁴¹). • Economic instability (UNDP, 2018⁴²) • Social unrest and conflict (ICG, 2019⁴³) • Forced migration and displacement (UNHCR, 2021⁴⁴) • Environmental degradation (UNEP, 2017⁴⁵)

³¹ AfDB (2022). <https://www.afdb.org/en/contacts>.

³² Eloff, P. (2021) Armed Conflict, Organised Violence, and Food Insecurity in South Sudan. World Food Programme. https://fscluster.org/sites/default/files/documents/03022021_conflict_and_fsl_fslc_final.pdf.

³³ World Bank (2021) Horn of Africa Regional Economic Memorandum. <https://www.hoainitiative.org/wp-content/uploads/2022/03/0-HoA-REM-Overview-combined.pdf>

³⁴ UNDP (2022) AFRICA'S PROMISE: The UNDP Renewed Strategic Offer in Africa. <https://www.undp.org/sites/g/files/zskqke326/files/migration/gw/207c01616c26cedb6b14d278c8523d5758d74fa4cb4688047e94db2f2746ddcf.pdf>

³⁵ IMF (2021) Regional Economic Outlook for Sub-Saharan Africa. <https://www.imf.org/en/Publications/REO/SSA/Issues/2021/10/21/regional-economic-outlook-for-sub-saharan-africa-october-2021>

³⁶ FAO (2018) Africa Regional Overview of Food Security and Nutrition 2018. <http://www.fao.org/3/ca2710en/ca2710en.pdf>

³⁷ WHO (2023) Public Health Situation Analysis. https://cdn.who.int/media/docs/default-source/documents/emergencies/greater-horn-of-africa-public-health-situation-analysis-january-2023.pdf?sfvrsn=ab0b51db_4&download=true

³⁸ FAO (2023) Food Policy Monitoring in the Near East and North Africa region. <https://www.fao.org/3/cc9189en/cc9189en.pdf>

⁴⁰ UN (2021) Promotion of durable peace and sustainable development in Africa. https://www.un.org/osaa/sites/www.un.org.osaa/files/docs/2109875_osaa_sg_report_web_new.pdf

⁴¹ World Food Programme. (2020). Horn of Africa: Desert Locust Crisis - Situation Report No. 10. Rome.

⁴² United Nations Development Programme. (2018). Regional Human Development Report for Africa: Accelerating Gender Equality and Women's Empowerment in Africa. New York.

⁴³ International Crisis Group. (2019). Climate Change, Ethiopia's Conflict and the Horn's Famine Threat. Africa Report No. 274. Brussels.

⁴⁴ United Nations High Commissioner for Refugees. (2021). Global Trends Forced Displacement in 2020. Geneva.

⁴⁵ United Nations Environment Programme. (2017). The Horn of Africa: Abundance amid Adversity. Nairobi.

	refugee crises (UNHCR, 2020 ³⁹).		
2.	Climate change and environmental degradation: (1) Prolonged drought events for example, the 2005-2006 over equatorial east Africa, 2008-2010 over Sudan, northern Kenya, and the Horn of Africa, the 2010-11 in the Horn of Africa, 2015-16 in Ethiopia, the October-December 2020, March-May 2021, October-December 2021 and March-May 2022 lean seasons that affected parts of Somalia, southern and south-eastern Ethiopia, and northern and eastern Kenya. (2) Heavy rainfall and severe floods (3) Prolonged dry spells etc.	• Inadequate finance and funding gaps (<i>80% funding gap as only 20% of the annual cost of adaptation in the region comes from public budgets</i>). • High indebtedness of the countries, suggest that addressing extreme climate events, would require additional adaptation finance from both development partners and the private sector • Access to climate finance for smallholder farmers and pastoralists • Disaster risk management (early warning systems and response, contingency planning, disaster risk reduction and climate change adaptation, and reduction of vulnerability to climate related hazards and natural disasters) • Coordination, institutional strengthening, and partnerships • Context-specific climate risks and vulnerability assessments. • Gender inequality in climate change adaptation.	• Increases heat stress: under prolonged heat stress depletion of carbohydrate reserves and plant starvation are observed • Increases the occurrence of pests/pathogens – cassava mosaic, leaf rust, coffee rust, berry borer, potato taste defect, brown rot disease etc. • Increases post-harvest losses; spoilage of animal products because of the inability to effectively process agricultural products and safely store seeds, grains and other perishable products. • Shifts in timing of planting/harvesting as a result prolonged drought period • Damage to road network caused by extreme weather events such as flooding, landslide, erosion etc., leading to decreasing access to finance, markets, and extension services. • Increased damage to water infrastructure and transportation networks • Unreliability of hydroelectricity production due to more variable river flows • Expanded ranges of disease vectors (e.g., mosquitos) and increased transmission of infectious diseases such as malaria and Monkeypox virus • Increasing loss of property and infrastructure associated with rising flood events • Increased incidence of diarrheal diseases (particularly cholera)
3	Fragility: All the five countries are highly exposure to risks and have insufficient coping capacities to manage, absorb or mitigate such risks. In the recent Fragile State Index (FSI)⁴⁶, Ethiopia, South Sudan, Kenya, Somalia and Djibouti rank 11th, 8th, 4th, 32nd, 2nd, and 49th respectively, with three out of the five countries (Somalia, South Sudan, and Ethiopia), classified as	• Linking peacebuilding, climate change adaptation and sustainable livelihoods. • Promoting context specific assessments tools used to inform national and regional efforts. For example, IDDRSI Strategy. • Coordination, institutional strengthening, and partnerships • Inadequate finance and funding gaps • Land fragmentation • High indebtedness of the countries, suggest that	• Political violence • Social unrest • Food and livelihood insecurities • Environmental-induced migration • Armed and resource conflicts • Rising poverty • Humanitarian disasters • Unhealthy investment environment inconducive for the growth of the private sector investors.

³⁹ UNHCR (2020) East and Horn of Africa and the Great Lakes.

https://reporting.unhcr.org/sites/default/files/ga2020/pdf/Chapter_EastHornGreatLakes.pdf

⁴⁶ The Fund for Peace (2022). <https://fragilestatesindex.org/wp-content/uploads/2022/07/22-FSI-Report-Final.pdf>.

	vulnerable to collapse.⁴⁷ This is due mainly to political violence, protests sparked by economic grievances, drought conditions, armed conflicts, resource conflicts among pastoralists and communities, rising poverty, and humanitarian disasters.	addressing fragility would require additional adaptation finance from development partners and the private sector • Enabling environment for private-sector led growth. • Gender inequality	
4	Conflicts: Human and resource conflicts play a major role in food and livelihood insecurity in the HoA region. For example, the 2017 and 2021 famines caused by conflicts in South Sudan hit over 2.5 million people living them on the brink of starvation and death. Similarly, violence is currently driving severe food insecurity in the Tigray region of Ethiopia	• Linking peacebuilding, climate change adaptation and sustainable livelihoods. • Promoting context specific assessments tools used to inform national and regional efforts. For example, IDDRSI Strategy • Coordination, institutional strengthening, and partnerships • Land fragmentation • Access to climate finance for smallholder farmers and pastoralists • High indebtedness of the countries, suggest that addressing conflicts, would require additional adaptation finance from development partners and the private sector • Programs linking peacebuilding, climate change adaptation and sustainable livelihoods. • Institutional strengthening and capacity gap deficiencies • Gender inequality • Support for inclusive technical dialogues on the evolving understanding of climate security	• Political violence and insecurity • Social unrest • Food and livelihood insecurity • Livelihood distortion • Environmental-induced migration • Armed and resource conflicts • Rising poverty • humanitarian disasters Unhealthy investment environment conducive for the growth of the private sector and investors.

31. Other complementary initiatives on which BREFOL builds and key lessons learned: There are several past (**Appendix 1.9a &b**) and ongoing (**Appendix 1.10**) climate change adaptation/mitigation programmes/projects in the HoA on which BREFOL builds. While the past initiatives were consulted to draw key lessons from their implementation, some of the ongoing initiatives were consulted either to avoid duplication of efforts and/or to identify existing gaps to be filled by BREFOL. As part of the Feed Africa strategy, AfDB implemented the regional Drought Resilience and Sustainable Livelihoods Programme (DRSLP) from 2012-2021, which aimed to reduce poverty and food insecurity and accelerate sustainable economic growth in the Greater Horn of Africa (HOA). BREFOL builds on the achievements and lessons learnt from DRSLP, in particular by capitalising on the infrastructure built

⁴⁷ <https://fragilestatesindex.org/>.

and managed. BREFOL also builds on IFAD's Inclusive Green Financing Initiative (IGREENFIN I): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) countries - Phase I. This initiative, which aims to support the greening of agricultural banks and the financial sector to foster climate resilient, low emission smallholder agriculture in Burkina Faso, Chad, Côte d'Ivoire, Djibouti, Eritrea, Ethiopia, Ghana, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan, provided very useful insights for the design of BREFOL. Similarly, *IGAD's Drought Disaster Resilience and Sustainability Initiative (IDDRSI) strategy and the Horn of Africa Regional Resilience Framework 3.0* were very instrumental in packaging the sub-investment activities of BREFOL. *The African Union's Draft Africa Climate Change Strategy - DACCS (2020 - 2030)* also played a key role in narrowing down BREFOL's sub-investments in line with the region's adaptation needs.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

- 32. Description of the problem:** The Horn of Africa is one of the most disaster-prone regions in the world, with crippling food insecurity resulting from multiple hazards in the region. Since the 1970s, several countries in the HoA have been threatened by famine at least once every decade. It is worth noting that the frequency of these hazards and disasters has recently increased to every 3-5 years. The development of programmes that can strengthen the resilience of communities in the region is of paramount importance. The Food Security and Nutrition Working Group (FSNWG), for example, predicts that between 18.6 and 21.1 million people in Ethiopia, Kenya and Somalia are likely to face crisis-IPC 3 or worse levels of acute food insecurity due to drought. The number of people in need of emergency food assistance to meet basic needs could rise to as many as 26 million people in the three countries by February 2023 if a fifth consecutive season of poor rainfall occurs between October and December, as early projections suggest is likely. As noted above, extreme poverty and food insecurity (**root cause #1**), climate change and environmental degradation (**root cause #2**), fragility (**root cause #3**) and conflict (**root cause #4**) are some of the key drivers of recurrent food crises, hunger and famine in the region. Unfortunately, the high indebtedness of these countries severely constrains public spending to address some of the key challenges facing the region. Similarly, their highly fragile nature (Somalia, South Sudan and Ethiopia are prone and vulnerable to armed conflict). Exposure to frequent conflict acts as a disincentive to the inflow of private financial resources: critical for the creation of self-sustaining green/circular economy activities, sustainable wealth creation and green job opportunities. For smallholder farmers and pastoralists, who account for more than 80% of all low-return subsistence farming activities, lack of access to financial resources often limits their ability to adapt. The end result is often violent conflict over land use and resources, environmentally induced migration and exposure to vicious poverty traps.
- 33. Preferred adaptation solution:** The preferred short-, medium- and long-term adaptation solutions to the above issues at the regional level, particularly for IGAD member states, are contained in the **IGAD Drought Disaster Resilience and Sustainability Initiative (IDDRSI) Strategy**⁴⁸. At the national level, preferred adaptation solutions are expressed in each country's legal framework or action plan on climate change (Table B.1.3). **BREFOL** is key to the implementation of the IDDRSI strategy at the regional and national levels in the short, medium and long term, including the realisation of the Fund's impact at the global level. Specifically, at the global level, BREFOL contributes by increasing the adaptive capacity and resilience of 1) vulnerable people and communities, 2) the health and well-being as well as food and water security of the target population, and 3) the environment and ecosystems and ecosystem services in the target areas. In terms of mitigation co-benefits, BREFOL contributes through reduced emissions from: 1) facilitating access to renewable energy (solar and bio-digesters), and 2) improved forest cover and land use change. At the regional and national levels, BREFOL contributes to 1) supporting, developing and rehabilitating climate resilient infrastructure (water resources, energy, market integration, cross-border health and research facilities, road networks, ICTs, etc.), and 2) supporting sustainable forest management.) and 2) supporting sustainable management of agro-pastoral production systems (**IDDRSI PIA 1**); 3) increasing access to digital information flows and advisory services, especially on markets and trade; 4) developing entrepreneurship skills and increasing access to finance for vulnerable groups, especially women and youth; and 5) increasing access to innovative financing for smallholder farmers, pastoralists and agribusiness MSMEs (**IDDRSI PIA 2**); 6) improve access to climate-smart innovations and technologies (**IDDRSI PIA 3**); 7) improve climate service infrastructure and extension services, and 8) promote climate risk finance and insurance to cushion vulnerable smallholder production systems (**IDDRSI PIA 4**); and finally, 9) support gender-sensitive climate regulatory framework reforms and institutional strengthening for climate-smart planning and development (**IDDRSI PIAs 5,6 & 7**).
- 34.** To support the implementation of the key activities of the '**BREFOL Preferred Adaptation Solution**', the AfDB will draw on the approaches developed under three of its ongoing special initiatives to support BREFOL in the HoA region. These initiatives include: 1) **The Technologies for African Agricultural Transformation (TAAT) programme**, which aims to help the continent realise its vast potential through the adoption of high-impact innovative technologies to increase agricultural production (see, for example, Appendix 2.1 for TAAT success stories in the continent); 2) **AFAWA (Affirmative**

⁴⁸ IGAD's Drought Disaster Resilience and Sustainability Initiative (IDDRSI) Strategy
<https://faolex.fao.org/docs/pdf/igad215119.pdf>

Finance Action for Women in Africa), which aims to empower women and bridge the \$42 billion financing gap facing women in Africa (see, for example, Appendix 2.2 for AFAWA success stories on the continent); and 3) The **AfDB-GCA (Global Centre on Adaptation) Africa Adaptation Accelerator Programme (AAAP)**, which specialises in scaling up access to digital extension services and technologies and related data-driven agricultural and financial services for at least 30 million smallholder farmers in Africa across 26 countries (see Appendix 2.3 for AfDB-GCA AAAP success stories). Figure B.2.1 shows the integration of AfDB special initiatives into the programme.

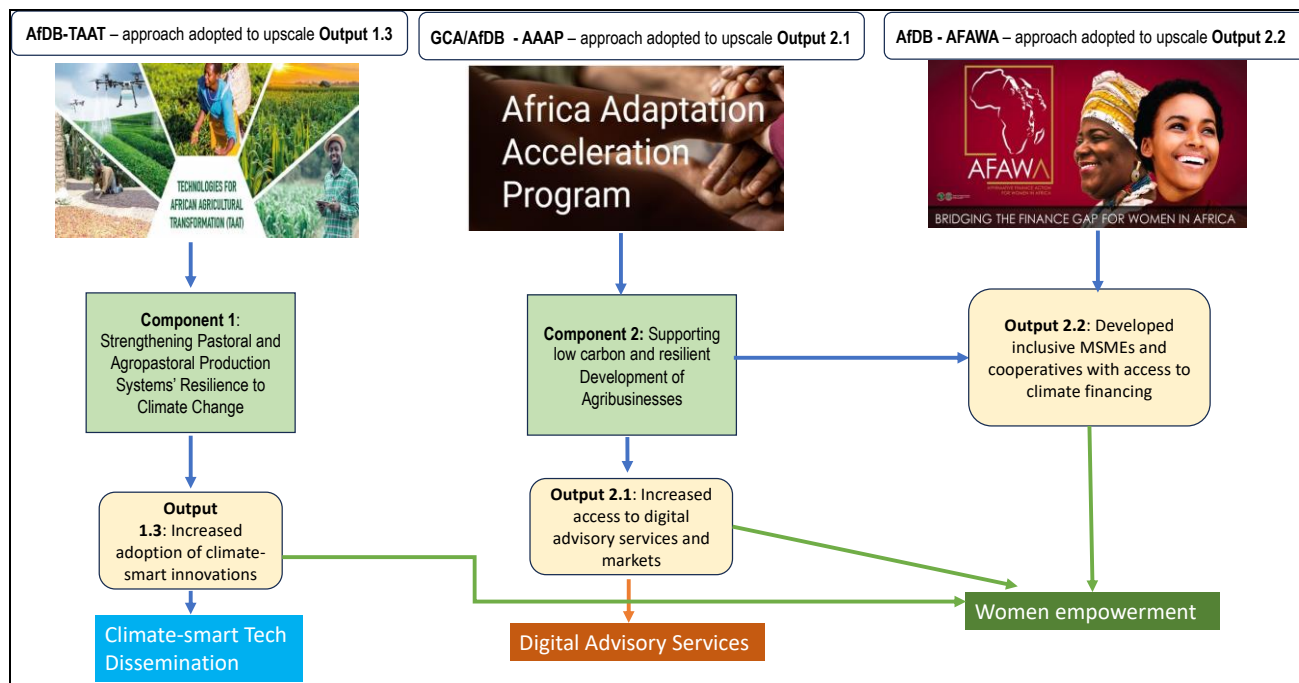


Figure B.2.1: Integration of AfDB Special Initiatives into BREFOL

- 35. Expected outputs of BREFOL:** The main expected outputs of BREFOL include: 1) construction/rehabilitation of 538 water availability and mobilisation infrastructures; 2) construction/rehabilitation of 5,980 ha of community irrigation schemes; 3) construction/rehabilitation of 152 community livestock infrastructures (livestock markets, livestock health centres, etc.); 4) rehabilitation and management of 102,000 ha of degraded rangelands through agroforestry and afforestation programmes; 5) introduction of climate-smart innovations (improved resilient seeds for crops and fodder) to more than 542,000 (278,000 + 64,000) smallholder farmers and pastoralists. to more than 542,000 (278,000 + 264,000) smallholder farmers and pastoralists to increase agricultural and fodder production; 6) Establishment and deployment of 5 digital extension services through PPP to benefit more than 1,142,000 smallholder farmers and pastoralists; 7) Construction and rehabilitation of 125 rural markets; 8) Installation and operation of 125 agro-meteorological stations; 9) Operationalisation of 3 cross-border Memoranda of Understanding to facilitate the control and management of transboundary animal diseases and zoonoses in border areas, including 3 cross-border veterinary laboratories to facilitate cost-benefit analysis of integrated pest management technologies (IPMT) to control crop and livestock pests and diseases such as fall armyworm, arboviruses in transhumant cattle and locust invasion; 10) Building resilience of 4.9 million direct beneficiaries and over 18.7 million indirect beneficiaries, their health and well-being, as well as food and water security, improved ecosystems and ecosystem services through improved land use and management practices; 11) improved access to renewable energy (co-benefit to climate change mitigation), resulting in reduced post-harvest losses and emission reductions through increased carbon sequestration of 14.1 million tCO₂e; 12) leveraged an additional USD 8.9 million through a blended finance approach in fragile states (Ethiopia and Djibouti); 13) promoting green/circular economy activities with a view to gender mainstreaming, leading to gender-sensitive sustainable wealth creation

and green job opportunities; and 143) strengthening operational capacity to build climate resilience at community, national and regional levels.

36. Potential Barriers addressed by BREFOL:

37. Barrier No 1. Lack of access to concessional finance for smallholder farmers and pastoralists to adapt to climate change: Timely access to finance is critical to cope with the debilitating effects of climate change and extreme weather events that characterise the HoA region. Not infrequently, smallholder farmers do not have access to credit facilities to purchase agricultural inputs such as climate-smart seed varieties, fertilizers and farm implements, and to adopt climate-smart agricultural practices to cope with sudden disasters such as seasonal droughts and flash floods. This is illustrated by statistics on access to finance from the World Bank's Global Findex Database, which shows that on average only 22% of smallholder farmers in the HoA have an account with a financial institution, less than 7% are able to borrow from a financial institution, while only about 6% of the active farming population is able to borrow to start, operate or expand a farm or business. With such low access to formal finance in an environment characterised by extreme climate events (droughts and floods), fragility, conflict and population explosion, building resilience at scale is almost impossible. The main constraints to accessing credit are: i) the high market interest rate (12-18%), ii) the limited capacity of smallholders to guarantee the credit requested due to the high risks of the agricultural sector, iii) the limited capacity of smallholders to finance the design of bankable projects, and iv) the limited management capacity of smallholders, which increases the risk of non-repayment of loans.

38. Barrier No 2. Political instability, Insecurity, and Weak Governance: Political instability, insecurity and weak governance are severely hampering efforts to achieve climate-resilient food and livelihood systems in the Horn of Africa, affecting Djibouti, Ethiopia, Kenya, Somalia and South Sudan. According to the Fragile States Index 2021, all five countries are among the world's most fragile states, with Somalia and South Sudan in the top ten. In 2021, conflict and insecurity displaced more than 5.9 million people in Somalia, while in South Sudan some 7.9 million people were in need of humanitarian assistance due to conflict-related displacement (UNOCHA, 2021). Weak governance exacerbates these challenges, with Djibouti, Ethiopia, Kenya, Somalia and South Sudan scoring below the global average on the World Bank's Worldwide Governance Indicators, particularly in terms of political stability and government effectiveness (World Bank, 2021). This combination of political instability, insecurity, and weak governance hampers the implementation of climate adaptation and mitigation measures, increases vulnerability to climate-related shocks, and undermines efforts to build resilience and ensure sustainable food and livelihood systems in the region.

39. Barrier No 3. Infrastructure, policy, and technologies gaps and limited access to climate information: Countries in the HoA face many challenges in terms of preparedness and readiness to respond to sudden climate shocks in the region due to infrastructure, technology and capacity gaps. Access to reliable and timely weather information in formats that are easily accessible to end-users such as farmers and pastoralists, especially in the event of sudden disasters such as drought and famine, is still limited in the region and needs to be widely promoted. Similarly, the availability of digital technology services, which have been shown to significantly improve the resilience of people and communities, also during the crisis, has not been widely promoted in the region, partly due to low private sector participation. Similarly, innovative financing mechanisms such as crop and livestock insurance, which can help farmers and pastoralists manage agricultural and livestock risks, especially during seasonal droughts and lean seasons, suffer from low enrolment rates due to weak institutions and capacity constraints. Capacity gaps also exist in terms of the extent of ministerial coordination on climate change interventions, adequate mainstreaming of climate change into policies and projects, and the development of bankable proposals that qualify for access to climate finance from international climate funds to scale up mitigation and adaptation activities.

40. Barrier No 4. Low private sector involvement in fragile states: As discussed under **Activity 5.2 of the DACCS (2020 – 2030)**: After most governments, the private sector is often the next biggest spender on the continent. For example, there are many private entrepreneurs with vested interests in lucrative renewable energy, energy conservation, waste recycling and other businesses for profit and not necessarily as climate change adaptation and mitigation or NDC interventions. Beyond these green/cradle-to-grave ventures, the private sector also offers various types of private finance such as

debt, equity, guarantees, insurance products, etc., which have great potential to bridge the adaptation financing gap in Africa. Unfortunately, they often lack the incentives and other enabling instruments to create a self-sustaining green/circular economy, sustainable wealth creation and jobs in the region. This is particularly true in fragile states, which attract even fewer private companies because of the high-risk profile associated with them. The creation of effective win-win private-private (2P), private-public (3P) and public-private-producer (4P) partnerships brings together combined expertise and resources to accelerate the achievement of results and increase the impact of green investment projects in the region, creating more wealth while contributing to the goals. Government budgets and donor funding have finite limits. However, there is virtually no limit to profit-seeking private capital. So, creating new profit opportunities for private capital through investments that address climate change opens the doors to this vast source of capital. Concessional green finance from the GCF, together with the **AfDB's Strategic Private Sector Investment for Fragile States project**, can help cascade effective win-win private-private and private-public partnerships for a strong **green/circular economy** in the HoA region.

41. **Barrier No 5. Weak implementation capacity risk at local and national levels:** Weak implementation capacity at both local and national levels poses a significant risk to achieving climate-resilient food security and livelihoods in Djibouti, Ethiopia, Kenya, Somalia and South Sudan. At the local level, limited resources, technical expertise and institutional capacity hinder the effective implementation of climate change adaptation and mitigation strategies, agricultural practices and disaster risk reduction measures. Local communities often lack access to information, technology, and financial resources needed to cope with climate-related shocks and build resilience (UNDP, 2018). Furthermore, inadequate coordination and collaboration among local stakeholders further hampers efforts to address climate change impacts and sustainably manage natural resources.
42. Taken together, this means that addressing the HoA problem through the combination of well-designed programme interventions discussed in Section B.3 will help trigger a paradigm shift towards low-carbon and climate-resilient development pathways in the HoA. However, this assumes that: (1) extreme climate events (**risk no. 1**); (2) insufficient participation of women and youth in programme activities; and (3) economic downturns due to COVID-19 and other pandemics (**risk no. 3**) are well managed.
43. The overall objective of this programme is to reduce the vulnerability of the region to climate-induced food insecurity by mobilising climate and other sources of adaptation finance needed to strengthen the resilience and adaptive capacity of the five IGAD countries in the HoA. The region will be equipped and better prepared to face the challenges of food insecurity caused by climate and other environmental hazards such as drought, floods, locust invasion, land degradation, etc. The highlight of the programme intervention is presented in the Theory of Change (ToC, Figure B.2.2).

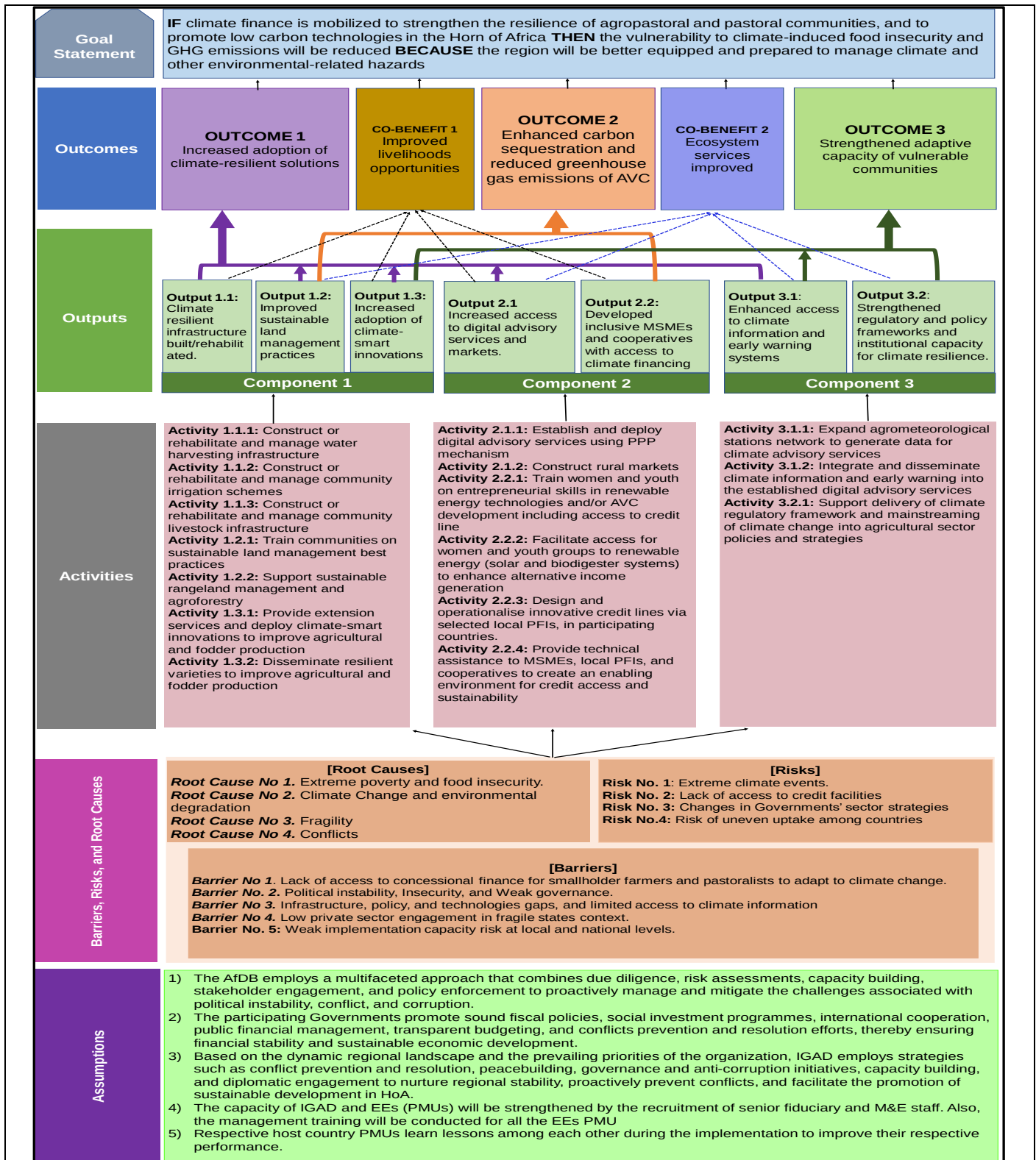


Figure B.2.2: Theory of change for BREFOL

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☒	☒	☒	☐	☒
Outcome 2	☒	☐	☐	☐	☒	☐	☐	☐
Outcome 3	☐	☐	☐	☐	☒	☒	☐	☐

If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1	☐	☒	☒	☐	☐	☐
Co-benefit 2	☒	☐	☐	☐	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

44. The proposed programme: **Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL)**⁴⁹, estimated at USD 335.30 million (AfDB + Country USD 184.29 million + GCF USD 151.0 million), will trigger a paradigm shift towards secure and increased agropastoral production and income generation in the HoA. Through the construction and rehabilitation of physical infrastructure and climate resilient assets (e.g., water mobilisation infrastructure, community irrigation schemes and areas, livestock markets, animal health facilities, rangelands and pastures); training and capacity building in agribusiness development, entrepreneurship and adoption of climate-smart agricultural technologies and innovations; improving access to climate finance, digital markets and digital extension services; access to climate information and early warning systems (CIEWS); and ensuring the creation of an enabling environment (institutional strengthening and regulatory and policy framework reforms); BREFOL is strategically positioned to **"catalyse impact"** beyond the initial duration of the AfDB and GCF investments.
45. The program mainstreams gender across the proposed initiatives economic advantages, such as women's empowerment (economically and financially) through access to credit facilities, including other gender-related benefits, such as the development and promotion of women-led agribusiness MSMEs. Women are known to demonstrate solid repayment behaviour, and have solid track record as savers, often through informal institutions such as unregulated Savings and Credit Cooperative Organizations (SACCOs) and Rotating Savings and Credit Associations (ROSCAs). Another compelling characteristic of women finance behaviour is that they typically reinvest up to 90% of their income in the education, health and nutrition of their family and community, compared to up to 40% for men. This means that investing in women's businesses can transform lives and societies. Moreover, it seeks to yield social benefits by improving women's economic independence, which may have positive social, cultural, and political implications in the region.
46. Environmental co-benefits are expected through the promotion of resilient agropastoral production systems by encouraging the adoption of best practices in sustainable land management and agroforestry, leading to improved ecosystem services and increased soil organic carbon. In addition,

⁴⁹ Please refer to Annex 17 for the multiple country program information

the programme envisages the installation of 4.8 MW of solar photovoltaic systems for solar irrigation, lighting and processing, as well as the potential generation of biogas energy from over 3,890 m³ of biogas digesters to be installed under the programme.

47. GCF support and participation in the BREFOL programme is therefore of paramount importance. First, concessional resources from the GCF will help to bridge the existing adaptation finance gaps for these countries. For example, Ethiopia's Enhanced NDC estimates that the country's adaptation and mitigation need in the six priority sectors will require a total investment of USD 40.5 billion and USD 281.3 billion, respectively. Similarly, that for Kenya highlights that the investment and costs required by the international community for both mitigation and adaptation would be over \$61.1 billion. Estimates for Djibouti and South Sudan are \$6.4 billion and \$50.0 billion respectively. Although an estimated 20% of the annual adaptation costs in the region will come from public budgets, this is clearly insufficient, with a financing gap of 80%. Given the high level of indebtedness of the countries, it is clear that the implementation of planned investments such as BREFOL would require additional funding from the GCF and other development partners. This was highlighted in the 5th Intermediate Outcome of the **African Union Draft Africa Climate Change Strategy - DACCS (2020-2030)** for these countries.
48. Second, as highlighted under **barrier #4**, the private sector has a key role to play in scaling up adaptation finance, especially in fragile contexts such as the HoA region. Unfortunately, the private sector often lacks the incentives and other enabling instruments needed to attract private capital to climate-sensitive sectors in the region. Again, concessional resources from the GCF are critical to unlocking private capital in the region. For example, for an initial investment of \$24.35 million from GCF concessional resources, BREFOL, through its Direct Credit Window (BDCW) to PFIs, would help leverage an additional \$5.49 million (23%) of private capital from local private financial institutions (LPFIs) in Djibouti and Ethiopia. This will go a long way towards addressing **barrier #1** (*lack of access to concessional finance for smallholder farmers and pastoralists to adapt to climate change*).
49. Third and finally, GCF funds will support the training of more than 4,395 cooperatives and SMEs in the region. These well-trained 4,395 cooperatives and SMEs will be able to mobilise additional resources from other commercial banks and microfinance institutions. In addition, we expect this critical mass of well-trained cooperatives and SMEs to serve as trainers of trainers (ToT) for other cooperatives and SMEs interested in accessing credit from local private financial institutions (PFIs).
50. Table B.3.1 presents the activity mapping of BREFOL (i.e., Baseline Investments versus GCF Financed Operations per country).

Table B.3.1. Activity Mapping for BREFOL Baseline and GCF Interventions

Component	Output	Djibouti	Ethiopia	Kenya	Somalia	South Sudan	IGAD
Component 1: Strengthening Pastoral and Agropastoral Production Systems' Resilience to Climate Change	Output 1.1: <i>Climate resilient infrastructure built/rehabilitated (Activities 1.1.1 to 1.1.3)</i>	AfDB + GCF	AfDB + GCF	N/A	AfDB	AfDB	N/a
		AfDB + GCF	AfDB	AfDB	N/A	AfDB	N/a
		AfDB	AfDB	AfDB	AfDB	AfDB	N/a
	Output 1.2: <i>Improved sustainable land management practices (Activities 1.2.1 & 1.2.2)</i>	AfDB + GCF	GCF	AfDB + GCF	N/A	AfDB + GCF	GCF
		AfDB + GCF	AfDB + GCF	AfDB + GCF	GCF	AfDB + GCF	AfDB
	Output 1.3: <i>Increased adoption of climate-smart innovations (Activities 1.3.1 & 1.3.2)</i>	AfDB + GCF	AfDB + GCF	AfDB + GCF	GCF	GCF	AfDB + GCF
		GCF	GCF	AfDB + GCF	GCF	GCF	N/a
Component 2: Supporting low carbon and resilient	Output 2.1: <i>Increased access to digital advisory services and markets (Activities 2.1.1 & 2.1.2)</i>	AfDB + GCF	AfDB + GCF	AfDB + GCF	AfDB	AfDB + GCF	AfDB + GCF
		AfDB + GCF	AfDB + GCF	GCF	AfDB + GCF	AfDB + GCF	N/a

Development of Agribusinesses	Output 2.2: Developed inclusive MSMEs and cooperatives with access to climate financing (Activities 2.2.1 to 2.2.4)	AfDB + GCF	AfDB + GCF	AfDB	AfDB	AfDB + GCF	N/a
		AfDB + GCF	AfDB + GCF	GCF	AfDB + GCF	GCF	N/a
		GCF	GCF	N/A		N/A	N/a
		N/A	AfDB + GCF	AfDB + GCF	AfDB	N/A	N/a
Component 3: Strengthening Rural Communities' Capacity to Adapt to Climate Change	Output 3.1: Enhanced access to climate information and early warning systems (Activities 3.1.1 & 3.1.2)	AfDB	AfDB + GCF	AfDB + GCF	N/A	AfDB + GCF	N/a
		AfDB+ GCF	AfDB + GCF	AfDB + GCF	N/A	AfDB + GCF	AfDB + GCF
	Output 3.2: Strengthened regulatory and policy frameworks and institutional capacity for climate resilience. (Activity 3.2.1)	AfDB+ GCF	AfDB + GCF	AfDB + GCF	GCF	AfDB + GCF	AfDB + GCF
Component 4: Program Coordination and Knowledge Management	Output 4.1: Project management and coordination (Activities 4.1.1 to 4.1.4)	AfDB+ GCF	AfDB + GCF	AfDB + GCF	AfDB + GCF	AfDB + GCF	AfDB + GCF
		AfDB	AfDB	AfDB	AfDB	AfDB	AfDB
		AfDB+ GCF	AfDB + GCF	AfDB + GCF	AfDB + GCF	AfDB + GCF	AfDB + GCF
		AfDB	AfDB	AfDB	AfDB	AfDB	AfDB

51. Description of Programme Components, Outputs and Activities

Component 1: Strengthening Pastoral and Agropastoral Production Systems' Resilience to Climate Change: This component aims to strengthen the resilience of pastoral and agropastoral production systems to climate change through improved production systems. It implements **the IDDRSI priority intervention area 1**, which aims to improve the *management of natural resources and the environment in the HoA*. The main expected outputs or results under Component 1 that will deliver on IDDRSI PIA 1 are:

52. Output 1.1: Climate resilient infrastructure built/rehabilitated.

53. Activity 1.1.1: Develop and rehabilitate the resilience of infrastructure (AfDB + GCF Financed): This activity will include the construction/rehabilitation and management of infrastructure to secure water for agropastoral production in different districts of the respective countries. In Djibouti, for example, the programme will include the construction of six (6) water resource mobilisation infrastructures tailored to climate change challenges. This will include the construction of micro-dams, related infrastructure and gender-sensitive boreholes, as well as the construction of rural drinking water distribution networks. In addition, efforts will be made to construct or rehabilitate 50 numbered small earth dams, boreholes, covered water pans, shallow wells and water distribution systems. In Ethiopia, the focus will be on the construction of 227 multipurpose, demand-driven community water sub-project distribution systems/networks, the rehabilitation of 239 existing small-scale, demand-driven water-related infrastructure, the construction of 25 multipurpose, demand-driven community water sub-project distribution systems/networks, the rehabilitation of 20 existing small-scale, demand-driven water-related infrastructure, and the implementation of 10 resource-rich small-scale irrigation schemes to develop 1500 ha of irrigable land. It will also support the operationalisation of 35 effective irrigation water user associations (WUAs) through the formulation of resource management guidelines in collaboration with the Ministry of Irrigation and Lowlands (MoIL), and the monitoring of physical infrastructure projects. In Southern Sudan, activities will include the construction of 42 water pans/small earth dams, 22 community berkedes (covered water pans), equipment/upgrading of 3 boreholes, training of water infrastructure management committees (with 30% women participation) and technical studies for water infrastructure projects.

- 54. Activity 1.1.2: Construct or rehabilitate and manage community irrigation schemes (AfDB + GCF Financed):** The main objective of this activity is to improve water control and efficiency through the construction/rehabilitation of community irrigation systems. This activity will focus on the development of 240 hectares of climate resilient irrigated perimeters in Djibouti, including the maintenance of hydraulic structures and the installation of six (6) drinking troughs and standpipes downstream of threshold structures for livestock watering and irrigation. In addition, efforts will be made to develop 390 hectares of climate-resilient irrigation perimeters, including multicultural farmland, through a public-private partnership (PPP). In Ethiopia, the programme will develop 900 hectares of demand-driven small-scale irrigation for crops and fodder. In Kenya, activities will include the rehabilitation and development of 800 hectares in seven irrigation schemes, including the rehabilitation of 100 kilometres of roads leading to irrigation schemes and the rehabilitation of 112 reservoirs, boreholes, shallow wells and sub-surface dams. In Southern Sudan, initiatives will focus on the construction of a 25-kilometre access road from Aweil town to a rice scheme and the installation of a chain-link fence for the rice irrigation scheme.
- 55. Activity 1.1.3 Construct or rehabilitate and manage community livestock infrastructure (AfDB Financed):** This activity will facilitate the development of community-driven infrastructure to improve the resilience of livestock to climate change. Specifically, AfDB funding will be used to conduct two technical studies on the management of community livestock infrastructure, including the procurement of animal health laboratory products and fisheries resource monitoring and assessment tools in Djibouti. In Ethiopia, AfDB funding will support the construction of 15 demand-driven animal health posts and 8 other livestock infrastructure facilities such as pastoralist training centres (PTCs), extension centres, veterinary clinics and storage facilities. In Kenya, the budget will cover the construction of 21 livestock markets, 28 hay sheds, 28 pasture rehabilitation projects, and the procurement of equipment for 7 diagnostic laboratories and holding offices. In Somalia, funds will be used to equip/upgrade 1 veterinary laboratory, purchase 12 mobile veterinary clinics, introduce 6 climate-smart improved fodder varieties, support the establishment of fodder banks, map regional rangelands, degraded areas for rehabilitation, prosopis control and livestock routes. Regional rangeland monitoring infrastructure and drought crisis/early warning systems will also be established. In Southern Sudan, funding will support the construction/rehabilitation of 15 veterinary clinics (in Southern, Western and Eastern Kapoeta), 18 livestock checkpoints (in Southern, Western and Eastern Kapoeta), 14 abattoirs (in Southern, Western and Eastern Kapoeta) and 2 livestock quarantine centres.
- 56.** The signing and implementation of the cross-border MoUs will be facilitated by IGAD through the IGAD Centre for Pastoral Areas and Livestock Development (ICPALD). The aim is to address transboundary livestock development challenges. ICPALD has implemented a number of livestock-related programmes that contribute to sustainable peace in cross-border border communities. These include: (1) Facilitated the signing of a cross-border MoU by ministers to improve the control of transboundary animal diseases (TADs). This has helped to synchronise vaccination and reduce outbreaks of TADs in cross-border areas, thereby contributing to sustainable cross-border livestock trade. The minimal or no loss of livestock due to the control of TADs has helped to sustain livelihoods and reduce the vulnerability of youth to extremist activities or misuse by community elites; (2) Enhanced livestock trade within Member States and international markets such as Middle East and North Africa (MENA) and Asian countries. Our interventions, such as supporting exporters to participate in the GULF International Food Fair for three years, helped improve promotion of products to importers and attracted new buyers for meat and live animal exporters in the IGAD region, resulting in increased sales of products. This has helped pastoralists and agro-pastoralists to sell more livestock and increase their income, thereby encouraging pastoralists who had stopped the practice to resume it in order to continue earning a living. (3) Facilitated and supported the development, signing and operationalisation of the transhumance protocol, which has been adopted in most of the cross-border areas in the region. Under this protocol and the Animal Health Production and Health MoU, communities on both sides of the border have participated in joint activities and shared rangeland resources, including water and grass and forage. These activities have helped to strengthen community relations and peace; and (4) ICPALD has supported cross-border communities to establish common livestock markets, which have facilitated local trade in livestock and livestock products and other household goods. These cross-border community interactions are the basis for peace and amicable conflict resolution.

57. Output 1.2: Improved sustainable land management practices.

58. Activity 1.2.1: Train communities on sustainable land management best practices (AfDB + GCF Financed): This activity aims to scale up the adoption of sustainable land management (SLM) best practices by farmers and pastoralists. This will be done through a demand-driven approach to support improved agropastoral and pastoral production systems through training and capacity building to improve land management systems and practices. The project will train approximately 31,000 farmers and pastoralists (50% women) in SLM best practices. In addition, 100 community-based pasture management approaches will be developed with the International Livestock Research Institute (ILRI), using participatory pasture management (PPM) toolkits for dissemination. In Ethiopia, for example, the programme will develop and deliver training of trainers (ToT) to over 100 regional and zonal stakeholders on sustainable rangeland development techniques and tools and provide 300 technical backstopping training sessions for district and grassroots staff. In Kenya, support will be provided to accelerate the process of organising, formalising and building the capacity of Community Land Management Committees and Platforms (CLMCPs). This will include sensitising communities on land rights and interests, organising outreach campaigns to train farmer organisations in small-scale irrigation technologies, and providing training to support the sustainable management of agropastoral lands covering 6000 hectares. In addition, degraded rangelands will be identified and mapped, covering 6000 hectares, and sustainable land management tools will be developed and disseminated through training sessions involving 50 stakeholders per country. Reseeding of degraded agropastoral land will also be carried out, covering 2000 hectares. In Southern Sudan, efforts will focus on developing local capacity to manage land degradation, organising stakeholder consultations, training local land management committees and providing communities with technologies for land resource use, rehabilitation and restoration. In the IGAD region, the focus will be on developing and disseminating tools and approaches for sustainable rangeland management, including training of trainers (ToT).

59. Activity 1.2.2: Support sustainable rangeland management and agroforestry (AfDB + GCF Financed): This investment activity aims to sustainably improve land and rangeland productivity. It will support various initiatives to improve rangeland management and regeneration at regional and national levels. In Djibouti, for example, AfDB financing will be used to carry out regeneration, enrichment and management activities on 25 kilometres of rangeland, including planting trees and shrubs, and establishing 1,000 hectares of community forestry. In addition, GCF funding will be used to regenerate, enrich and manage a further 500 hectares of rangeland, construct a green wall, establish nurseries for endangered species and develop anti-erosion structures in catchment areas. In Ethiopia, investments will be made in bush clearing on 31,500 hectares, soil and water conservation measures, reseedling and rangeland management, and the purchase of various seeds and agricultural inputs for 10,000 hectares. GCF funding will promote physical and biological soil and water conservation techniques, expand rangeland agroforestry and rehabilitate 600 hectares of rangeland affected by invasive plant species. In Kenya, activities will include identification and mapping of degraded rangelands, support for land use planning, construction of erosion control structures, support for spring protection and soil erosion control, and tree planting in degraded areas. In Somalia, rehabilitation and management of 3,500 hectares of rangeland will be carried out, along with dissemination and training activities with ILRI. In Southern Sudan, 1,000 kilometres of livestock routes will be constructed/rehabilitated, and 3,000 hectares of rangeland will be rehabilitated and managed using participatory approaches. In the IGAD region, efforts will focus on operationalising and institutionalising the IGAD Regional Rangeland Management Strategic Framework, improving the competitiveness of livestock value chains, and assessing cross-border veterinary laboratories.

60. Output 1.3: Increased adoption of climate-smart innovations.

Activity 1.3.1: Provide extension services and deploy climate-smart innovations to improve agricultural and fodder production (AfDB + GCF Financed): This activity will address a major barrier to the uptake of innovation in the region. This is mainly due to limited knowledge of CSA technologies and their applications. It will provide extension services and access to resilient improved varieties. According to the World Bank (2021), scaling up support for climate-smart agriculture (CSA) across the

agri-food value chain through robust policy and technology interventions can deliver robust triple-win benefits: increasing productivity, reducing greenhouse gas emissions, and improving resilience to climate-related shocks. In Djibouti, for example, under this activity, BREFOL will focus on: 1) developing and disseminating tools from the catalogue of technologies and good practices with the TAAT programme to producers, including women (<30%) and youth (<30%) in the context of climate change (AfDB), and 2) drafting an agreement between ISV and DAF to set up trials and multiplication of fodder seed production, and purchase and distribute 1,000 improved breeds of goats for restocking and livestock feeding (GCF). For Ethiopia, this activity will finance 1) fodder, crop and aquaculture production using innovative TAAT technologies (see Annex 2). 1 for TAAT achievements across the continent) and TAAT - Technical Assistance for Development of Catalogue of Innovative Technologies and Best Practices (AfDB), and 2) Procurement of 30 machinery/equipment to improve fodder production, establishment and strengthening of 30 hay storage facilities and fodder banks to stabilise fodder supply to agricultural MSMEs, establishment and strengthening of 30 fodder nurseries and support to 60 demand-driven and community-based small-scale irrigated drought-tolerant fodder seed production activities in selected locations (GCF). In Kenya, the AfDB will invest in capacity building of NARES officers, extension workers and farmer organisations on TAAT technologies, as well as access to and adoption of new genetics and associated technologies for the fish sector, including the production of high quality, low-cost fish feed and improved processing techniques. GCF funds will be invested in extension services and demonstrations of innovative technologies to reach 42,000 pastoralists and smallholder farmers, including capacity building of extension workers and farmer organisations on climate-smart fodder technologies. South Sudan, under this activity, will invest GCF resources in resilient fodder production systems in the absence of AfDB support by establishing additional fodder banks (with farm and fodder handling equipment, fodder storage infrastructure and fodder bank operating costs) to benefit 50,000 pastoralists and their 250,000 livestock. In addition, 50 tonnes of resilient fodder varieties will be provided. Finally, at the regional level, IGAD will invest AfDB resources to develop an interactive online tool that will be used by countries to identify the most appropriate and easy-to-adopt solutions for their agro-pastoralists, and to organise an interactive regional climate smart technology fair. Complementary GCF resources under this activity will support the development and dissemination of extension materials on innovative technologies, including support for the development of resilient forage systems.

61. Activity 1.3.2: Disseminate resilient varieties to improve agricultural and fodder production (GCF Financed for all countries and AfDB for Kenya): This activity aims to increase fodder production in the 5 countries to secure livestock production. It will mainly support commercial fodder production through the establishment of fodder banks in pastoralist areas, contributing to improved livelihoods and reduced livestock vulnerability. BREFOL will provide more than 264,000 pastoralists in the 5 countries with resilient seeds to increase fodder production. In Djibouti, through a partnership with the International Livestock Research Institute (ILRI), efforts will focus on extension services to train 40,000 pastoralists/agro-pastoralists in 50 districts on rangeland management and disseminate rangeland management plans (RMP). Similarly, in South Sudan, GCF funds will be used to provide extension services to 4,000 pastoralists and smallholder farmers. Beneficiaries in the other three countries include 120,000 pastoralists in Ethiopia, 50,000 in Kenya and 50,000 in Somalia.

62. Component 2: Supporting Low Carbon and Resilient Development of Agribusinesses: There is a growing emphasis on developing the agribusiness sector to enable vulnerable communities and people to adapt to climate change. Agribusiness development, such as access to digital technology and markets, entrepreneurship development and access to innovative financial instruments, has the potential to boost rural productivity, increase competitiveness and help rebuild better and stronger agriculture and food production systems, especially after COVID-19, and address climate change. This is an important change pathway to build resilience and address climate change issues through climate finance solutions in the HoA (**IDDRSI Priority Intervention Area 2**). This means facilitating market access, trade and financial resources. Thus, in line with **IDDRSI PIA 2**, Component 2 of BREFOL will deliver the following outputs:

63. Output 2.1: Increased access to digital advisory services and markets.

64. Activity 2.1.1: Establish and deploy digital advisory services using PPP mechanism (AfDB + GCF Financed). Promoting sustainable and resilient agro-pastoral value chain development requires timely access to various updated information on technologies, finance, market, climate information, Early Warning System (EWS), etc. for all stakeholders. This activity will therefore respond to these important needs for which the PPP is adopted to ensure sustainability. The implementation of this activity will build on the digital advisory services approaches developed by AfDB under the African Adaptation Acceleration Programme (AAP, see Appendix 2.3). The activity will support the development of digital solutions through public-private partnerships (see Annex 25 for more details).

65. Proposed Public Private Partnership Approach for scaling Digital Tools: Despite the significant growth of digitalization in Africa's agricultural sector, only 33 million of these rural smallholder farmers have been registered by the approximately 390 digital companies operating in Africa in 2019, according to the CTA/Dalberg Advisors report. However, the report recognizes that "digitalization could be a game changer in increasing productivity, profitability and resilience to climate change" and other economic stressors. Both the EU-Africa Task Force on Rural Africa (TFRA) report and the Global Forum on Food and Agriculture (GFFA) communiqué highlighted the power of digitalization to transform agriculture and rural economies. Figure B.3.1 illustrates the importance of digitizing the agricultural value chain.

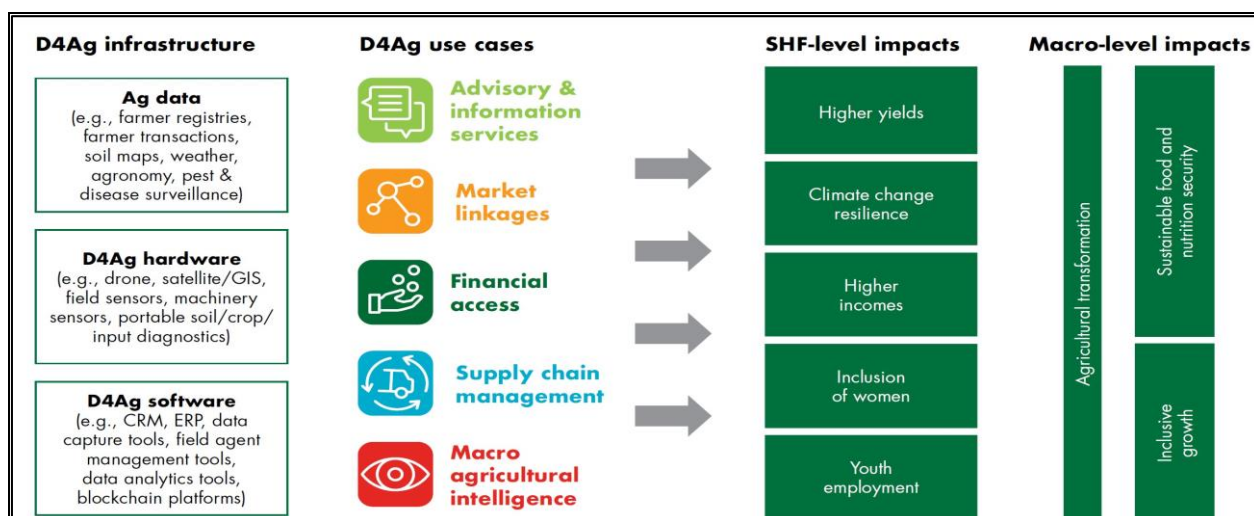


Figure B.3.2: Importance of digitalization on smallholder agriculture: Source CTA/Dalberg Advisors Report

66. One of the reasons for the low level of digitisation in African rural communities is the lack of appropriate public-private partnerships to scale up the adoption of digital solutions. Other challenges that have been identified with lack of traction in the adoption of digital solutions include the following

- The presence of many small agritech start-ups with stand-alone applications that address limited agricultural challenges in most project countries. In some countries, such as Kenya, the emergence of many agritech startups has led to fragmentation of the sector, and many of these startups have not been able to scale beyond a few hundred users.
- Most digital agricultural interventions are government and NGO driven, without a corresponding sustainability strategy to ensure the continued use or scaling of the digital solutions beyond the life of the projects that sponsored them.
- Lack of privately developed infrastructure to support agricultural services and general lack of public-private partnerships for business model innovation.
- Lack of market matching platforms for smallholder farmers and pastoralists.
- Limited critical information services for smallholder farmers and agropastoralists on climate-smart agronomic practices. Among the reasons for low levels of digitization among African rural communities is lack of appropriate public private partnership to scale-up the adoption of digital

solutions. Other challenges that have been identified with lack of traction in the adoption of digital solutions include the following:

- 67. Methodology for the PPP Business Model Innovation:** The methodology for the selection of the private sector partner and the subsequent development and deployment of the digital solutions shall include the following procedure, as shown in Figure B.3.2 (see Annex 25 for more details):

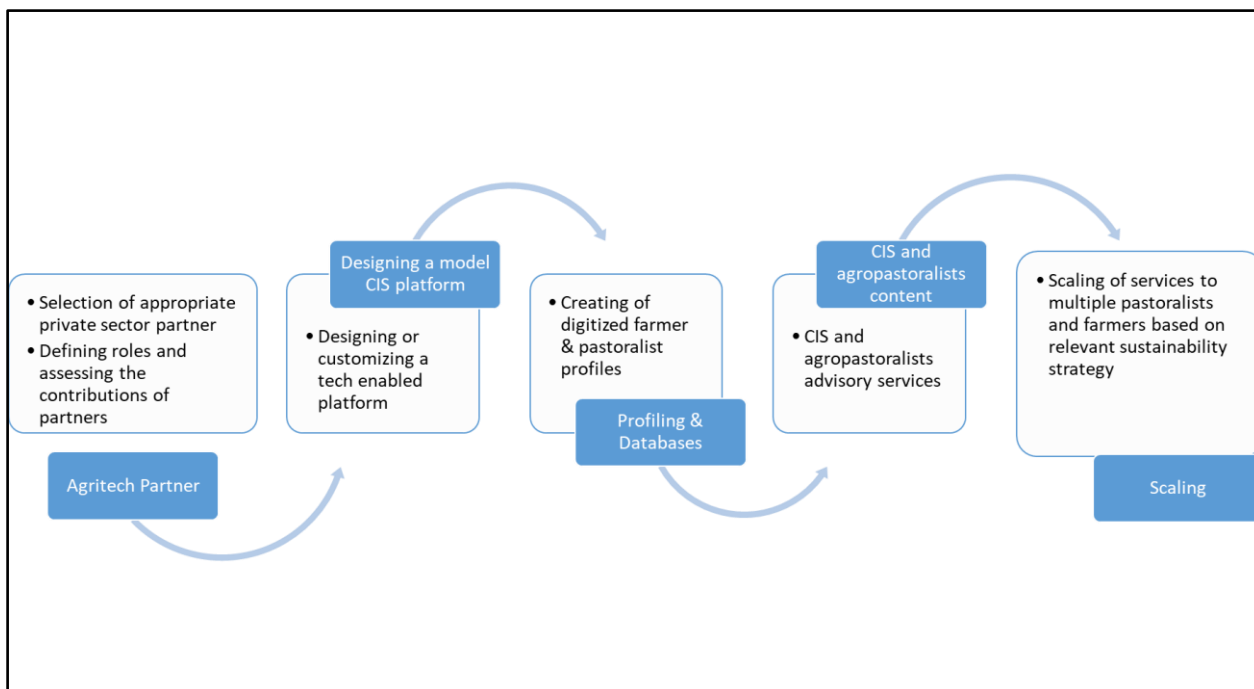


Figure B.3.3: Methodology for developing the PPP business model innovation.

- 68.** Under this activity, AfDB and GCF will: **1) Support over 1,142,000 smallholder farmers (women and youth), farmers organization and agribusiness MSMEs to benefit from extension on using digital advisory services;** and **2) Establish a regional innovation hub/data centre to coordinate and share knowledge among innovation partners, governments, and other stakeholders.**
- 69. Activity 2.1.2: Construct rural markets (AfDB + GCF Financed).** Limited access to markets is a major constraint for farmers and pastoralists to get the right price for agropastoral products. This activity will address this constraint through the construction and development of various market infrastructure components in different regions of the participating countries, with the main objective of facilitating trade in agropastoral products. In Djibouti, for example, efforts will focus on the construction of two community markets and the rehabilitation of rural market roads, as well as the development of livestock infrastructure in five community markets. In Ethiopia, efforts will include the construction of 60 km of access and feeder roads, six livestock market centres and 30 fodder market centres. Kenya will receive support for the establishment of a market with digital extension infrastructure and the construction of 14 demand-driven rural markets. In Somalia, activities will include the construction of 10 livestock market fences and shade structures, as well as technical studies and supervision of marketing infrastructure works, and the construction of 17 demand-driven rural markets. In Southern Sudan, activities will support the professionalization of agro-pastoral value chains/cooperatives, the development of smart maps for livestock routes and climate information, the establishment of a national e-registry of pastoralists and farmers, the facilitation of the provision of digital financial services, and the construction of 10 demand-driven rural markets.
- 70. Output 2.2: Developed inclusive MSMEs and cooperatives with access to climate financing.**
- 71. Activity 2.2.1: Train women and youth on entrepreneurial skills in renewable energy technologies and/or AVC development including access to credit line (AfDB and GCF Financed).** Low technical

and managerial capacity has been a major constraint for women and youth in developing SMEs for agricultural value chain (AVC) activities and job creation. Under this activity, BREFOL will train over 2,350 women and youth from 772 cooperatives and MSMEs on entrepreneurship in renewable energy technologies (solar and biogas) and agribusiness. The aim is to empower producer groups, including women's groups, through training, structuring and awareness raising on entrepreneurship, leading to the development of business plans for cooperatives and MSMEs. In Ethiopia, support will be provided to strengthen existing cooperatives through capacity building and technical assistance, as well as tailored training materials for women/youth led MSMEs in multiple languages. In Kenya and Somalia, the focus will be on supporting regulatory processes, developing a rural energy catalogue, and training youth and women on bio-digesters and solar energy systems. South Sudan will support youth and women's groups to develop sub-projects, support youth associations/cooperatives in beekeeping, and provide entrepreneurship training for women and youth MSMEs. Training will include business plan development and linkages to credit lines and commercial banks. This activity will build on the women's empowerment approaches developed under the AfDB's AFAWA programme (see Appendix 2.2).

72. **Activity 2.2.2: Facilitate access for women and youth groups to renewable energy (solar and biodigester systems) to enhance alternative income generation (AfDB + GCF Financed).** Limited access to energy causes significant post-harvest losses in agro-pastoral production. The fossil energy used is a source of greenhouse gas emissions. Under this activity, more than 10,680 women and youth entrepreneurs will be able to access gender-sensitive climate finance to invest in the introduction of renewable energy for staple food processing (4.8 MW solar and 3,890m³ (~0.12MW) biodigester systems) through the BREFOL Credit Window. In Djibouti, this activity aims to establish a gender-sensitive financing mechanism tailored to the needs of smallholder farmers, fishers and MSMEs, including guarantee funds and access to micro-credit for women, including 0.3 MW of solar for food processing to women/youth led MSME/IGA groups. In Ethiopia, the project will build community processing units for youth and women, strengthen existing cooperatives, facilitate capacity building for district agricultural offices to support MSMEs, purchase equipment for demand-driven processing units, coordinate the establishment of women/youth led MSMEs for renewable energy management, and establish 700 (3500m³) biodigesters for women/youth led MSMEs. In addition, access to renewable energy will be provided, including 1 MW of solar energy for food processing for women and youth-led MSMEs. In Kenya, inputs and access to renewable energy such as solar and bio-digesters will support alternative income generation for women and youth, including the provision of 1.5 MW of solar and 120 m³ of bio-digesters for food processing. In Somalia, renewable energy alternative income generation inputs will be provided, including 1.0 MW of solar food preservation systems for women and youth groups. South Sudan will receive support for 1 MW of solar and 270 m³ of bio-digester food processing systems for women and youth-led associations, as well as training and deployment support.
73. **Activity 2.2.3: Design and operation of innovative credit lines through selected public financial institutions (PFIs) in participating countries (GCF funded):** PFIs and the private sector in general do not invest significantly in AVC business due to the high-risk nature of the sector. As discussed under **barrier #4**, one way to attract private capital to climate-sensitive sectors is to provide the right incentives and an enabling environment suitable for private sector operations. To this end, BREFOL will use GCF concessional resources to support the establishment of a dedicated direct lending window (BDCW) to provide two local PFIs in Djibouti and Ethiopia with stable, lendable resources to adequately address the financing needs of smallholder farmers, pastoralists, agribusiness MSMEs, FBAs and other MSMEs, such as cooperatives, with an interest in green/circular economy development (Please see Annex 26 for more details of the LPFIs). A total of 1,985 farmer groups, cooperatives and MSMEs will benefit from the BDCW, of which at least 50% will be led by women and youth. A legal agreement will be signed at the level of the PFI and the beneficiaries (agribusiness MSMEs), without the involvement of the EE. During the last stakeholder consultation held from 4 to 14 March (see **Annex 7.1, section 3.4 for Djibouti and Annex 7.2.1, section 3.2 for Ethiopia**), two national development banks were contacted and agreed to participate in the programme. The two interest rates offered by the participating PFIs are relatively lower than the official lending rates in both countries (Table B.3.2). The establishment of these dedicated credit lines will not affect the business of the local IFIs, as they do not currently lend to this category of actors, and the eligibility criteria will limit access only to farmer and herder groups, cooperatives and SMEs.

Table B.3.2: Deposit vs. lending interest rates in participating countries

	Indicator	Deposit interest Rate	Lending Interest Rates by FEDE & DBE	BREFOL Maximum Proposed Rates
Djibouti	Rate (%)	1.70%	12% - 15%	6.00%
Ethiopia	Rate (%)	7.70%	12% - 18%	9.00%

74. In addition, under this activity, BREFOL will encourage existing climate insurance providers to scale up weather insurance (WII) coverage for livestock production. This will be done by coordinating the design of climate insurance schemes for targeted agri-MSMEs (cooperatives, enterprises, aggregators, etc.); developing frameworks that integrate climate insurance into climate finance strategies; optimising the use of concessional funds and grants (by providing targeted subsidies, de-risking investments of producers/enterprises/aggregators; and developing smart subsidy criteria for a climate insurance scheme for agri-MSMEs. Under this activity, TA support will be provided to establish or strengthen access to digital services for parametric/index insurance in Ethiopia. GCF resources will therefore complement AfDB investments to support the establishment of a **BEFOL Direct Insurance Credit Window (BDICW)** with the selected banks. This will help 3,600 smallholder and pastoralist groups, MSMEs, FBAs and other agribusiness MSMEs to mitigate climate-related shocks in the region. The criteria for the selection of sub-project beneficiaries under Activity 2.3.1 are summarised in Table B.3.3, while the sub-activities eligible for funding under the BDCW are listed in Table B.3.4 (subject to change on a case-by-case basis). In principle, the project will ensure that each beneficiary of the credit line is covered by a sustainability insurance under the BDICW.

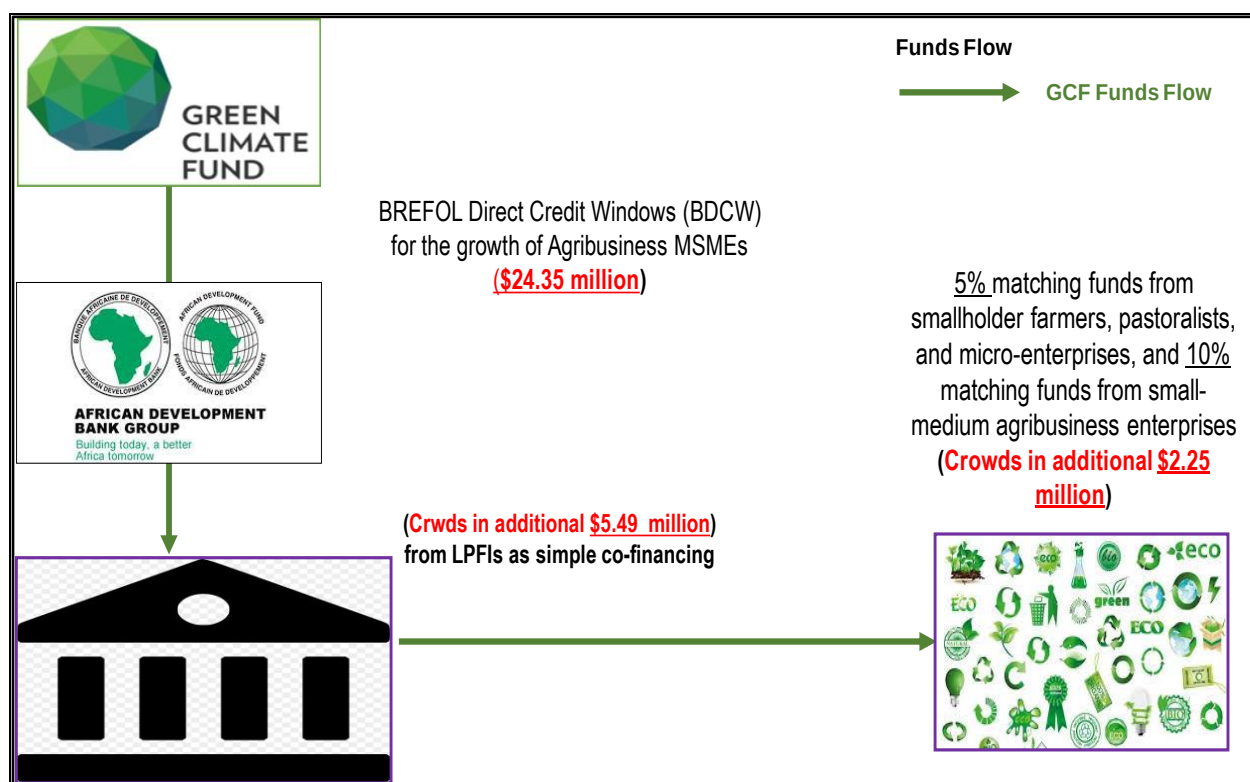


Figure B.3.3: BREFOL Direct Credit Window (BDCW) to attract Private Sector Investments in the HOA

75. It is important to emphasise here that the AfDB has been at the forefront of supporting SME development in Africa through credit lines, with notable success. For example, in 2016, the AfDB approved a \$150 million line of credit (LoC) to Nigeria's United Bank for Africa (UBA) Plc to finance infrastructure and SME projects, including women-owned enterprises in Nigeria (<https://www.afdb.org/fr/node/24>.) Still in Nigeria, in 2018, the AfDB approved a \$50 million LoC to Nigeria's Fidelity Bank Plc to support small and medium-sized enterprises and women-owned enterprises in selected transformative sectors, including nearly one hundred SMEs in manufacturing.

- health and education (<https://www.afdb.org/en/news-and-events>). In 2022, the AfDB approved a \$50 million LoC with Nigeria's First City Monument Bank (FCMB) to support access to finance for small and medium-sized enterprises and women-owned businesses. FCMB will use the credit line to expand lending activities that will support the country's post-pandemic economic recovery and promote more inclusive and climate-resilient growth. It will focus on companies in Nigeria's agribusiness, manufacturing, healthcare and renewable energy sectors (<https://www.afdb.org/en/news-and-events>).
76. As the AfDB will have an oversight role over the selected local PFIs, it will ensure that any changes to the selection criteria or the disbursement of any sub-loans to the selected beneficiaries are strictly in accordance with the sub-loan agreements between the local PFIs and the sub-loan beneficiaries.
77. *The financial context of the participating countries under the BDCW and the selected PFIs*
78. **Djibouti:** Djibouti's financial system is still very underdeveloped. It is dominated by the banking sector, which holds more than 95% of financial sector assets. As of July 2019, the financial landscape consisted of 11 banks with majority foreign ownership, including 8 traditional commercial banks and 3 Islamic banks. The non-bank financial sector consists of 20 licensed financial auxiliaries (foreign exchange bureaus and/or remittance offices), 3 microfinance institutions, 2 insurance companies, and an economic development fund specialising in SME financing⁵⁰. Access to financial services is constrained by several factors, including: (i) unemployment and poverty; (ii) low mobile phone and internet penetration; (iii) high borrowing costs; and (iv) underdeveloped payment and credit information systems.
79. Digital finance also remains underdeveloped, partly due to limited internet and mobile penetration and a lack of institutional and technical capacity (**barrier #3**). Access to finance for SMEs is a recurring problem in Djibouti. Most SMEs operate in the informal sector and do not have access to bank loans. SME loans account for only 12% of corporate loans. The limited scope of the credit reporting system and the unreliability of the financial information provided by SMEs partly explain the low level of business lending.
80. **Ethiopia:** Ethiopia's financial sector comprises 18 banking institutions, of which 16 are private and 2 are public. The total number of bank branches was 4,757, with more than a third (35.3%) located in the capital, Addis Ababa. The country has 37 microfinance institutions (MFIs), 17 insurance companies (16 of which are private) and 532 agencies. Most micro, small and medium-sized enterprises (MSMEs) and large companies have access to traditional financial services to manage their liquidity. However, access to credit is more limited, especially for MSMEs: only 30% have access, compared to 68% of large enterprises. Despite a vibrant digital financial subsector, barriers remain, including regulatory limits on daily transactions and mobile bank account balances⁵¹.
81. *Criteria for the selection of sub-project beneficiaries under activity 2.3.1:*
82. **Definition of Smallholder Farmers:** The definition of smallholder farmers varies from country to country. For the purposes of this programme, the most widely used definition of smallholder farmers has been adopted, namely farmers with a low asset base and less than 2 hectares of land under cultivation. Alternatively, they can be defined as farmers who operate under structural constraints, such as sub-optimal access to resources, technology and markets, or with limited resource endowments compared to other farmers in the sector. Farming less than 2.0 hectares.
83. **Farmer-based association (FBA):** These are associations or organisations established within a given territory to address the common needs of their members (smallholder farmers).
84. **Agricultural Cooperative Society (ACS):** Defined as a business association in which farmers pool their resources to increase revenues, reduce costs or share risks, depending on the type of cooperative.
85. **Agro-dealer (AD):** Defined as a locally based entrepreneur who sells seeds, fertiliser and agrochemicals to poor farmers in rural areas.

⁵⁰ [Djibouti | MFW4A - Making Finance Work for Africa](#).

⁵¹ [Ethiopia | MFW4A - Making Finance Work for Africa](#).

86. Producer Organisation (PO): In this context, a PO is defined as any legal entity formed by primary producers in the AFOLU and energy sectors that provides for the sharing of profits/benefits among its members. This could include, for example, any farmers' association (FBAs, WABEs, YABEs, fishermen, poultry, livestock, dairy, vegetable, palm, oil producers, weavers, etc.), cooperative, ESCO, etc.

87. Micro, small and medium enterprises (MSMEs): The definition of MSMEs varies from country to country. However, for this programme, the definition of MSMEs proposed by Afful-Koomson et al. (2014)⁵² will be used. This is based on the following quantitative variables, which are described in Table B.3.4 below:

Table B.3.4. Quantitative Variables for Defining MSMEs

Category	Quantitative Aspects
Medium Enterprise	Employs between thirty and hundred people; has a total annual turn-over in the range of US\$ 20,000 to \$ 50,000 and with fixed assets not exceeding \$1,000,000.
Small Enterprise	Employs between ten to twenty-nine people; has a total annual turn-over in the range of \$2,000 to below \$20,000, and with fixed assets not exceeding more than \$100,000.
Very Small and Micro Enterprises	Employs between one to nine people; has a total annual turn-over between \$500 and \$2,000, and with fixed assets not exceeding \$10,000.

Source: Adopted and modified from Afful-Koomson et al. (2014)

88. A formally registered agribusiness enterprise with more than 51% women's and youth's⁵³ ownership (more than 51% of the firm's assets holding or shares are owned by a woman or a youth. This will be assessed and validated against firm registration information from the relevant MDAs in charge, and the share of profits during the tenor of the sub-loans.

89. The criteria for selecting FBAs will be like that of the MSMEs as follows:

- A formally registered FBA/O with the Ministry of Agriculture or any other relevant body; The size of the Association must not be less than 5 members with at least 60% of the members being women or youth;
- Activities are mainly carried out along the agricultural and food systems value chains, including renewable energy;
- Track record of loan repayments;
- Operational bank account of more than 2 years;
- Records and bookkeeping;
- Agriculture land holding of between 5 and 15 ha; and
- Any other characteristics deemed necessary by the LPFIs

See **Annex 26** for possible eligible activities or investments under the Credit Line Facility.

90. Activity 2.2.4: Provide technical assistance to MSMEs, local PFIs, and cooperatives to create an enabling environment for credit access and sustainability ((AfDB + GCF Financed): Farmers, pastoralist groups and SMEs face financial and non-financial constraints in developing the agricultural value chain to add value and increase income. This activity aims to support these actors to ensure that they have access to credit, but also that they manage it efficiently to increase their income and repay the credit. It will therefore provide technical assistance and allocate funds to finance early-stage agripreneur start-ups in Ethiopia through grants and repayable grants via business development service organisations. It will also conduct a needs assessment to identify eligible agribusiness MSMEs, PFIs and cooperatives, coordinate ongoing dialogue, facilitate the review of legal and policy

⁵²Afful-Koomson et al. (2014). Economic and Financial Analysis of Small and Medium Food Crops Agro-Processing Firms in Ghana. United Nations University Institute for Natural Resources in Africa, Accra, Ghana.

⁵³ Defined as individuals between 18 and 35 years of age

frameworks for credit lines suitable for pastoral contexts, support local financial institutions, review existing legal and institutional frameworks, and establish six knowledge-sharing platforms with relevant stakeholders. In Kenya, the focus will be on capacity building for agro-pastoral communities on index-based livestock insurance schemes, sensitising the community on youth agribusiness plans for loan applications in collaboration with PFIs, creating linkages and networks with PFIs dealing with agricultural lending, and providing capacity building on business plan development to support agricultural investments. In addition, women and youth will be trained in agro-pastoral value chain (AVC) entrepreneurship skills and access to credit will be facilitated to reach 100 MSMEs with a total membership of 2,000. The activity also aims to develop a public-private partnership (PPP) framework and structure to incentivize private sector insurance companies to introduce index-based livestock insurance (IBLI) products, and to design and validate appropriate climate risk contracts for climate and livestock insurance in Somalia. A total of 4,395 farmer/pastoralist groups and MSMEs will benefit from capacity building in financial management and marketing to improve their management efficiency, thereby increasing their ability to repay the loan.

91. **Component 3: Strengthening Rural Communities' Capacity to Adapt to Climate Change:** Efforts to strengthen the capacity of agropastoral communities in the HoA to adapt to climate change are imperative due to the region's vulnerability to environmental changes exacerbated by climate change. The IDDRSI's Priority Area of Intervention (PAI) 4 emphasises disaster risk management (DRM) as central to enhancing resilience and food security in the HoA. Climate-related disasters such as droughts, floods and unpredictable rainfall patterns pose a significant threat to agricultural productivity and livelihoods in the region. By prioritising DRM, initiatives can focus on early warning systems, sustainable land management practices and community-based resilience building to mitigate the impacts of climate change on agropastoral communities. Such efforts aim to enhance adaptive capacity, promote sustainable resource management and ensure food security in the face of the challenges posed by a changing climate in the Horn of Africa. Thus, in line with IDDRSI PAI 4, BREFOL will deliver the following outputs to improve DRM in the HoA.
92. **Output 3.1: Enhanced access to climate information and early warning systems**
93. **Activity 3.1.1:** Expanding the network of agro-meteorological stations to generate data for climate advisory services - It is essential to expand the network of agro-meteorological stations to increase the availability of climate information for improving climate resilience, agricultural planning and livelihoods in the HoA. This activity will be key to improving climate resilience and agricultural productivity in Ethiopia, Kenya and South Sudan. The installation of agro-meteorological stations will significantly improve the capacity to generate accurate and localised weather and climate data, enabling more effective climate advisory services. In Ethiopia, twenty-five (25) automatic weather stations will be established to strengthen the early warning system and improve weather observation coverage, benefiting pastoralists and agro-pastoralists. In Kenya, fourteen (14) stations will be installed, with a focus on placing at least two in each county, to improve data generation for climate-informed extension services, enabling farmers to make informed decisions. Similarly, in South Sudan, a total of eight (8) automated agro-meteorological stations will be installed in project counties to strengthen the network infrastructure for digital climate-advisory services and support resilience-building efforts in vulnerable communities.
94. **Activity 3.1.2: Integrate and disseminate climate information and early warning into the established digital advisory services** – This activity will integrate and disseminate climate information and early warnings in Ethiopia, Kenya and South Sudan using digital solutions platforms established through the PPP mechanism (see Annex 25 on the digital solutions PPP model under Activity 2.1.1). The National Meteorological Services (NMS) of the three countries will work with qualified private sector entities providing digital agro-advisory solutions to update and integrate climate information into the digital solution model. Memoranda of Understanding (MoUs) will be signed with the NMSs to facilitate access to climate data and information for agropastoral and pastoral communities. The PPP mechanism will support coordination, capacity building and monitoring and evaluation efforts with NMS. The established PPP will facilitate the collection, synthesis and dissemination of information and create a business case. Specifically in Ethiopia, this activity will support the recalibration of the existing ATI

hotline service to provide tailored advice to the pastoral community, adding features and improving usability. In addition, one (1) mobile application for pastoral extension agents will be customised to disseminate climate information and early warnings to 700,000 pastoralists and smallholder farmers (an average of 100,000 per district). Unstructured Supplementary Service Data (USSD) and Interactive Voice Response (IVR) systems for climate information dissemination will also be developed and made operational. In Kenya, climate information and early warning systems will be deployed, and technical assistance will be provided to train and equip 500,000 pastoralists and smallholder farmers in the use of climate services. Similarly, in South Sudan, technical assistance support will be provided to train and equip 3,000 pastoralists and smallholder farmers in the use of climate services.

95. Output 3.2: Strengthened regulatory and policy frameworks and institutional capacity for climate resilience

96. Activity 3.2.1: Support delivery of climate regulatory framework and mainstreaming of climate change into agricultural sector policies and strategies – The enabling environment is critical to the achievement of the objective of BREFOL, especially for the mobilisation of the private sector in the development of agribusinesses for value addition to agro-pastoral products. The main constraint to investment for SME development is the lack of adequate regulation that can encourage private sector investment. This activity will further improve the existing investment and regulatory framework in the participating countries (see Annex 9). The activity will also support capacity building for participating countries on MRV-related activities. Under this activity, investments will be made to facilitate mainstreaming policy dialogues with key stakeholders on climate index-based insurance to advance actionable reforms in Ethiopia. Coordination will take place between the National Meteorological Agency (NMA), ATI, insurance service providers, the Ethiopian Disaster Risk Management Commission (EDRMC) and other key stakeholders to establish independent verification of index insurance and standardisation of insurance packages for the agro-pastoral and livestock value chain. In Kenya and Somalia, this activity will support the mainstreaming of climate change into sector policies and strategies, including dissemination in the agriculture, water, livestock and environment sectors. In South Sudan, the activity will support the National Agriculture and Livestock Extension Policy, the Integrated Natural Resources Research Policy, the Agriculture Sector Policy and the Pastoralism Policy. At the regional level (IGAD), socio-economic and environmental data on vulnerability and resilience will be regularly collected and processed. Documentation and sharing of best practices on adaptation and resilience will be supported, as well as mapping and assessment of existing climate resilience projects and initiatives. Facilitate dialogue on climate change resilience in the region and develop and disseminate a report on the state of climate and climate change impacts in the IGAD region. Raise awareness and facilitate stakeholder engagement on existing regional and national climate change and adaptation policies and strategies, including the Framework for Weather, Water and Climate Services, the IGAD Regional Climate Change Strategy and the Climate Change Adaptation Strategy. Support will be provided for mapping regional and Member States' requirements for mainstreaming climate change into sectoral policies, and for dialogue and platforms for sharing data and experiences on mainstreaming climate change into sectoral policies and strategies. A tracking platform on regional and national policies and strategies will be developed to support informed sector decisions. Given the high level of discrimination against women and girls in the HoA (Barrier No. 5) and the fact that this marginalised group plays a crucial role in the management of ecosystem services and food security in the region, the design and implementation of gender-sensitive pro-poor programmes and policies will contribute to the economic and financial empowerment of women and girls, in line with IDDRSI Priority Area 7. TA support is therefore requested from the GCF to conduct feasibility studies in the five IGAD countries to assess the feasibility of rehabilitating/developing new cooperative food security banks under the management of women-led organisations.

97. Component 4: Program Coordination and Knowledge Management (All countries and IGAD)

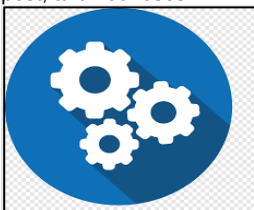
This component will serve as a cornerstone for effective programme management, evaluation and coordination and will include several key aspects. Firstly, it will involve the conduct of baseline surveys in various areas such as energy use and penetration, socio-economic conditions, assessment of the enabling business environment for agribusiness MSMEs, quality of life, agricultural baselines and food

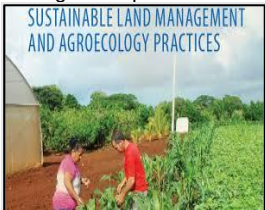
security. These surveys will provide essential data for assessing the impact and progress of the programme. Second, the component will cover project implementation (see Annex 5 for the detailed implementation schedule), management and coordination at country and regional levels to ensure smooth implementation of activities and alignment with programme objectives. Thirdly, it will address procurement and disbursement arrangements for goods, services and consultancy at both project and sub-project levels, ensuring transparency and efficiency in resource allocation (see Annex 10 for a detailed procurement plan). Fourth, the component will focus on the organisation of procurement and implementation at the level of the Executing Entities (EEs) and Project Implementing Units (PIUs) or Project Management Units (PMUs), streamlining processes and minimising bottlenecks. Fifth, it will oversee financial management arrangements at the level of EEs and PIUs, including preparation of audits, financial statements and reporting, ensuring compliance with financial regulations and accountability standards. Finally, the component will include monitoring and evaluation of project and sub-project performance measurement frameworks, using data from the M&E plan (Annex 11) to track progress, identify areas for improvement and ensure achievement of programme objectives. Overall, this component will play a key role in facilitating effective coordination, management and knowledge dissemination throughout the programme implementation process in order to achieve the outputs listed below.

98. **Output 4.1:** Project management and coordination: **Key activities to deliver on this output are:**
99. **Activity 4.1.1:** *Project management and Coordination* – This activity will focus on project management and coordination to ensure efficient implementation and alignment with programme objectives. Staff for the Project Management Unit in Ethiopia, Kenya, Djibouti and IGAD, and the Implementing Agency in Somalia and South Sudan will be recruited through a competitive process to ensure transparency and to select the most qualified person or entity for the task. Financial and procurement audits will be conducted regularly to ensure accountability and transparency, and to ensure that resources are used effectively and in accordance with established AfDB policies. This comprehensive approach to project management will facilitate smooth implementation, enhance oversight, and build confidence in the integrity and effectiveness of the project. Global coordination of the project will be provided by IGAD to ensure synergy in the implementation of the national and regional components.
100. **Activity 4.1.2:** *Technical and institutional capacity building* - This activity will focus on technical and institutional capacity building to enable the effective implementation of the project in the context of conflict, fragility and poor governance. In particular, it will focus on building the capacity of stakeholders to implement the Gender Action Plan, Environmental and Social Management Frameworks (ESMFs) and Environmental and Social Management Plans (ESMPs) (see Annex 6). These efforts will include training and support to stakeholders involved in these processes, including project staff and relevant institutions, to ensure compliance with environmental and social standards. In addition, gender mainstreaming will be emphasised throughout the project, integrating gender considerations into all aspects of design and implementation. A baseline survey will provide essential data for project design and monitoring. Dedicated roles such as project accountant, procurement specialist and project coordination support officer will be established to ensure efficient financial management, procurement processes and overall project coordination. This comprehensive approach to capacity building and staffing will support the successful implementation of the project while promoting environmental and social sustainability and gender equality.
101. **Activity 4.1.3:** *Monitoring and Evaluation* – This activity includes monitoring and evaluation through supervision missions carried out by national and regional focal points (see Annex 11 for detailed programme M&E plan). These missions will ensure that project activities are on track and meeting their objectives. Regular meetings and workshops will provide an opportunity for stakeholders to discuss progress, address challenges and share best practices. The project launch will mark the start of implementation and set the stage for effective coordination and cooperation. A mid-term review will provide a comprehensive assessment of progress and identify areas for adjustment or improvement where necessary. Finally, a project completion report will summarise achievements, lessons learned and recommendations for future projects, providing valuable insights for stakeholders and contributing to ongoing learning and development in the field. The costs of conducting the mid-term review and the final independent review are covered by the AE Fee (see Annex 12).

- 102. Activity 4.1.4: Knowledge Management and Communication** – This activity will focus on knowledge management and communication to facilitate project feasibility and knowledge sharing. This initiative will involve the establishment of robust systems for collecting, organising and disseminating project-related information and best practices. Stakeholders will be kept informed of project developments, milestones and results through effective communication channels. This proactive approach to knowledge management will enhance project viability by ensuring that decision-makers have access to timely and relevant information for informed decision-making. In addition, by promoting knowledge sharing among project team members and stakeholders, this activity will foster a culture of learning and innovation, ultimately improving the overall effectiveness and sustainability of the project.
- 103. Need for the proposed programme intervention:** Both the baseline investment and the GCF interventions have been carefully selected to ensure scalability. This means that fewer resources will be needed to achieve significant economic benefits. In particular, the programme design uses a blended finance structure, introducing innovative financing instruments such as credit lines, matching grants and simple co-financing with selected private sector investors and intermediaries (commercial banks). This approach will help to de-risk and crowd in private sector capital for resilience building in various energy and agriculture, forestry and other land use (AFOLU) value chains in the HOA.
- 104.** BREFOL will provide a credit line of USD 24.35 million to leverage an additional USD 8.72 million from two Public Financial Institutions (PFIs) in Djibouti and Ethiopia through the BDCW. These additional leveraged resources will help increase the PFIs' loan portfolio to agribusiness MSMEs and other programme beneficiaries. BREFOL will also assist farmer organisations and agribusiness MSMEs in preparing business plans for PFIs, and beneficiaries' access to the funds will be subject to a matching fund contribution of 5-10%: depending on the size of the agribusiness MSMEs.
- 105.** In total, the GCF requested amount of USD 24.35 million in credit lines will help to leverage 35.8% more capital, valued at USD 8.72 million as green blended finance.
- 106.** BREFOL will also deploy technological innovations as sub-projects to deliver critical climate-adaptation advisory services, market linkages, crop protocols, gender and rural/extension services through a variety of low- to high-tech channels, depending on the digital development of each country (Table B.3.6). It will also scale up the successful technology dissemination approaches of the **Technologies for African Agricultural Transformation (TAAT)** programme, which will increase the incomes of smallholder farmers and MSMEs in the HOA. Other key innovations to be scaled up include: the approaches of i) the **Global Centre on Adaptation's (GCA) Smallholder Adaptation Accelerator Programme (AAAP)** for digital advisory services, and ii) the **Affirmative Finance Action for Women in Africa (AFAWA) Technical Assistance (TA)**, which provides advisory services to financial institutions to ensure successful implementation of their product portfolios for women, and strengthens the capacity of women entrepreneurs through training to increase business productivity and growth.

Table B.3.6: Some key innovations introduced by BREFOL in the HOA

BREFOL Key Innovations	Component/ Sub-component	Type of Innovation	Purpose	Barrier(s) Addressed
Regional integration to improve livestock trading, and better control of transboundary animal diseases, pest, and zoonoses 	C-1; SC 1-1 & SC 1-3	Economic & Scientific Innovations	- Facilitate cross-border trading in livestock and staple food crops - Facilitate control of transboundary animal diseases and zoonoses, including pest control management - Facilitate cost-effectiveness analysis of integrated pest management technologies to fight against fall armyworm, arboviruses in transhumant cattle, and locust's invasion	Barrier No 5.

Integration of sustainable land management practices 	C-1, SC 1-2	Technical Innovation	- Promotion of resilient agroecological practices among smallholder farmers and pastoralists to support adaption while fostering ecological health synergies and mitigation co-benefits - Improve soil health and fertility - Minimize land conflicts associated with fragmentation and degradation	Barrier No 5
Scaling up and replication of proven agricultural technologies and innovations 	C-1, SC 1-2	Technology Innovation	- Scale up and disseminate proven agricultural technologies and innovations to millions of smallholder farmers - Improve soil health/fertility through improved farming practices - Minimize agricultural transaction costs	Barrier No 3
Integration of the Global Centre on Adaptation (GCA) Smallholder Adaptation Accelerator Program (AAAP) to scale up digital advisory services and markets in climate responsive planning  	C-2, SC 2-1	Digital & Market Innovations	Partner with GCA to: - Scale up the Smallholder Adaptation Accelerator Program (AAAP) - Facilitate access to digital advisory services and markets - Reduce the transaction cost of doing business in fragile and conflicts prone areas	Barriers No. 1, 3 & 4.
Integrate innovative financial instruments to help 'crowd-in' additional private sector finances in fragile states 	C-2, SC 2-1	Financial Innovations	Leveraging on GCF concessional finances to: - Leverage additional private sector investments in fragile states - De-risk the agriculture and other land use sectors in fragile states to attract private sector investors. - Design innovative financial instruments attractive to private investors in fragile states - Operationalize access to 'carbon credits' for smallholder farmers	Barriers No. 1, 3 & 4
Integrate gender-sensitive food security Banks to empower women economically and financially 	C-3; C 3-3	Technological, Planning, and Implementation	- Economically empower women and girls to fight against gender discrimination and income inequality. - Increased empowerment of women to play decision-making role in their households in vulnerable communities. - Increased small-scale or pilot adaptation initiatives adapted and accessible for women to build resilience of vulnerable	Barrier No. 2

			communities in climate resilient activities. - Reduced gender disparities in women's employment in vulnerable communities by sector.	
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107. In total, these measures will contribute towards the transition to a low-carbon climate-resilient sustainable development pathways.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

108. The Accredited Entity (AE): AfDB: The African Development Bank Group (hereinafter referred to as AfDB) is the Accredited Entity (AE). The AfDB is the premier pan-African development institution promoting economic growth and social progress across the continent. It is made up of three institutions, namely 1). The African Development Bank (AfDB), established in 1964, 2). The African Development Fund (ADF), a funding window of the AfDB, established in 1972, and 3) The Nigeria Trust Fund (NTF), established in 1976. It currently has 81 members, comprising 54 independent African countries (regional members) and 27 non-African countries (non-regional members). The AfDB focuses on five development priorities as part of its ten-year strategy (2013-2022) to promote inclusive and green growth. The development priorities, commonly referred to as the 'High 5s', are (i) Light Up and Power Africa, (ii) Feed Africa, (iii) Industrialise Africa, (iv) Integrate Africa, and (v) Improve the Quality of Life for the People of Africa. This programme is designed to support the realisation of the development priorities of "Feeding Africa" and "Improving the Quality of Life for the People of Africa". The AfDB has supported several transformative activities and investments in these priority areas. These include financing rural infrastructure (such as roads, irrigation, electricity, storage facilities and access to markets) to improve agricultural productivity and competitiveness. The AfDB has also supported several initiatives to strengthen agriculture and food security, particularly to improve the livelihoods of rural populations in Regional Member Countries (RMCs). As agricultural production in rural communities is largely dependent on rain-fed irrigation, they are highly vulnerable to extreme weather conditions and uncertainties. It will be unfortunate to pursue an integrated approach to improving agricultural production and food security without appropriate risk-sharing products, access to affordable finance, especially for women, and services to mitigate the impact of climate risks. The AfDB has financed several related projects and programmes in the HOA region. At present, the AfDB's portfolio in the HOA is estimated at about UA 2.4 billion and is distributed as shown in Table B.4.1.

Table B.4.1. AfDB's investment portfolio in the Horn of Africa

Country	Djibouti	Ethiopia	Eritrea	Kenya	Somalia	Sudan	South Sudan	Uganda	Total
Amount (UA Millions)	107.9	1,130.0	68.0	2 380.0	69.0	141.0	9.6	918	2,443.5

109. The AfDB, as the AE, will receive the GCF proceeds in trust and sign the Funding Activity Agreement with the GCF. It will sign loan and grant agreements with the Ministries of Finance of Djibouti, Ethiopia, Kenya, Somalia and South Sudan.

110. Executing Entities (EE): The Executing Entities (EE) are the Government of Djibouti, acting through the Ministry of Commerce, Economy and Finance (MoCEF) and the Ministry of Agriculture, Water, Fisheries and Livestock (MoAWFL); the Government of Ethiopia, acting through the Ministry of Finance, Ethiopia (MoFE) and the Ministry of Agriculture (MoA); the Government of Kenya, acting through the National Treasury and Economic Planning (NTEP), Kenya and the Ministry of Agriculture, Water, Fisheries and Livestock and Cooperatives (MoAWFLC); the Government of Somalia, through its Ministry of Finance, Somalia (MoFS) and Ministry of Livestock, Forestry and Range (MoLFR); the Government of South Sudan, through its Ministry of Finance and Planning, South Sudan (MoFP) and Ministry of Agriculture and Forestry (MoAF); and the Intergovernmental Authority on Development (IGAD), which is a separate legal entity. For Southern Sudan, FAO will act as a third-party implementing agency.

111. National Designated Authorities (NDAs). The Ministry of Housing, Urban Development and Environment (Djibouti), the Environment, Forestry and Climate Change Commission (Ethiopia), the National Treasury (Kenya), the Office of the Prime Minister (Somalia) and the Ministry of Environment and Forestry (South Sudan) are the respective GCF National Designated Authorities (NDAs) in these countries. They will ensure that the programme's activities in each country are aligned with national strategic objectives and priorities. The NDAs will be involved throughout the implementation of the Programme and will receive the Annual Performance Report (APR) detailing the status of project activities and impacts at country level.

- 112. Intergovernmental Authority on Development (IGAD):** The Intergovernmental Authority on Development (IGAD) in Eastern Africa was established in 1996 to replace the Intergovernmental Authority on Drought and Development (IGADD), established in 1986 to mitigate the effects of recurrent severe droughts and other natural disasters that caused widespread famine, environmental degradation and economic hardship in the region. Djibouti, Ethiopia, Kenya, Somalia, Sudan and Uganda took action through the UN to establish the intergovernmental body for development and drought control in their region. Eritrea became the seventh member after gaining independence in 1993, and South Sudan joined IGAD as the eighth member in 2011.
- 113.** In view of the emerging political and socio-economic challenges, the Assembly of Heads of State and Government, meeting in Addis Ababa in April 1995, decided to revitalise IGAD and broaden the areas of cooperation between Member States. The new and revitalised IGAD was launched at the 5th Summit of the IGAD Assembly of Heads of State and Government held in Djibouti on 25-26 November 1996. The Summit endorsed the decision to strengthen regional cooperation in three priority areas: food security and environmental protection; economic cooperation, regional integration and social development; and peace and security.
- 114.** IGAD is comprised of four hierarchical policy organs as follows:
- 115.** THE ASSEMBLY OF HEADS OF STATE AND GOVERNMENT is the supreme policy making and regulatory organ of the IGAD. It determines the objectives, guidelines and programmes for IGAD and meets once a year. A Chairman is elected from the Member States in rotation.
- 116.** THE COUNCIL OF MINISTERS is composed of the Ministers of Foreign Affairs and one other Focal Minister designated by each member state. The Council formulates policy, approves the work programme and annual budget of the Secretariat during its biannual sessions.
- 117.** THE COMMITTEE OF AMBASSADORS is comprised of IGAD Member States' Ambassadors or Plenipotentiaries accredited to the country of IGAD Headquarters. It convenes as often as the need arises to advise and guide the Executive Secretary.
- 118.** THE SECRETARIAT is headed by an Executive Secretary appointed by the Assembly of Heads of State and Government for a term of four years, renewable once. The Secretariat assists Member States in formulating regional projects in the priority areas, facilitates the coordination and harmonisation of development policies, mobilises resources to implement regional projects and programmes approved by the Council and reinforces national infrastructures necessary for implementing regional projects and policies.
- 119.** The Executive Secretary is assisted by Divisions responsible for:
- Agriculture and Environment;
 - Economic Cooperation and Regional Integration;
 - Health Social Development;
 - Peace and Security;
 - Administration and Finance; and
 - Planning Coordination and Partnerships
- 120.** In addition to the above-mentioned departments at the headquarters in Djibouti, IGAD has a number of specialised institutions and programmes hosted by other member states. These include the IGAD Conflict Early Warning and Response Mechanism (CEWARN), the IGAD Security Sector Programme (ISSP), the IGAD Centre for Pastoral Area and Livestock Development (ICPALD), the IGAD Climate Prediction and Applications Centre (ICPAC) and the IGAD Regional Aids Programme (IRAPP). In addition, due to the devastating drought that hit the IGAD region in 2010-2011, affecting more than 13 million people and exacerbating food insecurity to famine levels, IGAD launched the IDDRSI to end drought emergencies through the introduction of innovative sustainable development strategies, policies and programmes at member state, transboundary and regional levels aimed at building resilience to future climatic and economic shocks.
- 121.** At the country level, the following Ministries will act on behalf of the various governments as the EEs:
- The Government of Djibouti, acting through its Ministry of Commerce Economics and Finance (MoCEF), and Ministry of Agriculture, Water, Fisheries and Livestock (MoAWFL);
 - The Government of Ethiopia, acting through its Ministry of Finance, Ethiopia (MoFE) and Ministry of Agriculture, Livestock and Fishery Resources (MoALFR);
 - The Government of Kenya, acting through its National Treasury and Economic Planning (NTEP) and Ministry of Agriculture, Water, Fisheries and Livestock, and Cooperatives (MoAWFLC);
 - The Government of Somalia, acting through its Ministry of Finance, Somalia (MoFS) and Ministry of Livestock, Forestry and Range (MoLFR);
 - The Government of South Sudan, acting through its Ministry of Finance and Planning, South Sudan (MoFP) and Ministry of Agriculture and Forestry (MoAF);

- The Intergovernmental Authority on Development (IGAD).

122. Public Financial Institutions (PFIs): These are National Development Institutions (NDIs) owned and operated by the government or a public body. They differ from private financial institutions in terms of ownership (private financial institutions are owned and operated by private individuals or entities).

123. Program Implementation: The overall coordination of the programme, in particular of the regional activities, will be carried out by IGAD as the regional implementing agency, working in close liaison with the five national governments in the HoA region. While IGAD will be responsible for the regional coordination and management of the programme, the institutions of the IGAD Member States will be responsible for the implementation of the activities at national level. There will be a legally binding financing agreement (protocol of agreement) between AfDB and IGAD based on the grant amount and associated conditions. The IGAD Executive Secretariat (IES) and its specialised institutions, including ICPAC (IGAD Climate Prediction and Applications Centre) and ICPALD (IGAD Centre for Pastoral and Livestock Development), will provide the necessary technical backstopping for regional activities.

124. The IGAD Secretariat, through the IDDRSI Platform Coordination Unit (IPCU), will be responsible for coordinating the implementation of the Regional Programme. The Regional Programme Coordination Unit, established by the IGAD Secretariat, will be responsible for coordinating the programme. Under the supervision of a Regional Programme Steering Committee (RPSC), the IPCU will provide coordination support and overall guidance to ensure that the goals and objectives of the regional programme component are achieved. The mandates of the RPSC include the approval of programme budgets, programme activity reports and annual accounts related to the implementation of the overall programme. It will be chaired by the IES in accordance with the applicable instrument and its secretariat will be provided by a Regional IDDRSI Coordinator. During the implementation of specific regional activities under the programme, ICPAC and ICPALD will each provide technical assistance support within the framework and under the terms of a Memorandum of Understanding with the IGAD Secretariat. The MoUs will be signed by IGAD through ICPALD with relevant ministries or agencies of the participating countries sharing the border. The content of the MoUs will include the 3 IGAD border clusters for better control of transboundary animal diseases and zoonoses in border areas. It will also explore the possibility of establishing cross-border veterinary laboratories in the 3 IGAD border clusters and enhance the competitiveness of livestock value chains in the IGAD region by ensuring that stakeholders comply with the Agreement on the Application of Sanitary and Phytosanitary Measures in line with market research under the African Continental Free Trade Agreement. The Regional Programme Steering Committee (RPSC), which will include representatives from the RPCU and the National Programme Coordination Unit (NPCU), to ensure synergy and harmonisation in the implementation of the programme at country and regional levels.

125. At the national levels, the Executing Entities (EEs) are the Government of Djibouti, acting through The Ministry of Commerce Economics and Finance (responsible for the signing of loan and grant agreements with AfDB) and the Ministry of Agriculture, Water, Fisheries and Livestock (responsible for project implementation in Djibouti); the Government of Ethiopia, acting through The Ministry of Finance (responsible for the signing of the loan and grant agreements with AfDB) and the Ministry of Agriculture, Livestock and Fishery Resources (responsible for project implementation in Ethiopia); the Government of Kenya, acting through the Ministry of Finance (responsible for the signing of Loan and Grant Agreements with AfDB), and the Ministry of Agriculture, Water, Fisheries and Livestock, and Cooperatives (responsible for project implementation in Kenya); the Government of Somalia, acting through the Ministry of Finance (responsible for the signing of loan and grant agreements with AfDB) and the Ministry of Agriculture and Irrigation (in charge of project implementation in Somalia); and the Government of South Sudan, acting through the of Finance and Economic Planning (responsible for the signing of loan and grant agreements with AfDB) and the Ministry of Agriculture and Forestry (responsible for project implementation in South Sudan).

126. In line with the AfDB's financing of sovereign operations in member countries, each ministry responsible for project/programme implementation is required by law to establish a dedicated Project Management Unit (PMU) as part of the loan and grant "agreements" with the AfDB. The PMU must remain operational throughout the life of the programme. Among other things, the PMU coordinates directly with the EEs and project stakeholders and is responsible for supporting the implementation of day-to-day activities, the technical quality of project outputs, and effective stakeholder engagement. The PMU normally consists of a project coordinator, an administration, finance and accounting manager, an administration and finance assistant, an accountant, a procurement manager, a procurement assistant, a monitoring and evaluation manager, an E&S safeguards expert, an infrastructure manager, a private sector development and entrepreneurship expert, an agricultural value chain development and rural finance assistant, a communications assistant and support staff. Within the PMU, the

staff recruited under the project and those seconded by the Government will together form a project implementation team that will ensure sound and effective project implementation.

127. The PMU receives technical and advisory services from a **National Project Steering Committee (NPSC)** chaired by the State Minister of Agriculture or equivalent. The establishment of the NPSC or equivalent is a mandatory condition of the Loan and Grant Agreements signed between each country's Ministry of Finance and the AfDB. As part of the "Other Conditions" in the legal instruments for the programme - "Agreements", each country is required to provide evidence of the establishment of a National Project Steering Committee (NPSC) with a composition and terms of reference acceptable to the AfDB. The NPSC is only a steering committee to oversee the implementation of the programme in accordance with the Financing Agreement. There will be no legal agreement between the EE and the NPSC. The NPSC, which will meet at least twice a year, will provide implementation oversight and policy guidance, and ensure inter-ministerial coordination, harmonisation and support for alignment among donors providing related resilience-building programmes. The NPSC will also be mandated to review the progress of the project on a semi-annual basis and assess the work plans on an annual basis to ensure effective implementation of the project. It is also responsible for approving the country's budgets, activity reports and annual accounts. In addition, the NPSC will liaise with the IDDRSI Platform Coordination Unit (PCU) to ensure that all resilience-building activities implemented at country level receive the necessary technical guidance, oversight and coordination. The Project Coordinator will be the Secretary of the NPSC and the focal point for all coordination with the IGAD National Programme Coordination Team, including reporting, as shown in Figure B.4.1 below. The allocation of EEs to each activity is shown in **Appendix 3**.

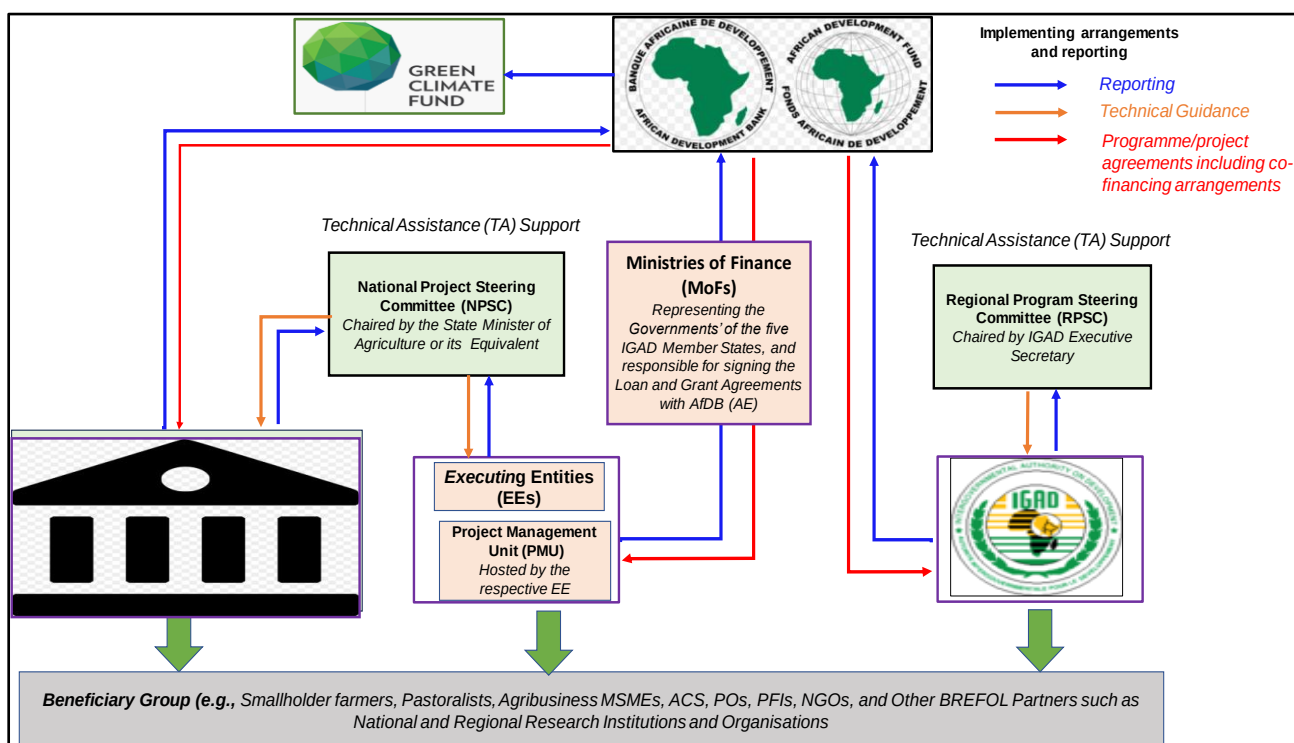


Figure B.4.1: BREFOL Implementation Arrangements and reporting. See **Appendix 4** for summary on financing source and assignments of Executing Entities.

128. Flow of funds: As shown in Figure B.4.2, the GCF resources held in trust will be received by the AfDB after the signing of the **Funded Activity Agreement (FAA)** and the fulfilment of all requirements for the execution and effectiveness of the FAA, as well as for disbursement according to the schedule. The AfDB will sign the Loan Agreement (2) and the Grant Agreement (6) with the respective Ministries of Finance (MoFs) of Djibouti, Ethiopia, Kenya, Somalia, South Sudan and the IGAD Secretariat in accordance with the AfDB's procedures and policies for financing sovereign operations and taking into account the terms and conditions of the Term Sheet for the BREFOL programme with the GCF. The Ministry of Finance of each country will disburse GCF non-repayable income for activities related to 1) sustainable agro-pastoral land management (C 1-2); 2) access to climate-smart innovations and technologies (C 1-3); 3) access to digital extension services and markets (C 2-1); and 4) improving access to climate services infrastructure (C 3-1), to the national executing entities (EEs) for the above interventions, in accordance with AfDB policies and procedures, such as for procurement and disbursement, ESS and gender, among others. The national executing agencies, through their project management units

(PMUs), will procure goods and services in accordance with AfDB procurement procedures and policies for the implementation of the programme.

- 129.** For the GCF repayable funds (i.e. **BREFOL Direct Credit Window (BDCW)** and **BREFOL Direct Insurance Credit Window (BDICW)**), the Ministry of Finance, with the written approval of the EE for each country (after the PFIs have met and complied with all eligibility criteria deemed satisfactory by the EE and the AE)⁵⁴, will sign loan agreements with the selected public financial institutions (PFIs) in Djibouti and Ethiopia and disburse the proceeds to the selected PFIs. Selected local PFIs will further sign loan agreements with selected beneficiaries (after the programme beneficiaries have met and fulfilled all eligibility criteria deemed satisfactory by the EE, AE and PFI) as shown in Figure B.4.2.

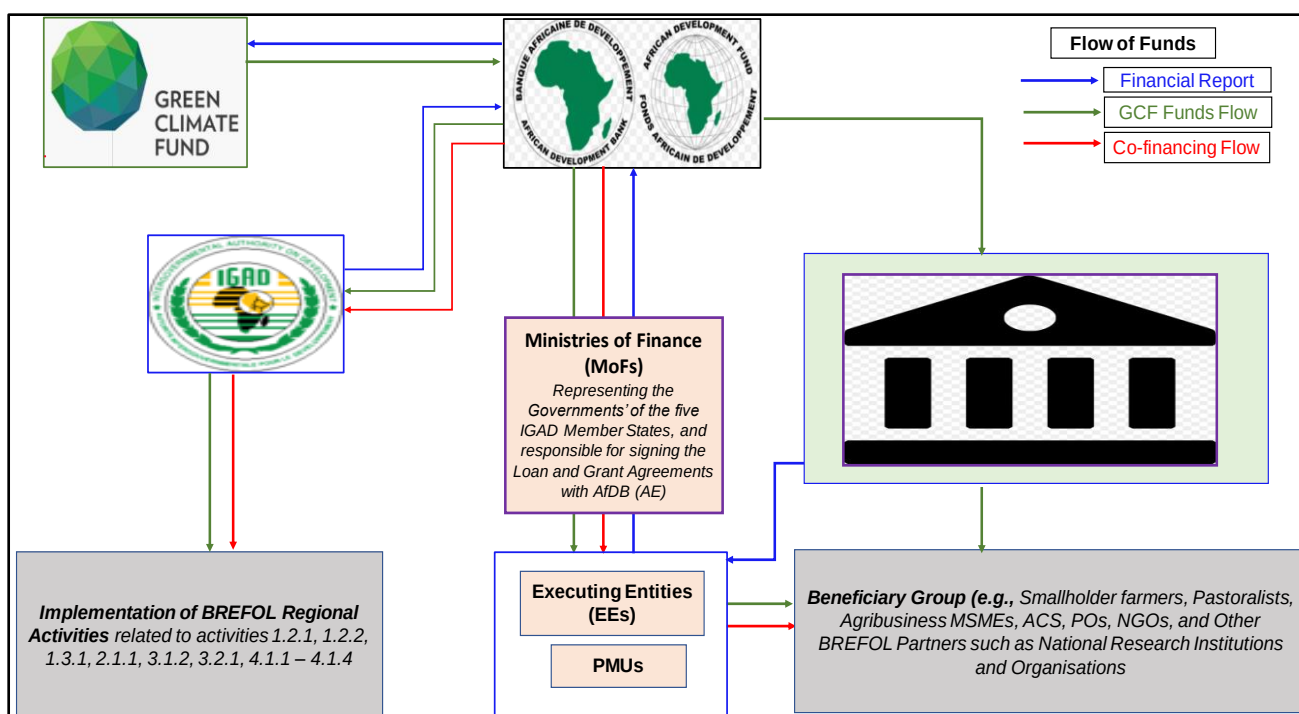


Figure B.4.2: BREFOL Flow of Funds

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

- 130.** The Horn of Africa is one of the most disaster-prone regions in the world, suffering from recurrent food crises caused by extreme poverty and food insecurity (**root cause 1**), climate change and environmental degradation (**root cause 2**), fragility (**root cause 3**) and conflict (**root cause 4**). Unfortunately, these countries' high levels of debt severely constrain public spending to address most of the key challenges facing the region. In addition, due to the fragile nature of the region, potential private sector investors that could drive green/circular economy jobs and sustainable wealth creation in the region are highly risk averse, putting additional pressure on already limited fiscal space. The end result is widening inequality, violence, social unrest, political conflicts over land use and resources, environmental migration and exposure to vicious poverty traps.
- 131.** In most cases, those typically affected are smallholder farmers, pastoralists and other small-scale producers and actors along various value chains, who are typically constrained by resource limitations (**Barriers Nos. 1 & 3**). More worryingly, the recent Covid-19 economic downturn has further affected investment activity in the region and target countries. As a result, institutional investors - particularly those interested in smallholder agriculture - are becoming more risk averse.
- 132.** The resources requested from the GCF will therefore play a critical role in supporting farmers and pastoralists to increase agropastoral production, incentivising and de-risking public and private institutional investment, and creating local and regional green business opportunities that will be

⁵⁴ The AfDB through its Financial Sector Department will support the respective PMU in carrying out due diligence of selected PFIs, in accordance with its mandate to put in place well-functioning financial systems that mobilize and allocate savings, supply the credit needs of economic agents, and allocate resources more efficiently while reducing intermediation costs. It will equally support the selected PFIs in the screening of sub-project loan applications to ensure strict adherence to BREFOL objectives.

attractive to private sector investment in the HoA. A key lesson from other ongoing initiatives in the countries and region is the centrality of green blended finance in catalysing a shift towards climate-resilient and low-emission development pathways for these countries. Without the GCF, leveraging additional private sector finance to scale up the priority adaptation actions mentioned above will certainly be difficult due to the rising public debt situation in these countries. This has severely constrained the fiscal space needed to implement relevant adaptation measures, which are essential for building the resilience of smallholder farmers and pastoralists. In addition, the fragile nature of some of these countries further discourages the private sector from investing in critical sectors that are essential for triggering high transformative impacts within the HoA.

- 133.** The key is to de-risk private sector finance. The GCF is an important player in this area due to the highly concessional nature of its funding instruments. In addition, GCF funding typically allows for sufficient grace periods and repayment periods that are equally attractive to the private sector. This is essentially what is needed to catalyse the priority adaptation inventions proposed in the *IDDRSI strategy* beyond the GCF-funded activities.

Rationale and level of concessionality of the GCF financial instrument

- 134.** Concessional resources from the GCF to BREFOL will help overcome critical barriers and risks to climate change adaptation and mitigation in the HoA region. In particular, the existence of large financing gaps (**Barrier #1**) and lack of access to finance for smallholder farmers and pastoralists (**Barrier #2**), as outlined in Section B.2, prevent the region from investing in and implementing the large-scale adaptation and mitigation programmes needed to build resilience in the region. Without additional funding from development partners and the private sector, as highlighted in the 5th interim outcome of the *African Union Draft Africa Climate Change Strategy - DACCS (2020 - 2030)*, scaling up the region's adaptation and mitigation efforts may be impossible. The mobilisation of GCF resources will be critical in overcoming these barriers to building resilience in the region.
- 135.** The private sector also has a critical role to play in scaling up adaptation finance to build resilience in the region. Unfortunately, due to the highly fragile nature of the region (Risk #2), most private sector investors are risk averse given the high-risk profile associated with investing in the region. In the latest 2022 Fragile State Index (FSI), Ethiopia, Sudan, South Sudan, Kenya, Somalia and Djibouti ranked 11th, 8th, 4th, 32nd, 2nd and 49th respectively, with four out of the five countries (Somalia, South Sudan and Ethiopia) classified as fragile. Thus, the requested concessional green financing from the GCF (**USD 90.71 million**), together with the *AfDB's Strategic Private Sector Investment for Fragile States Project*⁵⁵, will help to cascade effective win-win private-private and private-public partnerships for a strong green/circular economy in the HoA region.
- 136.** In addition, the high indebtedness of all five IGAD countries, limited access to credit and other financial instruments, and the existence of multiple poverty traps are significant constraints to adaptation finance in the region. Short funding horizons also make it difficult to fully integrate adaptation measures. Ethiopia's financing needs are significant, given its large physical and social infrastructure needs and low tax-to-GDP ratio, which is expected to average 10% from 2017 to 2022. The primary deficit plus debt service is estimated at close to 4% of GDP. As of June 2022, total public debt stood at around 57% of GDP, of which just over half was external. In Kenya, public debt increased from 61% of GDP in 2021 to 72% of GDP in 2022, mainly due to public investment in infrastructure, debt management challenges and the COVID-19 crisis. Kenya is now at high risk of debt distress according to the International Monetary Fund. The same goes for Somalia, where the IMF's 2022 debt sustainability analysis shows that even with interim debt relief, Somalia will remain in debt distress, with debt still at around 43.3% of GDP in 2024: above the 30% sustainability threshold. South Sudan's debt risk rating improved from debt distress to high risk in October 2022 due to the restructuring of the country's commercial debt with Qatar National Bank, which accounts for 46% of external debt. The debt restructuring and the clearance of arrears owed to the country also helped to reduce external debt from 38% of GDP in 2021 to an estimated 28.3% of GDP in 2022. Commercial loans accounted for 81% of total external debt in June 2022, followed by multilateral (8%) and bilateral (11%) loans. Based on the above analysis, 50% grant and 50% loan financing would be the best GCF instrument required to make

⁵⁵ AfDB (2022). <https://www.afdb.org/en/contacts>.

the BREFOL programme investment in the HoA viable. This is highly justifiable given that the baseline investment is split as follows: 76.6% grant financing and 23.4% loan financing.

- 137. A critical mass of GCF investment will crowd-in additional private sector co-finance at scale (35.81%), and equally help to train a critical mass of Agribusiness MSMEs (4395) in accessing climate finance**
- 138.** BREFOL seeks a mix of grant and loan financing from the GCF (i.e. \$60.30 million senior loan and \$90.70 million grant financing) to incentivise and de-risk additional private sector financing (13.2% of the requested GCF loan amount) in critical areas in various energy and AFOLU value chains for resilience building in the HoA, with high mitigation co-benefits (reduced emissions and enhanced carbon sinks totalling 14.1 MtCO₂).
- 139.** The requested amount will be channelled through a credit line (**\$24.35 million**) to help leverage an additional **\$6.48 million** in green blended finance from two LPFIs in Djibouti and Ethiopia, representing a **26.1%** increase in co-financing for the total credit line amount of \$24.35 million.
- 140.** The beneficiaries (smallholders, pastoralists, farmers' organisations and agribusiness MSMEs) will also provide matching funds of 5-10%, depending on the size of the enterprise, for a total of **\$2.25 million**.
- 141.** In total, the requested GCF credit line of **\$24.53 million** will help leverage an additional **\$8.72 million in green blended finance (over 35.8% of the GCF credit line investment)** and approximately **13.2%** of the total GCF investment in the programme (**\$151 million**), apart from the baseline investment of **\$184.30 million**.
- 142.** Finally, a critical mass of about 4,395 cooperatives and SMEs trained through GCF resources will be able to mobilise additional resources from other commercial banks by acquiring skills in business proposal development and credit access. Furthermore, we expect this critical mass of trained cooperatives and SMEs to serve as trainers of trainers (ToT) for other cooperatives and SMEs interested in accessing credit from commercial banks and other private financial institutions in these countries.

B.6. Exit strategy (max. 500 words, approximately 1 page)

- 143.** First and foremost, the use of different financial instruments in the programme is critical to ensure the continuity of the programme beyond GCF funding. As discussed in **section B.5**, these financial instruments will not only help to leverage additional resources from the private sector but will also raise awareness among PFIs and other private sector investors of the growing financial benefits associated with integrating climate change and green growth instruments into their business strategies and operations. As discussed under **barrier #2**, the private sector offers various types of private finance such as debt, equity, guarantees, simple co-financing and insurance products, etc., which have great potential to bridge the adaptation financing gap in Africa. Unfortunately, they often lack the incentives and other enabling instruments to create a self-sustaining green/circular economy, sustainable wealth creation and jobs in fragile states. Creating new profit opportunities for private capital from investments that address climate change, and the green/circular economy opens the doors to this great source of capital in the HOA now and in the future. Thus, concessional green financing from the GCF, together with the **AfDB's Strategic Private Sector Investment for Fragile States project**, can help cascade effective win-win private-private and private-public partnerships to create a self-sustaining green/circular economy, sustainable wealth creation and green jobs in the HoA region.
- 144.** Secondly, the programme integrates rigorous training and capacity building initiatives for 1) inclusive MSMEs and cooperatives with access to climate finance; 2) agribusiness MSMEs on renewable energy and AVC development; 3) women and youth on entrepreneurial skills in renewable energy technologies and AVC development, including access to credit lines; 4) climate change policy mainstreaming; 5) climate change, gender mainstreaming and conflict management, among many others. It is expected that this critical mass of trained beneficiaries will help to sustain the programme beyond the initial investment (**barrier #4**).
- 145.** Third, the AFAWA Technical Assistance Programme (ATAP) will contribute to women's economic empowerment and improve their creditworthiness. It will also strengthen the capacity of LPFIs to understand and respond appropriately to women entrepreneurs, who often lack the financial and

business acumen to respond to the needs of financial institutions. This will increase women's access to affordable finance, which is currently estimated to have a financing gap of \$42 billion across business value chains, including \$15.6 billion in agriculture alone. BREFOL, through the AGGP, will provide women with access to affordable finance to undertake activities along various value chains, leading to women's empowerment both financially and economically (**addressing barrier #5**). This will facilitate sustainable wealth creation and green jobs, especially for young girls and youth, who are often dependent on women. Thus, financial and economic empowerment of women is an important exit strategy for BREFOL in the HoA region.

- 146.** Fourth, ensuring that each eligible sub-loan beneficiary under BREFOL is subject to a 5-10% matching fund contribution to access the **BREFOL Direct Credit Window (BDCW)** with the two selected PFIs is very innovative for the long-term sustainability of the programme even after the GCF investment. Similarly, ensuring that the two participating PFIs in Djibouti and Ethiopia provide simple co-financing (20%) and matching funds (50%) to access GCF concessional resources will help to "crowd in" additional private sector resources (USD 6.48 million). The role of the private sector in increasing green finance and creating future green jobs can't be overstated. These measures, built into BREFOL, will ensure that additional private capital is injected into the programme beyond the GCF's initial investment.
- 147.** Fifth, BREFOL will enable smallholder farmers, agribusiness MSMEs, FBAs, ACSs, WABEs and POs currently excluded from financial markets to access digital advisory services and markets (financial literacy, bank accounts and transactions), enabling them to improve their production and incomes and thus compete locally and internationally. This is likely to be sustainable as they will receive formal training in business plan development prior to sub-project implementation.
- 148.** Finally, BREFOL will also use technological innovations to deliver critical climate adaptation advice, market linkages, crop protocols, gender and rural extension through a variety of low to high tech channels, depending on the digital development of each country. It will also replicate and scale successful TAAT COMPAC technologies and innovations that can increase the incomes of smallholder farmers and agribusiness MSMEs in the HoA region. Other innovations to be scaled up include: the Global Centre on Adaptation (GCA) Smallholder Adaptation Accelerator Programme (AAP), and the AFAWA TAP for financial and economic empowerment of women in the HoA. Taken together, these activities will contribute to the transition to low-carbon, climate-resilient, sustainable development pathways in the HoA.

C. FINANCING INFORMATION							
C.1. Total financing							
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount				Currency		
	151.00				million USD (\$)		
GCF financial instrument ⁵⁶	Amount	Tenor	Grace period	Pricing			
(i) Senior loans	60.30	30 years	10 years	0.5 %			
(vi) Grants	90.70						
(b) Co-financing information	Total amount				Currency		
	184.29 ⁵⁷				million USD (\$)million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority	
African Development Fund (ADF)	Senior Loans	\$42.90	million USD (\$)	30 years 10 years	0.5%	seniorsenior	
African Development Fund (ADF)	Grant	\$123.54	million USD (\$)				
Food and Agriculture Organization (FAO)	Grant	\$0.69	million USD (\$)				
Governments' Contributions	In kind	\$16.03	million USD (\$)				
Beneficiaries' Contribution	In kind	\$1.82	million USD (\$)				
(c) Total financing (c) = (a)+(b)	Amount				Currency		
	335.30				million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	Expected contributions from private sector financing The programme will leverage additional private sector blended finance to support its initiatives, including a simple co-financing contribution of \$5.49 from two development banks in Djibouti and Ethiopia. Smallholders and micro-enterprises will contribute a 5% matching fund, while small and medium-sized enterprises will contribute 10%, for a total of \$1.80. This combined effort results in a total green blended finance of \$8.72, demonstrating a collaborative approach to sustainable development.						
C.2. Financing by component							
149. Please refer to Annex 4 for the detailed program budget plan for BREFOL.							
Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount	Financial Instrument	Amount	Financial Instrument	Name of Institutions
			million USD (\$)		million USD (\$)		
C 1: Strengthening Pastoral and Agropastoral Production Systems' Resilience to Climate Change	Output 1.1: Climate resilient infrastructure built/rehabilitated	\$127,13	\$0,00	Grant	\$69,95	Grant	ADF/Gov't
			\$29,15	Senior Loan	\$28,03	Senior Loan	ADF
	Output 1.2: Improved sustainable land management practices	\$27,61	\$16,37	Grant	\$9,06	Grant	ADF
			\$0,00	Senior Loan	\$2,18	Senior Loan	ADF
	Output 1.3: Increased adoption of climate-smart innovations	\$32,97	\$23,56	Grant	\$8,60	Grant	ADF
			\$0,00	Senior Loan	\$0,80	Senior Loan	ADF

⁵⁶ Please refer to Annex 14 (Term sheet) for additional details on the GCF financing instrument

⁵⁷ Please refer to Annex 13 for the co-financing commitment letter

Sub-total Component 1		\$187,70	\$69,08		\$118,62		
C 2: Supporting low carbon and resilient Development of Agribusiness	Output 2.1: Increased access to digital advisory services and markets	\$39,24	\$19,46	Grant	\$12,62	Grant	ADF
			\$5,00	Senior Loan	\$2,16	Senior Loan	ADF
	Output 2.2: Developed inclusive MSMEs and cooperatives with access to climate financing	\$52,75	\$13,95	Grant	\$9,82	Grant	ADF
			\$26,15	Senior Loan	\$2,84	Senior Loan	ADF
Sub-total Component 1		\$91,99	\$64,56		\$27,43		
C 3: Strengthening Agropastoral Communities' Capacity to Adapt to Climate Change	Output 3.1: Enhanced access to climate information and early warning systems	\$8,76	\$3,58	Grant	\$4,05	Grant	ADF
			\$0,00	Senior Loan	\$1,12	Senior Loan	ADF
	Output 3.2: Strengthened regulatory and policy frameworks and institutional capacity for climate resilience.	\$17,90	\$5,81	Grant	\$9,79	Grant	ADF
			\$0,00	Senior Loan	\$2,30	Senior Loan	
Sub-total Component 3		\$26,66	\$9,39		\$17,27		
C 4: Program Coordination and Knowledge Management	Output 4.1: Project management and coordination	\$28,95	\$7,98	Grant	\$16,82	Grant	ADF
					\$0.69		FAO
			\$0,00	Senior Loan	\$3,46	Senior Loan	ADF
Sub-total Component 4		\$28,95	\$7,98		\$20,97		
Indicative total cost (USD)		\$335,30	\$151,00		\$184,29	-	-

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities? Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer? Yes ☒ No ☐

- 150.** Several categories of capacity development and training activities will be undertaken, including:
- Production, dissemination and application of agro-meteorological information;
 - Translation of early warning information into guidelines and warnings for government agencies, emergency services, aid agencies, agricultural NGOs, extension services, farmer organisations and the public.
 - Application of TAAT COMPAC technologies and innovations, Global Centre on Adaptation (GCA) and Smallholder Adaptation Accelerator Programme (AAAP).
 - Climate-resilient and high-quality gender-responsive business planning and development;
 - Promotion and adoption of climate-resilient and high-value practices and technologies;
 - Preparation of agribusiness development plans; and,
 - Accessing climate finance for local PFIs.
- 151.** Each category targets a different group of beneficiaries, including:
- Meteorological services, national and regional institutions and insurance providers;
 - Farmers and other local value chain actors, smallholder farmers, farmer-based associations (FBAs), agricultural cooperative societies (ACSs), agro-dealers (ADs), producer organisations (POs), micro, small and medium-sized enterprises (MSMEs), commercial banks (CBs) and microfinance institutions (MFIs);
 - Extension services, including extension officers from ministries of agriculture, water resources, forestry departments, district administrations, commune and village extension officers, and private extension providers, including NGOs;
 - Women-led agribusinesses and MSMEs; and,
 - Ministries and their sub-national departments.

152. Almost 20% of the requested GCF grant will be used to support capacity development of these beneficiary groups. The remaining 80% will directly support the construction of infrastructure and the acquisition and transfer of climate-resilient technologies and innovations to the beneficiary groups through the capacity development activities outlined above.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

153. BREFOL will contribute to **GCF Impact level E.2: Scalability, replicability and sustainability**. In terms of scalability, there are key innovations introduced by BREFOL with very high potential for scalability (see e.g. Table B.3.5). These include the **Global Centre on Adaptation's (GCA) Smallholder Adaptation Accelerator Programme (AAAP)** through digital extension services, the **Technologies for African Agricultural Transformation (TAAT)** programme through its innovative technology dissemination approach, and **AFAWA's** approach to women's empowerment. Similarly, in terms of replicability, the above novelty introduced by BREFOL comes with significant economic benefits even with less resource commitment. It's therefore very attractive for replication in other sectors, markets, geographical regions or countries. In terms of sustainability, four main outcomes are expected from the initial investments, which will ensure the sustainability of the programme.

154. Mitigation co-benefits: The programme is expected to install a total of 4.8 MW of solar PV for solar irrigation, solar lighting and processing systems, and to generate a total of 3,890 m³ (~0.12 MW) of methane for waste-to-energy (Table D.1.1), thereby reducing fuelwood and fossil fuel consumption. Taking into account the carbon sequestration potential of the 102,000 ha of rehabilitated rangelands and 29,800 ha of degraded lands/forests rehabilitated and managed under agroforestry, Table D.1.1 summarises the mitigation potential of the programme from two GCF outcome areas: 1) MRA1-Energy Production and Access; and 2) MRA4-Forestry and Land Use (see Annexes 22a-c for the underlying assumptions and calculations).

155. Table D.1.1. Summary of mitigation potentials of the program from two GCF Results Areas: Energy generations and access and forestry and landuse – Please refer to Annex 22a, 22b, 22c for detailed calculation and methodological note.

GHG reduction activity	Capacity / Unit	Yearly GHG emissions tCO ₂	GHG emissions avoided at mid term tCO ₂	GHG emissions avoided by end of programme tCO ₂	GHG emissions avoided over lifetime tCO ₂
Off grid solar PV for irrigation Systems, Lighting and Processing Systems - MW	4.8	6,046	9,686	26,614	135,434
Installation of Biodigester units - m ³	3,890	6,737	10,105	30,316	90,949
Rehabilitation of rangelands ha	102,000	320,405	274,287	1,226,939	8,010,125
Land under agroforestry - ha	29,800	236,513	183,941	806,312	5,912,829
Total		569,701	478,025	2,090,182	14,149,337

156. To ensure that mitigation results are properly recorded during the implementation period and throughout the life of the programme, the necessary resources have been allocated to track progress towards the mitigation targets and policies outlined in the Nationally Determined Contributions (NDCs) of the five countries. The project will support the operationalisation of a project-specific Monitoring, Reporting and Verification (MRV) mechanism during the project implementation period and will support the necessary institutions to collect the required information after the project period. For the PV and biodigester units, the responsibility for reporting on the operation of the installed systems will lie with the private contractors to whom

the contract will be awarded. As indicated in Annex 22b, the price per watt for the PV system and the price per m3 for the biodigester will also include the responsibility for monitoring the system parameters (installed capacity, energy produced and consumed) and reporting them to the Project Implementation Units, which will be trained in reporting. For agroforestry and rangeland rehabilitation, the PIUs will be trained to use the FAO EX Ante tool (version 9 or latest) to track the amount of GHG emissions avoided during the reporting period.

157. To ensure that MRV is in place for the project after the end of the implementation programme (after 7 years), capacity building will be provided during the project implementation period to set up a specific MRV mechanism for the project indicators (see Annex 4 - Budget under Component 3 and Component 4).

158. BREFOL will significantly improve access to and use of climate risk information and agro-advisory services to guide multiple project interventions to improve agropastoral land management. This will directly benefit more than 1,800,000 smallholder farmers, pastoralists and vulnerable communities in the HoA. The programme will also construct/rehabilitate some 450 water mobilisation infrastructures, including the operationalisation of 3 cross-border Memoranda of Understanding (MoU) to better control transboundary animal diseases and zoonoses in border areas. Other key achievements of the programme include Establishment of 3 cross-border veterinary laboratories to facilitate cost-benefit analysis of integrated pest management technologies to control fall armyworm, arboviruses in transhumant cattle and locust invasion; Restoration of degraded land by bringing about 102,000 ha of degraded agro-pastoral rangeland/pasture under sustainable land management practices; Providing about 1,200,000 smallholder farmers, pastoralists and communities with access to digital and non-digital climate-resilient innovations and technologies; facilitating carbon trading among smallholder farmers, pastoralists and communities, resulting in additional adaptation finance; and strengthening operational capacity to build climate resilience at community, national and regional levels.

159. BREFOL is expected to directly benefit 1.95 **million** (GCF plus baseline investments) smallholder farmers and pastoralists, who will need to be organised into Farm-based Associated (FBAs), Women in Agribusiness Enterprises (WABEs), Youths in Agribusiness Enterprises (YABEs), Agricultural Cooperative Societies (ACSS), Producer Organisations, Agro-Dealers and Agribusiness MSMEs. It will also indirectly benefit over 7.14 million people (GCF plus baseline investments), of which 50% will be women and 50% youth. Other beneficiaries include 10,495 women-led MSMEs/FBAs, youth-led MSMEs/FBAs ACS and other agribusiness MSMEs, and 2 PFIs in Djibouti and Ethiopia. In addition, around 7 Technologies for African Agricultural Transformation (TAAT) COMPACT technologies and innovative solutions will be transferred to support climate resilient, low emission development during the implementation of the programme. The breakdown of beneficiaries is summarised in Table D.1.2.

160. Table D.1.2: Summary of BREFOL Beneficiaries (see Annex 23 for details)

Programme Beneficiaries (Direct Plus Indirect)								
Region	Direct (ADF)	Direct (GCF)	Total D (ADF+GCF)	Indirect (ADF+GCF)	Total(D+I)	Male	Females	% of Total Population
IGAD Region	1,849,451.0	2,835,280.0	4,684,731.0	15,548,382.7	20,233,113.7	10,116,556.8	10,116,556.8	8.3%
% of Total Population	0.8%	1.2%	1.9%	6.4%	8.3%	50.0%	50.0%	8.3%

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

161. Potential for scaling-up and replication: As shown in Table B.3.4, a key feature of the programme is to increase private sector participation in climate change adaptation strategies in five IGAD member states in the Horn of Africa (HoA) region. This will be done through financial innovations that provide the right incentives and enable instruments that are attractive to private

sector investors, such as private-public partnerships (3Ps) and public-private-producer partnerships (4Ps). These models bring together combined expertise and resources to accelerate the delivery of results and increase the impact of green investment opportunities, wealth creation and green jobs, especially for the most marginalised (women, girls and youth). These models are strongly aligned with the GCF's goal of funding initiatives that catalyse climate impact beyond a one-off investment. Similarly, the vision of the **BREFOL Direct Credit Window (BDCW)** and **BREFOL Direct Insurance Credit Window (BDICW)** models is to use existing climate funds (GCF resources) to create an enabling environment that is attractive to private sector investors, such as agribusiness MSMEs and weather-based insurance promoters, with a vested interest in green/circular business opportunities.

162. With the initial investment of GCF funds and co-financing (**US\$ 335.29 million**), BREFOL aims to increase the baseline of private sector investor financing by **13.2% (US\$8.72 million)** to implement additional climate change adaptation initiatives along the energy, agriculture, forest and other land use (AFOLU) sectors in the HoA. BREFOL's financial innovations therefore represent a profound opportunity to use the 3Ps and 4Ps models to shift the status quo trajectory in five IGAD countries towards a low-carbon, climate-resilient pathway that will ensure that the energy and AFOLU sectors deliver critical services for decades to come. As a cornerstone investment, this GCF investment will already leverage an additional **US\$2.25** million from direct programme beneficiaries in the two participating countries. More importantly, BREFOL's financial innovation models will contribute to the larger adaptation initiative of the IDDRSI for the next 30 years and the *AfDB's Strategic Private Sector Investment for Fragile States Project* within the continent, further contributing to this potential paradigm shift through further scaling and replication. Technical and economic feasibility studies supporting past and proposed BREFOL initiatives in the five countries and the IGAD region are provided in Annexes 2.1-2.6.
163. If the financial innovations introduced by BREFOL are successful in the HoA, it is hoped that they can be replicated to support other AfDB-anchored programmes across Africa, such as *the Programme for Integrated Development and Adaptation to Climate Change in the Zambezi Watercourse (PIDACC-Zambezi)* and *the Programme to Strengthen Agriculture and Forest Value Chains for Climate Resilience and Reduce Deforestation in Congo and DRC (PRAFS)*. As such, the programme has significant scalability potential, as successful implementation can be scaled up immediately through the programme's critical monitoring, learning and evaluation components.
164. Another key feature of the programme worthy of up-scaling and replication is the integration of the **Technologies for African Agricultural Transformation (TAAT)** programme. TAAT specialises, inter alia, in the scaling up and dissemination of proven agricultural technologies and innovations to smallholder farmers on the continent. TAAT is currently being implemented in 27 low-income African countries. Under TAAT, the average productivity gains recorded by farmers who have received proven agricultural technologies for different commodities are 33% for small livestock, 38% for both cassava and maize, 40% for orange and flesh sweet potato, 42% for rice, 44% for aquaculture (tilapia), 64% for sorghum, 71% for fodder, 80% for millet, 100% for wheat and 113% for iron-rich beans, with an average across commodities of 62.5%. With the introduction of **7 new TAAT COMPAC technologies and innovations** in five IGAD member countries, successful implementation can be immediately scaled up and replicated in other countries in the region.
165. Similarly, the integration of **AFAWA (Affirmative Finance Action for Women in Africa)** to contribute to the economic and financial emancipation and empowerment of marginalised women in the HoA offers the greatest potential for scalability and replication in the HoA region: where gender discrimination is rampant. AFAWA's sole mandate is to bridge the gender gap in access to finance and unlock women's entrepreneurial capacity in Africa, particularly in the agricultural sector. This gap is currently estimated at **\$42** billion for African women across all business value chains, including **\$15.6** billion in agriculture alone. AFAWA therefore aims to unlock **\$1.3-2 billion** from local PFIs for on-lending to women entrepreneurs. Its integration into the BREFOL programme cannot be overstated in terms of AFAWA's scalability and replication potential.

- 166. Potential for knowledge sharing and learning:** BREFOL's potential for knowledge creation and learning can't be overemphasised. It is an integral part of several of the programme's planned activities and sub-activities. For example, an important activity of BREFOL is to **support the development of agribusinesses**. Key elements of this activity with deep knowledge sharing and learning capabilities are: 1) strengthening access to digital extension services and markets; 2) building project preparation, management and coordination capacities of key Ministries, Departments and Agencies (MDAs) in the five countries; 2) training women and youth in business skills development in renewable energy technologies (RETs) to help them generate alternative income from clean energy; and 3) designing innovative credit lines to local PFIs for on-lending to programme beneficiaries. BREFOL expects that many lessons will be learned and shared from its innovative credit line windows for other ongoing AfDB programmes in the Congo Basin and the Zambezi region of Southern Africa.
- 167.** Another key activity of BREFOL is to improve access to climate information and early warning systems (CIEWS) in the HoA region. This will be achieved through data collection, management and dissemination to different target audiences and programme beneficiaries. These strategies of generating, managing and disseminating climate advisory information to end-users will enable vulnerable people, communities and institutions involved in adaptation planning in the HoA to continue using climate information and adaptation knowledge after the programme ends. In particular, **strengthening the capacity of national MET services** to develop and maintain climate services for forecasting, drought monitoring and forecasting, crop and rangeland monitoring, seasonal forecasting, climate information and regular hazard early warnings in an actionable format, including support for the organisation of interactive seasonal user forums, will ensure that greater knowledge, appreciation and awareness of the identified climate change challenges in the HoA will improve future climate-smart planning, even after the programme ends. In addition, the programme will consolidate partnerships between private investors, government and national, regional and continental climate research institutions, including the **IGAD Climate Prediction Centre (ICPAC)** and **ClimDev-Africa**, to ensure the exchange of information, knowledge and best practices to translate early warning information into guidelines and warnings for government agencies, emergency services, aid agencies, agricultural NGOs, extension services, farmer organisations and the public. Overall, the knowledge management systems developed under this key activity will enable the National Hydromet Services of each participating country and other regional climate research centres to continue to provide training programmes on climate change adaptation for different types of users after BREFOL implementation.
- 168. Contribution to the creation of an enabling environment:** BREFOL will contribute to the creation of an enabling environment in three ways. The first is by addressing a critical challenge currently facing the African continent as a whole, and the HoA region in particular. This is increasing women's access to affordable finance to grow and feed Africa, with an estimated \$42 billion financing gap for African women across business value chains, including \$15.6 billion in agriculture alone. This challenge has been attributed to 1) **Finance**: Lending to women in agriculture is perceived as riskier, so they face prohibitive interest rates. In addition, women often lack traditional collateral and guarantees; 2) **Capacity**: Financial institutions lack the capacity to understand and respond appropriately to women entrepreneurs, who also often lack the financial and business acumen to respond to the needs of financial institutions; and 3) **Business environment**: In many African countries, legal and regulatory frameworks hinder women's full participation in private sector-led growth.
- 169.** The programme is specifically designed to improve access to finance for smallholder farmers to build their adaptive capacity, especially women (**Barrier #4**), and to help reduce gender discrimination (**Barrier #5**). The programme will provide women with access to affordable finance to carry out activities along the energy and AFOLU sectors, which will lead to women's empowerment. This will facilitate the creation of green jobs, especially for young girls and youth, who are often dependent on women. This will be done through a two-pronged approach. First, the programme will work with local private financial institutions (PFIs), in collaboration with the

AGG programme, to provide instruments that specifically target and incentivize lending to women agribusiness entrepreneurs, women-led cooperatives and smallholder women farmers. Secondly, the programme will support sub-activities that provide basic training programmes along the energy and AFOLU sectors. This will facilitate access to credit for the purchase of products, improve the creditworthiness of women and young girls, and increase the turnover of smallholder farmers. Similarly, sub-activities that provide incentives for women, young girls and youth to actively participate in food production using appropriate clean energy technologies will be particularly promoted to achieve the goal of creating employment for the most vulnerable groups in society.

170. The second enabling environment is the **BREFOL Dedicated Direct Credit Window (BDCW)**, specifically designed to provide local Private Financial Institutions (PFIs) with stable lendable resources to adequately meet the financing needs of agribusiness MSMEs, and the BREFOL Direct Insurance Credit Window (BDICW) model with local PFIs to attract private sector participation in promoting access to social protection mechanisms. The two financial innovations introduced in the programme will facilitate greater private sector participation in bridging the adaptation finance gap (**Barrier #1**). As discussed under **Barrier #2** of the TOC, climate resilience activities are often integrated into development interventions or business activities and therefore rarely stand alone. For this reason, the private sector has a critical role to play in financing adaptation. Beyond managing their own exposure to climate risks, different types of private finance - debt, equity, guarantees, insurance products - have great potential to bridge the adaptation financing gap on the continent. Unfortunately, the African continent lacks companies with the organisational capital to leverage economies of scale and specialisation to attract this level of investment for climate-smart planning. This is particularly true in the context of fragility, which attracts even fewer private companies, given the high-risk profile associated with the region due to potential conflict.
171. The third enabling environment created by BREFOL is: 1) facilitating access to digital extension services and markets in the region; 2) facilitating access to climate risk finance and insurance mechanisms to protect against the adverse impacts of climate shocks on food and agriculture systems; 3) strengthening national climate monitoring and reporting systems; 4) developing NDC implementation action plans for adaptation and resilience measures based on country commitments to implement the Paris Agreement; 5) constructing and rehabilitating 450 water mobilisation infrastructure (e.g. small earth dams, boreholes, covered water pans, shallow wells, water distribution systems); 6) development of 6,000 ha of irrigated land and rehabilitation of 300,000 ha of rangeland/pasture; 7) rehabilitation and development of 100 livestock feed, health and market access infrastructure (e.g. livestock markets, animal health posts, mobile clinics, veterinary laboratories (including cross-border laboratories in border areas), fodder banks or haystacks, feeder roads; 8) facilitating and operationalising 3 cross-border Memoranda of Understanding in 3 IGAD border clusters to better control transboundary animal diseases and zoonoses in border areas; and 9) establishing cross-border veterinary laboratories in 3 IGAD border clusters and increasing the competitiveness of livestock value chains in the IGAD region.
172. **Contribution to the regulatory framework and policies:** BREFOL is consistent and in alignment with *IDDRSI Strategy and the HoRN Regional Resilience Framework* as illustrated in Table B.1.4.
173. **Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans:** As discussed in the Programme ToC, addressing the HoA problem through the combination of well-designed interventions and packages discussed in Section B.2 will help to trigger a paradigm shift towards low-carbon and climate-resilient development pathways. However, this assumes that 1) extreme climate events (**risk no. 1**); 2) insufficient participation of women and youth in programme activities (**risk no. 2**); and; 3) economic downturns due to COVID-19 and other pandemics (**risk no. 3**) are well managed during programme implementation.

- 174.** BREFOL will reduce the negative environmental impacts of rangeland degradation by improving the behaviour and practices of farmers, pastoralists and communities. It will also have significant social benefits in terms of health, climate resilience and risk management. The baseline programme will therefore directly contribute to SDG 6 by benefiting an estimated 306,531 people (50% women, girls and youth) with access to clean and reliable energy, safe water and diversified livelihoods, including new access for 900,000 people through digital solutions and technological innovations (SDG 9). Climate-smart planning will be improved, including mainstreaming climate change and green growth tools in vulnerable communities in the HoA region. The programme will create more than 3.2 million green jobs and restore degraded rangelands and pastures, thereby reducing the impact of extreme climate events (SDG 13). In addition, the long-term sustainability of the programme will be ensured by empowering communities to maintain and manage their assets in the future. The restoration of degraded rangelands/pasturelands will contribute to the achievement of SDG15 (Halt and reverse land degradation and halt biodiversity loss) and SDG13 (Combat climate change through carbon sequestration and contribute to climate stabilisation). The programme will promote climate-resilient nature-based solutions that can help mitigate floods, droughts, erosion and landslides (hazards) and reduce vulnerability to climate change, while generating multiple benefits for the environment and local communities. This will promote gender equality in access, use and management of forests and trees and watersheds, and in the distribution of associated benefits, building on existing knowledge (SDG 5).
- 175.** It is important to note that climate change affects men and women differently, depending on their different roles and responsibilities at the household and community levels. Women play a more important role than men in managing ecosystem services and food security. They dominate the frontline in promoting action to tackle the impacts of a changing climate. However, women are the most exposed and vulnerable to climate change because they tend to be poorer, less educated and not involved in critical political and household decision-making processes that affect their lives and well-being. In some cases, cultural norms related to gender may limit women's ability to make quick and timely safety decisions in the event of disasters⁵⁸.
- 176.** BREFOL, through its focus on gender mainstreaming and women's economic empowerment through **AFAWA**, will support the reduction of entrenched gender inequalities in access to finance that hinder women's ability to manage current climate risks and adapt their livelihoods to long-term climate change trends. BREFOL has set targets: 50% of the loans/grants will go to women-led MSMEs/FBAs, WABEs, cooperatives and POs. The other 50% will be allocated to men, of which 50% will be allocated to youth-led MSMEs, YABEs, cooperatives and POs in which young women participate. Specific actions have been integrated into the programme (Activities 2.2.1 & 2.2.2) to strengthen the technical and managerial capacity of women, with the aim of providing them with appropriate tools to identify and develop bankable business plans and improve their creditworthiness with LPFIs.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

- 177.** Timely access to finance is critical to cope with the debilitating impacts of climate change and extreme weather events that are typical of the HoA region (**Root cause #2 and Barrier #1**). Frequently, smallholder farmers, especially women, including women in female-headed households, do not have access to credit facilities to purchase agricultural inputs such as climate-smart seed varieties, fertilizers and tools, and to adopt climate-smart agricultural practices to cope with sudden disasters such as seasonal droughts and flash floods. They also lack the skills and enabling environment to increase their productivity and production (**barrier #4**). The situation is further exacerbated by climate change and the recent global COVID-19 pandemic, which has caused further disruptions throughout the value chains of agriculture and food systems. Thus, without concrete measures to ensure greater access to finance and to strengthen the adaptive capacity of smallholder farmers, most of whom are women, to invest in climate-resilient livelihood options, the development of agriculture and food systems will

⁵⁸Nellemann, C., R. Verma, and L. Hislop (eds), Women at the frontline of climate change: Gender risks and hopes. A Rapid Response Assessment, 2011

continue in a "business as usual" manner and will only add to the multiple challenges facing the region (**root causes 1 - 4**).

- 178.** It is clear from the root causes that in order to successfully address the adaptation and mitigation needs of the region arising from climate extremes and non-climatic drivers, a significant number of resources and capacities will be required, far beyond what the actors in the region have been able to mobilise so far. Ethiopia's Enhanced NDC estimates a total investment requirement of \$40.5 billion and \$281.3 billion, respectively, to address the country's adaptation and mitigation needs in the six priority sectors listed in Table B.1.3. Similarly, for Kenya, the investment and costs required by the international community for both mitigation and adaptation would be over \$61.1 billion. Estimates for Djibouti and South Sudan are \$6.4 billion and \$50.0 billion respectively. Although an estimated 20% of the annual adaptation costs in the region will come from public budgets, this is clearly insufficient, with a financing gap of 80%. Given the high level of indebtedness of the countries, it is clear that the implementation of such planned investments and projects in the region would require additional financing from development partners and the private sector, as highlighted in the 5th Interim Outcome of the ***African Union Draft Africa Climate Change Strategy - DACCS (2020 - 2030)***.
- 179.** Given that the Horn of Africa is one of the most disaster-prone and food-insecure regions in the world, **building its resilience and improving its food security** is of continental importance: a key adaptation measure for the five IGAD member states in the region with a combined population of 187.6 million. However, this needs to be achieved through transformative planning, mobilising finance and supporting paradigm shifts in different value chains. Given the highly transformative nature of BREFOL, funding from the GCF will play a critical role in achieving this, given the current debt situation of these countries.

D.5. Country ownership (max. 500 words, approximately 1 page)

- 180.** A mission from the African Development Bank (AfDB) Group visited the five IGAD countries involved in the programme from 10 to 19 June 2019, to identify the components of the programme that could help build resilience for food and livelihoods in the Horn of Africa (HoA). The mission to identify the HoA programme followed the AfDB's announcement at the February 2019 roundtable on financing the Sahel Regional Climate Investment Plan (PIC-RS) for 2018-2030, and to support the implementation of the Priority Programme to Catalyse Climate Investments in the Sahel (PPCI) for 2020-2025.
- 181.** The work of the mission was carried out in close cooperation with the Secretariat and institutions of the Intergovernmental Agency for Development (IGAD), namely the Agriculture and Environment Department (AED), the Planning, Coordination and Partnerships Department (DCPP), the Economic Cooperation and Integration Department (DECI) and the Social Development Department (DSD), the IGAD Climate Prediction and Application Centre (ICPAC) and the ICPALD. The AfDB mission team also liaised with the relevant ministries responsible for agricultural development in the five countries involved, and subsequently secured buy-in from each country. See Annexes 1.1-1.5 for the No Objection Letters issued by the National Designated Authorities of the countries. Discussions were also held with the IDDRSI National Resilience Groups in Kenya, Djibouti and Ethiopia. The national meetings were facilitated by the IGAD IDDRSI National Focal Points. All the stakeholders met expressed their support for the new programme and their desire to see it implemented under the Bank's leadership.
- 182.** The IGAD Secretariat welcomed the Bank's initiative in preparing the HOA programme and provided guidance on the need to focus on structuring investments (agro-sylvo-pastoral investments, agribusiness, climate services, value chain and marketing) for the most vulnerable populations. They also stressed the importance of ensuring better synergy with ongoing strategies and actions at both regional and country levels.
- 183.** Consultations with countries highlighted the following considerations:
- The need for aggregation/integration of interventions; concentration of project activities
 - Strengthening the quality of project entry, which requires upstream techno-economic, environmental and social studies, as well as financial structuring studies and detailed designs.

- High staff turnover and inadequate capacity in some countries, which undermines project and programme continuity. This requires the recruitment of dedicated staff for the national PIUs.
- Synergy with existing programming to avoid duplication
- Land tenure security, which remains a challenge in agricultural and pastoral areas and needs to be integrated into programming.
- The need for a clear definition of responsibilities at national and sub-national levels, which will require Memoranda of Understanding between national and sub-national entities.
- Scaling up mechanisation and technology adoption at national level
- The need for platforms for private sector participation in value chains
- The need to initiate and support youth empowerment programmes; and
- The need to build the resilience of pastoralists.

184. Similarly, during the preparation of the feasibility study for the programme, high-level and grassroots consultations were held with all stakeholders, including women and youth. Workshops were held and individuals were contacted to identify the challenges and needs in each country. The large number of people mobilised during the consultations confirms that the countries and IGAD are highly committed to the programme and have high expectations of its ability to bring about improvements. The feasibility studies involved extensive consultation with beneficiary communities in all areas of likely intervention. In addition, the feasibility reports were validated through stakeholder workshops in each country. Consultation with beneficiaries and stakeholders - communities, women's groups, youth associations, civil society organisations (CSOs) and non-governmental organisations (NGOs) - will continue during the implementation phase.

185. During the programme preparation and appraisal missions, the Bank consulted widely with senior government officials, technical experts and development partners in the region. The final design of the programme reflects the views and recommendations of the stakeholders and the activities of the programme have been harmonised with other interventions in the region. This participatory approach at the design stage will be deepened during implementation. Representatives of the programme's beneficiaries will be members of the country project steering committee and will be closely involved in monitoring and evaluation. The capacity of beneficiary associations and relevant government institutions will be strengthened to better manage and coordinate programme activities.

186. As a follow-up to the various consultations held during the preparation of the baseline programme, additional consultations were held with countries and IGAD during the preparation of the full FP to ensure greater country ownership. Annex 7 provides more details on the additional consultative processes and the main inputs from each country. Some key highlights from the consultative meetings include the following

- Country support for the preparation of market research studies.
- Establishment of PMUs and drafting of ToRs for recruitment of staff.
- Stakeholder consultations with potential local private financial institutions (PFIs).
- Provision of no-objection letters in support of the programme.
- Identification of other potential partners such as UNCCD and NGOs.
- Contributions from the IGAD Climate Prediction and Application Centre (ICPAC) to improve the climate context of the programme.

187. Country summaries are indicated below:

188. *Djibouti:* The stakeholder engagement meetings (see Annex 7.1) brought together government officials, technical advisors to the Minister of Agriculture, representatives of the technical directorates and partners of the other ministries, the authorities of the intervention areas (Prefect and President of the Regional Councils), the Sub-Directors of Rural Development of the project areas, members of the communities, including pastoralists and agropastoralists,

representatives of farmers' associations and local investors/private financial institutions (PFIs). The meetings focused on the design and objectives of BREFOL, including the additional GCF funding. Based on the consultations, in summary, participants expressed the need for (i) demand-driven water harvesting infrastructure at the community level, (ii) rural markets, (iii) drilling of boreholes with solar-powered pumps as fuel is expensive to combat drought, (iv) seed production, (v) pasture conservation, (vi) strengthening their capacity to process agricultural products, (vii) hydro-agricultural infrastructure such as reservoirs, and (viii) empowerment of women and youth through agribusiness and access to finance/concessional loans. Discussions with LFIs focus on the management and administration of the credit line.

189. Ethiopia: Stakeholder consultation meetings were held with government officials at federal, regional and woreda levels, direct beneficiaries including pastoralists and agropastoralists, representatives of farmers' associations and MSMEs, local private financial institutions (PFIs) and a Development Bank in Ethiopia (see Annex 7.2.1). The meetings focused on the design and objectives of BREFOL, including the additional GCF funding. Based on the consultations, participants expressed the need for (i) community-level demand-driven water harvesting infrastructure, (ii) rural markets, (iii) drilling and rehabilitation of boreholes with solar-powered pumps, (iv) rangeland management, (v) institutional strengthening, (vi) access to credit/finance, and (vii) empowerment of women and youth, including access to finance/concessional loans (see Ethiopia market study in Annex 7.2.2). Discussions with the PFI and development bank in Ethiopia focus on the management and administration of the credit line.

190. Kenya: Stakeholder consultation meetings were held with government officials, direct beneficiaries including pastoralists and agropastoralists, and representatives of farmer associations and MSMEs (see Stakeholder Engagement Report in Annex 7.5.1). The meetings focused on the design and objectives of BREFOL, including the additional GCF funding. Based on the discussions, participants expressed the need for (i) demand-driven water harvesting infrastructure at the community level, (ii) fodder production and fodder banks, (iii) rangeland management, (iv) rural markets, (v) drilling and rehabilitation of boreholes with solar-powered pumps, (vi) empowerment of women and youth MSMEs, and (vii) capacity building and institutional strengthening (see Kenya market study in annex 7.5.2).

191. Somalia: Stakeholder consultation meetings were held with local government officials, direct beneficiaries including pastoralists and agropastoralists, and representatives of NGOs and MSMEs (see Annex 7.3.1). The meetings focused on the design and objectives of BREFOL, including the additional GCF funding. Based on the discussions, participants expressed the need for (i) rural markets, (ii) fodder production and fodder banks, (iii) rangeland management, (iv) drilling and rehabilitation of boreholes with solar-powered pumps, and (vi) empowerment of women and youth MSMEs and other local stakeholders (see Annexes 7.3.2 - 7.3.3). The discussions also presented gender-disaggregated household data in the project areas, list of main staples, average daily meal, main livelihoods/income sources, main shocks in the communities/areas, key priorities in the communities, fodder status and fodder production.

192. South Sudan: Stakeholder consultation meetings were held with government officials, direct beneficiaries including pastoralists/agropastoralists, and representatives of FAO, cooperatives, NGOs and MSMEs (see Annex 7.4.1). The meetings focused on the design and objectives of BREFOL, including the additional GCF funding. The main shocks experienced in the project area include drought/flood/erratic climate, increasing food and energy prices, pest infestation and conflicts related to cattle raiding (see Annexes 7.4.2 - 7.4.4). Based on discussions, key priorities include access to agricultural inputs and equipment (seeds, tools, ox ploughs and power tillers), livestock production, access to water, employment opportunities, rural markets, fodder production/fodder banks, pasture management, drilling and rehabilitation of boreholes with solar powered pumps, and empowerment of women and youth MSMEs.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

193. In terms of delivering the co-benefits of adaptation and mitigation as envisaged by the Fund, BREFOL is highly efficient and very cost-effective. For example, as shown in Table D.6.1, the total estimated cost of adaptation per beneficiary is about \$71.6. Of this adaptation cost, the GCF contributes only USD 28.8 per programme beneficiary, which represents a contribution of

40.1% and 59.9%, respectively, in terms of financing the adaptation potential for the co-financing mobilized by the AfDB and the GCF funding. Similarly, the total estimated cost per tCO₂e for the programme is about \$23.8. The GCF contributes only \$1.16 per tCO₂e achieved to these mitigation costs, representing 4.9% and 95.1% of the mitigation potential of the AfDB co-financing and GCF funding, respectively.

194. Table D.6.1: Summary of the efficiency and cost effectiveness of the program

Total Program Funding (mUSD)	\$335.3
GCF 33Funding (mUSD)	\$151.0
Program Direct Beneficiaries	4.7
Total Emissions Reductions in 25years (mtCo ₂ e)	14.1
Estimated Cost per Program Beneficiary	\$71.6
GCF Amount Allocated for Adaptation	\$134.7
Estimated GCF cost per Program Beneficiary	\$28.8
GCF Amount Allocated Mitigation	\$16.3
Estimated Programme Cost per tCO ₂ e	\$23.8
Estimated GCF cost per tCO ₂ e	\$1.16

195. In order to assess the financial and economic viability of the proposed interventions, including the effectiveness of the GCF investment in the Programme, an Economic and Financial Analysis (EFA) was carried out (see Annexes 3a&b for the EFA summary and Excel worksheets). A summary of the EFA results is presented in Tables D.6.2 & D.6.3 below. Note that the net benefits derived from the activity level models, in terms of incremental benefits relative to the baseline, are aggregated in both the financial and economic analyses, taking into account the scale of the project and its objectives (total area under cultivation in hectares), in order to assess the overall benefits generated by the proposed programme interventions. The derived benefits are compared with the project costs (estimated from the project budget) to assess the overall investment effectiveness indicators for different discount rates (DRs). These are the **Economic Net Present Value (ENPV)**, the **Economic Internal Rate of Return (EIRR)** and the **Modified Internal Rate of Return (MIRR)**. The financial and economic models of the crop and a summary of the economic analysis are provided in the attached Excel spreadsheets (Annex 3b).

196. The financial analysis (FA) has three objectives. 1) Firstly, to assess the financial soundness of the development interventions promoted under BREFOL. 2) Second, to examine the impact of BREFOL activities on the incomes of smallholder farmers, pastoralists, agribusiness MSMEs, Producer Organisations (POs), local PFIs and Farmers Based Associations (FBAs). 3) Thirdly, to provide a strong analytical framework for the economic evaluation of BREFOL in the region. Such an analysis is important for an overall assessment of BREFOL in terms of society rather than on an individual basis (FA).

197. Four scenarios are generally considered for each FA, as shown in the attached Excel worksheets (Annex 3b). These are 1) '**model assumptions**', which provide the general assumptions and parameters used for each FA; 2) '**without project**' (**WOP**) **scenario**, which is the baseline of the analysis; and '**with project**' (**WP**) **scenario**, which is the proposed activities and interventions of the programme. Note that the WOP is the business-as-usual (BAU) scenario, where yields are below potential and returns to family labour are expected to be lower, although not in all cases. Irrigation is an old farming technique and some traditional farmers, particularly those involved in intercropping in the region, have used it. In most of the analyses we have included irrigation as a '**drought factor**' for both scenarios, given the aridity of the region. If this assumption is relaxed in the WOP scenario, it is likely that yields would be below potential and returns to family labour would be lower in the WOP scenario. In addition, the use of the two scenarios allows us to identify the net benefits of the programme's 'incremental situation' at the individual and aggregate levels.

- 198.** The FA results show that most of the crop models are profitable from the farmers' perspective. The cash flows show that farmers will be able to cover the necessary operating costs. Furthermore, with the possible introduction of resilient varieties (TAAT COMPAC Technologies, livestock, etc.) into the programme, the potential yields of the selected crops will double, implying more cash inflows to farmers. The FA results also confirm that the proposed production packages are financially attractive to the participants and that the potential benefits to the beneficiary farmers of participating in the activities implemented under BREFOL are financially attractive. The cash flows also show that smallholder farmers will be able to meet the necessary operating costs, especially with the increased yields associated with the introduction of TAAT COMPAC seeds. Farmers may also decide to intercrop some of the horticultural crops (cabbage, lettuce and tomatoes) with staple crops or agroforestry to generate more income. However, as our analysis is conservative, this option is not considered here.
- 199.** At an aggregate level, the FA calculates an overall Financial Internal Rate of Return (FIRR) of 59.8% for the baseline scenario and average Financial Net Present Values (FNPVs) of \$477.48 million at a discount rate of 9.6%, \$732.13 million at a discount rate of 6%, \$586.76 million at a discount rate of 7.9% and \$186.11 million at a discount rate of 19.1%. Further sensitivity analysis was performed on the FIRR for various scenarios under the RCP 8.5, such as 10% and 20% cost overruns, benefit increases, benefit decreases and 1- and 2-year benefit delays. In all cases, as shown in the table below, the FIRRs were higher than the discount rates, and in most cases much higher. The FNPVs were also always positive. This shows that the programme is not only very robust from the perspective of the smallholder beneficiary but would also remain viable under a wide range of alternatives.

Table D.6.2: Summary table of the key performance indicators of the FA under different discount rates

		Key Performance Indicators for Different Discount Rates (DR)								
Financial Discount Rate (FDR)		9.6%	6.0%	7.9%	19.07%	9.6%	6.0%	7.9%	19.1%	
		FNPV				FIRR		MIRR		
Base Scenario		\$477.48	\$732.13	\$586.76	\$186.11	59.8%	21.7%	19.5%	20.6%	28.1%
costs +10%		\$1,667.55	\$465.48	\$574.28	\$176.20	54.0%	20.8%	18.7%	19.7%	27.2%
costs +20%		\$1,652.45	\$706.09	\$561.79	\$166.30	49.2%	20.1%	17.9%	19.0%	26.4%
benefits +10%		\$1,866.01	\$818.36	\$657.92	\$214.62	66.3%	22.4%	20.2%	21.3%	28.8%
benefits +20%		\$2,049.38	\$904.59	\$729.08	\$243.14	72.8%	22.9%	20.6%	21.8%	29.4%
benefits -10%		\$1,499.28	\$645.90	\$515.60	\$157.59	53.4%	20.8%	18.6%	19.7%	27.1%
benefits -20%		\$1,315.92	\$559.67	\$444.44	\$129.08	47.1%	19.7%	17.5%	18.6%	26.0%
benefits postipated 1 Year		\$1,600.37	\$665.13	\$523.43	\$139.57	41.3%	18.7%	16.6%	17.6%	24.9%
benefits postipated 2 Year		\$1,638.12	\$695.47	\$551.94	\$159.64	47.4%	13.9%	17.2%	18.3%	25.7%
* See Dashboard for NPV Value										
** FDR = Financial Discount Rate										

- 200.** With regard to the Economic Analysis (EA), the EA does several things. First, it assesses the economic viability and overall cost-effectiveness of the programme. This is usually done from the perspective of society, rather than at the individual and household level as in the FA. This is usually done by comparing the aggregate economic benefits with the economic costs of the programme and assessing the economic internal rate of return (EIRR). Secondly, it shows how sensitive the proposed investments may be to small changes in the model parameters used, such as prices, incomes, interest rates, discount rates, etc., due to risks and unforeseen factors. Details on sensitivity to model parameters, including EA model assumptions such as economic prices and their conversion factors, social discount rates, shadow prices for carbon and the Economic Programme Cost (EPC), can be found in the attached Excel spreadsheets.
- 201.** Three performance indicators were calculated for the EA. These are the **Economic Net Present Value (ENPV)**, the **Economic Internal Rate of Return (EIRR)** and the **Modified Internal Rate of Return (MIRR)**, as shown in the table below. On this basis, the EA calculates an overall Economic Internal Rate of Return (EIRR) of 28.0% for the programme's baseline

scenario, an overall Economic Net Present Value (ENPV) of USD 112.74 million at a discount rate of 11.0%, USD 133.92 million at a discount rate of 10.0%, USD 100.48 million at a discount rate of 7.9% and USD 238.11 million at a discount rate of 6.0%.

202. In order to test the stability of the EA model, we performed an additional robustness check on key variables of the model. That is, 10 and 20% cost overruns, benefit increases, benefit decreases, and 1- and 2-year benefit delays. The results are shown in the table below. In all cases, the EIRR and MIRR were higher than the discount rates used, and in most cases much higher. The NPVs were also always positive. This shows that the programme is not only very robust but would also remain viable under a wide range of alternatives.

Table D.6.3: Summary table of the key EA performance indicators for different discount rates

[illegible]

E. LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

- ☒ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Following on from the paradigm shifting potential of the project described in Section D.2, the scalability, replicability and sustainability of the proposed climate change adaptation and mitigation activities are presented here. Scalability, replicability and sustainability are threatened by the six barriers identified in this proposal. The trios are achieved here through the following outputs shown in this table.

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Current private sector involvement in climate change adaptation and mitigation is very limited, especially in the fragile countries of the IGAD region. This is due to the lack of innovative financial mechanisms that provide incentives and an enabling environment for investment in high-risk areas. Therefore, creating new profit opportunities for private capital based on climate change adaptation and mitigation investments opens the doors for private sector investment.	<u>Low</u>	The scaling-up of most of the results is particularly evident in the increased participation of the private sector in climate change adaptation strategies in five IGAD member states in the Horn of Africa (HoA). The enabling financial innovations, such as the BREFOL Direct Credit Window (BDCW) and the BREFOL Direct Insurance Credit Window (BDICW), are expected to help scale up sustainable land management for agropastoralism, climate-smart innovations, and climate risk finance and insurance mechanisms that will help strengthen the climate resilience of climate-vulnerable communities. These innovative financial mechanisms will provide the right incentives and enabling instruments that are attractive to private sector investors, such as private-public partnerships (3Ps) and public-private-producer partnerships (4Ps) in agribusiness MSMEs and weather-based insurance promoters with an interest in green/circular business opportunities in the energy, agriculture, forest and other land use (AFOLU) sectors in the HoA.	The project will contribute to realizing this target scenario through: • Output 1.2: Improved sustainable land management practices • Output 1.3: Increased adoption of climate-smart innovations - Output 2.3: Developed inclusive MSMEs and cooperatives with access to climate financing • Output 3.1: Enhanced access to climate information and early warning systems

Replicability	Currently, smallholder farmers and the private sector have limited access to innovative programmes of action and climate adaptation and mitigation technologies. Unlocking innovative financing mechanisms that help integrate climate smart technologies and climate information services could support smallholder farmers in the IGAD region. Due to the lack of human, technical and financial capacity, some projects are not replicable in other IGAD member states, such as the TAAT COMPAC technologies and innovations.	<u>Low</u>	Given the regional nature of the GCF proposal, it is designed to be replicable in IGAD member countries and beyond. Thus, almost all of the proposed activities are designed to be replicable at community, country, regional and continental levels (beyond IGAD/HoA countries), including 1) development and rehabilitation of climate-resilient infrastructure (water resources, energy, market integration, cross-border health and research facilities, road networks, ICTs, etc.); 2) support for sustainable agro-pastoral land management; and 3) increasing access to digital technologies.), 2) supporting sustainable agro-pastoral land management, 3) increasing access to digital extension services and markets, 4) developing entrepreneurship skills and increasing access to finance for women and youth, and 5) increasing access to innovative finance for smallholder farmers, pastoralists and agribusiness MSMEs; 6) improving access to climate-smart innovations and technologies; 7) improving climate services infrastructure and advisory services; and 8) promoting climate risk finance and insurance; and finally, 9) supporting gender-sensitive reforms of climate regulatory frameworks and institutional strengthening for climate-smart planning and development. For example, the 7 new TAAT COMPAC technologies and innovations are expected to be replicated from the pioneer country such as Kenya to other IGAD member countries. The holistic approaches ensure the successful replicability of the results.	The project will contribute to realizing this target scenario through: • Output 1.1: Climate resilient infrastructure built/rehabilitated • Output 2.2: Developed inclusive MSMEs and cooperatives with access to climate financing.
Sustainability	Due to the six barriers described in Section C, the adaptive capacity of smallholder farmers and pastoralists is not sustainable and vulnerable to the climate hazards identified in this proposal. Although various activities are being implemented,	<u>Low</u>	The sustainability of the proposed outputs will require effective training of trainers, capacity building, increasing access to digital advisory services and markets, developing entrepreneurship skills and increasing access to finance for women and youth, improving climate services infrastructure, providing climate risk finance and insurance mechanisms,	The project will contribute to realizing this target scenario through: • Output 2-1: Increased access to digital advisory services and markets • Output 2-2: Developed inclusive MSMEs and

	their sustainability is not guaranteed due to the lack of adequate supporting institutions, infrastructure and capacities. The sustainability of the proposed programmes could only be ensured by strengthening these institutions, where some of the proposed actions such as developing policy frameworks, capacity building and building robust knowledge management are critical.		strengthening operational capacity for resilience, and programme coordination and knowledge management will help ensure the sustainability of the target scenarios designed for scale-up and replication.	cooperatives with access to climate financing and access to RE Output 3-1: Enhanced access to climate information and early warning systems • Output 3-1: Enhanced access to climate information and early warning systems • Output 3-2: Strengthened regulatory and policy frameworks and institutional capacity for climate resilience. • Outcome 4: Program Coordination and Knowledge Management
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E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁵⁹	
Total GHG emission reduction	Core 1: GHG emissions reduced, avoided or removed/sequestered		0 Mt CO ₂ eq.	0.48 MTCO ₂	2.1 Mt CO ₂ over project period 14.1MTCO ₂ over lifetime	Implementation targets for solar PV, biodigesters, and agro forestry is met as per project timelines.
<u>MRA1 Energy generation and access</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Baseline, mid-term and end term assessment in each country, contract agreements,	0 Mt CO ₂ eq.	19k TCO ₂ avoided through the deployment of PV systems and biodigesters	56k tCO ₂ over project period avoided through the deployment of PV systems	Regulatory and legal Frameworks, and capacity in place for solar power generation and Biodigester installation, and operation and maintenance.

⁵⁹ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

		procurement documents, and bill of quantities.			and biodigesters 226kTCO2 over lifetime avoided through the deployment of PV systems and biodigesters	
<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.3: Installed renewable energy capacity</u>	Baseline, mid-term and end term assessment in each country, contract agreements, procurement documents, and bill of quantities.	0 MWp	At least 1 MW of solar PV installed, and a capacity of at least 0.12MW Biodigester systems installed	4.8 MW of solar and 0.12MW of Biodigester systems Installed	Regulatory and legal Frameworks, and capacity in place for solar power generation and Biodigester installation, and operation and maintenance
<u>MRA4 Forestry and land use</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Baseline analysis report, survey results and final reports	0 Mt CO2 eq.	0.46 MTCO2 through rangeland rehabilitation and agro forestry activities	2.0 Mt CO2 over project period through rangeland rehabilitation and agro forestry activities 13.9MTCO2 over lifetime through rangeland rehabilitation and agro forestry activities	Enabling environment to put in place through IGAD's Centre for Pastoral Area and Livestock Development (ICPALD), to enact behavioural changes in sustainable land management practices. Adequate training and sensitization programs result in behavioural changes in land use management. Beneficiaries are therefore willing to adopt sustainable land management practices

						such as agroforestry management practices with positive environmental benefits
<u>MRA4 Forestry and land use</u>	<u>Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under resoration and/or improved ecosystems</u>	Baseline analysis report, survey results and final reports	0 ha	9,000ha of pastureland placed under sustainable land management practices with agroforestry and afforestation program introduced	29,800 ha of pastureland placed under sustainable land management practices with agroforestry and afforestation program introduced	Enabling environment put in place through IGAD's Centre for Pastoral Area and Livestock Development (ICPALD), to enact behavioural changes in sustainable land management practices. Adequate training and sensitization programs result in behavioural changes in land use management. Beneficiaries are therefore willing to adopt sustainable land management practices such as agroforestry management practices with positive environmental benefits
Total number of programme beneficiaries	Core 2: Direct and indirect beneficiaries reached		0	1,405,419 (50% male and 50% female) of direct and 4,664,525 (50% male and 50% female) of indirect beneficiaries by end of year 3	4,684,731 (50% male and 50% female) of direct and 15,548,383 (50% male and 50% female) of indirect beneficiaries by end of year 7	Value chain actors, communities, and institutions etc., are willing to adopt climate resilient livelihood options & technological innovations, introduced by the programme.

<u>ARA1 Most vulnerable people and communities</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Minutes reports, baseline assessment reports, survey results and final reports	0	1,191,870 (50% male and 50% female) of direct and 4,056,565 (50% male and 50% female) of indirect beneficiaries by end	3,972,900 (50% male and 50% female) of direct and 13,521,883 (50% male and 50% female) of indirect beneficiaries by end of year 7	Small farm holders and pastoralists are willing to adopt regenerative agriculture and other climate-smart practices introduced
<u>ARA1 Most vulnerable people and communities</u>	<u>Supplementary 2.4: Beneficiaries (female/male) covered by new or improved early warning systems</u>	Baseline assessment reports, contract agreements, bill of quantities, pre- and post-intervention (qualitative and quantitative) survey results	0	At least 999,126 pastoralists and smallholder farmers received TA support for training and equipping with the use of climate-informed agro-advisory services and 2,317,080 indirect beneficiaries of which 50% are male and 50% are female	A total of 3,330,420 pastoralists and smallholder farmers received TA support for training and equipping with the use of climate-informed agro-advisory services and 7,723,600 indirect beneficiaries of which 50% are male and 50% are female	Farmers and pastoralists are trained on how to use climate information. There is adequate medium for communicating and disseminating climate-base agro-advisory services such as through users' friendly formats
<u>ARA1 Most vulnerable people and communities</u>	<u>Supplementary 2.5: Beneficiaries (female/male) adopting innovations that strengthen climate change resilience</u>	Minutes reports, baseline assessment reports, survey	0	Climate-resilient innovations adopted by 192,744 direct and 1,799,364	Climate-resilient innovations adopted by 642,489 direct and 5,997,883	Small farm holders and pastoralists are willing to adopt regenerative agriculture and other climate-smart practices introduced

		results and final reports		indirect beneficiaries of which 50% are male and 50% are female	indirect beneficiaries of which 50% are male and 50% are female	
<u>ARA2 Health, well-being, food and water security</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Baseline assessment reports survey results, and final reports.	0	213,550 direct and 211,950 indirect of which 50% females and 50% males have better food and water security	711,831 direct and 706,500 indirect of which 50% females and 50% males have better water and food security	Smallholder farmers and pastoralists have access to water mobilization infrastructure and tailored agro forestry management practices and climate resilient varieties to increase food security
<u>ARA2 Health, well-being, food and water security</u>	<u>Supplementary 2.2: Beneficiaries (female/male) with improved food security</u>	Baseline assessment reports survey results, and final reports.	0	121,590 direct and 211,950 indirect of which 50% females and 50% males have better food security	405,300 direct and 706,500 indirect of which 50% females and 50% males have better food security	Smallholder farmers and pastoralists have access to tailored agro forestry management practices and climate resilient varieties to increase food security
<u>ARA2 Health, well-being, food and water security</u>	<u>Supplementary 2.3: Beneficiaries (female/male) with more climate-resilient water security</u>	Baseline assessment reports survey results, and final reports.	0	91,959 direct of which 50% females and 50% males have better water security	306,531 direct of which 50% females and 50% males have better water security	Smallholder farmers and pastoralists have access to water mobilization infrastructure sensitive to gender ⁶⁰ (small earth dams, boreholes, covered water pans, shallow wells, water distribution systems) constructed or rehabilitated

⁶⁰ Gender-sensitive infrastructure is designed, planned, and managed on the basis of systemic consultations with the most vulnerable or disadvantaged (UN Women).

<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice</u>	Baseline assessment reports survey results, and final reports.	0	40,000 ha of rangeland/past ureland rehabilitated	102,000 ha of rangeland/past ureland rehabilitated	Smallholder farmers and pastoralists are willing to participate and adopt new sustainable land management practices on degraded rangeland/pastureland
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under resoration and/or improved ecosystems</u>	Baseline assessment reports survey results, and final reports.	0	40,000 ha of rangeland/past ureland rehabilitated	102,000 ha of rangeland/past ureland rehabilitated	Smallholder farmers and pastoralists are willing to participate and adopt new sustainable land management practices on degraded rangeland/pastureland

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation</u>	Currently climate smart agricultural and pastoral technologies are not widely disseminated and applied. The GCF will support to undertake the economics of integrated pest management technologies to fight against fall armyworm, arboviruses in transhumant cattle, and locust's invasion, and upscale	<u>low</u>	Techno-economic analysis integrated pest management technologies to fight against fall armyworm, arboviruses in transhumant cattle, and locust's invasion completed in 3 agroecological zones in the HOA countries, and Technologies for African Agricultural Transformation (TAAT) Program to over 278,000 pastoralists and smallholder farmers in the HOA upscaled. The TAAT are a high-impact technologies to boost agricultural output.	The GCF support in dissemination and technology transfer of the high-impact technologies to boost agricultural output.	<u>Multi-countries</u>

	Technologies for African Agricultural Transformation (TAAT) to over 278,000 pastoralists and smallholder farmers in the HOA.				
<u>Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u>	Government officials in the IGAD member countries lack adequate capacity to develop bankable proposals that can qualify for climate finance, mainstreaming climate risk management in policy documents and plans is very weak, and the countries don't yet have NDC implementation action plans on adaptation and resilience interventions building on respective country's commitments to the implementation of the Paris Agreement.	<u>low</u>	Government officials in IGAD member countries will receive training in strengthening their capacity to develop bankable proposals that can qualify for climate finance, mainstreaming of climate risk management into sectoral and national plans, policies and strategies in 3 countries, and NDC implementation action plans on adaptation and resilience interventions building on country commitments to the implementation of the Paris Agreement completed.	The GCF funding will contribute to strengthening the capacity of the policy makers, and climate risk shall be incorporated in national action plans and policies.	<u>Multi-countries</u>
<u>Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level</u>	Due to lack road infrastructure that hinders access to market, lack of entrepreneurship skills and increasing access to finance and markets for the most	<u>low</u>	About 152 animal feed, health, and market access infrastructure (e.g., livestock markets etc.) rehabilitated/developed), over 1,100,000 smallholder farmers (women and youth), farmers organization and	The GCF will build road infrastructure of increase access to market, enable to having access to digital advisory services so as to have access to their products and build their	<u>Multi-countries</u>

	marginalized groups in the society, and lack of digital advisory services and markets are some of existing conditions of the beneficiaries.		agribusiness MSMEs being supported in using digital advisory services of the market for their products, and over 1985 women and youth received inputs (entrepreneurship skills) and finance to generate alternative income from renewable energy sources.	entrepreneurship skills to have access to renewable energy technologies to improve their value chain for market.	
<u>Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u>	The target communities don't have effective knowledge generation and learning process, use of good practices, methodologies, and standards. The GCF funding is expected to address this challenge so as they could effectively increase their adaptation capacity.	<u>low</u>	Regional innovation hub/data centre to coordinate and share knowledge among innovation partners, governments and other stakeholders established and development of the regional capacity building and training of trainers on the gender action learning system (GALS) approach and methodology completed	Knowledge management systems and methodologies developed under this key activity will enable the national hydromet services for each involved country, and other regional climate research centres to continue providing training programmes on climate change adaptation to different types of users after BREFOL implementation.	<u>Multi-countries</u>

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	

Outcome 1: Increased adoption of climate-resilient solutions						
Outcome 1: Increased adoption of climate-resilient solutions	Number of agro pastoralists and pastoralists benefitting from the project activities		0	1,405,419 (50% male and 50% female) of direct and 4,664,525 (50% male and 50% female) of indirect beneficiaries by end of year 3	4,684,731 (50% male and 50% female) of direct and 15,548,383 (50% male and 50% female) of indirect beneficiaries by end of year 7	Value chain actors, communities, and institutions etc., are willing to adopt climate resilient livelihood options & technological innovations, introduced by the programme.
Output 1-1: Climate resilient infrastructure built/rehabilitated	Number of water mobilization infrastructure (e.g., small earth dams etc.) constructed or rehabilitated and managed	Baseline assessment survey results, and progress and final reports.	0	177 water mobilization infrastructure (i.e., small earth dams etc.) constructed	538 mobilization infrastructure (e.g., small earth dams etc.) constructed	Institutions and stakeholders' readiness to participate in the activities, and IGAD member states capability in place to effectively implement the regional activities.
	Area (hectares) of community irrigation lands developed and managed	Baseline assessment, feasibility studies, and quantitative analysis results	0	1,950 ha of irrigation lands developed	5,980 ha of irrigation lands developed	
	Number of animal feed, health, and market access infrastructure (e.g., livestock markets etc.) constructed or rehabilitated and managed (number of	Baseline assessment, feasibility studies, and quantitative analysis results	0	47 infrastructures set up	152 infrastructures set up	Community mobilization of women and youth in the IGAD member states and effective sensitization, identification, and engagement of women and youth at the community level

	infrastructure built or equipped)					to take up jobs in land and pasture rehabilitation
Output 1.2: Improved sustainable land management practices	Number of agropastoralists and pastoralists trained on sustainable land management best practices (disaggregated by male and female)	Baseline analysis report, survey results and final reports	0%	10,300 agropastoralists and pastoralists trained on sustainable land management best practices	31,300 agropastoralists and pastoralists trained on sustainable land management best practices	Enabling environment to put in place through IGAD's Centre for Pastoral Area and Livestock Development (ICPALD), to enact behavioural changes in sustainable land management practices.
	Number of Trainers trained at regional level on improved sustainable land management	Annual Progress reports	0	30 (50% male and 50% female)	100 (50% male and 50% female)	Enabling environment put in place through IGAD's Centre for Pastoral Area and Livestock Development (ICPALD), to train trainers

	Area (hectares) of degraded rangeland rehabilitated and managed	Baseline assessment, progress and final reports	0	32,600ha	102,000 ha	Community mobilization of women and youth in the IGAD member states and effective sensitization, identification, and engagement of women and youth at the community level to take up jobs in land and pasture rehabilitation
	Number of cross border rangelands (cluster) sustainably managed	Baseline assessment, progress and final reports	0	1	3	Enabling environment put in place through IGAD's Centre for Pastoral Area and Livestock Development (ICPALD), to enact behavioural changes in sustainable land management practices
	Area (hectares) of degraded land/ forest rehabilitated and managed under agroforestry and plantation	Baseline assessment, progress and final reports	0	9,000 ha	29,800 ha	Community mobilization of women and youth in the IGAD member states and effective sensitization, identification, and engagement of women and youth at the community level

						to take up jobs in land and pasture rehabilitation
Output 1.3: Increased adoption of climate-smart innovations	Number of small holder farmers and pastoralists benefiting from extension services (disaggregated by male and female)	Baseline assessment, progress and final reports	0	90,000 (50% male and 50% female)	278,000 (50% male and 50% female)	Community mobilization of women and youth in the IGAD member states and effective sensitization, identification, and engagement of women and youth at the community
	Number of trainers trained on tools for extension services on climate smart innovations/dissemination	Training reports progress and final reports	0	80 (50% male and 50% female)	250 (50% male and 50% female)	Enabling environment put in place to train trainers
	Number of beneficiaries receiving resilient varieties to improve agricultural and fodder production	Bill of quantities, contracts, quantitative and qualitative analysis reports, and final reports	0	83,750 (50% male and 50% female)	264,000 (50% male and 50% female)	Enabling environment are in place for scaling up TAAT COMPAC technologies and innovations
Outcome 2: Enhanced carbon sequestration and reduced greenhouse gas emission of AVC						
Outcome 2: Enhanced carbon sequestration and reduced greenhouse gas emission of AVC	Amount of GHG emissions avoided through project activities		0 Mt CO ₂ eq.	0.48 MTCO ₂	2.1 Mt CO ₂ over project period 14.1 MTCO ₂ over lifetime	Implementation targets for solar PV, biodigesters, and agro forestry is met as per project timelines.

Output 2-1: Increased access to digital advisory services and markets	Number of digital advisory services established through PPP mechanism	Baseline assessment, progress and final reports	0	5 (one each in Djibouti, Ethiopia, Kenya, Somalia, and South Sudan)	5 (one each in Djibouti, Ethiopia, Kenya, Somalia, and South Sudan)	Conducive environment in place in all 5 countries to enable signature of PPP for the setting up of digital advisory services.
	Number of small holder farmers and pastoralist benefiting from the established and deployed digital advisory services (disaggregated by male and female)	Baseline assessment reports, qualitative and quantitative assessment reports, and bill of quantities, contract agreements	0	500,000 (50% male and 50% female)	1,592,000 (50% male and 50% female)	Effective communication and dissemination mechanisms put in place for using digital advisory services
	Number of regional infrastructure (hubs) developed for supporting digital advisory services	Baseline assessment, progress and final reports	0	2	2	Enabling environment in place to facilitate IGAD's mission of coordinating and sharing knowledge among innovation partners, governments and other stakeholders established
	Number of rural markets constructed included access	Baseline assessment, progress and final reports	0	30 (2 in Djibouti, 6 in Ethiopia, 7 in Kenya, 10 in Somalia, 5 in South Sudan)	95 (7 in Djibouti, 37 in Ethiopia, 14 in Kenya, 27 in Somalia, and 10 in South Sudan)	Institutions and stakeholders' readiness to participate in the activities, and IGAD member states

				at the end of year 3	Sudan) at the end of year 7	capability in place to effectively implement the activities
Output 2.2: Developed inclusive MSMEs and cooperatives with access to climate financing	Number of women and youth groups and cooperatives trained on renewable energy and/or AVC development and having access to credit	Baseline assessment reports, contracts and direct purchase agreements, bill of quantities, pre-and post-intervention (qualitative and quantitative) survey results	0	123	410	Technical assistance (TA) support provided to women and youth on entrepreneurial development in renewable energy utilization and or AVC development
	Number of agribusiness MSMEs trained on renewable energy and/or AVC development and having access to credit	Budget, Baseline assessment reports, pre-and post- intervention (qualitative and quantitative) survey results	0	114 (50% male and 50% female owned)	362 (50% male and 50% female owned)	Credit lines are made available through local PFIs for on-lending to program beneficiaries
	Number of people including women in female headed households and youth with a target of 50% to receive inputs and finance to generate alternative	Budget, Baseline assessment reports, pre-and post- intervention (qualitative and quantitative) survey results	0	2,800 (50% male and 50% female)	10,680 (50% male and 50% female)	Budget, Baseline assessment reports, pre-and post- intervention (qualitative and quantitative) survey results Mid-term and final evaluation reports, contract agreements, bill of

	income from renewable energy sources such as solar and biodigesters					quantities, procurement documents, and survey results
	Capacity in MW of solar PV installed	Mid-term and final evaluation reports, contract agreements, bill of quantities, procurement documents, and survey results	0	2.8 MW (0.3MW in Djibouti, 0.5MW in Ethiopia, 1MW in Kenya, 0.5MW in Somalia, 0.5MW in South Sudan) at the end of year 3	4.8 MW (0.3MW in Djibouti, 1MW in Ethiopia, 1.5MW in Kenya, 1MW in Somalia, 1MW in South Sudan) at the end of year 7	Enabling environment created for private sector investment in renewable energy generation: 4.8 MW of solar and 3,890m ³ of Biodigester systems
	Capacity in cubic meter of biodigesters installed	Mid-term and final evaluation reports, contract agreements, bill of quantities, procurement documents, and survey results	0	2,160 m ³ (2000m ³ in Ethiopia, 60m ³ in Kenya, 100m ³ in South Sudan) at the end of year 3	3,890 m ³ (3500m ³ in Ethiopia, 120m ³ in Kenya, 270m ³ in South Sudan) at the end of year 7	Enabling environment created for private sector investment in renewable energy generation: 4.8 MW of solar and 3,890m ³ of Biodigester systems
	Number of small holder farmers, organizations and agribusiness MSMEs including women and youth headed households being eligible for getting access to innovative climate financing (credit) supported	Budget, Baseline assessment reports, pre-and post-intervention (qualitative and quantitative) survey results	0	660 (50% male and 50% female)	1,985 (50% male and 50% female)	Credit lines are made available through local PFIs for on-lending to program beneficiaries

	Number of small holder farmers, organizations and agribusiness MSMEs including women and youth headed households being covered by Climate Insurance	Budget, Baseline assessment reports, pre-and post-intervention (qualitative and quantitative) survey results	0	1,200 (50% male and 50% female)	3,600 (50% male and 50% female)	Climate insurances are made available through PFIs
	Number of agribusiness MSMEs and cooperatives including women and youth headed households being eligible for getting access to innovative climate financing (credit and insurance) supported	Budget, Baseline assessment reports, pre-and post-intervention (qualitative and quantitative) survey results	0	1,460 members of agribusiness MSMEs and cooperatives (73) including women and youth headed households have access to innovative climate financing	4,395 members of agribusiness MSMEs and cooperatives (219) including women and youth headed households have access to innovative climate financing	Credit line and Climate insurance are made available through PFIs
Outcome 3: Strengthened adaptive capacity of vulnerable communities						
Outcome 3: Strengthened adaptive capacity of vulnerable communities	Number of climate-related agricultural policies, regulations, or laws enacted with the use of climate information		0	5 climate-related agricultural policies, regulations, or laws enacted with the use of climate information	21 climate-related agricultural policies, regulations, or laws enacted with the use of climate information	Institutions and stakeholders readily engage with TA provided and show political will to streamline climate related policies
Output 3.1: Enhanced access to climate information and early warning systems	Number of agrometeorologic al stations installed and generating	Bill of quantities, contracts, quantitative and qualitative analysis reports, and final reports	0	25 (4 in Djibouti, 10 in Ethiopia, 7 in Kenya, 4 in South Sudan) at the end of year 3	56 (8 in Djibouti, 25 in Ethiopia, 14 in Kenya, 8 in South Sudan) at the end of year 7	Institutions and stakeholders' readiness to participate in the activities, and IGAD

	weather and climate data					member states capability in place to effectively implement the regional activities.
	Number of pastoralists and smallholder farmers (women and youth including women in female headed households) with a target of 40% having access to and use of climate-informed agro-advisory services	Baseline assessment reports, contract agreements, bill of quantities, pre-and post-intervention (qualitative and quantitative) survey results	0	500,000 (50% male and 50% female)	1,592,000 (50% male and 50% female)	Beneficiaries are willing to be trained in the use of climate-informed agro-advisory services through innovations introduced into the program for scaling up climate information services (CIS).
	Number of trainers trained on the use of climate services including early warning systems and data use	Training reports, progress and final reports	0	65 (50% male and 50% female)	200 (50% male and 50% female)	Enabling environment put in place to train trainers
	Number of tools for harmonisation early warning and climate services developed at regional level	Baseline assessment, progress reports, Midterm and final reports	0	2	2	Institutions and stakeholders' readiness to participate in the activities, and IGAD member states capability in place to effectively implement the regional activities.

Output 3.2: Strengthened regulatory and policy frameworks and institutional capacity for climate resilience.	Number of climate-related agricultural policies, regulations, or laws enacted or updated	Gazetted policies, regulations or laws, progress reports, Midterm and final reports	0	5	21	Institutions and stakeholders readily engage with TA provided and show political will to streamline climate related policies
	Number of regional climate risk mechanism designed and operationalised	Mainstreamed national plans and policies, contract agreements, and pre-and post-intervention (qualitative and quantitative) survey results	0	0	1	Enabling environment put in place for effective mainstreaming of climate risk management into sectoral and national plans, policies and strategies
	Number of trainers trained on mainstreaming climate change in policies	Training reports, progress and final reports	0	45 (50% male and 50% female)	140 (50% male and 50% female)	Enabling environment put in place to train trainers
	Number of farmers and pastoralist made aware of climate change, gender mainstreaming, conflict management, etc	Baseline assessment, progress reports, Midterm and final reports	0	110,000 (50% male and 50% female)	336,000 (50% male and 50% female)	Beneficiaries are willing to be trained in the use of climate-informed agro-advisory services through innovations introduced into the program for scaling up climate information services (CIS).
Cross-cutting - Program Coordination and Knowledge Management						
Output 4.1: Project management and coordination	Number of financial audit report	Progress reports, Annual performance	0	18	36	Effective coordination and management at the

FUND		reports, midterm and final reports				national and regional levels by the respective governments and IGAD Effectiveness of program M&E systems at the national and regional levels Effective and efficient M&E reporting
	Number of procurement audit report	Annual audit reports	0	18	36	
	Annual performance report	Annual performance reports	0	18	36	
	Number of capacity-building activities aimed at enhancing understanding and implementation of climate-smart agricultural practices and the program	Training reports, progress and final reports	0	36 (50% male and 50% female participation)	72 (50% male and 50% female participation)	
	Number of established M&E system		0	6	6	
	Number of knowledge products developed and disseminated	Reports, publications, and annual reports	0	18	36	
Project/programme co-benefit indicators						
Co-benefit 1	Number of jobs created for women and youth from rangeland and pastureland rehabilitation with women including women in female	Baseline assessment reports	0%	25,000 jobs created or sustained with at least 60% of jobs allocated to women and youth including women in	80,000 jobs created or sustained with at least 60% of jobs allocated to women and youth including women in	Number of jobs created for women and youth from rangeland and pastureland rehabilitation with women including women in female

	headed households involved in seed preparations for rehabilitation of lands			female headed households	female headed households	headed households involved in seed preparations for rehabilitation of lands
Co benefit 2	Ecosystem services improved	Annual Performance Report and national statistic records	0 ha	49,000 ha of rangelands / forests with improved ecosystem services	131,800 ha of rangelands / forests with improved ecosystem services	Activities related to rangeland rehabilitation and agro forestry are effectively implemented.

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 1.1.1: Construct or rehabilitate and manage water harvesting infrastructure Activity 1.1.2: Construct or rehabilitate and manage community irrigation schemes Activity 1.1.3: Construct or rehabilitate and manage community livestock infrastructure	The major activities include construction/rehabilitation of water mobilization infrastructure; development of irrigation lands and rehabilitation of rangeland/pastureland, rehabilitation of animal feed, health, and market access infrastructure, facilitating and operationalizing of cross-border memorandums of understanding to better control transboundary animal diseases and zoonoses in border areas.	N/A	• 538 mobilization infrastructure (i.e., small earth dams etc..) constructed, • 5980 ha of irrigation lands developed, • 152 animal feed, health, and market access infrastructure (e.g., livestock markets etc.) rehabilitated/developed,
Activity 1.2.1: Train communities on sustainable land management best practices	Introduction of pastureland sustainable land management practices with agroforestry and afforestation program, with community agroforestry and afforestation committees being set up and women including women in female headed households to serve as leaders and members of these committees, land loss and land restoration study, an assessment of land tenure patterns and corridors, design of tools, and the production of catalogues for land conflicts for	N/A	• 31,300 agropastoralists and pastoralists trained on sustainable land management best practices • 100 trainers trained at regional level on improved sustainable land management • 102,000 ha of rangeland/pastureland rehabilitated,

Activity 1.2.2: Support sustainable rangeland management and agroforestry	transformation to enhance resilience and adaptation to climate change, land planning for climate smart villages and local development plan at community, mapping exercise to identify smallholder farmers and pastoralists who must be organized into Farm-based Associated (FBAs). Enhancement, management and regeneration of rangelands at regional and national level will also be supported.		3 cross border rangelands (cluster) sustainably managed 29,800 ha of degraded land/ forest rehabilitated and managed under agroforestry and plantation
Activity 1.3.1: Provide extension services and deploy climate-smart innovations to improve agricultural and fodder production Activity 1.3.2: Disseminate resilient varieties to improve agricultural and fodder production	The three transformative areas under this activity include investing in the validation and cost-effectiveness analysis of integrated pest management technologies to fight against fall armyworm, arboviruses in transhumant cattle, and locusts invasion, developing regional guidelines on integrated pest management control and facilitating the signing of a regional MoU between animal health services and research institutions; and upscaling the Technologies for African Agricultural Transformation (TAAT) Program to pastoralists and smallholder farmers in the HOA. This activity will also support commercial production of fodder in pastoralists areas; contributing to improved livelihood and reduced vulnerability	N/A	278,000 small holder farmers and pastoralists benefiting from extension services 250 trainers trained on tools for extension services on climate smart innovations/dissemination 264,000 beneficiaries receiving resilient varieties to improve agricultural and fodder production
Activity 2.1.1: Establish and deploy digital advisory services using PPP mechanism Activity 2.1.2: Construct rural markets	This activity will support the development of digital solutions through public private partnership The initiative will also support smallholder farmers (women and youth), farmers organization and agribusiness MSMEs to benefit from extension on using digital advisory services; and establish a regional innovation hub/data centre to coordinate and share knowledge among innovation partners, governments and other stakeholders. This activity will also involve the construction and development of various market infrastructure components across different regions of the participating countries	N/A	5 of digital advisory services established through PPP mechanism 1,142,000 small holder farmers and pastoralist benefiting from the established and deployed digital advisory services 2 regional infrastructure (hubs) developed for supporting digital advisory services 125 rural markets constructed included access
Activity 2.2.1: Train women and youth on entrepreneurial skills in renewable energy technologies and/or AVC development including access to credit line Activity 2.2.2: Facilitate access for women and youth groups to renewable	Under this activity, AfDB and GCF will provide training to over 1985 women and youth on entrepreneurship in renewable energy technologies (Solar and Biogas). The focus is to empower producer groups, including women's groups, by providing training, structuring, and raising awareness of entrepreneurship, leading to the development of business plans for cooperatives and MSMEs. BREFOL will support the establishment of a Dedicated Direct Credit Window (BDCW) to provide local private financial institutions (PFIs) with stable lendable resources to adequately meet the financing needs of smallholder farmers, pastoralists, MSMEs, FBAs and other	N/A	410 groups women and youth and cooperatives trained on renewable energy and/or AVC development and having access to credit 362 agribusiness MSMEs trained on renewable energy and/or AVC development and having access to credit 10,680 people including women in female headed households and youth with a target of 50% to receive inputs and finance to generate

energy (solar and biodigester systems) to enhance alternative income generation Activity 2.2.3: Design and operationalise innovative credit lines via selected local PFIs, in participating countries. Activity 2.2.4: Provide technical assistance to MSMEs, local PFIs, and cooperatives to create an enabling environment for credit access and sustainability	agribusiness MSMEs with vested interest in green/circular economy development		alternative income from renewable energy sources such as solar and biodigesters • 4.8 MW of solar PV installed • 3,890 meter cube of biodigesters installed • 1985 small holder farmers, organizations, and agribusiness MSMEs including women and youth headed households being eligible for getting access to innovative climate financing (credit) supported • 3,600 small holder farmers, organizations, and agribusiness MSMEs including women and youth headed households being covered by Climate Insurance • 4,395 agribusiness MSMEs and cooperatives including women and youth headed households being eligible for getting access to innovative climate financing (credit and insurance) supported •
Activity 3.1.1: Expand agrometeorological stations network to generate data for climate advisory services Activity 3.1.2: Integrate and disseminate climate information and early warning into the established digital advisory services	The project will support to expand the agrometeorological stations network to improve climate resilience, agricultural planning, and livelihoods in the HoA. This activity will be pivotal in enhancing climate resilience and agricultural productivity in Ethiopia, Kenya, and South Sudan. By installing agrometeorological stations, the capacity to generate accurate and localized weather and climate data will be significantly improved, facilitating more effective climate advisory services. This activity will also integrate and disseminate climate information and early warnings in Ethiopia, Kenya, and South Sudan using digital solution platforms established through the PPP mechanism	N/A	• 128 agrometeorological stations installed and generating weather and climate data • 1,100,000 pastoralists and smallholder farmers (women and youth including women in female headed households) with a target of 40% having access to and use of climate-informed agro-advisory services • 200 of trainers trained on the use of climate services including early warning systems and data use • 2 tools for harmonisation early warning and climate services developed at regional level
Activity 3.2.1: Support delivery of climate regulatory framework and mainstreaming of climate change into agricultural sector policies and strategies	Investment will be directed to facilitate mainstreaming policy dialogues with key stakeholders on climate index-based insurance to advance actionable reforms in Ethiopia. There will be coordination among the National Meteorological Agency (NMA), ATI, insurance service providers.	N/A	• 21 climate-related agricultural policies, regulations, or laws enacted or updated • 1 regional climate risk mechanism designed and operationalised • 140 trainers trained on mainstreaming climate change in policies

			336,000 farmers and pastoralist made aware of climate change, gender mainstreaming, conflict management, etc
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E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

203. Monitoring of the programme

All projects financed under the Facility will be monitored by the AfDB's portfolio management team in accordance with relevant internal policies and procedures. The AfDB, as the accredited executing agency, will be responsible for directly monitoring implementation conditions and reporting periodically to the GCF under terms to be agreed between the AfDB and the GCF. All projects financed under the Facility will comply with AfDB standards and procedures for appraisal, approval, monitoring and supervision, involving representatives or all relevant teams (engineers, lawyers, project finance specialists, procurement experts, E&S specialists, gender specialists, climate finance officers, financial management officers, supervision and monitoring specialists). The implementation and monitoring of each stage of the project will be guided and managed by the AfDB Project Lifecycle Management Framework. The key task managers who will carry out due diligence, implementation supervision, risk monitoring and mitigation will be located in the relevant teams at Headquarters and in the IGAD regional and country offices.

204. Reporting

1) Reporting by project companies to the AfDB will be in line with the standard loan agreement and the AfDB will conduct a semi-annual supervision.

2) AfDB reporting to the GCF: The AfDB will comply with the relevant GCF policies (as specified in the AMA) in the reporting and evaluation arrangements for this framework. The AfDB will submit the Annual Performance Report (APR) to the GCF during the seven-year implementation period. In addition, a semi-annual activity report on the status of individual subprojects financed by the GCF will be provided during the life of the sub-loans. For the TA component, reports from the beneficiaries will be consolidated by the AfDB for reporting to the GCF. In addition, an inception report, mid-term evaluation report, final evaluation report, and financial information reports will be submitted (semi-annually throughout the life of the programme), as agreed under the AMA and FAA.

205. Evaluation

The evaluation arrangements for this framework will be in line with the relevant AfDB and GCF policies. Both the mid-term and final independent evaluations will be carried out by the AfDB's Independent Evaluation Department (IDEV). The AfDB's independent evaluation work will be guided by internationally accepted principles for the evaluation of development assistance, in particular the Organisation for Economic Co-operation

and Development Assistance Committee's (OECD DAC) Evaluation Guiding Principles and the Evaluation Cooperation Group's (ECG) good practice standards.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistle-blowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

Selected Risk Factor 1

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>

Description

206. This risk arises when agriculture, rangelands and food production systems are affected by many uncontrollable events, often related to weather, biology and the environment, including excessive or insufficient rainfall, extreme temperatures, diseases and zoonoses, and land degradation. This may also be due to low levels of technology adoption and access to finance. Key factors for this risk include natural conditions, biological and environmental hazards, technology levels, natural disasters, finance, demand and policy decisions.

Mitigation Measure(s)

207. The mitigation measures put in place through the program include:

- Increasing access to affordable finances to smallholder farmers, pastoralists, and other agribusiness MSMEs to adapt to climate change,
- Enhancing access to climate-informed advisory services in agriculture and food production systems
- Introducing sustainable land management practices such as agroforestry and carbon farming in degraded rangeland/pastureland.
- Facilitating cost-effectiveness analysis of integrated pest management technologies to fight against fall armyworm, arboviruses in transhumant cattle, and locust's invasion
- Increasing the preparedness and readiness of smallholder farmers, pastoralists, and institutions to adapt to climate change and climate extremes
- Promoting greater private sector participation in green/circular economic building activities.
- Integration of technological innovations in agriculture such as the **Technologies for African Agricultural Transformation (TAAT)** Program, and the **Global Centre on Adaptation (GCA) Smallholder Adaptation Accelerator Program (AAAP)** to scale up digital advisory services and markets in climate responsive planning
- Provision of access to improved seed varieties, pest control and surveillance services through cross-border MoUs.
- Scaling up and disseminate proven agricultural technologies and innovations to millions of smallholder farmers
- Introducing cross-border memorandums of understanding to better control transboundary animal diseases and zoonoses in border areas.
- Capacity building in agroecological systems that promote crop diversification and therefore reducing risks to crop diseases and weather shocks.
- Introducing social protection mechanisms such as weather-based crop/livestock insurance programs.

Selected Risk Factor 2

Category	Probability	Impact
<u>Credit</u>	<u>Medium</u>	<u>Medium</u>

Description

208. Timely access to finance is critical to cope with the debilitating impacts of climate change and extreme weather events that characterise the HoA region. Not infrequently, smallholder farmers do not have access to credit facilities to purchase agricultural inputs such as climate-smart seed varieties, fertilizers and farm implements, and to adopt climate-smart agricultural practices to cope with sudden disasters such as seasonal droughts and flash floods. This is illustrated by statistics on access to finance from the World Bank's Global Findex Database, which shows that on average only 22% of smallholder farmers in the HoA have an account with a financial institution, and less than 7% are able to borrow from a financial institution, while only about 6% of the active farming population is able to borrow to start, operate or expand a farm or business. With such low access to credit in an environment characterised by extreme climate events (droughts and floods), fragility, conflict and population explosion, building resilience at scale is almost impossible.

Mitigation Measure(s)

209. The mitigation measures put in place through the program include:

- Integration of innovative financial instruments to help 'crowd-in' additional private sector finances for resilience building in the region.
- Introducing carbon credit schemes to help increase access to finance and building of the adaptive capacity of smallholder farmers and pastoralists in the region.
- Establishing dedicated credit windows to increase access to finance to invest in different value chains.
- Training farmers and business holders on the principles of business and financial planning and management.
- Ensuring diversified source of incomes for the participants such as through carbon farming, use of renewable energy in food crops processing.
- Broadening climate finance access based for smallholder farmers, pastoralists, agribusiness MSMEs etc., from different financial sources such as from local PFIs, private investors, insurance promoters, and beneficiary groups.

Selected Risk Factor 3

Category	Probability	Impact
<u>Governance</u>	<u>Medium</u>	<u>Medium</u>

Description

210. This risk can arise from changes in government sector strategies and policies, the political and governance context of countries (insecurity and conflict), and non-compliance with the Bank's fiduciary guidelines, resulting in implementation delays. It may also be due to limited coordination between agencies and stakeholders, and weak implementation capacity within Project Management Units (PMUs).

Mitigation Measure(s)

211. The mitigation measures put in place through the program include:

- Building communities' resilience to climate change is a priority for the region and for the five participating IGAD countries. The involved governments are maintaining adequate political commitment to implementing the program and maintain their effort on promoting sector specific strategies and policies conducive for the implementation of BREFOL.
- IGAD has put in place a conflict resolution mechanism that might reduce the challenges of insecurity and conflicts. The Program is itself expected to foster peace among conflicting communities.
- BREFOL will build the capacity of the program staff and the program's beneficiaries. Lessons learnt from the DRSLP have been integrated in the program's design and the implementation plan,
- The Bank will continue to run fiduciary clinics to strengthen capacity.
- In some participating countries, the program will use PMUs that are already running DRSLP to be strengthened with the addition of specialized staff. Additional capacity building/training will be provided. AfDB will also conduct annual fiduciary clinics to address any remaining capacity gaps.

Bank staff will conduct regular supervision missions and audits and follow up closely to monitor adherence to Bank standards and procedures as a way of securing value for money.

- 212.** Additionally, a study has been conducted to determine how economic, institutional, political, cultural, and structural problems in the HoA countries may contribute to fragility and conflict (see Annex 24 for the Fragility assessment report). This assessment contributes to a better understanding of how vulnerable the host nations are in their particular setting and how to generate useful information that could be incorporated into projects, programs, and strategies for development.

Selected Risk Factor 4

Category	Probability	Impact
<u>ML/FT</u>	<u>Medium</u>	<u>Medium</u>

Description

- 213.** This risk can manifest itself in scenarios characterised by a lack of integrity among programme beneficiaries, the involvement of politically exposed persons (PEPs), the presence of sanctions, civil, criminal and regulatory backgrounds, and potential anti-money laundering and countering the financing of terrorism (AML/CFT) issues. In such situations, vulnerability to financial impropriety, illegal activities and compliance breaches becomes a significant concern, requiring thorough scrutiny and risk mitigation measures. An integrity assessment has been carried out on the executing agencies. This integrity assessment was conducted using the principles of integrity due diligence, namely beneficial ownership, politically exposed person risk, sanctions risk, civil, criminal and regulatory background, and AML/CFT & integrity policy assessment.

Mitigation Measure(s)

- 214.** The mitigation measures put in place through the program include:
- The Bank will constantly monitor its counterparties through its supervision activities. This process is also strengthened by the daily automated sanctions, PEP, and adverse media monitoring on Bank's counterparties.
 - All outgoing payments from the Bank will be automatically screened against sanctions lists through Swift.
 - A training program will be defined on an annual basis in consultation with the relevant Bank's units. Training and Capacity Building is an integral part of the Bank preventive activities around sanctionable practices and ML/TF. The main audience will be staff from Project Implementation Units and the training topics will focus mainly on the Bank's integrity framework and ML/TF issues in Project Management.
 - Specific integrity clauses are incorporated in all project documentation between the Bank and its counterparties. These clauses include including Audit & Inspection Rights, Adherence to the Bank Integrity Framework, Transposition of due diligence requirements for final beneficiaries.
 - The Bank's has different oversight mechanisms that will ensure that any indicator of sanctionable practices is reported to the Integrity and Anti-Corruption Office for investigation with a sanction mechanism for substantiated cases. The Bank also has a whistleblowing policy which provides several channels for raising concerns related to sanctionable practices and provisions to protect whistle blowers from retaliation.

Selected Risk Factor 5

Category	Probability	Impact
<u>Other-- Risk of uneven uptake among countries</u>	<u>-Medium</u>	<u>Medium</u>

Description

There may be challenges in the use of resources during implementation, which could lead to uneven uptake between countries. For example, if a country has a lower take-up rate, the funds allocated to that country may remain unused, which could affect the climate impact of the project.

Mitigation Measure(s)

The mitigation measures retained are

- Firstly, flexibility of resource reallocation in most secure areas of the country in case of conflict and
- Secondly, restructuring of the financing plan among countries, if necessary, as last option, after the mid-term review to transfer resource to most competitive countries.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

- 215.** BREFOL is a climate change adaptation operation that aims to improve the resilience of communities and agropastoral production systems with minimal negative environmental and social impacts. As a result, the Bank's integrated safeguards system has classified the programme as Category 2, meaning that site-specific programme activities are expected to have only moderate environmental and social risks and impacts that can be managed through the instruments proposed in the programme's environmental and social safeguards instruments, Environmental and Social Impact Assessments (ESIAs), Environmental and Social Management Plans (ESMPs) and Environmental and Social Management Frameworks (ESMFs). The ESIA/ESMP and ESMF reports were validated by the participating countries and disclosed both in-country and on the Bank's website in accordance with the 30-day disclosure requirement before the BREFOL was submitted to the AfDB Board of Directors for consideration.
- 216.** BREFOL will have a significant positive impact on the environment and ecosystems in the Horn of Africa. In particular, it will: (i) improve natural resource management at local, national and regional levels; (ii) improve access to infrastructure for water, animal health and access to markets; (iii) rehabilitate rangelands and pastures; (iv) improve productivity/production in the livestock, crop, forestry and pastoral sectors; (v) increase the number of farmers and pastoralists using climate services; and (vi) conserve rangelands and soils and prevent erosion.
- 217.** The disclosed ESMF (see Annex 6 for details) outlines measures to minimise and mitigate the potential negative impacts of the programme and to optimise its benefits. The implementation of environmental measures will involve various stakeholders, including governments, local authorities, communities, grassroots organisations, NGOs, local people, women, the private sector, research institutes, etc. The environmental monitoring of the programme will be carried out by the entity responsible for environmental and social safeguards in each country.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

- 218.** Information from the gender assessment reports and consultations with stakeholders during the assessment mission in the five participating countries indicate that women in male- and female-headed households in agropastoral communities face several gender-based barriers (see Annex 8). These barriers include harmful traditional practices, low levels of literacy and less power over intra- and extra-household decisions on productive resources. These conditions undermine women's well-being, economic empowerment and gender equality as a whole. In addition, women have less access to social infrastructure and services, and their low levels of literacy and marketable skills mean that they are less active in the labour market. If these challenges are not addressed during programme implementation, they could hamper women's participation in BREFOL, hinder women's economic empowerment and undermine gender equality. For this reason, BREFOL has budgeted for a gender action plan at regional and national levels. The plan uses the Gender Action Learning System (GALS) approach, which seeks to transform harmful and disadvantageous socio-cultural gender norms and practices that significantly hinder women's empowerment and gender equality. Based on the regional components of the programme and the GALS methodology, BREFOL will train national champions to support planned women's economic empowerment activities. In these activities, women will act as the main agents of change and men as their allies. An additional concern is the role of women in the agribusiness activities planned by the programme: this role could increase women's workload, as women are already responsible for household and family activities. To address this concern, BREFOL has budgeted for gender-sensitive infrastructure that will reduce women's workload and poverty, and remove barriers to childcare, by building children's playgrounds and crèches, and locating boreholes near villages and residential areas, solar energy systems and bio-digesters. Of the 1.6 million people who will benefit from climate-informed agricultural extension services, 50% will be women and 30% will be youth (males and females aged 18-35). In addition, a climate risk insurance scheme will be established through a public-

private partnership to de-risk existing and new agribusiness MSMEs in the project areas (see Annex 16 for proposed intervention areas).

219. 221. In the Bank's gender marker system, BREFOL is classified as GEN II. This category indicates that, by the end of the programme period, the programme is expected to have generated several positive outcomes for women. For example, it is expected to increase the incomes of women involved in agro-pastoral value chains (the output indicator for this outcome is the increase in annual per capita income for women in both male- and female-headed households) and to create jobs for women and youth. The programme is also expected to improve women's access to social services and infrastructure, which will reduce women's labour and time poverty and enable women to participate in more livelihood diversification activities. A regional gender analysis and the corresponding gender action plan with a regional budget allocation are provided as Annex 8 to the FP. In summary, it is envisaged that BREFOL will reduce women's gender-related challenges and empower women economically by improving their access to productive resources and gender-sensitive infrastructure and services.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

220. 222. AfDB's fiduciary and financial management measures for BREFOL in upstream engagement with programme countries during the preparation of Country Strategy Papers (CSPs), where fiduciary risks and country capacity were assessed in the Country Fiduciary Risk Assessment (CFRA). This is a key document for dialogue with the country on the risk assessment and agreed mitigation actions to address identified fiduciary risks, including corruption and AML/CFT and prohibited practices issues. The AE has also carried out an integrity assessment using the principles of integrity due diligence, namely beneficial ownership, the presence of politically exposed persons (PEPs), the existence of sanctions, civil, criminal and regulatory background and AML/CFT of the EEs.

221. Generally, the Financial Management Arrangement (FMA) of BREFOL, requires:

- **Accounting Management:** Legal agreements require BREFOL participating countries to: (i) maintain accounts and records and prepare financial statements for projects and subprojects financed from AfDB resources and resources administered by the AfDB; and (ii) establish accounting, administrative, and financial procedures in manuals. These accounts and records should justify the use of funds disbursed to the programme or project and be prepared in accordance with applicable accounting standards.
- **Accounting standards and practices:** Although accounting standards and practices may vary from country to country, the accounts of the programme or sub-project under BREFOL, irrespective of the system adopted, shall comply with the principles of transparent management and provide comprehensive information on the following aspects 1) accountability for the funds provided to the programme or sub-project; these funds may include the proceeds of the AfDB loan or grant, or other resources from co-financiers and counterpart funds; 2) disclosure of expenditure and specific information on components financed under the programme.

222. Programme Financial Statements: The programme or sub-projects financial statements prepared under BREFOL shall include:

- For all projects: i) Statement of Income (funds received from AfDB, counterpart funds and, where applicable, co-financiers) and Expenditure (expenditure incurred both for the current year and accumulated to date) showing separately AfDB funds, counterpart and co-financiers funds and cash balances; 2) Statement of Special Account(s); 3) Notes to the Financial Statements describing the applicable accounting principles used and a detailed analysis of the main accounts.
- For revenue generating sub-projects under BREFOL (in addition to the requirements above): i) a balance sheet showing the accumulated funds of the project, bank balances and other assets and liabilities of the project at the end of each financial year; ii) a profit and loss account (or an operating

account or an income and expenditure account or a profit and loss account); and iii) a cash flow statement showing the cash flows during each financial year.

- As an annex to the financial statements prepared under BREFOL, the following would be included as an annex to the financial statements of all projects under the programme: i) a reconciliation between the amount shown as "received from the Bank" and the amount shown as disbursed by the AfDB. The reconciliation should indicate the methods used for disbursement, i.e. special account, direct payment or reimbursement guarantee, and the reimbursement methods with those recommended in the Project Appraisal Reports (PARs) and the Disbursement Letter; and ii) a comprehensive list of all fixed assets acquired with dates, value and condition of the assets.
- **Financial reporting:** BREFOL participating countries are required to provide regular and comprehensive accounting and financial information on the programme and sub-projects under the programme. The information should detail the various uses of the loan and grant proceeds. Financial systems and financial communication reports may vary depending on the type of sub-project, lending instrument and country. The AfDB may not require a standard format but will ensure that reports submitted for review and consideration are sufficiently informative.
- **Retention and filing of documents:** Each country participating in the BREFOL programme is required to maintain and preserve programme/sub-project documents and supporting documents and to make them available to AfDB representatives as required.
- **Retention of documentation:** The AfDB's General Conditions require each country participating in the programme to retain all documents relating to the loan proceeds so that they can be audited by external auditors or the AfDB's representative. Documentation should be retained in a manner that is readily accessible.
- **Audit:** The general conditions of the loan and grant agreements require that all programme and sub-project accounts be audited annually in accordance with relevant generally accepted standards. According to the standard terms of reference (ToR) for external audits of AfDB-financed programmes or projects, the auditor is required, among other things, to examine relevant documents, review internal financial control mechanisms to identify deficiencies and weaknesses that could affect the efficiency of the project, form an opinion on the quality of the financial statements, and confirm that the funds granted to the project have been used for their intended purpose. Each country must have the accounts of the project or programme audited by qualified and independent auditors who meet AfDB requirements. In countries where the audit is entrusted to the Auditor General's Office (AGO) under the supervision of the government, the AfDB may approve the appointment of this office, provided that the matter is discussed during programme/project appraisal and loan negotiations, and the audit is conducted in accordance with the AfDB's standard terms of reference for the audit of AfDB-financed projects/programmes. The audited financial statements, auditor's report and management letter must be received by the AfDB no later than six months after the end of the financial year to which they relate. Failure to comply with this requirement will result in the immediate suspension of further replenishments to the Special Account.

223. Procurement and Disbursement Arrangements

- **Procurement Arrangements:** The procurement of goods (including services other than consultancy services), works and the acquisition of consultancy services financed under BREFOL will be carried out in accordance with the Procurement Framework for Bank Group-financed Operations, 2015 Edition, and in compliance with the provisions of the GCF FAA (see Annex 10 for the detailed procurement plan). More specifically, procurements will be made in accordance with the AfDB's Procurement Methods and Procedures (BPP), based on the relevant Standard Bidding Documents (SBDs) for goods and works contracts, as well as consulting services, for which the BPP is considered the most appropriate. Indeed, following an analysis of the procurement systems of the five BREFOL participating countries, the procurement systems and procedures of **Djibouti, Somalia and South**

Sudan were considered to be weak. On this basis, it was decided to apply the AfDB's procurement methods and procedures in **Djibouti, Somalia and South Sudan**. In Ethiopia and Kenya, BREFOL will use each country's procurement system for national competitive bidding and procurement of goods and works, subject to the conditions for Ethiopia and Kenya discussed above. When these two countries procure packages of goods, works and consultancy services through open international competitive bidding, the AfDB's procurement methods and procedures will apply.

224. Disbursement: The disbursement methods to be used to mobilise project funding are: (i) the direct payment method; (ii) the special fund/revolving fund method; and (iii) the reimbursement method. Direct payments will be made in respect of contracts for works, goods and services signed between borrowers and suppliers in accordance with AfDB procurement rules and procedures and national procurement laws. The special account method will be used to settle operating expenses, training costs, field mission costs, etc. It requires the opening of a special account with the central banks of the five participating countries, which will in turn transfer all funds received from the AfDB to the PMU account opened with a bank acceptable to the AfDB. The opening of the bank account will be a condition precedent to the first disbursement. The reimbursement method will be used when eligible expenditures of all project funds are pre-financed with the prior approval of the AfDB. Disbursements will be made in accordance with the List of Goods and Services and AfDB's policies and procedures as described in the Project Disbursement Manual.

225. Conditions Precedent to First Disbursement of Grants and Loan under BREFOL: The obligation of AfDB to make the first disbursement of the respective grants and loan shall be subject to the entry into force of the respective Protocol of Agreements (POA) and the Loan Agreement (LA), in accordance with the provisions set out in the POA and LA, and the submission of evidence, in a form and substance acceptable to AfDB, of fulfilment by the Recipient or Borrower of the following conditions:

- i. Submission of evidence of the recruitment/appointment of staff for the **Project Management Unit** (PMU) for each country, and a **Regional Program Coordination Unit** (RPCU) for IGAD with qualifications and terms of reference acceptable to the AfDB.

Other conditions:

226. The Borrower/ Recipient shall provide to the AfDB, within six months after the first disbursement:

Djibouti, Ethiopia, Kenya, Somalia, South Sudan:

- ii. Submission of the evidence of the establishment of the National Project Steering Committee (NPSC) with composition and terms of reference acceptable to AfDB.
- iii. Provision of Counterpart Contribution, as stipulated in the respective loan/grant agreements

IGAD:

- i. Submission of the evidence of the establishment of the Regional Program Steering Committee (RPSC), which will include representatives from the regional programme coordination unit (RPCU), IGAD Climate Prediction and Applications Centre (ICPAC), the IGAD Pastoral Centre for Livestock and Pastoral Development (ICPALD), the IGAD Food Security, Nutrition and Resilience Analysis Hub (IFRAH), and representatives from the National Program Steering Committees (NPSCs), within three months of the first disbursement of the Grant.

227. Undertakings:

The Borrower/ Recipients undertakes to:

- ii. carry out the Program in accordance with the Environmental and Social Management Plans ("ESMPs"), and the Bank's Safeguards Policies and the applicable national legislation in a manner and in substance satisfactory to the AfDB;

- iii. prepare and submit to the AfDB Project Reports, including quarterly reports on the implementation of the ESMPs including any deficiencies identified and the corrective measures thereto;
- iv. refrain from taking any action which would prevent or interfere with the implementation of the ESMPs, including any amendment, suspension, waiver, and/or voidance of any provision thereof, whether in whole or in part, without the prior written concurrence of the AfDB;
- v. each of the Borrower/Recipient shall cause its respective relevant sector ministry to provide coordination for implementation of country Projects; and
- vi. each of the Borrower/Recipient shall cause the establishment of a National Project Steering Committee (NPSC) which shall be responsible for respective country Project oversight, overall policy guidance, strategic direction, and review and approval of work plans and budgets.

228. Programme Monitoring and Evaluation (M&E): Simply refers to the process of procurement-related M&E throughout the programme (and procurement) cycle. The information to be monitored for the programme and sub-projects under the programme will be identified in the Programme Executing Agency Procurement Assessment (EAPA), drawing on sources such as the Country Procurement Assessment Report (CPAR). Capacity development as outlined in the Procurement Development Action Plan (PDAP) will also be subject to M&E throughout programme implementation. The AfDB's primary monitoring tools are post-procurement reviews (PPRs), procurement audits (PAs), independent procurement reviews (IPRs), and other miscellaneous reviews associated with AfDB programme or project-related supervision and monitoring.

229. In all five participating countries, including IGAD, the programme will be implemented over a period of 84 months. The M&E Officers to be appointed from the PMUs and the Regional Programme Coordination Unit (RCU) will be responsible for collecting all data, including gender-disaggregated data at all levels where possible, analysing and compiling information on physical achievements and financial implementation. To this end, he/she will have at his/her disposal a roadmap with quantitative indicators of the programme's progress, relating to the implementation of the sub-components. This monitoring will enable the preparation of quarterly progress reports in accordance with the Bank's format and procedures: physical objective, level of achievement, expected costs, actual costs, variances and explanations for any variances. This information will be used to prepare the quarterly, mid-year and annual performance programme reports by the respective EE together with the NPSC and RPSC. In addition, the countries will recruit the necessary expertise to develop a harmonised, gender- and conflict-sensitive monitoring and evaluation mechanism and prepare an implementation manual to operationalise the collection and processing of national and regional data and the reporting of results against the indicators set out in the logical and M&E frameworks. In addition to the review work of the Steering Committee, the AfDB will conduct two monitoring missions per year, in addition to a mid-term review mission. The mid-term review will be conducted in year 3. Upon completion of the programme, the AfDB and the respective governments will prepare a final report within the required timeframe.

G.4. Disclosure of funding proposal

Indicate below whether or not the funding proposal includes confidential information.

☐ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) **(template provided)**
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan **(template provided)**
- ☒ Annex 5 Implementation timetable including key project/programme milestones **(template provided)**
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
(ESS disclosure form provided)
 - ☒ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan **(template provided)**
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan **(template provided)**
- ☒ Annex 11 Monitoring and evaluation plan **(template provided)**
- ☒ Annex 12 AE fee request **(template provided)**
- ☒ Annex 13 Co-financing commitment letter, if applicable **(template provided)**
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

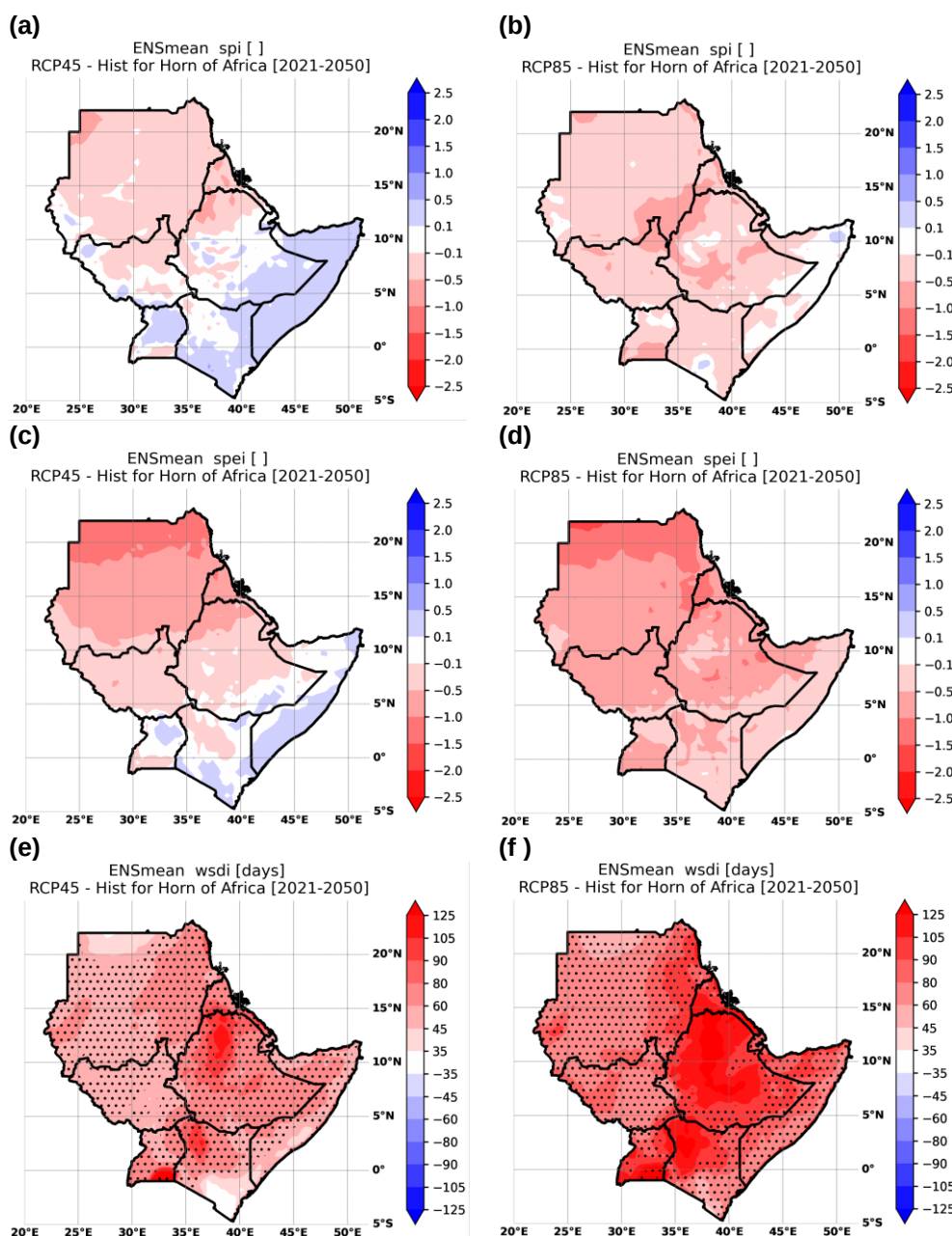
- ☒ Annex 15 Evidence of internal approval **(template provided)**
- ☒ Annex 16 Map(s) indicating the location of proposed interventions
- ☒ Annex 17 Multi-country project/programme information **(template provided)**
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)⁶¹
- ☒ Annex 23 BREFOL Beneficiary Estimate
- ☒ Annex 24 BREFOL GCF Fragility Assessments
- ☒ Annex 25 Digital Solution PPP Model for BREFOL

* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.

APPENDICES

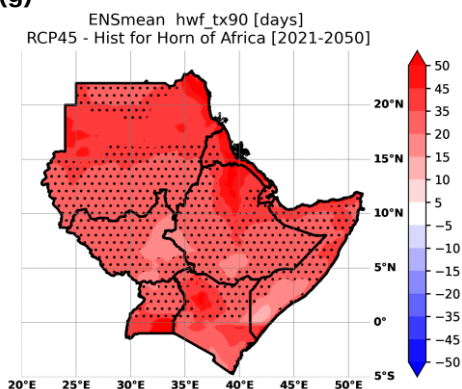
Appendix 1

Appendix 1.1: Projected change in some selected WMO Expert Team Sector-Specific Climate Indices: (a&b) standardized precipitation index (SPI); (c&d) standardized precipitation evapotranspiration index; (e&f) warm spell duration (days) and (g&h) heat wave frequency (days); in the GHA countries between the period 2021-2050 and reference time (1976-2005) under RCP4.5 and 8.5 emission scenarios. Stippling denotes regions where the difference is statistically significant at 0.05 level.

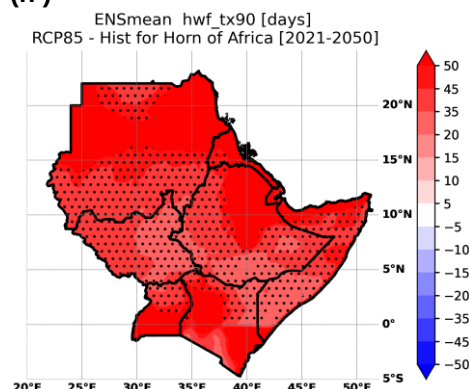


⁶¹ Annex 22 is mandatory for mitigation and cross-cutting projects.

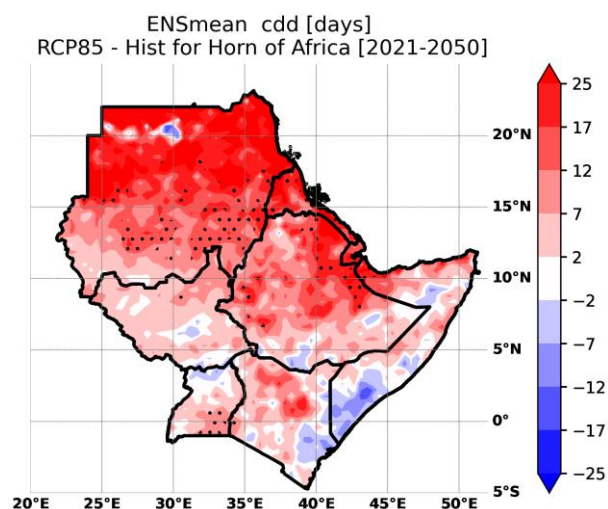
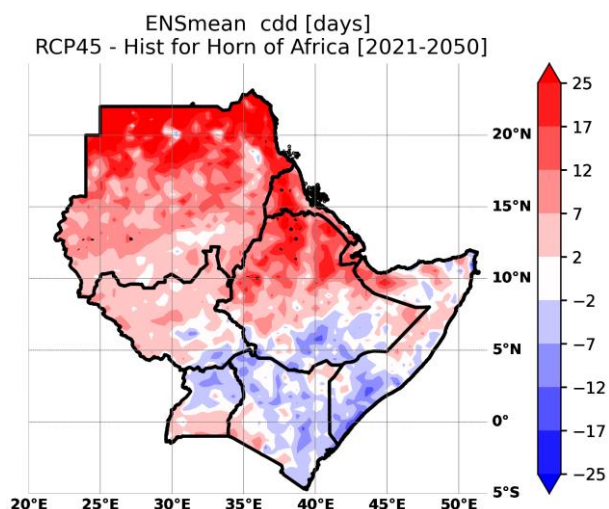
(g)



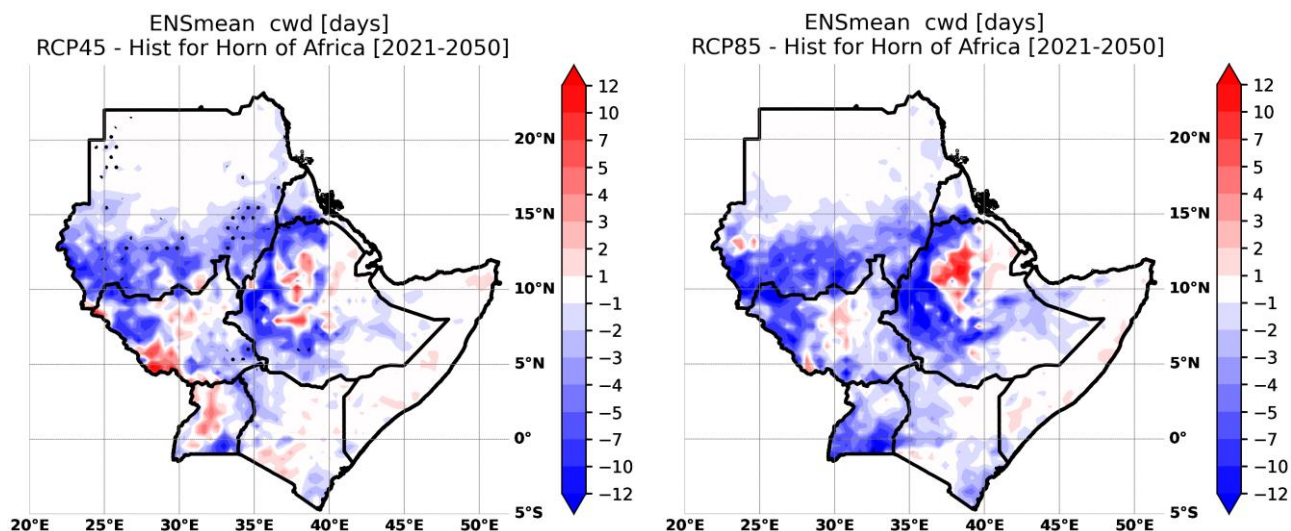
(h)



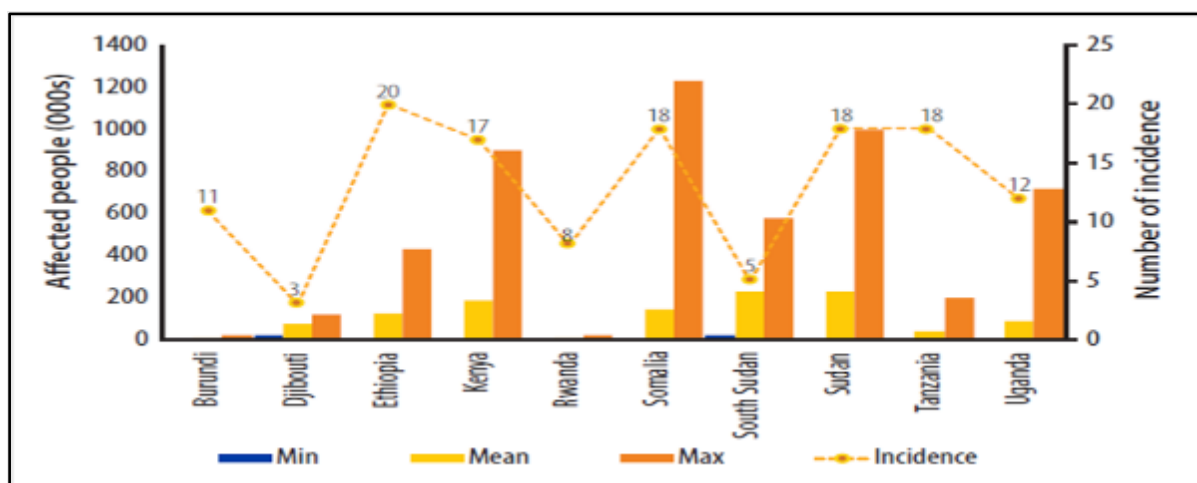
Appendix 1.2: Projected change in consecutive dry days (CDD) in the GHA countries between the period 2021-2050 and reference time (1976-2005) under RCP4.5 and 8.5 emission scenarios. Stippling denotes regions where the difference is statistically significant at 0.05 level.



Appendix 1.3: Projected change in consecutive wet days (CWD) in the GHA countries between the period 2021-2050 and reference time (1976-2005) under RCP4.5 and 8.5 emission scenarios. Stippling denotes regions where the difference is statistically significant at 0.05 level.

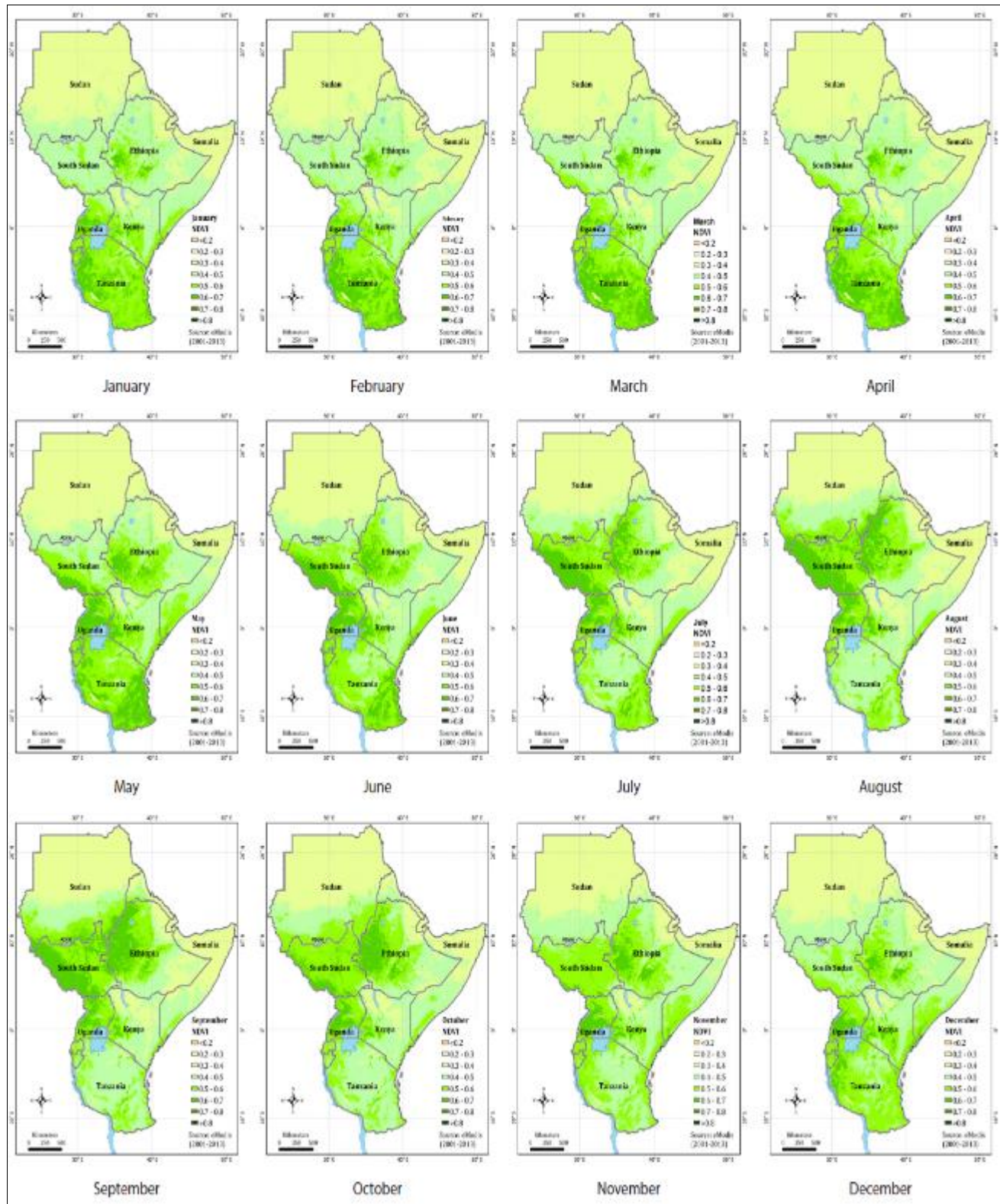


Appendix 1.4: Number of People Affected by Floods in the HoA Region beginning from 90s

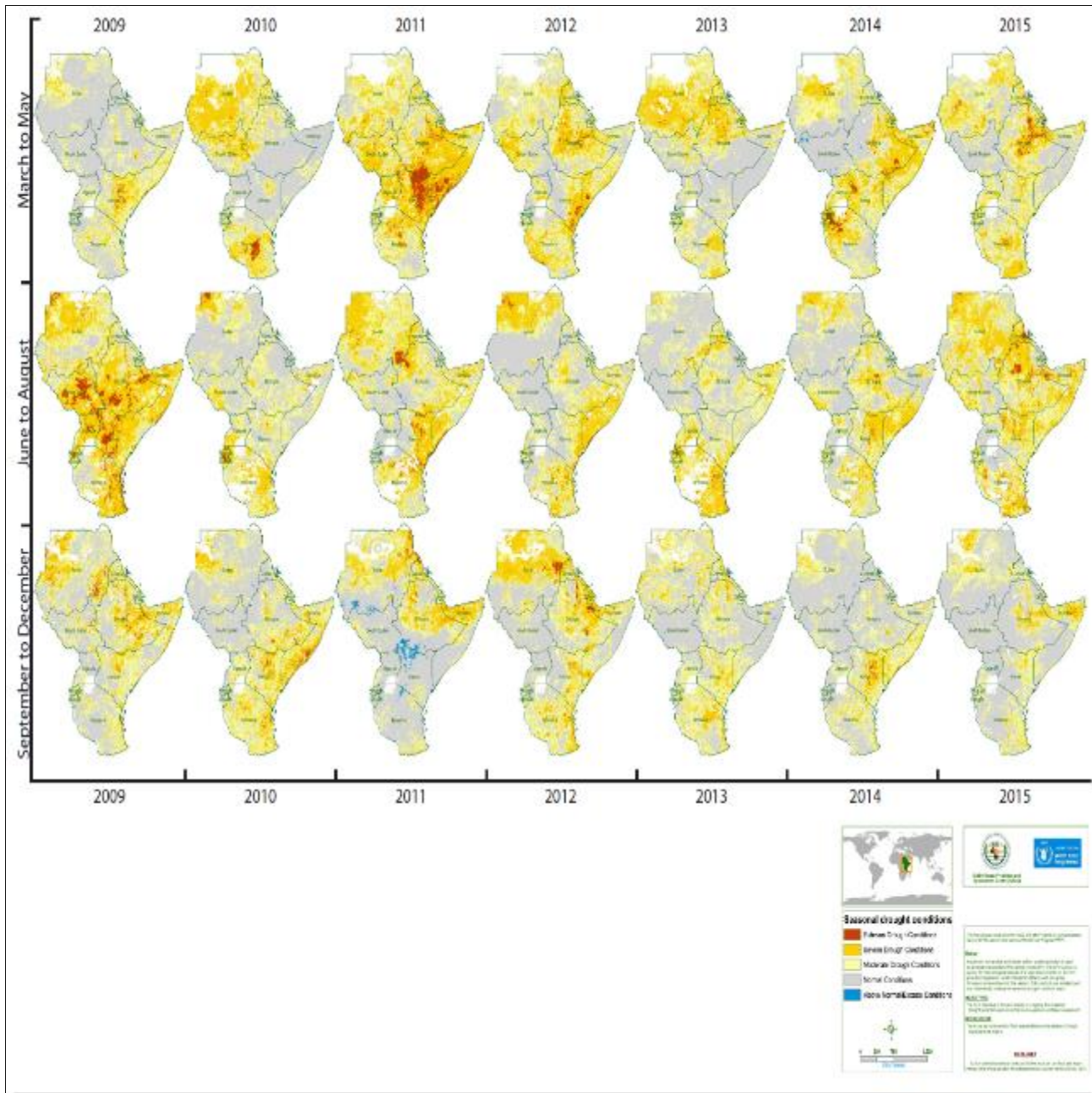


Source: Adapted from the Climate Risks and Food Security Atlas of the GHA based on EM DAT Database

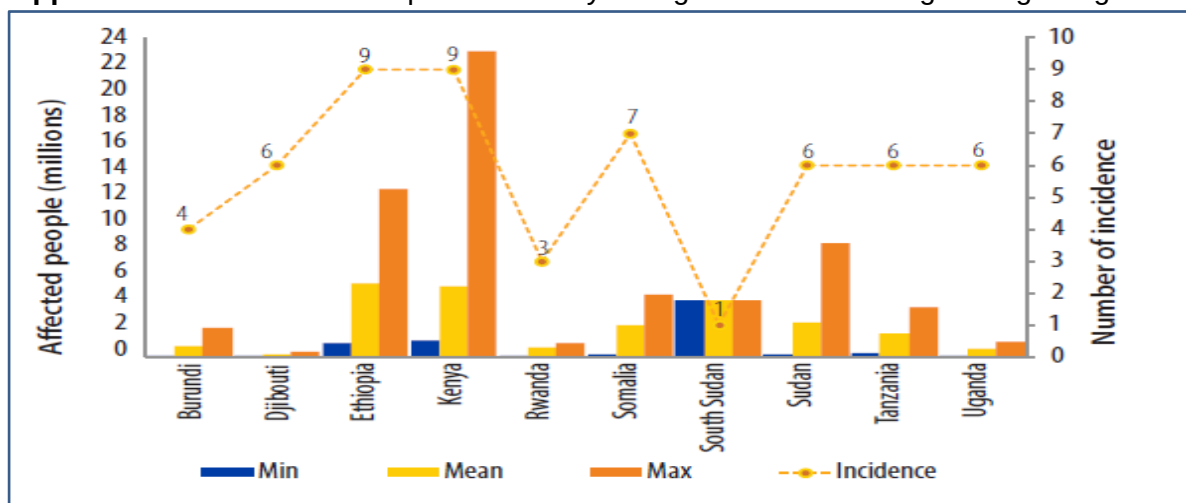
Appendix 1.5: Mean Monthly Vegetation Status (NDVI) in the GHA Region



Appendix 1.6: Combined Seasonal Drought Conditions in the HoA (2009 - 2015)

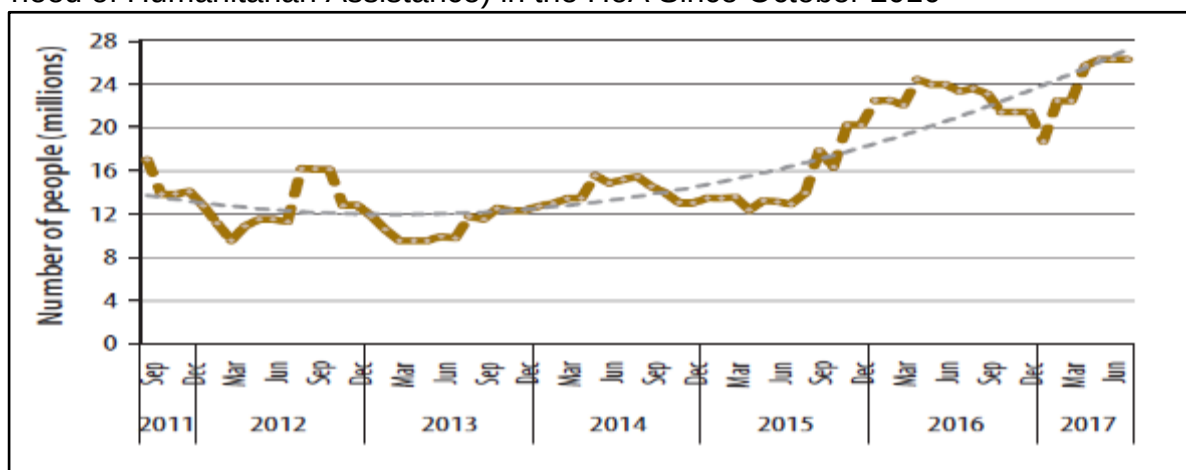


Appendix 1.7: Number of People Affected by Droughts in the HoA Region beginning from 90s



Source: Adapted from the Climate Risks and Food Security Atlas of the GHA based on EM DAT Database

Appendix 1.8: Trends in Number of Food Insecure People (IPC Phase 3 and above or in need of Humanitarian Assistance) in the HoA Since October 2010



Source: Adapted from the Climate Risks and Food Security Atlas of the GHA based on EM DAT Database

Appendix 1.9a: Past Climate Finance Projects in the HoA

Name of Project	Fund (Implementing agency)	Funding Approved (USD millions)	Date of project approval (GEF)/Implementation Dates (others)
Djibouti ^{102 103}			
RLACC - Rural Livelihoods' Adaptation to Climate Change in the Horn of Africa (PROGRAM)	GEF-5 (African Development Bank)	5.08	2016
Supporting Rural Community Adaptation to Climate Change in Mountain Regions of Djibouti	GEF-5 (UNDP)	5.4	2013
Implementing Adaptation Technologies in Fragile Ecosystems of Djibouti's Central Plains	GEF-5 (UNEP)	7.36	2013
Geothermal Power Generation Program	GEF-5 (World Bank)	6	2013
Developing Agro-Pastoral Shade Gardens as an Adaptation Strategy for Poor Rural Communities in Djibouti	Adaptation Fund	4.65	2013-2018
Eritrea ^{104 105}			
Mainstreaming Climate Risk Considerations in Food Security and IWRM in Tsilima Plains and Upper Catchment Area	GEF-6 (UNDP)	9	2016
Integrated Semenawi and Debubawi Bahri-Buri-Irrori-Hawakil Protected Area System for Conservation of Biodiversity and Mitigation of Land Degradation	GEF-5 (UNDP)	5.9	2014
Climate Change Adaptation Programme In Water and Agriculture in Anseba Region, Eritrea	Adaptation Fund	6.5	2012-2017
Ethiopia ^{106 107}			
CCA Growth: Implement-ting Climate Resilient and Green Economy plans in highland areas in Ethiopia	GEF-6 Least Developed Countries Fund (LDCF) (UNDP)	6.28	2016
Promoting Sustainable Rural Energy Technologies (RETs) for Household and Productive Uses	GEF-5 (UNDP)	4.1	2015
Ethiopian Urban NAMA: Creating Opportunities for Municipalities to Produce and Operationalise Solid Waste Transformation (COMPOST)	GEF-6 (UNDP)	6.7	2016
Geothermal Sector Strategy	Scaling-Up Renewable Energy Program for Low Income Countries (SREP)	24.5	2014
Responding to the Increasing Risk of Drought: Building Gender-responsive resilience of the most vulnerable communities	Green Climate Fund	50	2017-2022
Climate Smart Integrated Rural Development Project	Adaptation Fund (AF)	10	2017-2021

Appendix 1.9b: Past Climate Finance Projects in the HoA

Name of Project	Fund (Implementing agency)	Funding Approved (USD millions)	Date of project approval (GEF)/Implementation Dates (others)
Kenya ^{108 109}			
Building Resilience To Climate Change & Adaptive Capacity of Vulnerable Communities	Adaptation Fund (AF)	10	2014-2019
Concessional Finance Program for Geothermal Generation	Clean Technology Fund	30	2016
Energy Modernisation Project	Scaling-Up Renewable Energy Program for Low Income Countries (SREP)	7.5	2015
Promoting biogas as sustainable clean cooking fuel for rural households in Kenya	World Bank	4.6	2016
Kenya Climate Venture Facility	World Bank	4.9	2015
Kenya water security and climate resilience project	World Bank	155	2013
Additional financing water security and climate resilience project		58	
Coastal Region Water security and climate resilience project	World Bank	200	2014
Somalia ¹¹⁰			
Support for Integrated Water Resources Management to Ensure Water Access and Disaster Reduction for Somalia's Pastoralists	LDCF GEF-6	8.8	2017 (Concept Approved)
Enhancing Climate Resilience of the Vulnerable Communities and Ecosystems in Somalia	LDCF GEF-5	8	2014
South Sudan ¹¹¹			
Strengthening the Capacity of Government and Communities in South Sudan to Adapt to Climate Change	LDCF GEF-6	9	2017 (concept approved)
Sudan ¹¹²			
Enhancing the Resilience of Communities Living in Climate Change Vulnerable Areas of Sudan Using Ecosystem Based Approaches to Adaptation (EbA)	LDCF GEF-5	4.3	2016
Livestock and Rangeland Resilience Program	LDCF GEF-5	8.5	2014
Climate Risk Finance for Sustainable and Climate Resilient Rainfed Farming and Pastoral Systems	LDCF GEF-5 (UNDP)	5.4	2014

Appendix 1.10:

Some Selected Ongoing Experiences in the HoA

Initiatives / Projects /Programs	Indicative Amount	Intervention Areas
Program to Build Resilience for Food and Nutrition Security in the Horn of Africa (BREFONS) - 2021-2027	UA 149 million (AfDB)	Designed as a follow-up phase of the DRSLP with the sole objective of improving the living standards for women, youth, and the population in general in the HoA (Djibouti, Ethiopia, Kenya, Somalia, South Sudan, and Sudan).
"Agricultural Climate Resilience Enhancement Initiative	USD 6.8 million (Adaptation Fund)	The goal of the project is to develop and implement adaptation strategies and measures

(ACREI).” Targeted countries include Ethiopia, Kenya, and Uganda (2017)		that will strengthen the resilience of vulnerable smallholder farmers, agro-pastoralists, and pastoralists in the Horn of Africa to climate variability and change in line with the IDDRSI programme, the National Adaptation Programme of Action (NAPAs) and Development Strategies/Visions of participating countries.
Multinational (Somalia/Sudan) Rural Livelihoods Adaptation to Climate Change in the Horn of Africa II (RLACC II)”. 2017	US \$17 million (The Global Environment Facility-GEF)	RLACC II aims to enhance rural livelihoods by reducing food insecurity through climate adaptive measures in pastoral and agro-pastoral communities. It is a clear response to the devastating drought, environmental degradation, and poverty in the Horn of Africa.
Strengthening Climate Information Systems for Climate Change Adaptation in the Greater Horn of Africa through Regional Cooperation	US \$9.5 million. The Green Climate Fund (GCF)	The programme aims to strengthen the climate resilience of 8 targeted countries (all members of Intergovernmental Authority on Development) to the climate change impacts of extreme rainfall events, extreme droughts and floods that can be of transboundary nature and impact several countries at the same time.
The Horn of Africa Climate Change Programme (HoA-CCP)	€ 4.000.000 from the Netherlands Ministry of Foreign affairs (DGIS),	A leverage to attract other financial resources and sustain the regional programme
Financing Supports Regional Initiative on Communities Hosting Refugees - 2016	\$175 Million from the International Development Association (IDA), the World Bank’s fund for the poorest, and the Governments of Ethiopia, Uganda, and Djibouti	To mitigate the impact of forced displacement on refugee-hosting communities in the Horn of Africa.
Horn of Africa Regional Initiative - 2014	\$8 billion from World Bank, United Nations, African Union, European Union, African Development Bank, and the Islamic Development Bank	To help the Horn of Africa countries address the drivers of fragility and conflict, particularly in tackling forced displacement, supporting populations in borderlands, and facilitating regional economic integration.
Horn of Africa Initiative - 2021	\$83.6 million from the AfDB	To boost cross-border trade in electricity between Ethiopia and Djibouti and to deepen integration in the Horn of Africa sub-region.
COVID-19 Rapid Response Facility (CRF) - 2020	\$10 billion from AfDB	To bolster health systems and disease surveillance, enhance infection prevention and control, and improve regional coordination by the East African Community (EAC) and the Intergovernmental Authority on Development (IGAD) to contain cross-border transmissions.
Food and Nutrition Security Resilience Programme in the Horn of Africa - 2019	\$28 million - FAO	The Program focusing on addressing the cause-effect relationship between conflict and food insecurity in Somalia, South Sudan and Sudan, and seeks to set good examples of how to build food system resilience in protracted crises.

Resilient Landscapes and Livelihoods Project (FP136 - Ethiopia) – Approved in August 2020	\$296 million from GCF and World Bank	The Project aimed at improving climate resilience, land productivity and carbon storage and increase access to diversified livelihood activities in selected rural watersheds.
Cooling facility (FP177-Multi-country including Somalia)	\$879.8 million from GCF and World Bank	An innovative, multi-sector and multi-country programmatic financing mechanism to help address the key sustainable development challenge of providing access to cooling while minimizing negative climate impacts.
Inclusive Green Financing Initiative (IGREENFIN I): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) countries - Phase I (FP183 – Multi-country including Djibouti, Sudan and Ethiopia)	€177.482 million from GCF and IFAD	A cross-cutting programme will enhance access to credit and technical assistance for local farmers, farmers' organizations, cooperatives and micro and small sized enterprises.

Appendix 2:

2.1 Technologies for African Agriculture Transformation (TAAT)

TAAT is an AfDB flagship program, which offers a wide range of technology brokerage services to assist African countries. The services include the latest climate-smart technologies, their accompanying management practices, and post-harvest and value-addition interventions. Different strategies are offered for different commodities but are usually combined to strengthen national food systems. The overall goal of TAAT which is an integral part of AfDB's Feed Africa Strategy (2016–2025) is to radically transform African agriculture into a competitive sector by deploying high-impact, proven agricultural technologies to raise agricultural productivity in Africa; mitigate risks and promote diversification and processing in 18 agricultural value chains within eight priority intervention areas, namely: self-sufficiency in rice production; cassava intensification; food and nutrition security in the Sahel; transforming African Savannas into breadbaskets; revitalizing tree plantations; expanding horticulture; increasing Africa's wheat production and achieving self-sufficiency in inland fish production. These work alongside six enabler domains that address transversal issues such as soil fertility management, water management, and capacity development, policy support, attracting African youth in agribusiness, and fall armyworm response.

TAAT's partnership with the Consultative Group on International Agricultural Research (CGIAR) Centres and Advanced Agricultural Research Institutes, the world's largest global agricultural innovation network and provider of climate smart agricultural technologies. The CGIAR and Advanced Agricultural Research Institutes work in consortium with the National Agricultural Research and Extension Systems (NARES), Private sector, and Ministries of Agriculture. TAAT program has emerged as a cornerstone in the deployment of climate-smart innovations and technologies, particularly in the agricultural sector. This is achieved through a strategic focus on innovative farming solutions, large-scale deployment of proven agricultural technologies, and the promotion of Climate Smart Agriculture (CSA) technologies and innovations.

Climate-Smart Farming Solutions:

Soil and Water Conservation Techniques: TAAT spearheads the adoption of innovative soil and water conservation techniques such as Zai, Half-moons, and Contour Lines Technique. These soil and water conservation techniques, specifically demonstrated to boost sorghum, maize millet production in the Sahel region, and dry areas in East

Low-Cost Irrigation Technologies: Through its Water Enabler Compact (TAAT-WEC), TAAT promotes the use of cost-effective and easily deployable irrigation and water management technologies. This initiative not only improves water management and irrigation practices but also contributes significantly to climate-smart agriculture.

Maize activities and Impact on Productivity:

Drought-Tolerant Maize Varieties: The TAAT Maize Compact, led by the African Agricultural Technology Foundation (AATF), is disseminating climate-smart maize technologies such as the Water Efficient Maize Varieties and other breeding programs across 14 countries in Africa. These technologies are disseminated through a strong Public-Private Partnership (PPP) with significant participation of commercial seed companies, farmer groups, commodity associations, and National Agricultural Research Systems (NARS). When drought hit the Southern Africa region in 2019, TAAT deployed drought-tolerant maize varieties, which were cultivated by 5.2 million households on 841,000 hectares. As a result, farmers survived the drought, allowing maize production to expand by 631,000 metric tons of maize, with a value of USD 107 million. The TAAT Maize Compact also facilitated the establishment of 4,256 demonstration plots, conducted 757 field days, and distributed 84,321 free small-pack seeds to boost the scale-up of over 17,340 tons of climate-smart maize seeds.

Market Linkages and Strategic Partnerships: The Maize Compact facilitates linkages between farmers and input/output markets, ensuring a ready market for maize grain at fair prices. Collaborations with agro-processors, grain off-takers, and millers have resulted in value-added products such as maize flour and stockfeed. TAAT, through its innovative approach of aggregating farmers into groups, has

facilitated contractual partnerships with maize off-takers, creating a demand for certified drought-tolerant maize seed and ensuring a stable market for farmers.

TAAT's Impact on Wheat Production:

Wheat Seed Activities: In Ethiopia, following the recent successful and nationwide deployment of heat-tolerant wheat varieties, the irrigated wheat area was rapidly expanded from less than 5,000 ha in 2018/19 to 187,240 ha in 2020/21 and 650,000 ha in 2021/22. On average, yields of wheat increased on average from 2.0 tons/ha to 4 tons/ha. Ethiopia's wheat production increased by an additional 1.6 million metric tonnes in 2022, and the country achieved self-sufficiency in wheat production. Ethiopia says it is poised to export wheat for the first time in the coming wheat season. This model is being upscaled to other East African countries of Sudan, Tanzania, etc.

Strategic Initiative and Capacity Building: TAAT places a strong emphasis on capacity building, providing hands-on training for seed technicians, pioneer farmers, and youth engaged in community-based seed production. The program's approach involves the establishment of Innovation Platform (IP) sites, promoting the adoption of heat-tolerant wheat varieties with associated management packages.

Sustainable Livestock Production:

Comprehensive Livestock Technologies: TAAT's involvement in sustainable livestock production includes water storage facilities, feed mixers, slaughtering equipment, rotational grazing, eco-friendly production equipment. The program also emphasizes research on high-performance animal breeds with enhanced yield and quality and the soil mapping to support integrated soil fertility management usage in the production of livestock feed. The program supports the processing of livestock waste into compost, contributing to sustainable waste management practices. Private sector participation is encouraged through the Risk Sharing Facility (RSF), to finance private enterprises involved in livestock production.

Sustainable Horticulture and Vegetable Production:

Climate-Resilient Varieties and Precision Farming: TAAT through its compact led by WorldVeg focuses on introducing heat-tolerant, high-yield, and disease-resistant varieties in horticulture and vegetable production. TAAT promotes precision farming, changes in cropping patterns, and the use of early warning systems for sustainable horticulture and vegetable production. Moreover, soil mapping guides fertilizer use, and efficient water management policies are implemented to ensure sustainable practices in horticulture and vegetable production.

Sustainable Staple Crops Production:

Climate-Resilient Staple Crops production: TAAT focuses on sustainable intensification of adapted cassava, maize, rice, and soya seed varieties. The program advocates for the use of improved short-cycle, heat, and drought-tolerant rice, cassava, maize, and soybean varieties. The program emphasizes the production of early generation seeds (EGS) and the construction of Semi-Autotrophic Hydroponics technology (SAH) centers for rapid multiplication of seed and demo trials. Integrated rice-based farming systems, efficient irrigation, the use of climate-resilient cultivars, weather-proof farming equipment and post-harvest facilities contribute to sustainable staple crops production. The adoption of no-tillage practices, organic fertilizers, and precision farming techniques further enhances climate-smart agriculture.

Innovative Financing and Climate Forecasting: Innovative financing, including the Risk Sharing Facility (RSF), and crop insurance schemes are introduced to attract private sector participation in staple crops production. TAAT emphasizes the importance of climate forecasting and monitoring systems, leveraging remote sensing technologies for informed decision-making.

In summary, TAAT's multifaceted approach encompasses innovative farming techniques, adoption of climate-smart technologies, strategic partnerships, and capacity building. This comprehensive strategy is instrumental in deploying climate-smart innovations and technologies, ensuring increased productivity, improved income for farmers, and addressing challenges related to climate change in the agricultural sector. The program's success stories in various agricultural domains showcase its vital role in advancing sustainable practices across the African continent.

Monitoring and Evaluation

The Monitoring and Evaluation of Agri-Science Uptake in Research and Extension (MPRO) is a well thought out monitoring and evaluation system providing a new cutting-edge data collection architecture that uses analytical tools and latest methods for tracking progress and assessing performance results based on defined program indicators. Fully deployed in TAAT spanning in 31 countries with a powerface data collection interface, the tool uses innovative analytical tools to showcase implementation progress of ongoing field work and provides access to real time data highlighting how TAAT is positively impacting smallholder farmer behaviours that lead to transformative change. This innovative platform provides a holistic and comprehensive data collection, analysis, and visualization solution, offering valuable insights through an intuitive, interactive dashboard. The project's M&E system and related activities will be financed and supported by funds allocated to the M&E. Using the project's M&E planned budget, this existing digital tool MPRO will be applied to this project and used to inform progress on adaptation to the M&E systems in each of the participating countries. This involves ensuring data is correctly and timely gathered on adaptation related activities in each participating country. The MPRO Platform has been designed to encourage reshare and reuse of the M&E data for: (i) routinely monitoring the progress in real-time beyond indicators and reporting, (ii) continuously providing insightful evidence of achievement for management decision making to improve implementation, (iii) Interactive dashboards for data visualization and (iv) enhanced performance reporting. The platform automatically aggregates data and has information available in an organized space saving time, resources, and reduces the risk of errors when retrieving data. The system will empower the entire ecosystem of partners and stakeholders across all targeted countries in IGAD to make informed decision as a pivotal digitally enabled system in the new cutting-edge Tech enabled-M&E arena. Additionally, MPRO is interoperable with other Management Information Systems (MIS). In this case, no duplicate efforts will be needed both for AfDB and GCF internal M&E systems to receive data in real-time. The platform which is already effective, will be populated with results, and greater resources and expertise will be dedicated to M&E activities, particularly at the community level and to inform progress on adaptation.

2.2 Affirmative Finance Action for Women in Africa (AFAWA)

Affirmative Finance Action for Women in Africa (AFAWA) is a pan-African initiative to bridge the USD 42 billion financing gap facing women in all countries in Africa, with the aim of strengthening enabling environments and stakeholder capacity. Financing will be in-form of loans (concessional and market-rate). AFAWA adopts a holistic approach through three pillars. AFAWA finance leverages the AfDB's financial instruments and has two innovative solutions to drastically transform the banking and financial landscape in Africa and to create incentives for lending to women in business: (i) USD 300 million risk-sharing instrument to unlock USD 3 billion in credit for women businesses and enterprises in Africa. This mechanism will build upon the existing network of commercial banks and microfinance institutions to create structural change and lasting impact for women, and (ii) rating system to evaluate financial institutions based on the share and quality of their lending to women, and their socio-economic impact. Top institutions will be rewarded with preferential financing terms from AfDB. AFAWA technical assistance provides advisory services to financial institutions to ensure successful implementation of their product portfolios for women and strengthens the capacity of women entrepreneurs through training to enhance business productivity and growth. AFAWA technical assistance provides advisory services to financial institutions to ensure successful implementation of their product portfolios for women and strengthens the capacity of women entrepreneurs through training to enhance business productivity and growth. AFAWA enabling environment engages with African Governments and other key stakeholders to support legal, policy and regulatory reforms and strike down the structural barriers impeding women in business.

Through the Bank's commitment to fostering access to finance and technical support for Women-led Small and Medium Enterprises (WSMEs), AFAWA achieved significant milestones in 2023, a year of resilience and transformation. In collaboration with 96 financial institutions across 32 regional member countries, AFAWA unlocked a cumulative lending of about USD 1.6 billion by leveraging on the Bank's

investments and AFAWA risk sharing mechanism implemented by the African Guarantee Fund (AGF). As of end of 2023, AFAWA counts a cumulative amount of USD 54.5 million approved and signed technical assistance projects, which participate in strengthening Partners Financial Institutions' value propositions for women market; strengthening the women entrepreneurs' capacities; and shaping a conducive enabling environment for women's access to finance on the continent. AFAWA's deployment after two years has contributed to multiply the volume of the Bank's investments towards WSMEs benefiting a total of 7,193 vibrant women-led SMEs continent-wide.

Enhancing Visibility and Empowering Financial Institutions: Throughout 2023, AFAWA extended its reach beyond Tanzania and the Democratic Republic of Congo, organizing 4 additional AFAWA Finance Series, in partnership with the African Guarantee Fund (AGF) in Ghana (February), Angola (March), Nigeria (October), and Kenya (November). These events focused on advocating with Countries' Central Banks for crucial actions and reforms, resulting cumulatively in the training of over 500 bankers and Financial Ecosystem practitioners over two days, emphasizing smart-gender investing and building strong value propositions for WSMEs.

Strengthening the ecosystem and Championing Reform Advocacy: AFAWA has played a pivotal role in policy reform, currently supporting 20 member countries in developing gender-responsive policies for WSMEs' access to finance in partnership with the Alliance for Financial Inclusion (AFI). Simultaneously, the program is partnering with 10 Women Entrepreneurship Enablers to support over 8,000 WSMEs in accessing finance in 20 African countries.

On special projects, the Bank through AFAWA signed an agreement in February 2023 with *Femmes Afrique Solidarite* under the *African Union Empowering women in agriculture initiative* to capacitate women cooperatives in the cashew sector in Tanzania. The USD 280,000 grant will enable the women to increase their cashew production from 500 to 3,000 metric tonnes while meeting the quality standards required by cashew off-takers. During the month of March 2023, the Bank through AFAWA signed \$425 million Technical Assistance grant with *Banque du Commerce et d'Industrie de Mauritanie* to support its goals of supporting access to finance for WSMEs in Mauritania, Mali, Guinea, and Senegal. The grant follows a \$7 million trade facility to BCI signed in 2022. *With the support of AFAWA under the Guarantee for Growth (G4G) program, Zambia National Commercial Bank (ZANACO) launched the Zee Women's Banking program*, that is extending credit facilities to all women in the program with no collateral requirements. In the context of the launch, ZANACO hosted its inaugural Zee Women's Banking Awards, on March 07 in Lusaka, to celebrate the accomplishments of the women entrepreneurs who have made a positive impact in their areas of expertise. Awards were presented in categories such as: Entrepreneurs of the Year, CEO of the Year, Creative of the Year, Rising Star Youth, among others awarded to top performers in their sector.

The Bank together with AGF went on a business development mission to Djibouti in March 2024 to explore opportunities for AFAWA through the financial sector and potentially at government level with the recently approved project with the Alliance for Financial Inclusion (AFI). Following the mission colleagues from AGF are discussing with some of the financial institutions who have expressed initial interest in the AFAWA Guarantee Mechanism. AFAWA in partnership with the African Guarantee Fund and the African Bankers' Award has integrated a new award category, namely the *AFAWA Bank of the Year Award*, which goes to the bank that has greatly incorporated the development and growth of WSMEs as an integrated part of their SME strategy. Letshego Microfinance Bank Nigeria has allocated 63% of its 5,115 loans to support women-led small businesses through their innovative Step Up Loan solution.

Pathway towards Unlocking USD 5 billion and Bridging the Gender Financial Gap in Africa: Looking ahead, AfDB commitment to unlocking the targeted USD 5 billion remains strong. A comprehensive approach will guide AFAWA's efforts on fostering lasting partnerships and synergies while accelerating mainstreaming AFAWA into the bank's operations across the sectors and the regions. In 2024, AFAWA intends to enhance efforts in pursuing de-risking, empowering, and transforming the landscape for women-led SMEs in Africa while ensuring a greater impact and inclusive growth for years to come.

2.3 The Africa Adaptation Acceleration Program (AAP)

The AAP is an initiative to scale up adaptation efforts and mobilise USD 25 billion to support climate-resilient development in Africa. Led by the African Development Bank (AfDB) and the Global Centre on Adaptation, the programme focuses on strengthening the capacity of African countries to adapt to the impacts of climate change. It comprises several pillars addressing critical areas such as agriculture and food security, resilient infrastructure, youth empowerment for entrepreneurship and job creation in climate adaptation and resilience, and innovative financing initiatives.

Agriculture and food security pillar: Agriculture is vital to Africa's economies, employing more than half the population and contributing 25% to GDP. However, climate change poses significant challenges to the continent's agriculture, affecting food security and incomes, particularly for women and rural households. The use of digital technology offers potential solutions, with estimates of 40-70% increases in yields and incomes. However, some 250 million people, including smallholder farmers and pastoralists, lack access to digital agricultural services. The Agriculture and Food Security Pillar aims to scale up the adoption of climate-smart digital technologies and related data-driven agricultural and financial services for at least 30 million farmers in Africa by 2025, leading to improved food security for at least 10 million people in 26 least developed countries. Projects supported include Programme to Build Resilience for Food and Nutrition Security in the Horn of Africa (BREFONS, 331,914 euros). In this project, AAP's added value is evident through the provision of upstream technical assistance to ensure the integration of climate-smart digital technologies for adaptation into the project. This includes several completed tasks such as the preparation of Digital Adaptation Profiles for Kenya & Ethiopia, technical reports on agricultural insurance support, and national multi-stakeholder workshops to identify key digital adaptation solutions. The project aims to achieve several outcomes, including benefiting 1.3 million farmers and pastoralists and 15 million livestock through the use of gender-responsive climate services. It aims to create 55,000 jobs for youth and women, while cultivating 170,000 hectares using digital climate-smart solutions.

National Integrated Livestock Development Programme in Senegal Phase 1 (PNDIES P1, €300,000). The objective of the programme is to contribute to the development of competitive, climate-resilient and job-creating value chains for youth and women. In pursuit of this goal, the AAP has provided several value-added activities. These include conducting an evidence-based assessment of PNDIES' climate vulnerability and identifying strategic options for integrating digital and smart agriculture to address climate change. In addition, AAP has analysed the climate vulnerability of livestock value chains to inform adaptation strategies. It has also identified constraints and opportunities for private sector investment in digital tools for adaptation, facilitated multi-stakeholder dialogues and actions to increase commitment to climate adaptation in the livestock sector, and conducted capacity building for livestock professional associations on the use of digital tools to build resilience in the livestock sector.

Programme to Build Resilience to Food and Nutrition Insecurity in the Sahel (P2 P2RS, €409,163). The programme aims to contribute to the improvement of living conditions and food and nutrition security in the Sahel and West Africa region. AAP added value to the programme through the dissemination of climate information services, in particular by promoting public-private partnerships to facilitate the sustainable production of hydro-climatic information and services, thereby creating employment opportunities for young people. In Niger, the project supported the organisation of a regional fair on best practices and technological innovations for climate change adaptation, mitigation and resilience, including digital solutions. In The Gambia, the project focused on upgrading the electronic farmer registry and integrating project beneficiaries, providing digital/mobile extension services through agritech providers, and supporting the development and implementation of national frameworks for climate services. Through these initiatives, the project strengthened regional cooperation and used digital technologies to promote climate resilience and adaptation in the target countries and regional entities covered by the project.

Programme for Integrated Development and Adaptation to Climate Change in the Zambezi River Basin (PIDACC Zambezi, EUR 417,937). In this African Development Bank investment, the AAAP has made significant progress on several fronts. It carried out mapping exercises to identify digital technologies suitable for climate adaptation and to highlight their applicability in different sectors. It has also deepened understanding of the challenges and opportunities associated with using digital tools for climate adaptation. By providing actionable recommendations and facilitating engagement opportunities, the AAAP has strengthened climate adaptation efforts through digital technologies. It has also developed a comprehensive database cataloguing existing technologies, their applications in agriculture and digital solutions for climate adaptation. AAAP also facilitated training sessions on key topics such as water accounting, integration of remote sensing and hydro-meteorological data for sustainable water management, and climate adaptation strategies, including regional climate insurance and weather risk pooling.

Climate Resilient Wheat Value Chain Development Project (CREW, €150,000). AAAP's contribution to the CREW project is to introduce climate adaptation measures and integrate digital adaptation solutions throughout the wheat value chain. This includes assessing local climatic conditions and food security, developing micro-regional climate risk maps, and facilitating stakeholder workshops to identify priority digital extension mechanisms. In addition, AAAP supports capacity building of extension agents and stakeholders involved in the implementation of digital climate advisory services. These efforts aim to increase the resilience of Ethiopia's wheat production system, improve wheat production and productivity, and promote national wheat self-sufficiency, regional trade and exports.

Mkondvo Ngwavuma Water Augmentation Programme (MNWAP, €249,952). The overall objective of MNWAP is to improve food security, poverty reduction and climate change resilience by supporting the development of water and irrigation infrastructure in Eswatini. The project will contribute to poverty reduction in the target region. Increased production will lead to higher incomes for producers and improve their social and economic well-being. AAAP will contribute to increasing agricultural productivity and growth in the agricultural sector, and significantly improve food and nutrition security and incomes by providing climate risk assessments to improve the project. The project will transform approximately 9,400 hectares of uncultivated land into diversified commercial cash crops. The programme aims to create 100,000 direct jobs and many more indirect employment opportunities.

Technical assistance: Stakeholder workshops were conducted in all regions of Africa to prioritise digital extension mechanisms and promote the adoption of Digital Climate Advisory Services (DCAS). These trainings aimed to inform and strengthen the capacity of selected public and private actors, enabling them to share knowledge, best practices and experiences, and explore innovative approaches to promote the adoption of DCAS in their respective regions. The training focused on the design and implementation of DCAS projects tailored to reach the last mile, thereby empowering farmers to improve food security and climate resilience. To date, more than 500 participants have been trained in South, East, West and Central Africa.

Appendix 3:

Table A.4: Assignment of EEs

Component	Output	Activity	Financing Source	Executing Entity (EE)
Component 1: Strengthening Pastoral and Agropastoral Production Systems' Resilience to Climate Change	Output 1.1: Climate resilient infrastructure built/rehabilitated	Activity 1.1.1: Construct or rehabilitate and manage water harvesting infrastructure	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoLFR, Somalia; MoAF, South Sudan
			Gov't Contributions	
			GCF Loan	
		Activity 1.1.2: Construct or rehabilitate and manage community irrigation schemes	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoAF, South Sudan
			GCF Loan	
		Activity 1.1.3: Construct or rehabilitate and manage community livestock infrastructure	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan
			ADF Loan	
	Output 1.2: Improved sustainable land management practices	Activity 1.2.1: Train communities on sustainable land management best practices	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoAF, South Sudan; IGAD
			ADF Loan	
			GCF Grant	
		Activity 1.2.2: Support sustainable rangeland management and agroforestry	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan; IGAD
			ADF Loan	
			GCF Grant	
	Output 1.3: Increased adoption of climate-smart innovations	Activity 1.3.1: Provide extension services and deploy climate-smart innovations to improve agricultural and fodder production	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan; IGAD
			ADF Loan	
			GCF Grant	

		Activity 1.3.2: <i>Disseminate resilient varieties to improve agricultural and fodder production</i>	ADF Loan	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan
			GCF Grant	
Component 2: Supporting low carbon and resilient Development of Agribusiness	Output 2.1: Increased access to digital advisory services and markets	Activity 2.1.1: Establish and deploy digital advisory services using PPP mechanism	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan; IGAD
			ADF Loan	
			GCF Grant	
		Activity 2.1.2: Construct rural markets	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan;
			GCF Grant	
			GCF Loan	
	Output 2.2: Developed inclusive MSMEs and cooperatives with access to climate financing	Activity 2.2.1: Train women and youth on entrepreneurial skills development in renewable energy technologies	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan;
			ADF Loan	
			GCF Grant	
		Activity 2.2.2: Facilitate access for women and youth groups to renewable energy (solar and biodigester systems) to enhance alternative income generation	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan;
			GCF Grant	
		Activity 2.2.3: Design and operationalise innovative credit lines via selected local PFIs, in participating countries.	GCF Grant	MoAWFL, Djibouti; MoA, Ethiopia
			GCF Loan	
		Activity 2.2.4: Provide technical assistance to MSMEs, local PFIs, and cooperatives to create an enabling environment for credit access and sustainability	ADF Grant	MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia;
			ADF Loan	
			GCF Grant	
Component 3: Strengthening Rural Communities' Capacity to Adapt to Climate Change	Output 3.1: Enhanced access to climate information and early warning systems	Activity 3.1.1: Expand agrometeorological stations network to generate data for climate advisory services	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoAF, South Sudan;
			ADF Loan	
			GCF Grant	

		Activity 3.1.2: <i>Integrate and disseminate climate information and early warning into the established digital advisory services</i>	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoAF, South Sudan; IGAD
			ADF Loan	
			GCF Grant	
	Output 3.2: <i>Strengthened regulatory and policy frameworks and institutional capacity for climate resilience.</i>	Activity 3.2.1: <i>Support delivery of climate regulatory framework and mainstreaming of climate change into agricultural sector policies and strategies</i>	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan; IGAD
			ADF Loan	
			GCF Grant	
Component 4: Program Coordination and Knowledge Management	Output 4.1: <i>Project management and coordination</i>	Activity 4.1.1: <i>Project management and Coordination</i>	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan; IGAD
			ADF Loan	
			GCF Grant	
		Activity 4.1.2: <i>Technical and institutional capacity building</i>	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan; IGAD
			ADF Loan	
		Activity 4.1.3: <i>Monitoring and Evaluation</i>	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan; IGAD
			GCF Grant	
		Activity 4.1.4: <i>Knowledge Management and Communication</i>	ADF Grant	MoAWFL, Djibouti; MoA, Ethiopia; MoAWFLC, Kenya; MoLFR, Somalia; MoAF, South Sudan; IGAD

No-objection letters issued by the national designated authority(ies) or focal point(s)

REPUBLIQUE DE DJIBOUTI
UNITE – ÉGALITE – PAIX

MINISTRE DE L'ENVIRONNEMENT
ET DU DEVELOPPEMENT DURABLE

LE SECRÉTAIRE GÉNÉRAL

N° 170

Djibouti, le 09 APR 2023



جمهورية جيبوتي
الوحدة - المساواة - السلام

وزارة البيئة
والتنمية المستدامة

الأمين العام

رقم
جيبوتي في

To: The Executive Director
Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by the African Development Bank regarding Program to Build Resilience and Food Security in the Horn of Africa

Dear Madam, Sir,

We refer to the programme titled *Program to Build Resilience and Food Security in the Horn of Africa (BREFONS)* in Djibouti as included in the funding proposal submitted by African Development Bank to us on 17 December 2021.

The undersigned is the duly authorized representative of Ministry of Environment and Sustainable Development, the National Designated Authority of Djibouti.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Djibouti has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Djibouti;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programmes included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Dini Abdallah Omar
Secretary General
Ministry of Environment and Sustainable Development



FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA
MINISTRY OF PLANNING AND DEVELOPMENT

Ref.No. PDM/25/G-3/687

Date: 25 APR 2023

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by the African Development Bank regarding Program to Build Resilience and Food Security in the Horn of Africa

Dear Madam, Sir,

We refer to the programme titled *Program to Build Resilience and Food Security in the Horn of Africa (BREFONS)* in Ethiopia as included in the funding proposal submitted by African Development Bank to us on 17 December 2021.

The undersigned is the duly authorized representative of Ministry of Planning and Development, the National Designated Authority of Ethiopia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Ethiopia has no-objection to the programme as included in the funding proposal;
- (b) The Programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Ethiopia;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Abas Mohammed Ali
NDA for GCF projects
Ministry of Planning and Development
Ethiopia





REPUBLIC OF KENYA

THE NATIONAL TREASURY AND ECONOMIC PLANNING

Telegraphic Address: 22921
FAX NO. 310833
Telephone: 2252299

THE NATIONAL TREASURY
P O BOX 30007 – 00100
NAIROBI

When Replying Please Quote

Ref: TNT/CONF/36/021/C/ (96)

Date: November 16, 2023

Ms. Mufalda Duarte
Executive Director
Green Climate Fund
G-Tower, 24-4 Songdo-dong
Yeonsu-gu
Incheon City, Republic of Korea

Dear 

Re: Funding proposal for the GCF by the African Development Bank - Agriculture Research, Production and Sustainability Division - regarding Financing the Program to Build Resilience and Food Security in the Horn of Africa.

We refer to the programme titled ***“Program to Build Resilience and Food Security in the Horn of Africa (BREFONS)”*** as included in the funding proposal submitted by the Africa Development Bank on November, 13 2023.

The undersigned is the duly authorized representative of the National Treasury, the National Designated Authority of Kenya.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

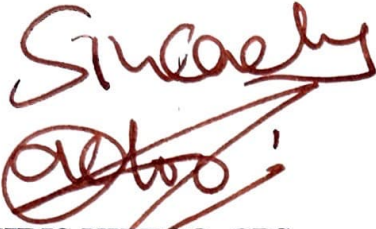
- (a) The Government of Kenya has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Kenya;

- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed. Further, we confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours



DR. CHRIS KIPTOO, CBS
PRINCIPAL SECRETARY/NATIONAL TREASURY

Copy to: **Prof. Njuguna Ndung'u, CBS**
Cabinet Secretary
The National Treasury and Economic Planning
NAIROBI

Laouali GARBA
Division Manager
Agriculture Research, Production and Sustainability Division (AHAI.2)
African Development Bank

FEDERAL REPUBLIC OF Somalia
MINISTRY OF ENVIRONMENT & CLIMATE CHANGE
Office of the Minister

Ref: MoECC/0130/2023

Date: 29/04/2023

Re: Funding proposal for the GCF by the African Development Bank regarding Program to Build Resilience and Food Security in the Horn of Africa

Dear Madam/Sir,

We refer to the programme titled *Program to Build Resilience and Food Security in the Horn of Africa (BREFONS)* in Somalia as included in the funding proposal submitted by African Development Bank to us on 17 April 2023.

The undersigned is the duly authorized representative of Ministry of Environment and Climate Change, the National Designated Authority of Somalia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- The Ministry of Environment and Climate Change (MoECC) of the Federal Government of Somalia has no objection to the programme as included in the funding proposal, subject to the fulfilment of subsequent points (b and c).
- MoECC is the Executing Agency (EA) of the project, point of contact and will be responsible for the overall coordination of the project.
- Considering Somalia's security and humanitarian situation, MoECC will either designate or competitively recruit a Third-Party Implementing Agency (IA).
- The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Somalia.
- In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the program and in accordance with MoECC's continuous oversight as enlisted above.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

H.E Amb. Khadija Mohamed Almaxhoum
Minister, Ministry of Environment and Climate Change
Federal Republic of Somalia





REPUBLIC OF SOUTH SUDAN MINISTRY OF ENVIRONMENT AND FORESTRY

Office of the Undersecretary of Environment

Date: 4 April 2023

To: The Green Climate Fund ("GCF").

Re: Funding proposal for the GCF by the African Development Bank regarding
Program to Build Resilience and Food Security in the Horn of Africa

Dear Madam, Sir,

We refer to the programme titled: **Program to Build Resilience and Food Security in the Horn of Africa (BREFONS)** in South Sudan as included in the funding proposal submitted by African Development Bank to us on 17 December 2021.

The undersigned is the duly authorized representative of Ministry of Environment and Forestry, the National Designated Authority of South Sudan.

Pursuant to **GCF decision B.08/10**, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of South Sudan has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of South Sudan, and

c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Joseph Africano Bartel
Undersecretary,
Ministry of Environment and Forestry,
The Republic of South Sudan, Juba.



Email: bartel64@yahoo.com Tel: +211 921 706 605

Bilpam Road , North of Seventh Day Adventist Road Round About , Juba

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL)
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	African Development Bank (AfDB)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Djibouti, Ethiopia, Kenya, Somalia, and South Sudan
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Friday, February 2, 2024
Language(s) of disclosure	Djibouti: French Ethiopia: English, Amharic Kenya: English Somalia: English, Somali South Sudan: English
Explanation on language	Due to the African Development Bank's multicultural status and depending on the country's preference, all communications are made through the use of one of the two official languages, English and French. In Ethiopia and Somalia, where the official languages are Amharic and Somali, respectively, we have also published the documents in these languages.
Link to disclosure	<u>Djibouti</u> English: https://www.afdb.org/sites/default/files/djibouti_cges_drl_sp_ii_vf_30.09.21_-_version_anglaise.pdf French: http://www.maepe-rh.dj/index.php?option=com_content&view=article&id=91&Itemid=374 <u>Ethiopia</u> English: https://www.afdb.org/sites/default/files/esmf_main_report_final_21_september_2021_ethiopia_0.pdf Amharic: http://moa.gov.et/images/2023/BREFONS_ESMF.pdf <u>Kenya</u> English: https://www.afdb.org/sites/default/files/kenya_esmf_final-revised.pdf

	<u>Somalia</u> English: https://www.afdb.org/sites/default/files/environmental_and_social_management_framework_-_brefons_-_somalia.pdf Somali: https://molfr.gov.so/wp-content/uploads/2024/01/Somali-Version-ESMF-report.pdf <u>South Sudan</u> English: https://www.afdb.org/sites/default/files/ssd_environmental_and_social_management_framework_ss-_hoa_project_em2.pdf
Other link(s)	<u>AfDB</u> https://www.afdb.org/en/documents/multinational-program-build-resilience-food-and-nutrition-security-horn-africa-p-z1-c00-073 <u>Ministry of Agriculture, Livestock, and Fisheries of Djibouti</u> French: http://www.maepe-rh.dj/index.php?option=com_content&view=article&id=91&Itemid=374 <u>Ministry of Agriculture of Ethiopia</u> English: http://moa.gov.et/images/2023/Ethiopia_ESMF_version_GCF24Jan2024.pdf Amharic: http://moa.gov.et/images/2023/BREFONS_ESMF.pdf <u>Ministry of Agriculture, Livestock, Fisheries and Cooperatives of Kenya</u> English: https://kilimo.go.ke/wp-content/uploads/2024/01/ENVIRONMENTAL-AND-SOCIAL-MANAGEMENT-FRAMEWORK-ESMF-FOR-BREFONS.pdf <u>Ministry of Livestock Forestry & Range of Somalia</u> English: https://molfr.gov.so/wp-content/uploads/2024/01/Somalia_ESMF-version_GCF25Jan2024-1.pdf Somali: https://molfr.gov.so/wp-content/uploads/2024/01/Somali-Version-ESMF-report.pdf

	<u>Ministry of Agriculture and Food Security of South Sudan</u> English: https://mafs.gov.ss/wp-content/uploads/2024/01/South Sudan ESMF version GC F24Jan2024.pdf
Remarks	A framework for an ESIA consistent with the requirements for a Category B programme is contained in the Environmental and Social Management Framework (ESMF) for each of the target countries.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Friday, February 2, 2024
Language(s) of disclosure	English, Amharic, French, and Somali
Explanation on language	Due to the African Development Bank's multicultural status and depending on the country's preference, all communications are made through the use of one of the two official languages, English and French. In Ethiopia and Somalia, where the official languages are Amharic and Somali, respectively, we have also published the documents in these languages
Link to disclosure	<u>Djibouti</u> English: https://www.afdb.org/sites/default/files/djibouti_cges_drl_sp_ii_vf_30.09.21_-_version_anglaise.pdf French: http://www.maepe-rh.dj/index.php?option=com_content&view=article&id=91&Itemid=374 <u>Ethiopia</u> English: https://www.afdb.org/sites/default/files/esmf_main_report_final_21_september_2021_ethiopia_0.pdf Amharic: http://moa.gov.et/images/2023/BREFONS_ESMF.pdf <u>Kenya</u> English: https://www.afdb.org/sites/default/files/kenya_esmf_final-revised.pdf <u>Somalia</u> English: https://www.afdb.org/sites/default/files/environmental_and_social_management_framework_-_brefons_-_somalia.pdf Somali: https://molfr.gov.so/wp-content/uploads/2024/01/Somali-Version-ESMF-report.pdf <u>South Sudan</u> English:

	https://www.afdb.org/sites/default/files/ssd_environmental_and_social_management_framework_ss_d_hoa_project_em2.pdf
Other link(s)	<u>AfDB</u> https://www.afdb.org/en/documents/multinational-program-build-resilience-food-and-nutrition-security-horn-africa-p-z1-c00-073 <u>Ministry of Agriculture, Livestock, and Fisheries of Djibouti</u> French: http://www.maepe-rh.dj/index.php?option=com_content&view=article&id=91&Itemid=374 <u>Ministry of Agriculture of Ethiopia</u> English: http://moa.gov.et/images/2023/Ethiopia_ESMF_version_GCF24Jan2024.pdf Amharic: http://moa.gov.et/images/2023/BREFONS_ESMF.pdf <u>Ministry of Agriculture, Livestock, Fisheries and Cooperatives of Kenya</u> English: https://kilimo.go.ke/wp-content/uploads/2024/01/ENVIRONMENTAL-AND-SOCIAL-MANAGEMENT-FRAMEWORK-ESMF-FOR-BREFONS.pdf <u>Ministry of Livestock Forestry & Range of Somalia</u> English: https://molfr.gov.so/wp-content/uploads/2024/01/Somalia_ESMF-version_GCF25Jan2024-1.pdf Somali: https://molfr.gov.so/wp-content/uploads/2024/01/Somali-Version-ESMF-report.pdf <u>Ministry of Agriculture and Food Security of South Sudan</u> English: https://mafs.gov.ss/wp-content/uploads/2024/01/South_Sudan_ESMF_version_GCF24Jan2024.pdf
Remarks	An ESMP consistent with the requirements for a Category B programme is contained in the Environmental and Social Management Framework (ESMF) for each of the target countries.
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A

Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity's website	N/A
N/A	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, February 2, 2024
Place	Djibouti Ministry of Agriculture, Water, Fisheries and Livestock (MoAWFL), Zone Boulaos – BP: 4250, Djibouti – République de Djibouti Ethiopia Ministry of Agriculture (MoA), PO Box 62347, Addis Ababa, Ethiopia Kenya Ministry of Agriculture and Livestock Development, Cathedral Road, Nairobi, P. O. Box 30028-00100 Kenya Somalia Ministry of Livestock, Forestry and Range, Warta Nabada District Mogadishu, Somalia South Sudan Ministry of Agriculture and Food Security, Ministries Complex, Juba, South Sudan.
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	Monday, September 30, 2024
Date of GCF's Board meeting	Monday, July 15, 2024

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP239

Proposal name:	Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL)
Accredited entity:	African Development Bank (AfDB)
Country/(ies):	Djibouti, Ethiopia, Kenya, Somalia and South Sudan
Project/programme size:	Large

I. Overall assessment of the Secretariat

- The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The programme presents innovative solutions to improve, at scale, the adaptive capacity of communities in the Horn of Africa through improved production and productivity of local agricultural and food systems and improved management of rangeland and agropastoral landscapes.	As three of the five target countries are currently affected by conflict, and social and institutional fragility, there are additional risks for a timely and synchronized implementation of the programme. Several mitigation measures, including separate disbursement per country and reallocation clauses, have been included in the design.
The focus on agribusiness development, especially youth and women-led micro, small and medium-sized enterprises (MSMEs), and the deployment of innovative financial instruments including credit lines and simple co-financing with selected private investors and intermediary financial institutions will help to stimulate investment for building climate change resilience through transformation of the agricultural and food systems and management of rangeland ecosystems in the Horn of Africa region.	
Implementation through both national and regional institutions is commendable, because it will enable building on the existing capacities, structures and governance mechanisms in the Horn of Africa, while continuing to build the region's capacity at all levels.	

2. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the accredited entity, and, if considered appropriate, subject to the conditions set out in Annex II of document GCF/B.39/02.

II. Summary of the Secretariat's assessment

2.1 Programme background

3. The Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL) programme is a large-scale intervention aimed at reducing climate-induced food insecurity in the Horn of Africa, which is one of the world's most disaster-prone regions, with debilitating food-insecure conditions arising from various hazards in the region. The objective of the programme is to enhance the capacity of vulnerable communities across five countries that are part of the Intergovernmental Authority on Development, the region's economic bloc), namely Djibouti, Ethiopia, Kenya, Somalia and South Sudan. The programme aims to enable these countries to prepare adequately for and manage food insecurity challenges caused by climate and other environmental hazards by fostering solutions for mobilizing the financial resources needed to strengthen the resilience and adaptive capacity of the vulnerable communities in the target countries.

4. The Horn of Africa region is home to approximately 250 million people who are highly dependent on the local ecosystems for their survival, 64 per cent of whom live in areas prone to extreme food shortages, where the frequency of the threat of famine has increased in the recent past, and the need of emergency food assistance to meet basic needs continues to rise. Moreover, the region has experienced high inter-annual rainfall variability, along with a consistent regional warming trend over the past 40 years (1981–2020), posing a threat to crop productivity. This has resulted in farmers experiencing declining crop yields and some areas becoming less suitable for production of major crops, thereby further affecting food systems, well-being and local livelihood strategies in the region.

5. The programme will overcome these challenges by helping to transition the local food systems to diversified agroforestry systems, with more robust species and mix of varieties compared with current systems, and by supporting communities to manage and produce an array of food and tree crops targeting 4.68 million direct beneficiaries and 16.37 million indirect beneficiaries. Through capacity-building and the establishment of the new agroforestry systems, communities will be more resilient, better able to cope with shocks and damage, and will also benefit from access to finance through a network of local financial institutions to help reinvest in and diversify the local food systems.

6. The programme will deliver mitigation co-benefits in the form of reduced and avoided emissions that will be achieved through activities aimed at increasing energy access through improved vegetation regeneration, improved power generation, and improved forest cover and land-use conditions.

7. The programme is requesting USD 151 million of GCF financing, structured in USD 90.70 million grant financing and USD 60.30 million loan financing. The total cost of the programme, including co-financing, is USD 335.30 million, with co-financing primarily in the form of grants and loans from the accredited entity (AE), the African Development Bank (AfDB).

8. In terms of environmental and social safeguards (ESS), this proposal has been categorized as environmental and social (E&S) risk category I-2. The review by the Secretariat confirms the E&S risk category assigned by the AE, although it recommends that the programme be re-categorized as E&S risk category B.

2.2 Component-by-component analysis

Component 1: Strengthening pastoral and agropastoral production systems' resilience to climate change (total cost: USD 187.70 million; GCF cost: USD 69.08 million)

9. Component 1 will strengthen the resilience of pastoral and agropastoral production systems to climate change by (i) supporting the development of climate-resilient infrastructure; (ii) supporting the sustainable management of agropastoral land; and (iii) improving access to climate-smart innovations and technologies. This component clearly demonstrates the complementarity between the GCF and AfDB investments in developing an integrated approach (infrastructure – natural resources management – climate-smart agriculture inputs and technologies nexus) to transforming pastoral and agropastoral production systems in response to climate change.

Component 2: Supporting low-carbon and resilient development of agribusinesses (total cost: USD 91.99 million; GCF cost: USD 64.56 million)

10. Component 2 will support the development of agribusinesses by increasing access to digital advisory services and markets, and supporting the development of inclusive MSMEs and cooperatives with access to climate financing. Agribusiness opportunities will provide significant avenues for livelihood diversification as a strategy for building resilience, especially for groups with limited access, such as women and youth. Investments in agribusinesses are also expected to greatly contribute to improved and more resilient agricultural production, further contributing to the food security of the region. In addition, channelling GCF loans through local financial institutions is complemented by technical assistance to these financial partners, to ensure a long-term transformation of increased access to finance for smallholders and other underserved groups.

Component 3: Strengthening agropastoral communities' capacity to adapt to climate change (total cost: USD 26.66 million; GCF cost: USD 9.39 million)

11. Component 3 will support the strengthening of agropastoral communities' capacity to adapt to climate change by enhancing the climate services infrastructure, strengthening operational capacities for climate regulatory framework reforms, and institutional strengthening in the target countries.

Component 4: Programme coordination and knowledge management (total cost: USD 28.95 million; GCF cost: USD 7.98 million)

12. Project implementation and monitoring and evaluation costs have been adequately provided for, and specific activities have been planned and integrated within the design of the programme to ensure that project implementation and monitoring and evaluation processes will be effectively coordinated.

Gender and environmental and social management framework aspects

13. A costed environmental and social management framework (ESMF) was developed for each of the programme's five target countries, outlining specific activities that will sufficiently ensure that ESS are put in place and observed throughout the programme's lifespan. A well-designed gender assessment and action plan was also developed, laying out costed plans of actions that will be undertaken in the five programme countries to ensure that gender equality, equity and safeguards will be upheld throughout the lifespan of the programme.

III. Assessment against investment criteria

3.1 Impact potential

Scale: High

14. The programme presents an innovative structure that will help unlock resilience-building opportunities and benefits for vulnerable poor communities in the Horn of Africa region. The programme will support 4.68 million direct beneficiaries and 16.37 million indirect beneficiaries. The communities will be supported to access climate-resilient infrastructure and climate-smart innovations and technologies that will help to transform their local food systems and livelihoods. The communities will also be supported to develop agribusiness skills and capacity to be able to spread their food production and livelihood strategies in a manner that will allow for sustainable management of the agropastoral lands on which they depend.

15. The programme will also deliver greenhouse gas emission reductions amounting to 10.7 megatonnes of carbon dioxide equivalent (Mt CO₂ eq) in 25 years, particularly from activities that will also have the benefits of increasing community access to clean energy and enabling communities to attain enhanced ecosystem services from sustainable forest and land management practices.

3.2 Paradigm shift potential

Scale: High

16. BREFOL is an innovative and strategic approach to promoting and increasing the participation of the private sector in the development and implementation of climate change adaptation strategies in the five target countries. The approach is scalable and can be practically implemented across the Horn of Africa region, building on lessons and outputs and outcomes of the programme.

17. The programme will adopt a strategic approach to enhancing the adaptive capacity of vulnerable communities in the Horn of Africa region by focusing on promoting green investment, wealth creation and job opportunities, especially for the most marginalized groups: women, girls and youth. This can help to ensure the long-term security and sustainability of livelihoods by fostering income levels that can cushion communities from current and future climate shocks and impacts. The livelihood opportunities and diversification will further help to improve production, productivity and management of land and other natural resources in the target areas, thereby improving food and water security in the region.

18. The programme promotes an incentive-based mechanism to create an enabling environment for attracting private sector actors to pool expertise and resources through a combination of innovative tools such as credit lines, weather-based insurance mechanisms and investment vehicles (e.g. MSME agribusinesses). Through this approach, the programme will generate an increase in the baseline of financing from private sector investors by up to USD 8.72 million of green blended finance, which will catalyse investment in climate change adaptation initiatives across the energy and agriculture, forest, and other land-use sectors in the Horn of Africa.

19. The paradigm shift potential is well embedded across all three components and is well designed to be influential over the long term.

3.3 Sustainable development potential

Scale: High

20. The sustainable development potential is embedded in all components of the programme and places vulnerable communities at the centre of efforts to strengthen livelihoods

and the sustainable management of rangeland and agropastoral ecosystems and, through this, the creation of jobs, rural entrepreneurship and social benefits.

21. The programme contributes directly towards efforts to achieve the Sustainable Development Goals (SDGs), in particular SDG 5 (gender equity), SDGs 6 and 7 (access to clean energy and water), SDG 9 (technological innovation), SDG 13 (climate action) and SDG 15 (halting and reversing land degradation and biodiversity loss).
22. The economic, social and environmental co-benefits that the programme offers stem from the capacity-strengthening that the programme will provide to communities to enhance their ability to create, diversify and sustain livelihood strategies and opportunities for themselves.
23. The livelihood opportunities will help to boost production and income, which will in turn contribute towards improved productivity and sustainable management of rangeland and agropastoral ecosystems by local communities, thus offering communities excellent opportunities to create value and economic possibilities across the value chain of the numerous products of both livestock and crop-farming systems. This will help to establish resilient rangeland and agropastoral landscapes across the target areas, with the potential for scaling up to the entire Horn of Africa region. These livelihood security and enhanced ecosystem management outcomes will further build resilience in the target communities.
24. The climate resilient and smart infrastructure, financing mechanisms and agribusiness approaches that the programme will promote will provide the incentives and architecture to foster added value to key commodities and products derived from the livestock and crop-farming systems and bring innovation and technology into local food systems in the Horn of Africa region. These factors will help communities generate, process and transform key products, with enhanced access to surrounding markets and thereby further generate security and resilience for livelihoods.

3.4 Needs of the recipient

Scale: High

25. The programme targets one of the world's most disaster-prone regions, where debilitating food-insecure conditions arise from various hazards which are compounded by the high political fragility of the region.
26. The communities are facing severe challenges from climate change threatening their current livelihood strategies and yet they lack the financial resources and access to credit facilities to purchase farm inputs such as climate-adapted seed varieties, fertilizers and farm implements, and to deploy climate resilient agricultural practices to cope with the frequent hazards and the recurrent food crises, hunger and famine that characterize the region. The current local livelihood and food systems are not able to withstand the pressure from climate change. The proposed interventions, through agribusiness development and the credit lines and weather-based insurance mechanisms, will help communities transform their livelihood options, local food systems and ecosystems to become resilient and diverse and to foster long-term food systems and landscapes, thereby building resilience and adaptive capacity.
27. The target countries are heavily indebted and do not have sufficient resources to invest in the actions to safeguard their populations from climate-induced hazards and food insecurity. The countries and beneficiaries are currently struggling to access finance to support the transition they need to secure their well-being, livelihood and food systems. Without access to finance and investments, the local food systems will not be able to provide vulnerable communities in the region with the safety nets they need. The programme will fill an important gap through working with key regional, national and local institutions in both the private and public sectors, and will help to foster long-term solutions through financial innovation, building

access for communities to business skills, local financial institutions and markets. In addition, agribusinesses will create pathways for communities in the region to reinvest in the local food systems and ecosystems in ways that promote sustainable rangelands and agropastoral landscapes.

3.5 Country ownership

Scale: High

28. Country ownership is well embedded in the programme, which is fully aligned with the climate change and broader development aspirations, strategies and policies of all five target countries. The programme will promote the development of capacity and structuring of agro-silvo-pastoral investments, agribusinesses, climate services, value chain and marketing mechanisms for the most vulnerable communities in the target countries. The programme also places an important focus on ensuring better synergy with ongoing regional and national strategies and actions being implemented in the target countries.

29. The programme is aligned with the Nationally Determined Contributions (NDCs), National Adaptation Plans (NAPs), National Adaptation Plans of Action (NAPAs), and the GCF country programmes of the five target countries. Furthermore, the programme has been developed with the strong participation and commitment of the governments of the five target countries, as well as the participation and commitment of other national- and local-level stakeholders in the countries, involving both high- and grassroot-level stakeholder groups, including women and youth.

30. The AE, AfDB, has a long-standing presence in the region and has a wealth of experience working with stakeholders and dealing with the development and environmental challenges in all five target countries. AfDB offers a unique structure as well as experience and partnerships that will facilitate meaningful engagement and involvement of all relevant stakeholders in the target countries in the successful implementation and scaling up of the programme activities and outputs.

31. The architecture of the programme consists of the governments of the five target countries acting through their relevant ministries as the executing entities (EEs). This will bolster the integration of local contexts and ownership into the programme's activities and outputs.

32. The programme is built on well-designed stakeholder consultations that were conducted throughout the programme development process, ensuring local knowledge, innovation and sustainability, and participation is fully embedded in the design of the programme.

3.6 Efficiency and effectiveness

Scale: Medium to high

33. The programme will directly support poor and vulnerable communities in the Horn of Africa region to strengthen, secure, diversify and transform their livelihood and local food systems. It will target 4.68 million direct beneficiaries and 15.55 million indirect beneficiaries and will drive innovation and transformation by providing access to finance, agribusiness techniques and climate resilient infrastructure.

34. The funding amount is well justified to provide the support needed to unlock the transition of livelihoods and local food systems and to provide innovative approaches and lessons for transforming the agricultural and food systems of the entire Horn of Africa region. The programme is fostering an enabling environment for private and public sector investments through the structuring and deployment of various innovative financial instruments and mechanisms including credit lines, weather-based insurance, and other green blended finance,

and through collaboration with MSMEs, intermediary financial institutions and other private sector actors to propel capacity-building, technology adoption and market access to and for a wide variety of key products and value chains in the proposed enhanced agricultural and agribusiness systems. The programme thus fosters an innovative approach to help create long-term financial innovation and sustainability in the Horn of Africa region.

35. The programme also has an innovative exit strategy that will enable scaling up its approach to finance across the Horn of Africa and will build access and partnership between communities, intermediary financial institutions and other private sector actors for targeting local markets with innovative products and commodities from the local agricultural and food systems.

36. The programme has a proposed budget of USD 335.30 million, with GCF funding representing 45 per cent of the estimated cost at USD 151 million and co-financing at 55 per cent or USD 184.30 million. The GCF financing includes a USD 90.70 million grant (27.1 per cent of the total funding) and a USD 60.30 million senior loan (18 per cent of total funding). AfDB will provide USD 42.9 million in senior loans at a 0.5 per cent interest rate with a 30-year tenor and a 10-year grace period (12.8 per cent of the total funding) and support the programme with a USD 123.54 million grant (36.8 per cent of total funding), while the Food and Agriculture Organization of the United Nations will provide a grant of USD 0.69 million (0.21 per cent of the total funding). The governments and beneficiaries will also make in-kind contributions of USD 16.03 million (4.8 per cent of the total funding) and USD 1.82 million (0.54 per cent of the total funding), respectively.

37. Since the proposed programme involves multiple interventions in different countries, it was not possible to conduct a full economic and financial appraisal (EFA) of all the subprojects at the ex ante analysis stage. However, a detailed EFA was carried out on a selection of indicative subprojects, as outlined in annex 3 of the funding proposal. As per the AfDB policy guidelines, all other interventions intended for funding under the funding proposal will be thoroughly scrutinized to assess their financial and economic viability. Only subprojects that meet the prescribed economic net present value (NPV) and internal rate of return (IRR) thresholds will be financed by the programme.

38. The overall aggregate EFA results modelled over 25 years at an average financial discount rate of 9.6 per cent and an average economic discount rate of 11 per cent revealed the ex ante estimates for financial aggregate results (table 1) and the economic individual and aggregate results (table 2).

Table 1. Overall appraisal of financial results of indicative projects

Overall aggregate financial discount rate	9.6%	
	Financial NPV (USD thousands)	Financial IRR (%)
Base scenario (WP scenario)	447.55	64.2
<i>costs +10%</i>	1 518.06	57.7
<i>costs +20%</i>	1 489.37	52.3
<i>benefits +10%</i>	1 730.12	71.3
<i>benefits +20%</i>	1 913.48	78.4
<i>benefits -10%</i>	1 363.39	57.0
<i>benefits -20%</i>	1 180.02	49.9
<i>benefits postponed 1 year</i>	1 464.47	42.8
<i>benefits postponed 2 years</i>	1 494.67	47.1

Abbreviations: IRR = internal rate of return, NPV = net present value, WP = worst possible

Table 2. Overall appraisal of economic aggregate results of indicative projects

Overall aggregate economic discount rate	11.0%	
	Economic NPV (USD millions)	Economic IRR (%)
Base scenario (WP scenario)	91.44	28.3
<i>costs +10%</i>	78.43	24.4
<i>costs +20%</i>	65.42	21.3
<i>benefits +10%</i>	113.59	32.5
<i>benefits +20%</i>	135.74	36.9
<i>benefits -10%</i>	69.28	24.1
<i>benefits -20%</i>	47.13	19.9
<i>benefits postponed 1 year</i>	67.07	21.0
<i>benefits postponed 2 years</i>	82.04	24.4

Abbreviations: IRR = internal rate of return, NPV = net present value, WP = worst possible

39. The EFAs were conducted on the basis of guidelines provided by GCF on a “per indicative project” basis for each country of interest. The analysis shows that the proposed investments are financially feasible and economically viable. The EFAs suggest that the investment is justified, with positive financial and economic NPVs, and financial and economic IRRs higher than the chosen discount rates of 9.6 per cent and 11 per cent, respectively. Additionally, the EFA sensitivity analyses results indicated that the estimated profitability and sustainability of proposed interventions are relatively insensitive to changes in discount rates, increase in costs or decrease in benefits, as well as when benefits are postponed by one or two years. This in turn suggests that the financial and economic results obtained are robust.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

40. **Programme overview.** BREFOL aims to help build climate resilience and food security in the five Intergovernmental Authority on Development member countries of the Horn of Africa. The programme includes components to strengthen the resilience of pastoral and agropastoral production systems such as the development and rehabilitation of water and irrigation infrastructure for croplands and pasturelands, including for animal feed, health, and market access infrastructure. It also aims to promote sustainable land management practices such as agroforestry and afforestation programmes and deploy climate-smart innovations and upscaling of technologies to pastoralists and smallholder farmers in the Horn of Africa. The programme also seeks to facilitate access to digital advisory services and markets and provide entrepreneurial skills development in renewable energy technologies. It also aims to design innovative credit lines to local participating financial institutions to finance (i) investments in the crop, aquaculture, and livestock value chains; (ii) woodlots for fuelwood; and (iii) the restoration of forest and degraded lands. The environmental and social co-benefits of the programme include reduced air pollution and improved air quality resulting in better health and well-being of the communities. The programme will also contribute to the improvement of water supply and security, and will enhance the overall ecosystem and ecosystem services through improved forest cover and land-use changes in the target areas.

41. **Environmental and social risk category and safeguard instruments.** The AE has classified the programme as Category I-2 in accordance with its integrated safeguards system classification. This is equivalent to GCF's Category B classification given that the activities to be supported will involve development and rehabilitation of water and market access infrastructure, and financing of a broad range of value chain activities from production to marketing, which will target smallholder beneficiaries. The activities foreseen could have potential limited adverse environmental and/or social risks and impacts that individually or cumulatively, are few, generally site-specific, largely reversible, and readily addressed through mitigation measures. The Secretariat confirms the risk category and that this is within the AE's accreditation level. The AE has submitted an Environmental and Social Management Framework (ESMF) for each of the target country (Djibouti, Ethiopia, Kenya, Somalia, and South Sudan).
42. **Compliance with GCF's environmental and social safeguards (ESS) standards.** The paragraphs below provide a description of the programme's compliance with the GCF ESS.
- (a) **ESS1 (Assessment and Management of Environmental and Social Risks and Impacts).** Each participating country has prepared an environmental and social management framework (ESMF). The ESMFs provided the procedures to assess and manage individual activities (subprojects) of the programme in each country. While the general overview of the potential risks and impacts of the programme in each of the countries has been described, more detailed analysis of the E&S impacts will be carried out in all target locations. The AE's guidelines will be applied in the conduct of these assessments. Participating financial institutions will also be required to have operational environmental and social management systems (ESMS) whose standards will meet the requirements of the AE's integrated safeguards system.
 - (b) **ESS2 (Labour and Working Conditions).** The ESMFs identified that the target countries are mostly signatories to International Labour Organization conventions and will comply with the core principles of these fundamental conventions. The ESMFs also identified risks in relation to occupational health and safety and basic worker's rights, non-discrimination in employment and benefits, prohibition of child labour and forced labour. The frameworks require access of programme and subproject workers to grievance redress mechanisms. It is also recommended that the AE and its executing entities ensure that labour issues in the primary supply chain are adequately assessed and managed as part of the due diligence of the programme's subprojects.
 - (c) **ESS3 (Resource Efficiency and Pollution Prevention).** The programme is generally geared towards efficient use of resources, particularly of water. Hence, the infrastructures that are to be developed and rehabilitated under the programme will be designed to achieve this. Some of the activities are also designed to complement the objectives of establishing the hard infrastructure such as by distributing drought tolerant crops to maximize scanty water resources. Nevertheless, the programme's activities will have potential pollution impacts as identified in the ESMFs. These include the possibility for soil erosion, noise, air and water pollution impacts due to construction activities from earthmoving, spillage and generation of wastes. There is also the potential for unsustainable water use and water loss from water harvesting structures/reservoirs through evaporation. The potential for siltation of water reservoirs due to erosive land use in the upstream areas can also be expected. As regards processing and livestock/dairy facilities, there is potential for air and water pollution due to the operation and use of chemicals and discharges of waste from laboratories and facilities. The ESMFs prescribe the development of waste management plans when these issues are identified in site-specific environmental and social impact assessments of the activities.

- (d) **ESS4 (Community Health, Safety and Security).** The following hazards are potentially relevant based on the current situations of the programme countries. These include construction site and route accident hazards, infrastructure collapse hazard for dams and water impoundments, and other natural hazards such as dust/sandstorms, locust infestation, and severe drought. There are also labour influx-related hazards such as potential for cultural conflicts, increased incidence of gender-based violence and/or sexual exploitation, abuse and harassment (SEAH), and spread of communicable diseases and water and vector-borne diseases. There is also the potential for wild animal encounters and incidence of wildfire in rehabilitated rangeland, forests and facilities. The agricultural activities could lead to pesticide poisoning and food safety hazard from processing facilities supported by the programme. Given that the target countries have a significant history or active cases of conflicts, there is potential for the exacerbation of conflict as a result of the programme's activities. Security arrangements may also have adverse interactions with the host communities. To mitigate these risks and prevent these types of conflicts, the programme will organize public consultations under the guidance of the Intergovernmental Authority on Development. During the conduct of the detailed E&S assessment studies, conflict sensitivity analyses will also be undertaken to determine the levels of fragility and likelihood of conflict occurrences on the ground, and that appropriate management measures will be developed.
- (e) **ESS5 (Land Acquisition and Involuntary Resettlement).** The ESMFs have assessed that land acquisition cannot be ruled out but the risks are low. At least two ESMFs (Djibouti and Kenya) have also indicated that subprojects with significant displacements will not be funded. The infrastructure envisaged under the programme are expected to be small scale and other interventions requiring significant area of lands can be sited in government owned lands or beneficiary lands (communal lands), hence land acquisition by expropriation would be rare. However, these lands may still be occupied by some families/tribes or may have been traditionally accessed by groups of people who are not necessarily the beneficiaries of the programme. Thus, during the conduct of site-specific environmental and social impact assessments, Resettlement Action Plans which will detail land acquisition and resettlement arrangements including on compensation will be prepared should this standard be triggered. The programme should also ensure that the ESMFs of participating financial institutions shall also cover impacts of land acquisition or use by their respective investments.
- (f) **ESS6 (Biodiversity Conservation and Sustainable Management of Living Natural Resources).** The ESMFs have identified risks related to loss or fragmentation and degradation of habitat, severance of animal migration routes and pathways (e.g. due to reduction in water supply, changes in water flow and drainage, soil erosion, pollution of water, soils or air), and loss of plant biodiversity as a result of introduction of invasive species such as *Prosopis*. There is also the potential loss of vegetation and/or natural habitat during earthworks and construction activities. The impacts and risks from the construction activities can be avoided by application of strict siting criteria. Among the strategies to be implemented by the programme are (i) proper inventory of flora and fauna; (ii) establishment of buffer zones around conservation and ecologically sensitive areas and avoidance or minimization of activity within these zones; (iii) rehabilitation of cleared areas with native species; (iv) soliciting advice from biodiversity authorities/wildlife specialists, indigenous knowledge experts, traditional natural resources management practitioners on appropriate management measures; and (v) supporting government initiatives on biodiversity conservation.
- (g) **GCF Indigenous Peoples Policy & ESS7 (Indigenous Peoples).** The scope of the programme and activities have potential to impact Indigenous Peoples who are found in the countries where the programme will be implemented and thus trigger application of

the requirements of the GCF Indigenous Peoples Policy noting that, regarding potential impacts, elements related to Indigenous Peoples safeguards will need to be reinforced in the ESMFs for the participating countries. The timeline for provision of the revisions to the ESMFs can, exceptionally, be considered no later than effectiveness of the funded activity agreement. In line with its roles and functions, the GCF Indigenous Peoples Advisory Group is available to provide advice to the AE, which is also encouraged to share emerging good practices and success stories with the Advisory Group.

- (h) **ESS8 (Cultural Heritage).** The potential impact on cultural heritage sites in the programme areas was identified as one of the risks under the programme. However, activities that will adversely impact cultural heritage will be excluded from financing. Further, to manage the risk in situations where there is discovery of physical cultural resources particularly during construction activities, the ESMFs have developed a Chance Find Procedure that outlines the actions required if culturally significant resources are encountered during project construction or operation.
43. **Sexual exploitation, abuse and harassment (SEAH) safeguarding.** The SEAH safeguarding approach proposed by the AE is considered inadequate by the Secretariat to comply with the SEAH provisions pursuant to the GCF Environmental and Social Policy. Although the AE has noted that the programme may exacerbate the pre-existing level of gender-based violence and SEAH in the target countries and has devised SEAH safeguards as part of the country-level ESMFs, only the ESMF for Kenya identifies the potential SEAH risks, mitigation measures and the related due diligence process. The Secretariat recommends that the AE take a consistent approach in assessing and addressing the potential SEAH risks that can be applied in all the target countries. The Secretariat also suggests a condition that the AE provide further revised ESMFs for Djibouti, Ethiopia, Somalia and South Sudan, incorporating the necessary SEAH risk screening, risk mitigation measures, the SEAH due diligences process for subprojects, and the procedures for addressing potential SEAH incidences through the grievance redress mechanisms (GRMs) per country.
44. **Implementation arrangements.** Each country covered by the programme will have a Project Management Unit or Project Coordinating Unit which will be responsible for the day-to-day operations of the programme. The Project Management Unit/Project Coordinating Unit usually will have E&S safeguard specialists/experts who will manage and monitor implementation of the ESMF. Consultants will also be engaged as needed, to provide technical review and advice on E&S management measures. At the subnational or region/district levels, E&S focal persons will be designated to ensure implementation, monitoring and reporting of the management measures. Programme staff will also undergo training on E&S safeguards.
45. **Stakeholder Engagement and Information Disclosure.** The programme has undergone consultations in each country, often being introduced as a second phase or follow-up of a Drought Resilience and Sustainable Livelihood Programme. These consultations were conducted with government officials at the national, subnational and local levels. Plans for further consultations have been outlined in the ESMFs. Detailed and updated stakeholder engagement plans will also be prepared during implementation. Stakeholder engagements will be very important given the fragile and conflict-affected situations of most of the programme areas.
46. **Grievance redress mechanism (GRM).** The ESMFs have described GRMs to be established in the country for the programme. The mechanisms often involve the setting up of independent Grievance Redress Committees at central and region/district levels. The Grievance Redress Committees will have representatives from the project-affected persons, municipal government officials, local civil society leaders, as well as women and youth groups. These are designed to ensure that concerns are proactively identified, apprehensions from impacts of the interventions are adequately clarified, and that issues raised by complainants can be resolved in

an efficient and timely manner. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples focal point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

47. The AE has provided a programme-level gender assessment and gender action plan and therefore complies with the requirements of the GCF Gender Policy.

48. The gender assessment was conducted through desk reviews and some stakeholder consultations with the ministries responsible for women's affairs and gender experts, but this did not cover all five participant countries. The stakeholder consultation for development of the gender assessment and gender action plan is limited in scope and in the range of categories of stakeholders engaged.

49. The gender assessment indicates that countries have favourable environments to promote gender equality and women's empowerment. Many of the countries have enshrined equality between women and men in their constitutions, legal, institutional and political frameworks as well as having programmes and projects in operation. Many already have established ministries responsible for women's affairs with devolved structures to support the gender mainstreaming agenda. However, although many initiatives exist to address gender issues, it is notable that the initiatives and commitments are yet to result in tangible changes in the lives of women and there remains a need for much more action in order to meet the commitments that have been set. In some instances, progress is impeded by the existence of widespread violence against women and girls, high levels of female unemployment and low levels of employability among women, who mostly operate in the informal sector in roles that are at subsistence level and are not very sustainable. It is also a reality that financial inclusion is not prioritized by financial institutions, particularly for women entrepreneurs, which is further exacerbated by the slow changes in sociocultural practices.

50. Women have limited ownership or no access to land, productive resources, access to finance and credit, or technical support to improve agricultural practices. Access to information for women is limited in general, and particularly for climate-related information that is needed for making informed decisions or being part of decision-making processes. Literacy rates among women are particularly low, given that overall literacy rates across the countries' populations are low. The sociocultural practices maintain women and girls in roles that inhibit their contribution to the overall household economy, and their personal growth and development. The agriculture sector is affected by such realities where the productivity gaps of women-owned farms are 20–30 per cent compared with farms of their male counterparts. The challenges in accessing agriculture, climate-related services, inputs, advisory services and digital and non-digital information for women are much higher than for men, affecting their productivity, nutrition and food security. In addition, it is well-known that much of the day-to-day work and engagement of women (be it in agriculture, entrepreneurship activities, women-led businesses, farmer associations etc.) rely heavily on natural resources. The detrimental impacts of climate change continue to negatively affect opportunities for women in terms of their sources of livelihood, income, day-to-day sustenance, and achieving returns on their engagements in various farmers' groups and farm-related activities.

51. The gender assessment recognizes that women-headed households and women in polygamous households in particular face more constraints than women in households headed by men. The constraints result in women lacking security, facing higher levels of poverty, higher levels of workload, and children not being able to attend school. However, women-headed households seem to have more economical use of resources at their disposal. Overall, the five countries are rated very low on gender-related indices.

52. The gender action plan provides activities that will address the challenges to women identified in the gender assessment and will invest in actions to build on the existing roles of women in adapting to climate change. Both regional and country-specific gender action plans include a host of activities with corresponding indicators, targets, timelines, gender expertise and budget.

53. The gender action plan focuses on both cross-cutting regional gender-related issues and country-specific gender issues. It should be noted that the country-level gender assessment and gender action plans will be further refined during project implementation for each of the countries.

54. The regional gender action plan, which covers all five participant countries, focuses on capacity-related issues through (i) the development and adaptation of Gender Action Learning System toolkits and methodologies, thus improving gender relations and promoting the empowerment women by addressing unequal sociocultural norms; (ii) mentorship support to the training of trainers for conducting gender-related research; (iii) cross-country learning exchanges and documenting best practices; and (iv) the collection of gender-disaggregated data.

55. The country gender action plans include very similar activities aimed at developing the leadership as well as technical skills of women; ensuring the active participation of women in land-related issues; improving the role and contribution of women in water management committees, management groups for land, demonstration sites and agricultural cooperatives; and engaging women as entrepreneurs with access to finance and related services. The action plans also address other constraints faced by women by making labour-saving and climate-smart agriculture technologies accessible to women (e.g. energy-saving cookstoves, solar panels, and biodigesters). These technologies are aimed at addressing the time constraints and workload of women and enabling women's engagement and participation in decision-making in (water) committees, which have a considerable impact on the quality of life as well as fulfilment of women's roles and responsibilities. Activities will focus on sectors where many women in the countries are already working, such as enhancing productivity in poultry rearing (i.e. increasing the value chain).

56. The Secretariat understands that the current assessments and gender action plans need further and deeper consultations at all levels; and notes that the gender assessment (to be conducted six months after the inception of the programme) needs to provide sufficient opportunities for various stakeholders to engage and influence the design of the country gender action plans. The AE needs to make sure that the consultations with women and groups representing women's interests are an integral part of the country-level gender assessments, including ensuring equal representation and voice within consultations and on the formulation of the GRM. GRM design, use and implementation needs to be responsive to local contexts, and the needs of women and vulnerable people.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

57. The Secretariat considers the overall programme as medium risk.

4.3.2. Accredited entity/executing entity capability to execute the current programme (medium risk)

58. The AE, AfDB, demonstrated its capability by reference to its existing portfolio of projects and programmes in the Horn of Africa of comparable size or above. The AE is rated AAA stable by Moody's, Standard & Poor's and Fitch.

59. Each participant country has designated a ministry as an EE. The AE carried out institutional assessments of the EEs and recommended measures to reduce fiduciary risks incorporated as conditions for GCF disbursements in the term sheet.

4.3.3. Programme-specific execution risks (medium risk)

60. Co-finance: the Secretariat acknowledges the alignment of interests between the AE and GCF, with co-financing in the form of grants and loans from the AE (ratio 1:1.22). If co-financing does not materialize then this may impede programme implementation; however, the risk is adequately mitigated by official approval from the AfDB board in 2021–2022 and by conditions for GCF disbursements in the term sheet.

61. Credit risk: the AE will channel GCF loans to the government ministries acting as EEs in Djibouti and Ethiopia, totalling USD 60 million or 40 per cent of total GCF financing. GCF is the sole lender under the BEFROL direct credit window and the BEFROL direct insurance credit window, and assumes the credit risk of intermediate lenders (local private financial institutions, insurance providers). Credit risk is high, given the sub-investment grade rating profile of the two countries (e.g. Ethiopia Moody's Caa3/ Standard & Poor's CCC). In addition, inflationary pressure in Ethiopia may lead to a depreciation of the local currency's exchange rate against the United States dollar and increase the cost of servicing a USD-denominated GCF loan. However, the highly concessional loans from GCF reduce the financial risk for the countries.

62. Programme viability and concessionality: GCF concessional funding is expected to crowd in private sector financing and catalyse investment in climate change adaptation initiatives, which would not have been possible otherwise. The Secretariat notes that the high concessionality from GCF loans will be passed down to end-borrowers at an agreed margin range and considering the prevailing market-based interest rates in the target countries (Djibouti, Ethiopia).

4.3.4. Compliance risk (medium risk)

63. This programme addresses food production and security in several countries within the Horn of Africa region. Much of this region is challenged by low capacities to instil financial integrity controls. However, for this programme, the AE has assessed the risks and has proposed a plan to monitor and mitigate such risks. With the imposition of these controls, the compliance risk in the programme is measured as "medium".

4.3.5. GCF portfolio concentration risk (low risk)

64. In case of approval, the impact of this proposal on the GCF portfolio concentration in terms of result area and single proposal is not material.

4.3.6. Recommendation

65. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall programme	Medium
Accredited entity (AE)/executing entity (EE) capability	Medium
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance	Medium

4.4 Fiduciary

66. AfDB will be the AE of this programme. There are six EEs of this programme, as follows:

- (a) The Government of Djibouti, acting through its Ministry of Commerce Economics and Finance, and Ministry of Agriculture, Water, Fisheries and Livestock;
- (b) The Government of Ethiopia, acting through its Ministry of Finance, and Ministry of Agriculture, Livestock and Fishery Resources;
- (c) The Government of Kenya, acting through its Ministry of Finance, and Ministry of Agriculture, Water, Fisheries and Livestock, and Cooperatives;
- (d) The Government of Somalia, acting through its Ministry of Finance and Ministry of Agriculture and Irrigation;
- (e) The Government of South Sudan, acting through its Ministry of Finance and Economic Planning and Ministry of Agriculture and Forestry; and
- (f) The Intergovernmental Authority on Development.

67. Loan pricing will be equal for all borrowers. Both the GCF loan and AfDB loan are *pari passu*. Although accounting standards and practices may differ from country to country, the accounts of the projects and subprojects under BREFOL shall, irrespective of the system adopted, comply with the principles of transparent management and provide exhaustive information on: (i) accountability of funds provided to the project or subproject (such funds may comprise proceeds from the AfDB loan or grant, or other resources from co-financiers and counterpart funds); and (ii) disclosure of expenditure and specific information on components financed under the programme.

68. All projects financed under the Facility will be monitored by the AfDB Portfolio Management team as per the relevant internal policies and procedures. The AfDB – as the accredited entity – will be responsible for direct monitoring of implementation conditions and reporting periodically to GCF under the terms to be agreed between AfDB and GCF. All projects financed under the Facility will comply with the AfDB appraisal, approval, monitoring and supervision standards and procedures.

69. The financial reporting should detail the various uses of the loan and grant proceeds. The audit arrangements require that all project and subproject accounts are audited each year in accordance with the relevant generally accepted standards. The auditor, in accordance with the standard terms of reference for external audits of AfDB-financed programmes or projects, is required, among other things, to examine pertinent documents, review internal financial control mechanisms to identify deficiencies and weaknesses that could affect the efficiency of the project, form an opinion on the quality of the financial statements and confirm that the funds granted to the project have been used for their intended purpose. Each country must have the accounts of the project or programme audited by qualified and independent accountants that meet AfDB requirements. In countries where auditing is entrusted to the Auditor General's Office under the supervisory authority of the state, AfDB may approve the designation of this Office provided that the matter is discussed at the programme/project appraisal stage as well as during the loan negotiations and that the audit is conducted in accordance with AfDB standard terms of reference for the audit for AfDB-financed projects/programmes. The audited financial statements, the auditor's report and management letter must be received by AfDB no later than six months after the end of the financial year to which they relate. Non-compliance with this requirement will result in the immediate suspension of further replenishments to the Special Account.

70. The Terms and Condition for the sub-loans is not yet available. To ensure the concessionality is benefited the final borrowers, this will be a condition for the first disbursement of the GCF proceeds on the TS.

4.5 Results monitoring and reporting

71. As a cross-cutting initiative, the programme aims at generating results across GCF mitigation results areas 1 and 4, and adaptation results areas 1, 2 and 4. The programme is expected to reach 4.68 million direct beneficiaries and 15.55 million indirect beneficiaries, corresponding to 0.8 per cent and 2.9 per cent of the national populations, respectively. At least 50 per cent of the beneficiaries across both categories are expected to be women. The expected mitigation outcome is an emissions reduction of 10.7 Mt CO₂ eq over the programme lifespan.

72. The Secretariat has worked with the AE over seven rounds of reviews to refine the programme theory of change and significant improvement has been noted. The theory of change diagram identifies co-benefits and risks/barriers relevant to the programme.

73. The logical framework has been developed in accordance with the GCF integrated results management framework and follows the recommended template. A significant paradigm shift potential has been identified for the programme. The AE has provided indicators at the project outcome level to facilitate performance management. However, in the latest round of review, the AE has been provided with recommendations for improving suggested indicators under Outcomes 1 and 3, and providing clarity on means of verification for indicators for Outcomes 1, 2 and 3. The AE has defined indicators for assessing the co-benefits listed under the programme's theory of change.

74. Suggestions have also been provided to the AE in the latest round of review to augment annex 11 (Monitoring and Evaluation Plan) in line with the latest logical framework. In annex 23 (Beneficiary Estimation), it has been recommended that the AE should provide further details on classification of beneficiaries as direct/indirect and arrangements for avoiding double-counting.

4.6 Legal assessment

75. The Accreditation Master Agreement was signed with the AE on 8 November 2017 and it became effective on 7 November 2019.

76. The AE has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the programme.

77. The proposed programme will be implemented in the following countries in which GCF is not provided with privileges and immunities. This means that, among other things, GCF is not protected against litigation or expropriation in these countries, which risks need to be further assessed. The countries are as follows:

- (a) **Republic of Djibouti:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Republic of Djibouti on 26 October 2017. The latest communication was sent by the GCF Secretariat to the Government of the Republic of Djibouti on 05 October 2018.
- (b) **Federal Democratic Republic of Ethiopia:** The Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Federal Democratic Republic of Ethiopia most recently on 26 April 2016. The latest communication was sent by the GCF Secretariat to the Government of the Federal Democratic Republic of Ethiopia on 03 November 2017.

- (c) **Republic of Kenya:** The Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Republic of Kenya on 30 September 2015. The latest communication was sent by the Government of the Republic of Kenya to the Secretariat on 5 October 2018.
- (d) **Federal Republic of Somalia:** The Secretariat and the Government of the Federal Republic of Somalia had a meeting on 8 October 2018. The Secretariat has not yet dispatched a draft agreement on privileges and immunities and a background note to the Government of the Federal Republic of Somalia hence, negotiations have not yet started.
- (e) **Republic of South Sudan:** The Secretariat has not yet dispatched a draft agreement on privileges and immunities and a background note to the Government of the Republic of South Sudan hence, negotiations have not yet started.

78. The Heads of the Independent Redress Mechanism and Independent Integrity Unit have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by GCF are made only after GCF has obtained satisfactory protection against litigation and expropriation in the countries, or has been provided with appropriate privileges and immunities.

79. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Submission by the Accredited Entity to the Fund of a certificate or legal opinion, in form and substance satisfactory to the GCF Secretariat, within 180 days after Board approval, confirming that the Accredited Entity has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed programme;
- (b) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval, or the date the Accredited Entity has provided a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later; and
- (c) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP239

Proposal name:	Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL)
Accredited entity:	African Development Bank (AfDB)
Country/(ies):	Djibouti, Ethiopia, Kenya, Somalia and South Sudan
Project/programme size:	Large

I. Assessment of the independent Technical Advisory Panel

1. This is the second submission of the funding proposal for Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL) under the standard proposal approval process.¹ The assessment by the independent Technical Advisory Panel (iTAP) is based on the funding proposal package presented to the iTAP on 3 May 2024 and informed by a set of written questions and answers further discussed in an online meeting between the iTAP and the accredited entity (AE) on 21 May 2024.²
2. The total budget remains at USD 335 million, with a total of USD 151 million (USD 90.7 million as a grant, and USD 60.3 million as a loan) requested from GCF. It also remains a cross-cutting programme, expecting to contribute to both mitigation and adaptation results areas, with an implementation period of seven years.
3. The AE for the programme is the African Development Bank (AfDB).
4. The executing entities (EEs) of the programme comprise:
 - (a) The governments of the five countries (Djibouti, Ethiopia, Kenya, Somalia and South Sudan), acting through their respective ministries responsible for finance and agriculture, respectively; and
 - (b) The Intergovernmental Authority on Development (IGAD), the regional organization for the governments of the [Greater] Horn of Africa Region.³
5. The proposed programme has four components, as listed in Table 1 below, with their respective indicative costs and share of total financing.

¹ The funding proposal was first submitted to the independent Technical Advisory Panel (iTAP) on 15 December 2023. The funding proposal was then titled "Program to Build Resilience and Food Security in the Horn of Africa". It was not endorsed by the iTAP at that time but has now been revised and resubmitted, guided by a set of recommendations from the iTAP (limited distribution document titled "Assessments and responses of funding proposals not endorsed by the independent Technical Advisory Panel"). Prior to the submission of the funding proposals, a concept note was presented in April 2022. (<https://www.greenclimate.fund/document/program-build-resilience-and-food-security-horn-africa>).

² The funding proposal package from 3 May 2024 consists of the funding proposal (clean and track-change versions from Secretariat discussions), the questions and answers (Q&A) from the previous submission, a draft Secretariat assessment, and 21 annexes. Most annexes contain sub-annexes relating to participant countries, resulting in a package with 50 attachments in total. The funding proposal was updated with new greenhouse gas (GHG) emission calculations on 22 May 2024, complementing the package with a marked and updated funding proposal (on GHG issues) and updated versions of annexes 17 and 22, along with the Q&A from the discussion between the accredited entity and the iTAP on 21 May 2024.

³ The Horn of Africa Region [proper] consists of Djibouti, Eritrea, Ethiopia and Somalia. The Greater Horn of Africa includes also Kenya, Sudan, South Sudan and Uganda.

Table 1: Overview of BREFOL components, outputs, and financing

Components and outputs		Amount (USD million)	Share of total
Component 1: Strengthening pastoral and agropastoral production systems' resilience to climate change			
	<i>Output 1.1: Climate-resilient built/rehabilitated infrastructure</i>		
	<i>Output 1.2: Improved sustainable land management practices</i>		
	<i>Output 1.3: Increased adoption of climate-smart innovations</i>		
Subtotal component 1		187.70	56%
Component 2: Supporting low-carbon and resilient development of agribusiness			
	<i>Output 2.1: Increased access to digital advisory services and markets</i>		
	<i>Output 2.2: Developed inclusive micro, small and medium-sized enterprises and cooperatives with access to climate financing</i>		
Subtotal component 1		91.99	27%
Component 3: Strengthening agropastoral communities' capacity to adapt to climate change			
	<i>Output 3.1: Enhanced access to climate information and early warning systems</i>		
	<i>Output 3.2: Strengthened regulatory and policy frameworks and institutional capacity for climate resilience</i>		
Subtotal component 3		26.66	8%
Component 4: Programme coordination and knowledge management			
	<i>Output 4.1: Project management and coordination</i>		
Subtotal component 4		28.95	9%
Total project		335.30	100%

Source: Adapted from funding proposal, Annex 4 "Budget by component" BREFOL Budget Revised 15042024.

6. Component 1 receives 56 per cent of the total budget (46 per cent of the GCF contribution and 64 per cent of the co-financing). It contains the programme's single largest output (output 1.1) on infrastructure construction and rehabilitation, in itself accounting for 38 per cent of the total budget (19 per cent of the GCF contribution and 53 per cent of the co-financing). Output 1.1 will be funded with a USD 70 million grant and USD 28 million loan from the African Development Fund (ADF),⁴ complemented by a USD 29 million loan from GCF. The activities under this output are geared towards the construction, rehabilitation and management of water harvesting infrastructure, community irrigation schemes, and livestock infrastructure. Component 1 also houses outputs 1.2 and 1.3 towards improved land management and the adoption of climate-smart innovations, operating with GCF grants (USD 16 million and 24 million, respectively).

7. Component 2 supports agribusinesses and other enterprises in accessing markets and climate credit. The related activities involve extension of digital services, training of women and youth on entrepreneurship, renewable energy technologies/solutions, and the extension of credit lines. The financing is a mix of grant and loan financing, amounting to a total of 43 per cent of the GCF contribution and 15 per cent of the co-financing.

8. The extension of credit, however, is limited to two of the beneficiary countries; BREFOL's Direct Credit Window (BDCW) is to be rolled out through two local private finance institutions in Djibouti and Ethiopia. Furthermore, BREFOL will promote existing climate insurance providers to scale up weather-based insurance coverage for livestock production. GCF resources are to complement AfDB's investments to support the establishment of a BREFOL

⁴ The ADF is a funding window for AfDB. It is the concessional arm of the AfDB that provides grants and concessional loans to low-income African countries.

Direct Insurance Credit Window with the selected banks. In principle, each beneficiary of the credit line is to be covered by insurance.

9. A smaller share of the programme resources are destined for component 3 (Strengthening Agropastoral Communities' Capacity to Adapt to Climate Change), under which the programme supports regulatory frameworks and enhances access to climate information and early warning systems.
10. An outcome and the main pursuit of component 3 in the previous submission was strengthened institutional capacity and climate regulatory frameworks. However, the expected results in this area have been toned down (and the previous outcome was replaced with 'Strengthened adaptive capacity of vulnerable communities'). This toning down of institutional capacity enhancement is entirely congruent with the limited effort and budget afforded to such activities. In essence, the funding proposal has chosen to focus on infrastructure and credit lines.
11. Component 4 (Programme coordination and knowledge management) has a funding allocation of USD 29 million, which constitutes 9 per cent of the total budget. This component also addresses technical and institutional capacity-building, monitoring and evaluation (M&E), as well as knowledge management and communications.
12. The iTAP has reviewed the funding proposal against the six investment criteria of GCF, commenting on the overall structure as well as the specifically proposed interventions planned.

1.1 Impact potential

Scale: Medium

13. This cross-cutting programme is to contribute to five GCF results areas; two mitigation results Areas (MRAs) and three adaptation results areas (ARAs). These are: MRA 1 – Energy generation and access; MRA 4 – Forestry and land use; ARA 1 – Most vulnerable people and communities; ARA 2 – Health, well-being, food and water security; and ARA 4 – Ecosystems and ecosystem services.⁵
14. According to the theory of change illustration, the three components contribute in different ways to the programme's outcomes and co-benefits. These are:
 - (a) **Outcome 1:** Increased adoption of climate-resilient solutions;
 - (b) **Outcome 2:** Enhanced carbon sequestration and reduced greenhouse gas emissions from [agricultural value chains];
 - (c) **Outcome 3:** Strengthened adaptive capacity of vulnerable communities;
 - (d) **Co-benefit 1:** Improved livelihood opportunities; and
 - (e) **Co-benefit 2:** Ecosystem services improved.
15. The need for greater clarity and logical flow in how activities, outputs, outcomes and co-benefits contribute towards the overarching goal and results areas was a major issue in the first submission of this funding proposal package. The overall consistency has improved but the logical flow and the application of the theory of change remains problematic. The iTAP is concerned with this design insufficiency and suggests improving the logic and clarity through a carefully and comprehensively updated logical framework to guide the implementation and monitoring (see condition (i) in the concluding section).
16. The project will support two activities related to MRA 1 (Energy generation and access). These are off-grid solar photovoltaics and the installation of biodigester units. The funding

⁵ Despite 38 per cent of the total budget going into infrastructure construction and rehabilitation, ARA 3 (Infrastructure and built environment) is not a results area BREFOL anticipates contributing to.

proposal claims a mitigation impact potential of 56,930 tonnes of carbon dioxide equivalent (t CO₂ eq) by the end of the seven years of implementation and 226,383 t CO₂ eq by the end of the 25 years lifespan. The mitigation impact potential of the off-grid solar photovoltaics was quantified using a clean development mechanism methodology for small-scale electrification, and the mitigation impact potential of the biodigester units was quantified using the International Renewable Energy Agency methodology for small-scale biogas capacity production.

17. The project also claims a mitigation impact potential from MRA 4 (Forestry and land use), specifically from activities for the rehabilitation of rangelands and for agroforestry. The mitigation impact potential was quantified using the Ex-Ante Carbon-balance Tool (EX-ACT) tool developed by the Food and Agriculture Organization of the United Nations. The anticipated mitigation results are 2,033,251 t CO₂ eq by the end of the implementation period and 13,922,954 t CO₂ eq over the lifespan of the project.

18. The iTAP assesses the overall estimation of 2,090,182 tCO₂eq by the end of the implementation period and 14,149,337 tCO₂eq by the end of the project lifespan as sound, clear and well justified. However, there is still a lack of clarity about how the climate impact of the project will be integrated into the M&E system and/or in the measurement, reporting and verification (MRV) system of the participating countries (see condition (i) in the concluding section).

19. On adaptation impact potential: The funding proposal indicates an expected adaptation outcome with 4.7 million direct beneficiaries. Indirect beneficiaries are assessed as 15.5 million. The funding proposal document suggests that at least 50 per cent should be women, while the calculations (see Annex 23) indicate exactly 50:50 male/female. In order to reach at least 50 per cent women – or even nearly 50 per cent – significant and proactive measures are needed (see discussion under section 1.3 (paras. 34–36) and condition (ii) in the concluding section).

20. Annex 23 (Estimation of Beneficiaries) includes different ways to estimate the number of beneficiaries, related to different activities. An update of the project's monitoring framework should review the beneficiary estimation methodologies and explore the necessary qualitative delineations required to assess and monitor how many can be said to have benefited from the programme. Even the use of more qualitative indicators can be explored as a way of drawing attention to the logical pathways for how specific activities are meant to contribute to specific outputs and broader outcomes (see condition (i) in the concluding section).

21. All BREFOL outcomes are to contribute to ARA 1 ([Enhanced resilience of the] most vulnerable people and communities). This might be the case given the general fragility and vulnerability of the region. It should be noted, however, that specific methodologies for reaching the most vulnerable [within the region] are lacking. The beneficiary contributions for accessing credit may be effective in targeting those able to successfully manage credit without specific support, but it may also bar the most vulnerable from accessing the concessional climate finance facilitated by the programme.⁶

22. Indirect beneficiary estimates are by definition imprecise, and rough estimates of hectares covered or percentages of the population need to be accepted as reasonable. One area where a more precise indirect beneficiary calculation would make great sense, however, relates to activity 3.1.2 'Integrate and disseminate climate information and early warning into the established digital advisory services.' An early warning system could very reasonably be expected to reach all people in a given geographic area with climate-related early warnings. The present monitoring framework gauges the number of pastoralists with access to climate-informed agro-advisory services (activity 3.1.1) but in relation to EWS, only the number of

⁶ The iTAP notes that the beneficiary estimation (or project/programme-specific indicators listed in section E5 of the funding proposal) does not count the number of individuals or businesses that actually acquire loans and insurance, but only how many are eligible (including women and youth).

trainers trained and the number of tools developed are to be monitored. This could be revisited along with the update of the M&E system, as outlined in condition (i) in the concluding section.

1.2 Paradigm shift potential

Scale: Medium

23. In the GCF context, paradigm shift potential is defined as the degree to which the proposed activity can catalyse impact beyond a one-off project or programme investment. To the extent that the BREFOL programme can be effectively implemented, there is a good potential for catalysing change beyond the one-off investment.

24. If the construction/rehabilitation of water harvesting, irrigation and livestock infrastructure is of high quality and communities are interested in using and maintaining such infrastructure, then these investments should provide benefits relating to water and food security for many years to come (the total lifespan indicated is 25 years).

25. Extension services and training for improved land management and climate-smart solutions in agriculture, rangelands and forestry should help communities maintain soil fertility, use a better mix of crops, and protect livestock from disease. If successful, this stands to greatly enhance livelihoods and adaptive capacity, which should be lasting qualities that improve the lives of pastoralists and agropastoralists.

26. Entrepreneurship training and the extension of credit – as connected with climate-smart businesses and insurance coverage – may also have lasting positive effects on the local community and its ability to cope with climate change and also make mitigation action more compatible with sustainable livelihoods.

27. Related to the credit lines, there are great opportunities for GCF to help overcome specific barriers to financing. More financing is needed in the region. However, the potential of the programme for helping new or vulnerable groups access finance is uncertain. With regard to gender, this may depend on the level of success of BREFOL in taking lessons and insights from:

- (a) The reportedly highly successful pan-African Affirmative Finance Action for Women in Africa (AFAWA); and
- (b) The AfDB's commitment to fostering access to finance and technical support for Women-led Small and Medium Enterprises.

28. The funding proposal is very vague on ways to strengthen regulatory and policy frameworks and/or institutional capacities for climate resilience.⁷ It seems there will be policy dialogues in some of the countries, but with different topical orientations. Policy and regulatory frameworks may otherwise be the most direct way of triggering action beyond the project or 'one-off investment', but this is not the path or 'paradigm shift potential' opted for by BREFOL.

29. The funding proposal correctly notes that conducive regulatory frameworks are critical for climate action as well as for business development. It is hoped that other related initiatives by the five country governments do work on the strengthening of institutional capacities for climate resilience and the creation of an enabling business environment for climate-smart entrepreneurship. Such developments beyond the programme would be important for making

⁷ The funding proposal (para. 97) suggests that activity 3.2.1 "will further enhance existing investment and regulatory framework in the participating countries (see Annex 9)". Annex 9, titled 'Legal due diligence', refers to the conditions for loan disbursement, although reference is also made to financial regulations in the beneficiary countries, drawing on information from the United Nations Conference on Trade and Development Investment Policy Hub (<https://investmentpolicy.unctad.org/>). Annex 7, including its 13 sub-annexes accounting for Stakeholder Engagement and Market Assessments in the beneficiary countries, contain some important research and insights on financial regulations and access to credit in some of the countries. The iTAP could not find any synthesis or analysis of how BREFOL seeks to enhance policy frameworks, but notes that the funding proposal, compared to the previous submission, now refers to and relies more on the contents of Annex 7 and its sub-annexes.

BREFOL's investments into tools, trainings and infrastructure more effective and sustainable in the long run. It would have been desirable for BREFOL to have a stronger plan for activity 3.2.1 'Support delivery of climate regulatory framework and mainstreaming of climate change into agricultural sector policies and strategies.'

1.3 Sustainable development potential

Scale: Medium to high

30. The theory of change figure (funding proposal, page 32, figure B.2.2) indicates co-benefits of 'improved livelihood opportunities' and 'ecosystem services improved'. These are fully consistent with the activities proposed across the programme.

31. BREFOL also seeks to contribute towards the 2030 Agenda for Sustainable Development and the Sustainable Development Goals (SDGs). Significant contributions should go towards SDG 2 (Zero hunger), SDG 5 (Gender equality), SDG 6 (Clean water and sanitation), SDG 7 (Affordable and clean energy), SDG 9 (Industry, innovation and infrastructure), SDG 11 (Sustainable cities and communities), SDG 13 (Climate action through carbon storage and contribution towards stabilizing of climate) as well as the restoration of degraded rangeland/pastureland as a significant contribution towards SDG 15 (Life on land).

32. The wider (co-)benefits of the proposed programme may be substantial, very much depending on the level of success in reaching the most vulnerable people and communities and enhancing their health, well-being, and food and water security.

33. The economic and social co-benefits may be the greatest, not least by way of enhancing women's access to credit and enabling further income-generation. If the focus on women is successfully maintained throughout the layers of implementation, the gender and social co-benefits could contribute to women's economic independence, which in turn may have beneficial social, cultural and political spin-offs.

34. However, the Horn of Africa faces serious gender inequalities, which may compromise the success of the programme as these hamper the freedom and well-being of women and girls. Annex 8 (Gender Analysis and Action Plan) suggests that the key gender inequalities faced by women and girls include the limited ownership of, access to, and control over productive resources along with extra work burdens relating to caregiving and the gathering of fuelwood, fodder, grass for construction and the otherwise reduced mobility outside the home. Other factors include low literacy levels leading to low skills, poor salaries and low levels of formal employment and reduced representation in political, legal and policymaking institutions. These inequalities are rooted in sociocultural patriarchal norms, kinship rules and traditional practices. As further pointed out in Annex 8, this affects the way women participate in livelihood diversification activities, and will thus have a bearing on the success of the proposed project.

35. The gender action plan (see Annex 8) is specific, budgeted and well elaborated. It is however not sufficiently integrated into the funding proposal's mainstream activities. Moreover, the rationale for having activities 2.2.1 and 2.2.2 focus specifically on women and youth, while none of the other activities have this focus, is unclear. The prevailing gender discrimination could motivate any other activity to (also) have a more specific gender focus and gender-responsive methodologies.

36. The iTAP notes that consultations with beneficiaries and stakeholders – communities, women's groups, youth associations, civil society organizations (CSOs), and non-governmental organizations – will continue in the implementation phase (see para. 185 of the funding proposal). This provides the AE with suitable opportunities for developing an updated gender action plan that is integrated into the mainstream activities (see condition (ii) in the concluding section).

37. Environmental co-benefits, as indicated in paragraph 46 of the funding proposal, are anticipated through the promotion of resilient agropastoral production systems by promoting the adoption of sustainable land management best practices and agroforestry, leading to improved ecosystem services and increased soil organic carbon content. Furthermore, BREFOL envisions the installation of 4.8 megawatts of solar photovoltaic systems for solar irrigation, lighting, and processing, alongside the potential generation of biogas energy from over 3,890 cubic metres of biodigester capacity that will be installed.

38. One factor which could limit environmental benefits, or even do harm, is the risk of overextraction of groundwater. In all five countries, stakeholder consultations came up with the idea of “drilling and rehabilitation of boreholes fitted with solar powered pumps” (see paragraphs 189–193 of the funding proposal). With the shift from “costly” diesel to “free” solar photovoltaics, the iTAP would like to point to the possible risk of overpumping and overextraction of groundwater. This risk is heightened where institutional capacities and regulatory frameworks are weak, and groundwater monitoring may be absent.

1.4 Needs of the recipient

Scale: Medium to high

39. The Horn of Africa is one of the world’s most disaster-prone and food-insecure regions. Conflict and climate change-intensified drought and flooding, compounded by economic downturn, continue to affect millions of citizens, including refugees, internally displaced persons and their host communities. Discrimination and threats of violence hamper access to the resources needed for the income generation that is essential for accessing food and adapting to climate change adaptation. Millions of children suffer from acute malnutrition. There is no question about the vulnerability and financing needs of the beneficiary countries and populations.

40. The proposed programme also sets out to support the most vulnerable people and communities and address gender aspects. These are critical intentions that need even further emphasis and critical attention, as highlighted above.

41. The fragility of the region is discussed in relation to governance risks (see paras. 211–213 of the funding proposal). It is critical for the successful implementation of the programme that the level and risk of conflict and lack of capacity be continuously monitored and addressed in project management and oversight. Due consideration of fragility and conflict risk also needs to be brought into project management and methodologies for programme delivery. Annex 24 (Fragility and Resilience Assessment) contains important and insightful information about the sociopolitical situation of the beneficiary countries, and would be worth updating. Such an update could provide opportunity for weaving the insights of the assessment into the methodologies for implementing the BREFOL activities (see condition (iii) in the concluding section).

42. Updated country-level strategies for managing fragility concerns in an adaptive manner during project implementation will be key for the implementation of activities 3.2.1 (Support delivery of climate regulatory framework and mainstreaming of climate change into agricultural sector policies and strategies) and 4.1.2 (Technical and institutional capacity building).

43. There is also a clear need for climate-smart agrifinance in the region. The BDCW to the Djibouti Economic Development Fund and Development Bank of Ethiopia will address the much-needed financing needs in the climate sensitive agricultural sector.⁸ This facility will enable the two local private finance institutions to demonstrate how to structure the financing and derisk the potential investment activities in the climate-smart agriculture sector. Given the prevailing market interest rates, ranging from 12 per cent to 15 per cent in Djibouti, and 12 per

⁸ See paragraph 73 of the funding proposal, and activity 2.2.3.

cent to 18 per cent in Ethiopia, the BDCW can offer more accessible loans with maximum rates of 6 per cent (in Djibouti) and 9 per cent (in Ethiopia), and this includes a parametric insurance premium for the loan.⁹ The lessons to be learned and best practices to be developed can form the basis of good lending procedures and financing structures that can also be replicated in the other programme countries in future. This should eventually spur some interest in the private sector (i.e. local financial institutions and insurance/guarantee companies).

1.5 Country ownership

Scale: Medium to high

44. The five participating countries – Djibouti, Ethiopia, Kenya, Somalia and South Sudan – have all provided no-objection letters and will contribute in-kind co-financing. The initial concept note for the present endeavour (then titled Building Resilience for Food and Nutrition Security) covered seven of the eight IGAD countries.¹⁰ Uganda excluded itself, and Sudan was unable to participate due to conflict in the country. Hence, five IGAD member States are involved in the presently proposed programme.

45. Additional consultations with the participating countries after the first submission of the funding proposal to GCF also led to changes in the activity setup between the countries. It appears, from AfDB's responses to the iTAP's questions, that several countries were not able to identify suitable financial institutions to roll out the credit lines. Hence, activities related to the BDCW and the related Direct Insurance Credit Window are limited to Djibouti and Ethiopia.

46. At the regional level, this whole endeavour is a follow-on from IGAD's Drought Resilience and Sustainable Livelihoods Program (DRSLP). Lessons learned from the DRSLP have been integrated in the programme's design and the implementation plan. In some participating countries, the programme will use project management units that are already running DRSLP, which are to be strengthened with the addition of specialized staff. The proposed programme aligns with IGAD's Drought Disaster Resilience and Sustainability Initiative Strategy and the HoRN Regional Resilience Framework 3.0. Also, the African Union Draft Africa Climate Change Strategy (2020–2030) played a pivotal role in narrowing down the BREFOL subinvestments, consistent with the adaptation needs of region.

47. At the national level, the proposed programme aligns with the respective countries' national strategies and action plans on climate change. All of them have submitted at least a first (indicative) nationally determined contribution.

48. The participating countries' capacity to implement funded programmes is essential. The ministries and regional organization acting as EEs all operate in critically challenging environments and have important capacity needs. Nevertheless, they are also the most suitable entities to take on a project of this nature, and their involvement should enable them to develop their capacity further. The nationally owned implementation also holds the key to future sustainability.

49. The iTAP hence commends the approach of working through regional and national entities, and trusts that the AE has incorporated necessary and sufficient means to support and develop the implementation capacity of the executing and implementing entities.

50. There has been engagement with CSOs and other stakeholders. The funding proposal (para. 185) suggests that "the large number of people mobilized during the consultations confirms that countries and IGAD are highly committed to the programme and have high expectations of its ability to bring improvements."

⁹ See paragraph 74 of the funding proposal, and activity 2.3.1.

¹⁰ <https://www.greenclimate.fund/document/program-build-resilience-and-food-security-horn-africa>.

1.6 Efficiency and effectiveness

Scale: Medium

51. The total programme funding of USD 335.3 million comprises USD 151.0 million of requested GCF funding, and USD 184.3 million in AE/ADF and government co-financing. The co-financing is hence larger than the amount requested from GCF, demonstrating high commitment from the AE and the respective country governments. Table 2 provides an overview of the funding by country.

Table 2: Overview of BREFOL financing by financier and receiving country

Country/entiti ty financed	GCF financing (USD million)				Co-financing: ADF + FAO + Governments + beneficiaries (USD million equivalent)					Total BREFOL	
	Grant	Loan	Total GCF	% of GCF	Grant	Loan	In-kind	Total Co-fin	% of Co-fin	All (USD million)	Per cent (%)
<i>Djibouti</i>	14.02	14.00	28.02	18.6%	21.45			21.45	12%	49.47	14.8%
<i>Ethiopia</i>	25.64	46.30	71.94	47.6%	38.89			38.89	21%	110.83	33.1%
<i>Kenya</i>	20.10	0.00	20.10	13.3%		42.90		42.90	23%	63.00	18.8%
<i>Somalia</i>	8.82	0.00	8.82	5.8%	21.45			21.45	12%	30.27	9.0%
<i>South Sudan</i>	14.64	0.00	14.64	9.7%	36.44			36.44	20%	51.07	15.2%
<i>IGAD</i>	7.50	0.00	7.50	5.0%	5.32			5.32	3%	12.82	3.8%
					Governments:			16.03	9%	16.03	4.8%
					Beneficiaries:			1.82	1%	1.82	0.5%
Grand total	90.71	60.30	151.01	100%	123.55	42.90	17.85	184.30	100%	335.30	100%

Source: Compiled from funding proposal, Annex 4, 'Budget Summary' BREFOL Budget Revised 15042024.

Abbreviations: ADF = African Development Fund; FAO = Food and Agriculture Organization of the United Nations; BREFOL = Building Climate Resilience for Food and Livelihoods in the Horn of Africa; IGAD = Intergovernmental Authority on Development.

52. The estimated government co-financing (in-kind) is around 9 per cent of total BREFOL resources. It seems mostly geared towards activity 1.1 on infrastructure construction and rehabilitation.

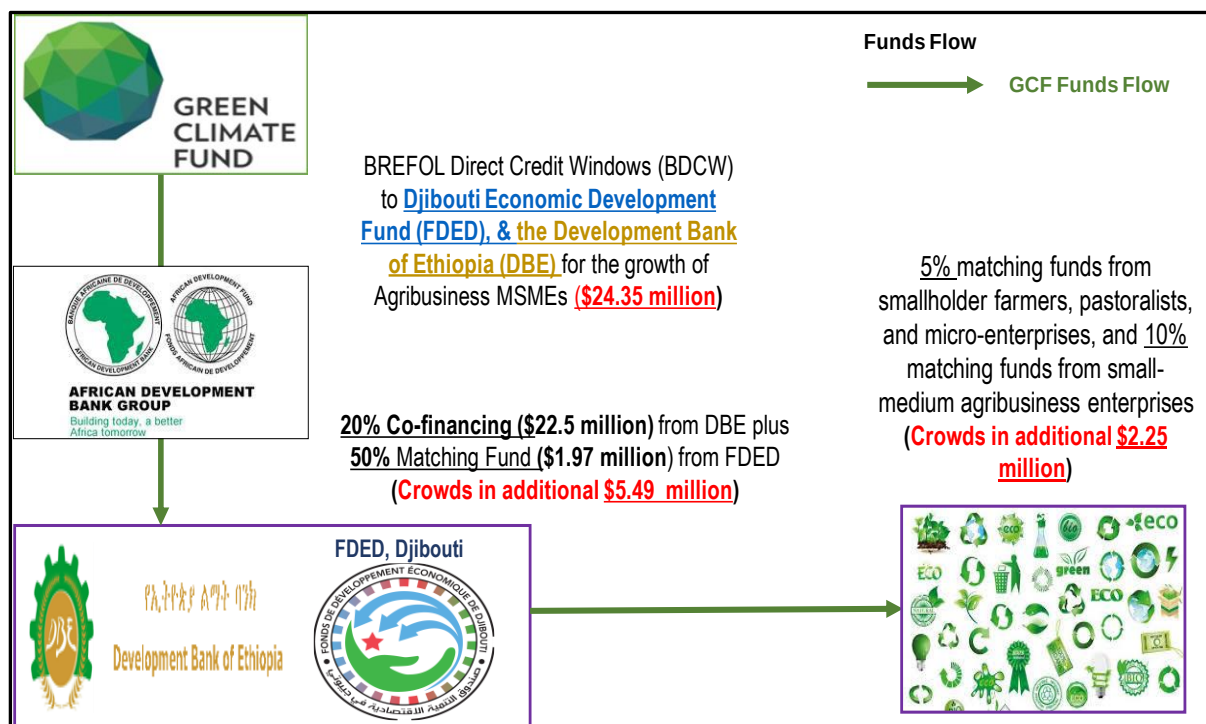
53. The programme anticipates leveraging additional private sector co-financing to the tune of USD 9 million (see Table 3). It is noted that most of the co-financing is from the Development Bank of Ethiopia. Smaller amounts of funding are provided by other sources of matching funds, including from small-medium enterprises, smallholder farmers and micro-enterprises.

Table 3: Expected contributions from private sector financing

Contributions from Private Sector Financing Channels	Amount (USD million)
20% co-financing from the Development Bank of Ethiopia	5.49
50% matching fund from Djibouti Economic Development Fund	0.99
5% matching fund from smallholder farmers and micro-enterprises	0.45
10% matching fund from small-to-medium-sized enterprises	1.80
Total Green Blended Finance	8.72

Source: Adapted from funding proposal, page 50, table B.3.5.

Figure 1 – Ways for the BREFOL Direct Credit Window to attract private sector investments



Source: Funding proposal, page 43, figure B.3.3.

54. The arrangements for the private sector contribution are illustrated in Figure 1 above. Co-financing is to be blended alongside the baseline investment and GCF resources to increase the lending portfolio for the BDCW and the BREFOL Direct Insurance Credit Window.
55. On the organization of the credit lines, the lessons learned on key factors contributing to the success of AfDB's ongoing AFAWA programme would still (as recommended also in previous iTAP reviews) need to be more clearly incorporated into the rolling out of the innovative credit lines (see condition (ii) in the concluding section).
56. The iTAP notes the close connection between credits, advisory services, regulatory reforms, as well as blended finance and potential guarantees, and would recommend the AE to develop these linkages in a similar manner to ensure cross-programme coherence and solid contributions to the enabling environment for continued beneficiary financial prowess and economic independence. This was recommended also in the previous review, and is reiterated in the proposed condition (i) in the concluding section.
57. On cost-effectiveness, the GCF cost per beneficiary is USD 29 and the cost per t CO₂ eq of avoided/reduced emissions is USD 31. The calculations are outlined in Table 4.

Table 4: BREFOL summary of the efficiency and cost-effectiveness of the programme

Total programme funding (in USD million)	335.3
GCF funding (in USD million)	151.0
Programme direct beneficiaries (million)	4.7
Total emissions reductions in 25 years (million t CO ₂ eq)	10.7
Estimated cost per programme beneficiary	71.6
GCF amount allocated for adaptation (in USD million)	134.7
Estimated GCF cost per programme beneficiary	28.8
GCF amount allocated for mitigation (in USD million)	16.3

Estimated programme cost per t CO ₂ eq	31.3
Estimated GCF cost per t CO ₂ eq	1.5

Source: Funding proposal, page 73, table D.6.1.

II. Overall remarks from the independent Technical Advisory Panel

58. The iTAP finds that the proposed initiative BREFOL is well aligned with the updated Strategic Plan for the GCF 2024–2027 by accelerating climate action for the most vulnerable.

59. The iTAP also recognizes that the AE, EEs and country teams are committed to addressing the important goal of enhancing climate resilience among agropastoral and other communities in the Horn of Africa. It is hence of great importance that GCF supports this kind of initiative in one of the most climate-affected and vulnerable parts of the world.

60. A general finding (from both the first and second submissions), however, is that the way that the programme is structured does not do justice to the needs of the region and the many important endeavours proposed. The iTAP hence recommends a set of measures, formulated in the conditions below, aimed at enhancing the clarity and manageability of the proposed initiative.

61. Hence, the iTAP recommends that the Board approve this funding proposal with the following conditions to be fulfilled prior to the first disbursement of GCF proceeds:

Conditions precedent to the first disbursement of GCF proceeds:

- (a) The Accredited Entity shall provide to the GCF, in form and substance satisfactory to the Secretariat:
 - (i) an updated logical framework that is consistent with the theory of change; and
 - (ii) a detailed monitoring, evaluation and learning plan (MEL Plan) that:
 - (1) Clearly outlines milestones towards achieving the Funded Activity outcomes and final targets, disaggregated by sex;
 - (2) Sets out the definition of indicators and data gathering methods for progress in mitigation of, and adaptation to, climate change of specific systems proposed in each Host Country;
 - (3) Includes the methods for analysing and interpreting the data;
 - (4) Includes the description per Host Country of the process to integrate the monitoring and evaluation information to the reporting system of the United Nations Framework Convention on Climate Change (M&E system for adaptation and MRV system for mitigation); and
 - (5) Sets out the steps and responsibilities for knowledge management and exchange.
- (b) The Accredited Entity shall deliver to the Fund, in form and substance satisfactory to the Secretariat, an updated gender action plan that explains how to achieve the stated goal of 50 per cent female and 50 per cent male split among direct and indirect beneficiaries of the Programme, considering the presently unequal access to resources such as credit, land titles or livestock, information and means of communication, as well as the ability to autonomously make financial decisions; and
- (c) The Accredited Entity shall update, in form and substance satisfactory to the Secretariat, the information and analysis contained in Annex 24 to the Funding Proposal, and

prepare corresponding country-level strategies for managing fragility concerns in an adaptive manner during project implementation in the Host Countries.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP239)

Proposal name:	Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL)
Accredited entity:	African Development Bank (AfDB)
Country/(ies):	Djibouti, Ethiopia, Kenya, Somalia and South Sudan
Project/programme size:	Large

Impact potential

Thank you for your insightful comments and recommendations. Yes, we agree to prepare and submit to the GCF Secretariat prior to first disbursement:

- (a) an updated logical framework that is consistent with the theory of change; and
- (b) a detailed monitoring, evaluation and learning plan (MEL Plan) that:
 - (i) Clearly outlines milestones towards achieving the Funded Activity outcomes and final targets, disaggregated by sex;
 - (ii) Sets out the definition of indicators and data gathering methods for progress in mitigation of, and adaptation to, climate change of specific systems proposed in each Host Country;
 - (iii) Includes the methods for analysing and interpreting the data;
 - (iv) Includes the description per Host Country of the process to integrate the monitoring and evaluation information to the reporting system of the United Nations Framework Convention on Climate Change (M&E system for adaptation and MRV system for mitigation); and
 - (v) Sets out the steps and responsibilities for knowledge management and exchange.

Paradigm shift potential

These iTAP comments are well noted. As suggested and described in the funding proposal, BREFOL leverage lessons and insights from existing African Development Bank (AfDB) initiatives such as the highly successful pan-African Affirmative Finance Action for Women in Africa (AFAWA). The AfDB is committed to fostering access to finance and technical support for Women-led Small and Medium Enterprises.

Furthermore, BREFOL will encourage the alignment with other related initiatives by the five country governments on the strengthening of institutional capacities for climate resilience and the creation of an enabling business environment for climate-smart entrepreneurship. We also agree that such developments beyond the programme would be important for making BREFOL's investments into tools, trainings and infrastructure more effective and sustainable in the long run.

Sustainable development potential

As rightly noted by iTAP consultations with beneficiaries and stakeholders including communities, women's groups, youth associations, civil society organizations (CSOs), and non-governmental organizations, will continue in the implementation phase. We also agree that this process will provide suitable opportunities for developing an updated gender action plan that is integrated into the mainstream BREFOL activities.

To address the possible risk of overpumping and overextraction of groundwater, site specific geophysical assessment will be performed to ascertain that the selected sites/acquifer can support and sustain the proposed drilling or rehabilitation of boreholes. The boreholes will not be located on the same location but will be scattered in the target communities based on demand. Note that this is the only possible source of water which will induce development interventions within the communities. In addition, the solar-powered submersible pumps which will be installed in the boreholes will only be pumping water for a short period of time during the day into overhead tanks, which will have a cut-off mechanism, and the pump will not be working for the rest of the day and at night, which will allow groundwater recharge.

Needs of the recipient

Thank you for your comments and for highlighting the unique challenges in the region and the need for the proposed BREFOL Direct Credit Window (BDCW).

The AfDB usual practice applicable to BREFOL will be to employ multifaceted approach that combines due diligence, risk assessments, capacity building, stakeholder engagement, and policy review to proactively manage and mitigate the challenges associated with fragility and conflicts. We agree to regularly update the country-level strategies for managing fragility concerns in an adaptive manner during project implementation to facilitate the implementation of activities 3.2.1 (Support delivery of climate regulatory framework and mainstreaming of climate change into agricultural sector policies and strategies) and 4.1.2 (Technical and institutional capacity building).

During BREFOL appraisal, the Bank's fragility and resilience department will update assessment in line with the AfDB "Strategy for Addressing Fragility and Building Resilience in Africa (2022-2026)".

Country ownership

Thank you for your commendation. The AfDB (AE) will continue to provide necessary support to develop the implementation capacity of the executing and implementing entities at regional and national level.

Efficiency and effectiveness

These comments are well noted. As recommended for the organization and rolling out of the innovative credit lines, BREFOL will leverage on the lessons learned on success of AfDB's ongoing AFAWA programme.

The AE take note of these iTAP recommendations and will ensure clear linkage between the proposed credits, advisory services, regulatory reforms, as well as blended finance and potential guarantees, and ensure cross-programme coherence and solid contributions to the enabling environment for continued beneficiary financial prowess and economic independence.

Overall remarks from the independent Technical Advisory Panel:

Firstly, the AfDB (AE) would like to use this medium to sincerely appreciate the efforts of iTAP and insightful comments during the assessment of the BREFOL proposal package. This has really helped to improve the quality of the funding proposal.

Secondly, we accept the following:

“Conditions precedent to the first disbursement of GCF proceeds:”

1. That the Accredited Entity shall provide to the GCF, in form and substance satisfactory to the Secretariat:
 - (a) an updated logical framework that is consistent with the theory of change; and
 - (b) a detailed monitoring, evaluation and learning plan (MEL Plan) that:
 - (i) Clearly outlines milestones towards achieving the Funded Activity outcomes and final targets, disaggregated by sex;
 - (ii) Sets out the definition of indicators and data gathering methods for progress in mitigation of, and adaptation to, climate change of specific systems proposed in each Host Country;
 - (iii) Includes the methods for analysing and interpreting the data;
 - (iv) Includes the description per Host Country of the process to integrate the monitoring and evaluation information to the reporting system of the United Nations Framework Convention on Climate Change (M&E system for adaptation and MRV system for mitigation); and
 - (v) Sets out the steps and responsibilities for knowledge management and exchange.
2. The Accredited Entity shall deliver to the Fund, in form and substance satisfactory to the Secretariat, an updated gender action plan that explains how to achieve the stated goal of 50 per cent female and 50 per cent male split among direct and indirect beneficiaries of the Programme, considering the presently unequal access to resources such as credit, land titles or livestock, information and means of communication, as well as the ability to autonomously make financial decisions; and
3. The Accredited Entity shall update, in form and substance satisfactory to the Secretariat, the information and analysis contained in Annex 24 to the Funding Proposal, and prepare corresponding country-level strategies for managing fragility concerns in an adaptive manner during project implementation in the Host Countries.

Lastly, we thank the iTAP, GCF Secretariat, and GCF Task Team (Manager and Support) for the invaluable efforts and time taken through out the funding proposal review cycle.

Gender documentation for FP239

BUILDING CLIMATE RESILIENCE FOR FOOD AND LIVELIHOODS IN THE HORN OF AFRICA (BREFOL)

Djibouti, Ethiopia, Kenya, Somalia, and South Sudan

Annex 8. Gender Analysis and Action Plan



BUILDING CLIMATE RESILIENCE FOR FOOD AND LIVELIHOODS IN THE HORN OF AFRICA (BREFOL)

GENDER ANALYSIS AND ACTION PLAN

DJIBOUTI

I. Introduction

The BREFOL Project in Djibouti is classified as *Category 2* in the Bank's Gender Marker System (GEN II), which means that the project contributes significantly through several of its outcomes and objectives to promoting gender equity and women's empowerment.

The following activities were discussed with all the technicians and experts of the project, including the Coordinator, the Monitoring and Evaluation Officer and the Gender Expert for the Ministry of Agriculture team; the Program Project Manager and the Djibouti Country Project Manager within the Bank. Each of the following activities was deemed already integrated or integrable within previously planned activities, including:

- (i) the implementation of guidelines for gender-sensitive agricultural infrastructure (within the budget line for technical manuals to be developed by the same firm),
- (ii) consultations with gender-sensitive beneficiaries (with <50% women),
- (iii) general gender training for all stakeholders, including those on the prevention of risks related to the transmission of HIV-AIDS, STDs but also the risks of gender-based violence (GBV) or sexual exploitation, harassment and abuse of power (SEAH) within the project.
- (iv) the deployment in the field of the innovative GALS (Gender Action Learning System) methodology under the supervision of IGAD and Oxfam at regional level,
- (v) the recruitment of a National Gender Expert.

The reworked table of gender activities in virtual session was shared with all participants with also a specific gender questionnaire to collect the country information required during the assessment mission, addressed to the Gender Expert, answered with the support of members of the Ministry of Agriculture and other national structures (including the Ministry of Women and Family). Finally, bilateral meetings between the Gender Experts (AfDB/Djiboutian party) and the project team were organized to finalize the project gender activity plan.

II. Gender analysis

Regarding the Africa Gender Equality Index (AGI) developed annually by the AfDB and UNECA, Djibouti is one of the three countries on the continent not to have because of the lack of too many indicators. It is therefore not possible to measure or compare the differences in Djibouti's performance vis-à-vis the continental average and other countries from this index. This presupposes a very weak consideration of gender issues in national development statistical studies, with ***¹a significant lack of sex-disaggregated data.***

According to the Mo Ibrahim Index of Governance in Africa (IIAG) 2020, ***at the gender level, Djibouti occupies the 52nd place out of 54 in Africa*** with a Gender score of 31.8/100 and an

¹ 51 of the 54 African countries are included in this index, using data from national sources, and by imputing values for data Missing. Djibouti, Equatorial Guinea and South Sudan are not included due to their lack too much important indicators. AGI 2019, methodological report, p.25.

improvement of 2.3 points since 2010.² This indicator measures and aggregates data on (i) women's political participation and representation, for which Djibouti scores 50.08/100, (ii) equal civil liberties for women, 36.2/100, (iv) equal access to public services for women, where Djibouti scores 9.4/100, as well as (v) laws on violence against women for which no score is available, and above all, (iii) socio-economic opportunities for women, sought by this agricultural project, whose score is 31/100.

Djiboutian women have limited socio-economic opportunities compared to men. Although Djibouti is experiencing high economic growth (8.4%), unemployment is massive and affects 48.2% of the active population, particularly affecting young people and **women** (62.8% and 65.6% **unemployment rate**). In the absence of significant private sector development, 41% of the workforce work in government or public services. The State remains the country's largest employer. However, the administration recruited twice as many men from 2009 to 2017 so that the unemployment rate for men even decreased three times more (from 54.6% to 38.7%) than that of women (from 68.6% to 63.4%) between 2002 and 2017. Accumulating inequalities rooted in traditions and customs limiting their socio-economic role, women experience a low level of employability. This leads them to turn mainly to the informal sector. From 2010 to 2015, the **rate of women working in the informal sector even increased significantly (from 31.7% to 74.3%)**.

Women's entrepreneurship in Djibouti is paradoxically weak and underdeveloped

In 2014, despite a recognized dynamism of Djiboutian women, **only 16% of new companies created were created by women** (Ouloufa Abdo Ismaël, Director of ODPIC, 2018). While it was almost absent ten years ago, "**female entrepreneurship has developed little by little and today represents a force for development**". The Republic of Djibouti has set itself a target of 35% of businesses created by women by 2021 by counting on support for the formalization of existing activities and support activities for women entrepreneurs affected by unemployment and precariousness (ibid.). According to Aicha Hamadou Hamid, it is an economic lever for the entire Djiboutian population but also "**a way to fight unemployment and the marginalization of women in the labor market**", the overwhelming majority of whom work in the informal sector. Women work every day to feed their families and **are very present in the subsistence economy**, small trade, clothing, cosmetics or handicrafts. **They are penalized by the lack of funding and infrastructure**. Several issues surrounding the start-up or expansion of women-owned businesses and the identification of socio-cultural problems and barriers faced by women entrepreneurs as well as ways and means to overcome them.

Financial inclusion for farmers is very limited, particularly for women farmers. Government efforts to combat the exclusion of people from the banking system with the implementation of microfinance tools, particularly for women traders known as "charcharis", have led to a significant increase in the number of credit beneficiaries and their volume. The number of women beneficiaries of credit has increased fifteen-fold (from 800 in 2010 to 12,500 in 2015, source: MPFPF, National Gender Policy 2011-2021; Republic of Djibouti, SCAPE 2015-2019, p.17), and the volume of credits granted by FD9 (from FDJ 57 million to FDJ 500 million, ibid.). However, **agricultural activities are ineligible for credit from microfinance institutions**. The majority of women and girls continue to face many constraints that prevent them from taking advantage of Djibouti's growth potential by remaining at the stage of "income-generating" activities, which are not very sustainable, with unchanged access to other means of production.

² Source: <https://iiag.online/data.html?meas=GEND&loc=DJ&view=overview>, Retrieved 09/22/21.

This stagnation is due to cultural factors. In rural areas, the girl prepares herself from an early age to take on household chores and take charge of the maintenance of the home. Mothers thus value the reproductive role of the woman and will sometimes prefer to prepare the little girl for the function of future mother and housewife. These factors constrain access to land, livestock or inheritance, as well as training factors. Indeed, even ***though the Family Code guarantees equal land rights for women*** and men (Article 101), ***in practice women have little access to land*** and are often even deprived of their inheritance in favour of the men in their family (AFD, 2020).

Unpaid work and burden of caregiving. Like many African countries, the weight of socio-cultural norms confers the majority of unpaid work on women. Djiboutian society is patriarchal. With regard to family education and the recognition of the common responsibility of men and women to bring up their children, the traditional perception that these responsibilities are attributed to the mother still prevails in the Republic of Djibouti. "Men leave this burden on women's shoulders. Indeed, it is rare for men to stay at home to care for children" (UN/CEDAW, 2010). Mothers thus value the reproductive role of the woman and will sometimes prefer to prepare the little girl for the function of future mother and housewife. This is especially observed ***in rural areas where the girl will be prepared from an early age to take on household chores and take charge of household maintenance***. While the Family Code Act has put an end to a large number of forms of discrimination, the persistence of patriarchal attitudes and deep-rooted and deep-rooted stereotypes concerning the roles and responsibilities of men and women in society that may discriminate against women are prevalent in Djibouti. The husband is considered the head of the family, relegating the wife to a subordinate role in decision-making concerning the family. These stereotypes still perpetuate conventional wisdom that women's role should be limited to the private sphere or, even more so, that it is up to them to look after the well-being of the home and family, even if they pursue a career. Considerable efforts have been made in the area of information and awareness-raising in urban areas. The educated couple recognizes that child-rearing is a shared responsibility of both men and women, and misconceptions about the roles of women and girls begin to evolve within educated families. On the other hand, ***Djiboutian rural society still has difficulty freeing itself from the weight of these social constructs*** (UN/CEDAW, 2010). In sum, despite notable top-down advances, pockets of conservatism encouraged by customs persist. Work on standards seems essential to change mentalities and provide a better balance in the division of labour.

Training for women contributes little to their employability. Enrolment in Djibouti's general and technical education is growing, but the proportion of girls is stagnating or decreasing, at the risk of perpetuating women's poverty. With a literacy rate that is struggling to increase (from 43.7% in 2011 to 48.2% in 2017), more women are deprived of assets in the labour market, especially ***rural women whose literacy rate is 16.2%***. This result is well below the 40 points of increase targeted for 2016. The goal of parity has ***not been achieved at the primary level, with a gender parity index that has even declined*** (from 0.98 in 2009 to 0.86 in 2017); which jeopardizes the achievement of parity at the average (0.82), secondary (0.83), technical (0.72) and higher (0.61) levels of education given the increase in gender gaps with the years of schooling. This decline in Djibouti is the result of a slow evolution in girls' enrolment (net enrolment rate (NER) from 65.7% in 2011 to 68.7% in 2017) and retention (gross enrolment rate (GER) from 68.8% in 2011 to 78.9% in 2017). The construction of gender-sensitive infrastructure and school canteens has had tangible effects on the retention of rural girls (GER from 21.9% in 2011 to 58.7% in 2017) but has still failed to significantly influence girls' dropouts. ***The main reason for girls dropping out and dropping out of school is related to marriage*** or the need to go to work for the family.

In the agricultural sector, ***smallholder farms run by women face a productivity gap of 20-30% compared to their male counterparts.*** In a context of resource scarcity, a third of women own land/farms and women herders own a quarter of the livestock (small ruminants). The effects of climate change are worsening the situation of women, even destroying their means of production and livelihood. Agriculture in Djibouti is limited to very small areas, due to the lack of water for irrigation, and is almost exclusively for self-consumption.

Djibouti also faces challenges in accelerating nutrition and food security. From 2010 to 2013, stunting stagnated at 29.7% and acute malnutrition decreased from 10.9% to 17.8% among children aged 0-5 years. Given the current health system, 8.98% of the Djiboutian population over 55 years of age is in a fragile situation according to the World Bank. The wasting of women (extreme thinness) of childbearing age remains high. Inadequate nutritional practices are the causes with micronutrient deficiencies. Women's lower access to knowledge, assets and power within the household is a barrier to seeking care. Thus, universal coverage of health services remains a challenge, penalizing ***poor and rural women*** in particular. Despite increased mobilization of public resources to improve the population's care in terms of access to health care, ***health inequalities persist***, and many challenges remain to consolidate gains and continue to expand access to care for populations, particularly in rural and disadvantaged urban areas. The maternal mortality rate is very high, at 383 deaths per 100,000 live births (PSDs), a figure almost five times higher than the regional average (78/100,000) (World Bank, 2013). Gender gaps also exist to the detriment of boys who experience more stunting (32% vs. 27% for girls), more acute malnutrition (19% vs. 16%), and greater health vulnerability. According to UNAIDS data, in Djibouti, in 2019, there were ***36,800 adults living with HIV (among them 56% women and 44% men over 15 years old)*** and between 500 and about 1,000 children (UNAIDS). This prevalence among adults aged 15 to 49 years would be 0.8% with a downward trend. In terms of family planning and protection against sexually transmitted diseases (STDs), the country missed PNG's target of a five-point increase target for 2016 with 17% of unmet needs.⁴

According to a study by another bank project in the Dikhil and Ali Sabieh area (INSUCO, ESIA, 2019), the population (men and women) knows that the HIV/AIDS virus exists but does not necessarily know the transmission channels, prevention mechanisms, or even their own HIV status. In rural areas, 56% of non-single women reported knowledge of AIDS (up from 66% in 2002), but ***85% of women with knowledge of AIDS did not know that not using condoms was a risk of transmission.***

The Covid-19 situation is exacerbating gaps in access to health to the detriment of the elderly and women. Indeed, ***Djibouti has the highest Covid-19 prevalence rate in East Africa***, recording 1,300 cases of Covid-19 per million inhabitants as of April 28, 2020, according to the African Union's Center for Disease Control and Prevention (CDC). At the level of the project area, with regard to health infrastructure, the feasibility study of this project mentioned the absence of a hospital structure ⁵in Defaineitou, and the request for the installation of a dispensary was made during the consultations of the populations.

13% of female-headed households do not have access to water compared to 10% of their male counterparts and 8% of female-headed households use traditional wells to fetch water.), and

³ UNAIDS, Djibouti, <https://www.unaids.org/fr/regionscountries/countries/djibouti>, consulté on 30/01/21.

⁴ Report on the national response to the HIV/AIDS epidemic, 2014; Country factsheets, HIV/AIDS Estimates, 2017.

⁵ Source: Le Monde, April 29, 2020, Covid 19 : [U.S. military isolates base in Djibouti, lemonde.fr](https://www.lemonde.fr)

the use of improved toilets for which the significant inequalities between urban and rural households are based. **Only 19% of rural households have access to improved toilets compared to 85% for urban households.** Depending on the gender of female heads of household, there is a four-point difference, to the detriment of those headed by women).

Moreover, it should be emphasized that Djibouti has the specificity of the existence of a **very vulnerable population, the nomads, in a situation of extreme vulnerability.** In 2014, more than 98% of nomadic households belonged to the two poorest quintiles of the population, and nomads accounted for the largest percentage of food insecure households. Since 2010, massive displacements of nomads have been grouped at sites in the north of the country, including Ali Sabieh. These populations, who have lost all their livestock, are entirely assisted by aid programs from the WFP, the State or donations from philanthropists in the Gulf countries. 42% are severely food insecure and 32% moderately food insecure (AGVSAN, 2014, WFP).

Climate change leads nomads to a frantic search for pastures for their livestock and a concentration of livestock on watered parts of the country, resulting in overconsumption of pastures with its corollary, the destruction of soil. The decrease in grazing areas leads nomadic populations to sell or see the death of part of their livestock, animal capital constituting their main source of income.

The digital divide hits rural women hardest. In terms of access to information and communication technologies (ICTs), strong inequalities prevail to the detriment of female-headed households and between rural and urban households. Inequalities emerge to the detriment of female-headed households for access to the Internet (20% vs. 25% for those with a man at the head), mobile phones (67% vs. 78%), and at least one means of information and communication technologies (68% vs. 79%). Despite the 2014 "Zero Women Without Telephone" campaign in rural areas, the digital divide remains characterised by inequalities related to gender and/or place of residence. Only 2% of rural households headed by women have access to the internet (2% compared to 25% of similar urban households), 20% to a mobile phone (compared to 80%) and 22% to at least one medium NICT (compared to 80% of urban households headed by women).

The provision of inclusive financial services is expanding in Africa to help women smallholder farmers overcome some of the constraints mentioned, with insurance, for example, to de-risk agricultural financing for women farmers or yield insurance pooling agricultural loans to reassure banks and encourage them to support small farms. Africa is the continent most vulnerable to climate change and is already prone to a wide variety of natural disasters and women in particular do not have collateral for loans. The African Development Bank (and several foundations) is encouraging investment in digital financial inclusion in Africa. This requires investments in digital infrastructure, policy and regulation, digital products (SMS providing agricultural advice), innovation and capacity building with a gender-centric approach.

III. Gender Analysis within the context of BREFOL Djibouti

The objective of the Programme is to contribute to the improvement of living conditions, including for women and youth, food and nutrition security, increased resilience, and peace and security in the Horn of Africa.

The watersheds selected in the project area are located in the regions of:

- Ali Sabieh: Assamo & Ali-Addé & Holl-Holl & Oboley & Daasbiyo, 513 km²

6500 inhabitants,

- Dikhil : Hanlé & As-Eyla & Harou & Mouloud & Arwo, 1060.8 km², 3500 inhabitants,
- Weima : Assa-Gaila, Adoyla, Guirror, Gaoura, Defaynatou, Ripta ; Randa, Bankoualé, Terdo, Debné, Daimoli, 1949 km², 1500 habitants,
- Tadjourah, 1949 km², 1500 inhabitants,
- Djibouti-city Ambouli Zone (population not specified),
- And Arta: Damerjog, Douda, Attar, 850 km², 950 inhabitants.

This corresponds to a total of 25 localities and a total population of more than 12,400 beneficiaries, including about 6225 women, not counting the nomadic, migrant or refugee population who can also carry out subsistence or cash agricultural activities.

Data from a survey conducted by INSUCO (ESIA, 2019) highlight the following facts for some localities concerned by this project:

- the unemployment situation is greatly exacerbated with **75% of people of working age unemployed** in the Ali Sabieh region, and 77% in the Arta region.⁶
- this situation **particularly affects women (70% of women)** while 58% of men are in this situation.⁷

The needs expressed during the feasibility study of women farmers residing in the project area to improve their productivity were as follows:

- ✓ The need to reduce dependence on fuel to power the well engine (wish to have solar panels to power these engines).
- ✓ Have agricultural gardens (although very active in their community, women have less than 50% of agricultural gardens).
- ✓ The construction of wells.
- ✓ The granting of pesticides for plants.
- ✓ The construction of protection against the wadi of water in case of rain
- ✓ The construction of hydraulic structures, water reservoirs, and micro-dams.
- ✓ Logistical means (cars to transport their crops to markets).

IV. Actions taken by the Government

At the level of Djibouti's legal, institutional and political framework, significant institutional progress in favor of gender equality can be noted on the part of the Djiboutian government with the inclusion of the right to gender equality in the national constitution, the ratification of international protocols on the promotion of gender equality, as well as the updating of a national gender policy (PNG) for the period 2011-2021. Progress on gender equality and the empowerment of women and girls is manifested in the prohibition of early marriage, forced marriage, repudiation and against moral and sexual harassment, although pockets of conservatism encouraged by customs persist.

However, national, political and budgetary instruments to implement these provisions remain insufficient (i.e. although recognized as a crime, violence against women, including female genital mutilation (FGM), is significant). This is important to note given that in any project, the influx of external male labour represents a major impact with the increase in GBV and

⁶ These data are taken from EDESIC 2015 and Djiboutian Research on Employment, the Informal Sector and Consumption and the Djiboutian Household Survey ; of the EDAM 2017, Household Survey, and were collected as part of the ESIA, Environmental and Social Impact Assessment (ESIA) of the project.

⁷ Source: Survey conducted by INSUCO in 2019 as part of the ESIA.

exploitative sexual relations (women and girls, especially migrants, finding themselves forced into prostitution).

With regard to the economic opportunities of Djiboutian women, at the initiative of the^{1st} Lady of the Republic of Djibouti, UNFD and ODPIC (Djiboutian Office of Industrial and Commercial Property), **a plan to promote women's entrepreneurship** has been launched.

Many initiatives were then born, such as the Centre for Leadership and Entrepreneurship (CLE), inaugurated in 2019, set up in partnership with the Ministry of Women and Family, which offers women entrepreneurs support in the process of setting up projects, developing business plans but also seeking funding. This incubation center also offers young people (men and women) an equipped and free space for a period of one year for the creation and experimentation of start-ups; Also, the Club des Jeunes Entrepreneurs Djiboutiens (CJED) offers a space for the exchange of experience of women entrepreneurs and women wishing to undertake, provides assistance in the form of support to strengthen the entrepreneurial skills of learners, promotes the coordination of institutions in charge of women's economic empowerment and offers expertise for the evaluation of applications of project leaders during competitions organized by the institutions in charge of promoting entrepreneurship. Finally, the Women Like Djibouti antenna, supported by ODPIC, supports women working in the informal sector to formalize their activities.

The "Djibouti Vision 2035" focuses on renewable energies for the modernization of rural areas and the Djiboutian State has undertaken a vast energy infrastructure program with a view to improving the living conditions of the population. However, large-scale capital-intensive infrastructure projects and other initiatives that create jobs or promote technical training have so far benefited women little. This project, which aims to target women and youth, has the potential to be a game changer.

The vision of "**Djibouti 2035**" aims for a situation where "**35%** of women are economically independent".

V. Project logical framework and gender-sensitive indicators

Instead of a logical framework (proposed at the regional programme level), a table presenting the project components was proposed to the project team with the general gender-sensitive project indicators:

	<i>Project Appraisal Report Objective</i>
RESULT 2: Increase revenues from agro-sylvo-pastoral value chains	
<i>RESULT INDICATOR: Increase in annual per capita income (M/F)</i>	<i>+30% (M/F) to be precisely defined in the baseline study</i>
<i>RESULT INDICATOR: Creation of additional jobs for youth</i>	<i>+500 jobs (30% of which are women)</i>
RESULT 3: Strengthen the adaptive capacity of populations to better prepare for and manage climate change risks and variations.	
<i>RESULT INDICATOR: Increased access to safe land for youth and women.</i>	<i>2000 (of which 50% of women)</i>
RESULTS STATEMENT 1: Strengthen the resilience of drought-prone areas and pastoral and agro-sylvo-pastoral production systems to climate change.	

RESULT INDICATOR 1.1: Construction or rehabilitation of climate-resilient and gender-sensitive infrastructure.	
RESULT INDICATOR 1.1.1 - Gender-sensitive agricultural and pastoral infrastructure / Irrigation land developed of which % intended for groups of women and youth farmers. 8	35/280ha (of which atleast 30% for women and youth)
RESULT INDICATOR 1.1.2- Managed pastoral land	45 /2000 ha
RESULT INDICATOR 1.1.3 - Alternative economic diversification infrastructure activities for women.	175 (40%)
RESULT INDICATOR 1.2: Sustainable agro-pastoral land management.	
RESULT INDICATORS 1.2.3: Secure land area for youth, women and vulnerable groups.	To be defined in the baseline study <(50%)
RESULT INDICATOR 1.3.1- Access of farmers and pastoralists to innovative technologies and practices	1000 (including atleast 30% women and youth)
I RESULTS STATEMENT 2: Supporting agribusiness development	
RESULT INDICATOR 2.2: Inclusive entrepreneurship development	
RESULT INDICATOR 2.2.1- Creation or strengthening of cooperatives and SMEs in agro-pastoral value chains, including SMEs and SMIs and youth-led SMEs (under 35 years of age).	50 (35%)
RESULT INDICATOR 2.2.2 - Access to innovative finance for cooperatives, vulnerable groups, SMEs and youth-led SMEs.	50 (35%)
RESULT STATEMENT 3: Strengthening capacity to adapt to climate change	
RESULT INDICATOR 3. 3.2- Development of tools and reports/Number of stakeholders trained (M/F)	10 / <50% women beyond those trained
OUTCOME STATEMENT 4: Project Coordination and Management	
Recruitment of a full-time Gender Expert within the project team	1
Implementation of gender action plan (% disbursement)	100%

VI. BREFOL Djibouti Gender Action Plan

The Bank's Gender Strategy focuses on improving access to finance and technical assistance, strengthening technical skills, building gender-sensitive infrastructure, and key external partnerships of the Gender Strategy 2021-2025. Bank management and the operations team demand that gender-specific challenges be taken into account in any Bank-financed project.

In order to address some of the gender gap issues raised in the gender analysis of this project, and in line with the ongoing efforts of the Government of Djibouti to improve gender equality and equity, it is recommended that the project include the following gender interventions to ensure that women and men benefit equitably from the project:

- promotion of financial inclusion and microfinance instruments for rural women,
- verbal, numerical and functional literacy
- support for the structuring of women's agro-pastoral cooperatives and the collective acquisition of land and livestock in field schools,
- extension and access to ICTs and renewable energy programmes targeting rural women,
- gender-sensitive agricultural infrastructure,
- extension of gender-sensitive agricultural and energy technologies,
- inclusion of gender analysis in all project studies with sex-disaggregated data collection and targeted recommendations,
- specific awareness-raising against gender-based violence.

⁸ A gender-responsive infrastructure will systematically include: consultations with the most vulnerable (ensuring disadvantaged voices are taken into account in design, planning and management, UN Women).

Details of the approved activities of the Gender Action Plan, as well as their implementation costs, are shown in the table below.

PROJECT TITLE	Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL) <i>Objective: To contribute to the improvement of living conditions, and food and nutrition security in the Horn of Africa</i>					
CATEGORY SMG (GMS)	GEN II					
Components	Activities	Budget in USD	Target(s)	Indicator	Responsible(s)	Timeline
Component 1. Strengthening Pastoral and Agropastoral Production Systems' Resilience to Climate Change	Consult communities, especially on infrastructure design, groups of women (50%) and men (50%).	Project (to be done by national gender expert)	50% women, 50% young people. (nb. to be determined from the baseline study).	Nb. of women, nb. of young people who were consulted	ERAP / Technical partners (Dir. Agriculture) / local elected officials / Gender and Community Development Expert	Year 1
	Produce a manual/guide with guidelines for the implementation of <i>gender-sensitive agricultural infrastructure</i> (manual that will be required in the deliverables of the infrastructure technical studies).	Project (to be done by national gender expert)	1 Guide available to project engineers to build gender-sensitive infrastructure	Availability of a guide/Community satisfaction (men/women)	ERAP / Gender and Community Development Expert	Years 1-2
Component 1. Strengthening Pastoral and Agropastoral Production Systems' Resilience to Climate Change	Realize agricultural infrastructures taking into account the gender dimension: wells, hygiene (improved toilets), security, lighting, and including small agricultural technologies (e.g. pick-up, solar panels, solar pumps, pesticides, protection against water wadis, flour grinders, home irrigation, chicken coops, community markets) and gardens (6 minimum) for women (<35% minimum of the proposed infrastructure budget).	Project (to be done by national gender expert)	100% gender-sensitive infrastructure30% women-specific infrastructure1 women's agricultural garden per watershed	Proportion of gender-sensitive infrastructure. Share of infrastructure budget dedicated to agricultural and garden technologies for women	ERAP / Gender and Community Development Expert / Gender Focal Points (PFG)	Year 2-6

	Acquisition of 1000 hens, equipment and food and veterinary products for one year, exclusively reserved for women farmers (100% women) and with 100% female management.	112 536	Number of women or groups of women to be defined in the baseline study	Nb. of women benefiting from poultry houses	ERAP / Gender and Community Development Expert	Year 1-2
	Community agroforestry (50% women and youth)	Project (to be managed by national gender expert)	120	Nb. of hectares	ERAP / Gender and Community Development Expert	Year 1-3
	Develop health and safety measures that take into account the needs of women.	Project (to be managed by national gender expert)	100% of women's groups satisfied.	Share of women's groups satisfied.	ERAP / Business / Gender and Community Development Expert	Year 1-2
	Train the women beneficiaries of the project in leadership and public speaking following an agreement signed with the Regional Directorates (at the beginning of the project and then spread over the duration of the project once a month).	Project (to be managed by national gender expert)	70% of women beneficiaries of the project	Number of women trained in leadership and speaking (baseline study)	Ministry of Health, Community Relays / Gender and Community Development Expert	Years 2-6
Component 2. Supporting the Development of Agribusinesses	Support farmers in financial inclusion (<50% women) in the localities of Tadjourah (6), Ali Sabieh (3), Dikhil (2), Djibouti-city (Ambouli) (50 people per locality, per year, 1st registered). Establish an agreement with the Ministry of Education for the free use of classrooms.	3 718	2400 people trained among the beneficiaries of the project, including at least 1200 women	Nb. of training and nb. of persons benefiting from this specific training	ERAP / Ministry of Education / UNFD / PFG localities - regions	Years 1-6

	Carry out a gender analysis on the acquisition of land rights with specific recommendations for actions and budgeted, integrated into the baseline study (TDRs with budget extension + \$ 20,000). NB: a questionnaire submitted to the project team includes a series of questions for which data are missing but will need to be integrated into the baseline study as well as this extension focused on land issues (see part 4 of Annex B13).	20 000	1 roadmap to support farmers in acquiring their AOS1 list of actions proposed, budgeted and validated for a phase 2	Availability of a roadmap to support farmers in acquiring their land cover certificates (AOS) with validated and budgeted actions for a phase 2 (or additional funding)	PIU / Prefectures / Gender focal points / Local development and gender expert / Ministry of the Interior	Years 1-2
	Establish partnerships with the engineering university, business schools and vocational technical high schools on trade, energy to create opportunities for collaboration and business with the project farmers (role of the Gender Expert and the Project Coordinator).	-	4 training institutes associated with the project / Partnership agreements	Nb. of training structures involved in the project (school support projects, volunteering and business case studies)	PIU (Coordinator)/ Gender and Community Development Expert / Ministry of Education	Years 1-7
	Carry out a particular activity with the communities/cooperatives benefiting from the project for a simplified procedure (e.g. sending requests by email and possibility of representation) with the Prefectures in partnership with the PFGs of the Capitals for the formulation of requests and obtaining certificates of agricultural occupation of the sub-projects (role of the Gender Expert and the PFG).	-	1 Prefectural agreement for a simplified procedure (email or document)	Existence of a facilitated procedure to set up this approach for the project	Gender and Community Development Expert / ERAP / Prefectures	Year 2-4

Component 2. Supporting the Development of Agribusinesses	Organize extension sessions on women's and men's land rights (access to rights of use and ownership of reclaimed land to ensure sustainability after the project) at the start of the project, with a view to supporting the most vulnerable (20%) and cooperatives (100%) in the acquisition of land rights (certificates of agricultural occupation).	Project (to be managed by national gender expert)	Situation at the end of the project +20% land held by men and women.	Baseline % land owned by men/women.	Project trainers (agents) / Gender focal points (PFG) localities and capitals / Gender expert and Dvp. Co.	Years 1-6
	Establish an agreement with the Ministry of the Interior for the acquisition of national identity documents for refugees, immigrants and stateless persons working on arable land.	(to be managed by national gender expert)	1 Convention is implemented during the project	Existence of a convention	ERAP / Ministry of Interior / ONARS / IOM / National Expert on Gender and Community Development	Year 1-2
	Deliver entrepreneurship training to business plan and SME development (<40% women).	Project (to be managed by national gender expert)	40% women entrepreneurs, 40% women-led businesses among the beneficiaries of the project	Proportion of women entrepreneurs and women-led businesses among beneficiaries	Project trainers / PFG localities and capitals / Gender and Community Development expert	Year 2
Component 2. Supporting the Development of Agribusinesses	Develop community sub-projects to provide access to innovative technologies to improve the yield of agropastoral activities, e.g. renewable energy (solar home systems' kits, pumping), improved cooking stoves. Based on demand (but at least <35% women).	Project (to be managed by national gender expert)	35% of sub-projects managed by women 50% of sub-projects for the benefit of women supported	Proportion of projects developed and led by women supported by the project Proportion of sub-projects for the benefit of women (expressed needs: i.e. solar panels, etc.) Number of women including women in female headed households and youth accessing clean, reliable and affordable solar home systems through	ERAP / Project Trainers / Gender and Community Development Expert	Years 2-5

				the PAYGo system of pay to own		
	Establishment of a gender-sensitive financing mechanism adapted to the needs of small farmers, fishermen and SMEs (Guarantee Fund) (35% women) Promote access to micro-credits, particularly through GEAs (Entry and Affinity Groups) and other women's caisse initiatives (CTEC, etc.).	Project (to be managed by national gender expert)	35% of the budget for women entrepreneurs - entrepreneurs; or £90,000 minimum.	Percentage of bank guarantee/micro-credit facility budget for women entrepreneurs or women-led businesses	ERAP / Gender and Community Development Expert	Year 2
	Policy dialogue with banking institutions: inclusion of rural women, LoC, support for the formalization of jobs in the sector.	Project	1 high-level workshop with recommendations	Nb. of workshops on financial inclusion and access to land rights for farmers in Djibouti	ERAP / Gender and Community Development Expert	Years 1-4
Component 2. Supporting the Development of Agribusinesses	Study on business opportunities in the agricultural value chain for domestic and cross-border trade, including a chapter on women's trade and recommendations for policy dialogue.	Project (to be done by national gender expert)	1 study on gender-responsive business opportunities	Existence of a study on business opportunities, cross-border trade	ERAP / Gender and Community Development Expert	Years 1-2
	Establish partnerships with the engineering university, business schools and vocational technical high schools on trade, energy to create opportunities for collaboration and business with the project farmers (role of the Gender Expert and the Project Coordinator).	(to be done by national gender expert)	4 training institutes associated with the project / Partnership agreements	Nb. of training structures involved in the project (support projects, volunteering and business case studies)	PIU (Coordinator)/ Gender and Community Development Expert / Ministry of Education	Years 2-5

Component 3. Strengthening Agropastoral Communities' Capacity to Adapt to Climate Change	Training of trainer(s) (12) or refresher on climate-smart agricultural practices (CFS) and nutrition, ensuring that 50% of women are trained (2 more women).	Project (to be managed by national gender expert)	14 Ministry of Agriculture officers trained or refreshed in CSA and nutrition (including 12 who will be trainers)	Number of community workers trained or upgraded	MORE	Year 1
	Training in CSA for women's farmers' groups (50% women or women's groups) throughout the project.	Project (to be managed by national gender expert)	50% women among the beneficiaries of the project (baseline)	Number of project beneficiaries trained, including 50% women 70% of project beneficiaries	Project Trainers / ERAP	Year 2-6
Component 4. Project coordination and management	Recruit a national expert in gender and community development through competitive recruitment for the ERAP full-time, bilingual in English: (i) responsible for the implementation of the PAG and its integration into the final logical framework of the project, (ii) training of the project team and gender stakeholders, (iii) organization of trainings, (iv) relationship with external partners (administrations: Prefectures, Ministries, military camps; design office, companies, NGOs, National Union of Djiboutian Women (UNFD), (v) coordination of PFGs of capitals and localities, (vi) supervision in the field, (vii) participation in the complaints management committee, COPIL and COTECH of the project, (viii) periodic reports on gender to the Coordinator, drafting of the gender part of the AfDB's periodic reports, (ix) Gender and CSO Focal Point at the AfDB on the project.	420 000	1 competent national gender and community development expert recruited throughout the duration of the project	An experienced and bilingual national gender and community development expert in English	MORE	Year 1 - 7

	Train the 12 trainers of the project (30% women) on the theme of female leadership and gender (including gender-based violence (GBV), female genital mutilation, FMG), women's economic empowerment, sexual exploitation and abuse, through the National Union of Djiboutian Women (UNFD), as well as the 4 Deputy Directors.	17 310	12 trained agents including at least 4 women	Number of trainers trained, including at least 30% women	MFF / UNFD / Gender and Community Development Expert	Years 1-2
	Training Awareness/mobilization of populations (50% women) and all project stakeholders (companies, external workers, ministries, traditional leaders) on gender and women's empowerment.	Project (to be managed by national gender expert)	According to the baseline study	Number of persons (M/F) and stakeholders trained	ERAP / Business / Gender and Community Development Expert	Year 2-4
	Further train the entire project team in gender (gender mainstreaming within agricultural projects, in climate resilience) and present PAG activities to the entire team at the start of the project. (3 half days).	-	MORE (100%)	Number of project employees with more in-depth gender training	Gender and Community Development Expert	Year 1
Component 4. Project coordination and management	Designate a full-time Gender Focal Point per locality (12) (50% women) to play a role of facilitator and close follow-up : (i) financial inclusion courses (opening of bank accounts), (ii) coordination (with trainers, the Prefecture, the Gender Expert), (iii) establishment of the list of people in need of CNI, certificates of agricultural occupation, people to register for training, etc., (iv) report on the activities of his locality to the PFG of the Capital, (v) followed trainings to become a trainer, (vi) writing a monthly report, (vii) training and training groups in the GALS methodology.	Project (to be managed by national gender expert)	12 Gender and community development focal points within localities including at least 6 women	Existence of a gender focal point (PFG) per locality, 50% of whom are women.	ERAP / Prefectures / Gender Focal Points / Gender and Community Development Expert / UNFD	Year 1

	Designate a Gender Focal Point per Region (4) part-time in the Sub-Directorate of each capital (Dikhil, Ali Sabieh, Tadjourah, Djibouti-city of the 6 watersheds) whose mission will be to: (i) report to the Gender Expert, (ii) facilitate travel and coordination with the prefectures, (iii) follow, support the activities of the localities, (iv) make supervision - monitoring of the field, (v) write a bi-monthly report to the Ministry of Agriculture (Gender Expert), (vi) train and groups in the GALS methodology.	Project	4 Focal points of the kind of capitals	Nb. of PFG by Capital	Gender and community development expert / Associations, local NGOs	Year 1
	Training of trainers on the Oxfam GALS methodology (Gender Action Learning System) by IGAD (6 people, 1 Ministry representative, 5 days). Necessary to include the Gender Expert (as well as the 4PFG of the capitals and the Project Coordinator).	IGAD	7 people trained including 3 women minimum	Number of people trained to be SALG trainers	IGAD / ERAP	Years 1-2
	Train gender focal points in localities (12) and regions (4) in the GALS methodology with tools translated into French and deployed in local languages (Somali, Afar) to be able to assist GALS trainers (by the Gender Expert).	14 338	12 Gender focal points of localities formed	Nb. of gender focal points of localities formed	Gender and Community Development Expert / UNFD	Years 1-2
	Train 30 people (15 couples) in local language per locality in the GALS methodology for 2 years (8 days of training in total), i.e. 360 people.	34 560	360 people trained, including at least 180 women	Nb. of persons trained at the GALS	PFG of the Capitals (trained) and Gender and Community Development Expert	

Component 4. Project coordination and management	Actively participate in the complaints management mechanism to ensure the representation of women (50%), monitoring of firms/companies in charge of the work, company.	PGES	100% of cases resolved	Number of cases referred to the Complaints Management Committee	Gender and Community Development Expert	Years 1-7
	Study of the baseline situation with sex-disaggregated indicators and a detailed gender analysis on the project area - collection and monitoring of sex-disaggregated indicators throughout the project with the Monitoring and Evaluation Expert	Project (50,000)	1 complete baseline taking full account of gender aspects and indicators	Existence of a gender-sensitive baseline and ToRs including this aspect	ERAP / Monitoring Evaluation Expert / Design Office / Gender and Community Development Expert / PFG	6 Months after inception
	Training/prevention workshops on GBV, FGM, GBV/SEAH, hygiene, HIV-AIDS with support from reference institutions on the issue and the signing of a Charter of Commitment at the start of the project by the Ministry of Women and Family (MFF) and partners. Additional sessions by the Gender Expert as needed	56 268	70% of the beneficiaries of the project (to be defined reference study) including 50% of women	Number of people who participated in the training workshops (<70% of project beneficiaries, <50% women).	MFF / Ministry of Justice / Police / Gender and Community Development Expert	Years 2-3
	Organization of exchange workshops, and national and international communication on the good practices of the project in terms of gender mainstreaming, participation in the sharing of lessons learned between the countries of the program (photos, videos, social networks, articles, interview, workshop, etc.).	5 000	2 virtual exchange workshops on gender communication objects on Djibouti's PAG	Nb. of virtual gender exchange workshopsNb. communication objects on the implementation of Djibouti's PAG	Gender and National Community Development Expert / IGAD	Years 3-6
	Integration of gender sensitivity into the procurement plan	Project (to be managed by national gender expert)	100% of PAG budget disbursed	Percentage of PAG disbursement	MORE / BAD	Year 1

TOTAL		733 720				
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In the event of additional funding or budgetary reserves during project implementation, the following activities are desirable:

- Support for the acquisition of land documents at the Prefecture, coverage of logistical costs (travel to prefectures or sub-prefectures: capitals and very remote localities).
- Verbal, digital, functional and digital literacy (min. 50% women) in the localities of Tadjourah (6), Ali Sabieh (3), Dikhil (2), Djibouti-city (Ambouli) following an agreement with the Ministry of Education (for the free use of classrooms).
- Training of gender focal points (trainers) of localities in training in verbal, digital, functional and digital literacy (12) by UNFD
- Extension of the tasks of the Gender Focal Points of the localities (50% women) to also deliver digital and functional literacy courses and establish the list of people in need of CNI and certificates of agricultural occupation.
- Strengthening of the early warning system with the establishment of a network of field assessors of climate vulnerability equipped (50% women/men) and the representation of women on the Committee set up by the Ministry of the Interior.
- Participatory governance within Climate Smart Villages (CSV) (planned for phase 2) by strengthening the participation of women and youth (50%).
- Equipment for home irrigation ("home water-pumps") and renewable energy solutions (solar panels) to intensify agricultural production and/or lighten women's domestic burden.
- Strategy for the inclusion of vulnerable populations (migrants, nomads) among the beneficiaries of the project with measures to combat their food insecurity,
- Specific support for women's entrepreneurship, for example with the establishment of an incubator in national priority agricultural value chains with comprehensive entrepreneurial and technical training in these sectors.

ETHIOPIA

Context

Ethiopia is a low-income country with a population of 112,078, 730 of which 50% are women according to the World Bank Data for the country. It is interesting to note that the country has made some tremendous progress towards actions to gender equality. According to the global Gender Gap Index of 2020 by the World Economic Forum, Ethiopia ranks 82nd in the 2020 rankings showing that it has closed 70.5% of its gender gap to date. Ethiopia has also achieved 16th position globally in political empowerment with about almost half (47.6%) of its ministers being women and 38.8% of parliament seats occupied by women⁹. It has also operationalized various policy and legal instruments designed to promote gender equality for example mainstreaming gender within the key national development plans such as the Growth and Transformation Plan I and II (GTP); instituting gender budgeting, increased political participation by women with a cabinet of 50% women and increased representation of women at regional and councils but though lower at federal level; registration of considerable improvements in reproductive health and some moderate strides for women's land rights with about 40% of women owning land and 25% owning through joint land certification for women in male headed households. Despite these remarkable results women are still marginalized in the fields of health services, education, labour market¹⁰ and economic participation and realization of their full potential. For example, Ethiopian women are worse off in agricultural productivity, business revenues and hourly wages than men due to differential access to productive resources like credit, inputs, education, and assets¹¹.

Specific Gender Gaps have resulted in challenges and barriers to women's economic empowerment and achievement of gender equality in Ethiopia. These include: Limited access to basic services like education Gender Parity Index (GPI) is low in secondary and tertiary levels. Females are less likely to complete Grade 8 (59.7percent against 64.4percent for male) and continue with secondary schooling compared to their male counterparts and more likely to perform poorly than boys on grade 10 examination. Apparently, there are fewer females enrolled in higher institutions of learning compared to males in undergraduate (34percent) and postgraduate (18percent) levels. Mean years of schooling for females is just 1.6, less than half of the already low mean years of schooling for males, at 3.7 years.

Low BMI among women in the reproductive age: Around 22.3 percent of women in the reproductive age (15-49 years) are underweight (18.5 >BMI) because of inadequate energy intake and/or diseases. The highest proportion of underweight in women is observed in Afar (39.1 percent), followed by Tigray (34 percent), Gambella (31.8) and Somali (31.2 percent). On the other hand, the prevalence of overweight and obesity among non-pregnant women of reproductive age (15-49) stands at 29.4 percent in Addis Ababa, while about one-in-five women in Dire Dawa and Harar are overweight or obese¹² (WFP).

⁹ World Economic Forum (2020). The Global Gender Gap Index 2020

¹⁰ World Bank (2019). Ethiopia Country Diagnostic Note

¹¹ IGAD (2018). Gender Analysis of the IGAD Drought Disaster Resilience and Sustainable Initiative and Regional Programming Paper (RPP), Volume I.

¹² WFP (2019), Comprehensive Food Security and Vulnerability Analysis (CFSVA)

Harmful traditional practices – FGM and Early Marriages: Around 65 percent of Ethiopian women aged 15-49 are victims of FGM. Somali and afar regional states have the highest FGM rates with 99 percent of women aged 15-49 having been circumcised. The FGM prevalence has decreased from 74 percent in 2005 with a notable decline among younger women (aged 15-19) which currently stood at 47 percent in 2016, showing a 24 percent decline over ten years¹³. Women in rural areas are more likely to be circumcised than their urban counterparts (68 percent and 54 percent respectively). Early marriage is a common practice in Ethiopia, particularly in some parts of the country, impacting on the school attendance of girls. The median age at first marriage for women is one of the lowest on the continent: 17.1 years. Among women aged 30–34, 27.3 percent had married by age 15.69¹⁴. More than 58 percent of girls between the ages of 15 and 19 give birth sometimes causing physical damages (HDR Ethiopia). A USAID study found that an estimated 36,000 to 39,000 women in Ethiopia live with obstetric fistula, and over 3,000 additional new cases occur each year¹⁵.

Gender Based Violence: More than one-third of ever-married women reported that they have experienced physical, emotional, or sexual violence from their husband or partner at some point in time. As per the 2016 DHS, almost 1 in 4 women reported experience of emotional violence and physical violence from a husband or partner. Experience of physical, emotional, or sexual violence is higher among older women 40-49 (38 percent), formerly married women (45 percent), those living in rural areas (36 percent vs 11 percent for urban). Comparably women in Oromia (39 percent), Harari (38 percent), and Amhara (37percent) had higher proportion of women experiencing domestic violence. Experience of spousal violence decreases with increasing educational level and household wealth. Women from the richest quintile are least likely to report ever experiencing sexual violence (4.5percent), while 12.3percent of women in the poorest quintile report having experienced it ¹⁶. The new Civil Servants Proclamation No. 1064/2017 addressed sexual harassment in the workplace prescribing serious disciplinary measures.

Barriers to economic participation – limited access to productive resources: Women fare worse off in agricultural productivity, business revenues, and hourly wages than men due largely to differential access to productive resources like credit, assets, inputs, and education. Women's low access to formal credit is partly linked to their limited ownership/control of physical assets that serve as collateral. Women experience lower returns on investments due to structural issues that perpetuate the broader social norms, market failures, and institutional constraints. The Government has prioritized increasing women's access to agriculture extension services. The Women Development and Change Package highlights a set of agricultural extension services from which women may benefit. The Growth and Transformation Plan II (GTP II) also has set the target of increasing the percentage of rural women farmers who are benefiting from extension services to 30 percent. It was reported that in 2019 around 2.8 million women 2,792,600 (32percent female headed) women farmers were reached through the agricultural extension services. Though women make up more than 40 percent of the agricultural labor force and head approximately 25 percent of all farming households, they have less access to land and other factors of production than men. On average male-headed households own 2.2 hectares of land compared to 1.7 hectares for female-

¹³ FDRE (2019), Fifth National Report on Progress made in the Implementation of the Beijing Declaration and Platform for Action

¹⁴ World Bank (2019), Ethiopia Gender Diagnostic Report .

¹⁵ Ethiopia Fistula foundation.

¹⁶ Central Statistical Agency (2016), Ethiopia Demographic and Health Survey.

headed households.¹⁷ Compared to male farm managers, female farm managers control smaller plots of land, cultivate fewer crops, use fewer inputs, and are less likely to access extension programs and formal credit. Use of inputs tend to involve cash investment and mobility that cash-constrained and homestead bound women may ill afford. Thus, productivity for female farm-managers is 23percent lower than their male counterparts¹⁸.

Disproportional overrepresented of women in informal sector as workers: According to estimates made using 2015/16 HICES data, at least 52.6percent of women and 47.5percent of men in the major towns are self-employed¹⁹. It is also estimated that 36 percent of women's work in the informal sector²⁰. In this category, women account for about half of the operators in the wholesale and retail subsector. As informal workers, women face challenges including a lack of working premises and shortage of working capital.

Women's underrepresentation and undervaluation in the labour market: As reported in the AfDB Africa Gender Index, female labour force participation rate was 74.3 (86.6 for male) in 2018²¹. Unemployed female youth in Ethiopia outnumber unemployed male youth, at 30percent and 19percent respectively²². In urban areas female unemployment rate (26.4 percent) is more than double as compared to male (12.2).²³ There are also significant gender gaps in wages received. Males receive 50percent more on average than women. Females earn 69 percent of what males earn in the whole economy and 63 percent in industry sector. In manufacturing, even after accounting for education and experience, men earn 23 percent more on average²⁴.

Under representation of women in High skill jobs and in managerial positions:

Close to 60 percent of the female production workers are unskilled (as opposed to 28 percent for males). The proportion of females who work as production managers or department heads is less than 4 percent, whereas that for males is about 13 percent. Women make up only 20 percent of science and engineering professionals, 21 percent of business services agents, and 30 percent of information and communication technologies (ICT) technicians²⁵. Women's share of highly qualified professions, senior positions and management is estimated at 30percent²⁶

Impact of COVID 19 on women and girls: Ethiopian women, like elsewhere, are vulnerable to infection due to their productive roles as first line health workers, health facility service staff (e.g., cleaners, laundry) and reproductive roles (as primary care providers). Overwhelmed health services may also hamper women's access to reproductive health, especially crucial pre-natal services, and access to family planning, exacerbating the already high level of maternal deaths in the country. With the slowing down of economic activities, more Ethiopian women will be particularly vulnerable as many of them depend on subsistence employment and the hardest hit self-employment sector, which is characterized by informal business. Increased household stress

¹⁷ World Bank (2019), Ethiopia Gender Diagnostic Report .

¹⁸ GIZ (2019), Employment and Labour Market Analysis (ELMA) in Ethiopia.

¹⁹ AfDB (2020), Growth, Inequality and Poverty Reduction and Job Creation in Ethiopia.

²⁰ UNDP (2018), A Study On Women Manufacturing In Ethiopia.

²¹ AfDB (2019), Africa Gender Index

²² FDRE (2020), Ten Year Perspective Development Plan (2021 – 2030).

²³ Ethiopian Central Statistical Agency (2018), The 2018 urban employment unemployment survey.

²⁴ UNDP(2018), ETHIOPIA | National Human Development Report.

²⁵ UNDP(2018), ETHIOPIA | National Human Development Report.

²⁶ GIZ (2019), Employment and Labour Market Analysis (ELMA) in Ethiopia.

over economic shocks, prolonged periods of coexistence in (small) living quarters, and reduced access to critical protection services and support networks during this pandemic could exacerbate cases of intimate partner violence in the country. In the absence of a vigorous back-to-school campaign, girls' school dropout is also expected to rise from the current 17.3 percent (elementary level). The need for family members to consistently wash their hands to protect themselves from infection, and keep homes and other establishments clean adds a new dimension to the water collection burden of women.

Lessons learned from Phase I (DRSLP I and DRSLP II)²⁷ of the project, data collected through key informant interviews, consultations with gender focal points in the relevant Ministry of Agriculture, and non-governmental organizations during the feasibility study period for this project in selected project regions indicate that pastoral and agro-pastoral women in Ethiopia are doubly marginalized; by virtue of their gender and through their migratory lifestyle making access to services difficult. A bigger percentage of the population is found in the rural areas with women providing most of the manual labour in agro pastoral communities. For example, while men migrate from one place to another with livestock, women and girls are left at home to take care of families during prolonged periods of drought and with very limited resources to support their families. Due to their socio-cultural gender norms, they are forced to walk for long distances in search of water, fuel wood, and grass for construction of houses. This activity pattern puts an extra constraint on their time budgets and reduces the amount of time available daily to engage in small income earning businesses (IGAD, 2016).

Gender disaggregated information on roles and access to assets and productive inputs by men and women remains scant making analysis and targeting difficult. However, consultations and research indicate that pastoral women face considerable barriers in accessing land and credit. Where land is owned by female heads of households, the parcels of land are much smaller than those jointly owned by women and men in male headed households. At the community and household level, patrilineal system of land ownership and inheritance based on customary laws and practices override the established laws, institutions and legal frameworks on land ownership such that preference is always given to sons versus daughters. Women in agro pastoral communities also have no access to credit in formal microfinance institutions and rely on traditional saving schemes to operate small businesses that are not economically viable. They are also often excluded from vital information for market and value chains and are not able to access information related to climate change resilient agricultural practices due to their low literacy levels and limited participation in decision making in community level institutions. This in turn raises their vulnerabilities to shocks related to climate change compared to men. (IGAD, 2015; UN Women, 2014)

The major beneficiaries of the proposed program will be pastoral and agro pastoral communities with 30% to 50% being women based on government commitment and number of women reached, and their local institutions in 30 districts across 4 administrative regions in Ethiopia. The primary beneficiaries of the proposed project interventions are expected to number about 2,267,736 million people (about 441,194 households). These beneficiaries will be selected individuals, households

²⁷ AfDB (2014). Drought Resilience and Sustainable Livelihoods Program in the Horn of Africa – Project II (DRSLP II). Ethiopia Appraisal Report.

(male headed and female headed households), and communities residing in agro pastoral areas in four Lowlands regions. They include pure pastoralists (camel, cattle, sheep, and goat owners) and agro-pastoralists in the targeted rangelands; those opting out of pastoralism and/or interested in taking up alternative forms of livelihood; and the populations in the selected woredas and kebeles.

Specifically, this project targets to build climate resilience for food and livelihoods among agropastoral communities and mainstream gender along the project components, thereby reducing the gender gap, achieving economic empowerment and gender equality. One of the main project outcomes is to increase income from agro pastoral value chains with increased per capita income among women beneficiaries and creation of jobs for women and youth. Additionally, the project targets to increase access to productive resources for youth and women like land, innovative and digital financial services tailored to meet the needs of the two marginalized groups, biofortified foods to women (in male headed and female headed households) and men within their households, participation in sustainable management of agro pastoral lands, and development of agribusinesses and agro pastoral value chains for women led and youth led cooperatives. Furthermore, the project will also strengthen women's decision-making role in the household and their community and their representation in the membership and leadership of grassroots institutions and organizations.

Therefore, the current project aims to mainstream gender in all components as well as address specific issues through targeted activities as outlined in the GAP. The project seeks to empower women both within the project areas and in program implementation and aims to achieve 30% to 50% of those benefiting to be women. It will encourage committees to include a 30% quota for women in management and training activities in line with GoE regulations. The GAP activities are supposed to address the persistent challenges and barriers that continue to negatively affect women to close the gender gap and achieve economic empowerment for women and youth. A gender expert within the PIU will ensure that the project is implemented in a way that addresses the gender issues among agro pastoral communities. Continuous monitoring and evaluation will be carried in a timely manner throughout the project period with gender mainstreamed indicators of change as outlined in the project log frame.

Identification of specific gender-based risks and mitigation measures to address identified gender gaps, inequalities, or discriminatory practices in the regions targeted for program implementation in Ethiopia.

- 1) Project activities that are dominated by men for example in design of infrastructure like boreholes, roads and rehabilitation of such structures can tend to limit the participation of women due to socio norms. This will be minimized through training of women in public speaking and making sure that at least 30% of representation in these forums are women with again equal representation in leadership positions.
- 2) that the project will ensure that women are not underrepresented in all forums across the project components (water committees, sustainable management of agro pastoral lands, climate smart villages with demonstration sites, access to digital technology on climate resilience practices and services etc.) with at least 30% representation. Also engaging with norms will involve transforming the mind-sets of both men and women and their traditional and customary leaders especially through gender transformative activities like community

dialogues. In addition, the GALS Approach and methodology should be implemented in all regions covered by the project. If it is not possible, then organized tours to the region where such methodology has been implemented should be arranged for other community members to learn. In cases where norms persist and hinder women's participation in the presence of men, then women only groups should be encouraged so that women can share their views on their needs, concerns, and interests with respect to implementation and management of the project activities so that they are not disadvantaged.

- 3) Markets for agro pastoral value chains especially livestock can have the tendency to be dominated by men and thus reduce the bargaining power of women with respect to prices of livestock and other agro pastoral value chain products. These challenges can be addressed through capacity building in public speaking for women, having digital platforms packaged and simplified to meet the market needs of women especially providing weekly market price targets on agro pastoral value chain products across the project regions.
- 4) Other challenges related to this project could arise due to time poverty constraints based on women's heavy workload. Alternative technologies like renewable energy – solar and biogas systems will be made available to women to reduce their time poverty constraints imposed on them by their productive and reproductive roles. Access to household water will be enhanced through provision of boreholes within short distances of homesteads and villages so that women spend less time travelling to fetch water. For water, adequacy during the dry season and accessibility through shorter distance travelled should be tackled as priorities. Women will have quotas (at least 30% participation) in reseedling programs and in land rehabilitation programmes
- 5) Women have low literacy levels and therefore could face challenges in understanding livestock and crop production technologies in agro pastoral value chains. The project should support simplified and packaged crop production technologies and extension service training manuals and modules targeting low literacy levels of women to enhance food production and nutrition. Such modules and technology manuals should be packaged in the local languages to meet the needs of women. Training should be conducted near villages and homesteads where women are comfortable to avoid the fear of traveling for long distances away from their homesteads based on socio cultural gender norms which limit women's mobility to faraway places without permission from the male heads of households and due to their heavy workload. Training periods of time and dates of training should be agreed upon based on consultations between women and education extension service providers.
- 6) Based on intrahousehold control of specific value chains by men and control of income from such value chains due to social cultural gender norms based on patrilineal systems in the agro pastoral communities, it is advisable to invest in women only value chains targeting women to make sure that women have the opportunity to earn income from value chains that have no value to men based on gender norms and traditions. Such value chains include poultry, goat and sheep fattening commonly termed as small stock value chains. Studies have shown that when a certain value chain becomes profitable, there is the possibility of control by men. Therefore, the project implementers at the community level

should put in place safeguards to ensure that markets, and market prices and products from women only value chains are not taken over by men. Such safeguards can include women organising into cooperatives for poultry, sheep and goat keeping and marketing their products as a group and thereafter sharing the income. Processing plants for produce from women only value chains are more secure from male dominance if they are operated as women only cooperatives for economic empowerment of women

- 7) Women are not homogeneous, and care must be taken not to assume that the interests, needs and concerns of women are the same. Such an assumption could result in biases in identification and participation of women beneficiaries in the project activities. There are women from male headed households and female headed households and younger women. Women from female headed households like widows and single mothers should be involved in all project activities targeting women using a quota system of representation to make sure that all groups and categories of women benefit equally from the project activities through representation at all planned activities across the project components.
- 8) Harmful traditional and customary practices can result in low participation of women in the project. Community mobilization to create awareness against such practices should be encouraged through gender transformative action plans for example through community wide campaigns against female genital mutilation and early marriages, reproductive health awareness and use of capacity building materials simplified and packaged to target different local community groups on behaviour change and use of gender sensitive and diversified behaviour change communication strategies to promote women and girls empowerment.
- 9) There could be gender biased implementation of project components by field staff if they are not trained on gender mainstreaming into the project cycle activities. The gender expert in the PIU gender expert and coordinator should build the capacity of field staff and gender focal points at the local level so that gender is mainstreamed into the project activities to achieve empowerment for women and close in the gender gaps identified in agro pastoral communities.

PROJECT TITLE	Resilience Building for Food and Nutrition Security (BREFOL) Programme in the Horn of Africa (HOA) <i>Objective: To contribute to the improvement of living conditions and food and nutrition security in the Horn of Africa</i>					
CATEGORY SMG (GMS)	GEN II					
Components	Activities	Budget in USD	Target(s)	Indicator	Responsible(s)	Timeline
Component 1. Strengthening the resilience of drought-prone areas and pastoral and agro pastoral production systems to climate change	Community consultations including women/men groups during design and site identification for infrastructure development activities	50,000	50	# of community consultations for development of infrastructure Community stakeholder consultation plan developed	Ministry of Agriculture/Ministry of Women and Social Affairs/ PIU	Ongoing
	Leadership training for women including public speaking training to participate fully in the Climate Change adaptation planning and range land management.	10,000	500	# of women trained	Ministry of Agriculture/Ministry of Women and Social Affairs/ PIU	
	Facilitated dialogues with women/community groups during site identification for sustainable range land rehabilitation.	20,000	20	# of dialogues with community groups on sustainable range management	Ministry of Agriculture/Ministry of Women and Social Affairs/ PIU Gender Expert in PIU	Year 2
	Payam Land Council, Land Committees, and clan leaders land management training with representations of at least 30% women	50,000	20	# of training sessions with PAYAMs # of women who participate in land management committees	Ministry of Agriculture/Ministry of Women and Social Affairs/ PIU	Year 2
	Replicate the Farm Africa gender inclusive bylaws in woredas to increase women's participation in conservation efforts , range land management, agroforestry and decision making.	100,000	20	# of woredas that adopt bylaws to accelerate women's participation in conservation efforts, range land management and decision making	Gender Expert in PIUs	Year 2-4
	Participation of youth and women groups in the sustainable land	Project budget (100,000)	50,000	# of women and youth who participate	PIU	Ongoing

	management practices in agro pastoral lands/			sustainable land management		
	Development of water infrastructure	9,299.4	200 5000	Number of water infrastructure developed to bring water close to communities # of households who have access to water sources closer to their homes	PIU	Ongoing
Inclusive Entrepreneurship Developed	Tailor made Business Development Services for women with low literacy levels	20,000	500	# of women trained	PIU	Year 2-6
	Dialogue between traditional leaders and district gender units to guarantee women's access to land resources beyond the project	20,000	20	# of communities where traditional leaders facilitate women's access to land	PIU Gender Expert	Year 2-6
	Gender analysis of selected value chains and identification of entry points for women's participation.	30,000	3	Gender Analyses reports	PIU	Year 1-2
	Facilitating the organisation of women farmers into cooperatives and supporting them to achieve some level of formalisation to benefit from financing. Capacity building on cooperative education: bylaws, leadership, management, and maintenance	50,000	5000	#of women who are registered members of cooperatives # of cooperatives trained on governance and Agri business	PIU Gender Expert	Year 2-4
	Provision of renewable energy solutions inputs and finance (energy saving cookers/solar panels for women and female-headed households)	500,000	500	# of women who adopt use of renewable energy solutions	PIU	Year 3-4

	Building capacity for women to access resources related to renewable energy at the group/cooperative level –	20,000	150	# of women who access finance for renewable energy technologies		Year 3-4
	Training and support for women to access bio digester units and solar systems	20,000	200	# of women trained	PIU. Gender Expert	Year 3-4
Gender Responsive Adaptive Capacity to Climate Change	Training of pastoralists and smallholder farmers on the use of climate-informed agro-advisory services (at least 30 % target are women)	Project budget (100,000)		# pastoralists using climate informed agro advisory services (at least 30% women)	PIU	Year 2-4
	Network of climate vulnerability assessors equipped with early-warning systems targeting women	Project budget		# of women who use early warning technologies	PIU	Year 1-2
Gender Responsive Project Coordination	Recruitment of a Gender Expert in PIU for an effective and efficient coordination of the GAP	420,000	1	Gender Expert recruited	PIU	6 Months
	Gender assessment of the project to obtain baseline data and inform project implementation	50,000	1	Gender assessment Report Updated gender Action Plan	PIU Gender Expert	6 Months after inception
	Capacity building on Gender mainstreaming of key stakeholders including on GBV/SEAH prevention	20,000	500	# of stakeholders trained	Contractors	Year 2-7
	Production of Gender Knowledge products	10,000	3	# number of reports produced	Gender Expert	Year 2-7
	Collection of gender disaggregated data for Monitoring, Evaluation and Learning					
	Facilitation of gender transformative forums for men and women: Sub-national Gender Action Learning Systems (GALS) Approach and methodology targeting women/men/traditional leaders across all the target locations	100,000	10	# of training sessions on the GALSS approach	PIU Gender Expert	Year 2-6

TOTAL		1,699,299. 4				

KENYA

Context

Kenya is a developing country whose development efforts are increasingly being hampered by the adverse impacts of climate change such as droughts and floods, which have increased in both frequency and intensity over the past decades. These impacts are felt across all sectors including agriculture, health, energy, transport, and water. Kenya's National Climate Change Action Plan (2018-2022) identified major climate hazards in Kenya, including the increasing frequency and intensity of extreme climate events, heat waves, droughts, floods, and landslides. The arid and semi-arid regions of Kenya have endured three severe droughts in the last decade (2010-2011, 2016-2017 and 2020-2022). The 2020-2022 drought was reported as the most severe and longest with widespread livelihood losses and massive displacement of populations.

In economic terms, each drought reduces the Gross Domestic Product by an average of 3.3 percent with the average annual cost of the damage estimated at \$ US 1.25 billion for drought alone. This has led to loss of lives, diminished livelihoods, reduced crop and livestock production, and damaged infrastructure, among other adverse impacts. Over the past 20 years, Kenya's population has doubled. Although Kenya has sustained population growth, it has both high birth and infant mortality rates. There has been a marked improvement in life expectancy, particularly in recent years. This is consistent with Africa overall. The United Nations estimated the Kenyan population to be at 54 million (51% women, 49% men) as at July 2022. Most of the population is mainly rural and relies on climate vulnerable nature-based resources.

Gender and Climate Change

Gender, together with other aspects of identity, shapes the lives, opportunities, and experiences of people across the globe. It also influences vulnerability to climate impacts. Despite growing recognition of the differential vulnerabilities as well as the unique experiences and skills women and men bring to development and environmental sustainability efforts, women still have less economic, political, and legal capabilities. Consequently, they are more exposed to and less able to cope with the adverse effects of climate change. Equally true is the reality of women as powerful agents of change, who despite structural and sociocultural barriers are making significant contributions to sustainable development.

In Kenya, patriarchal norms that assign gender roles place responsibility for household reproduction in women and girls. As primary caregivers they are charged with securing water, food, and fuel for cooking and heating while also being responsible for taking care of household members including children, the elderly, and disabled. In addition to reproduction activities, women also actively engage in livelihood production, undertaking the bulk of agricultural work and contributing to productivity for consumption and for sale. Nonetheless, women earn only a fraction of the income generated and own a nominal percentage of assets. Barriers to information, technologies, and financial services coupled with limits to women's full participation in household decision-making, markets and formal institutions contribute to these unequal outcomes between men and women. Despite women's participation and contribution to the economy, their work is under-rated, and women and girls' work is often unpaid, underpaid, or invisible. The majority of

women work in the informal sector, with only 34 percent working in the formal sector. For those that do work in the formal labour market they generally earn less than men; for every KSh100 earned by a man, a Kenyan woman only earns KSh 62.5. This wage gap cumulatively translates to different outcomes for both men and women, economically, politically, socially and culturally. Poverty is prevalent among low-income earning women, rural communities, and the urban poor, concentrated in increasing urban settlements. 30.2 per cent of female headed households are poor compared to 26 per cent of their male counterparts.

Women's dependence on natural resources for both reproductive and productive activities underscore their vulnerability in the face of increased climatic uncertainty. Yet, this same dependence also makes them critical stakeholders in adaptation and mitigation efforts. Therefore, implementation towards Kenya's NDC as well as efforts towards low carbon and climate resilient development requires inclusive and gender responsive approaches that address the factors that maintain inequalities, constrain women's opportunities, and limit their ability to plan and adapt to climate change.

Climate change risks lead to the loss of livelihood opportunities and caused considerable stress for men and boys. This is because of the socially ascribed role of men as the providers for the household. These have led to psycho-social illness and low self-esteem among men, while families are dislocated, and new gender roles emerge. Men are more likely than women to be affected by parasitic and infectious diseases during droughts and famines because they engage more in outdoor activities during severe weather events. However, women are more vulnerable to injury and fatality in climate change-induced hazards, such as flooding. This is because women have fewer opportunities to learn life skills such as swimming and tree climbing which are required in the event of severe flooding. Climate change affects the youth in diverse ways without any notable variations. During extreme weather events, youth experience unemployment, and some engage in substance abuse, suffer depression, and drop out of school. In addition, some of the female youth suffer from unwanted pregnancies and early marriages. In other cases, the youth engage in illegal activities such as drug addiction, robbery and join terrorist groups.

A quick overview of the context in the priority sectors of water, agriculture and energy highlights the following realities:

Water: Climate change is leading to a reduction in water resources which makes the management of this critical resource more difficult, affecting energy production and agricultural systems and increasing the likelihood of conflict. Currently, clean water coverage in Kenya stands at 60 percent leaving approximately 40 percent of Kenyans without access to clean and safe drinking water. iv In times of water scarcity, such as during droughts, it is women and girls who must travel long distances in search of water, with the result that they have less time for other activities including critical ones such as education and/or paid work. Prolonged drought leads to migration by pastoralist communities in search of pasture and water. They sometimes establish new settlements near other communities dependent on the same water source, which often lead to tensions and violent conflicts with unemployed youth becoming involved in different types of violent behavior, including livestock raiding. Additionally, because of strained livelihoods, intra-household relationships are strained, leading to tensions and gender-based violence (GBV).

Agriculture: Gender norms that dictate roles in agricultural underpin the disparities that men and women face in production systems. Women have less access to the means of production including land, finance, inputs, modern technologies, extension services, and marketing facilities while also having less say in decision making. Land title deeds are an important form of collateral for accessing agricultural credit, and hence the low rates of women's ownership of land in Kenya hampers their ability to obtain loans from financial institutions. These barriers will further constrain women's ability to adapt to climate change, with the result of an increased gender gap in agriculture coupled with more threats to families' food security. More needs to be done to ensure women's ownership of factors of production with land being key.

Energy: In Kenya, the energy sector plays an important role as the infrastructure enabler for sustained economic growth, development, and poverty. Access to modern energy is an enabler for women's empowerment as it plays a key role in meeting practical gender needs of women such as cooking, food processing and water hauling as well as strategic gender needs like street lighting for safety and power for enterprise development. While access to energy services would not necessarily guarantee gender equality, it increases efficiencies around household tasks such as cooking and allows time for education and other productive activities. Across all sectors, the impacts of climate change are often compounded for older women, women with a disability or for those with responsibility for other disabled or sick family members. Indigenous minorities lacking influence and access to decision-making are highly likely to face barriers to accessing solutions.

Key barriers to integrating gender considerations into climate compatible development include existing gender inequalities, patriarchal culture, and asymmetrical power structures. These need to be reflected in strategies and interventions. It is important to also underline that gender is not just about women and that a narrow focus on women in climate change adaptation results in narrow interpretation of complex contexts and this leads to shortsighted policies and interventions.

Policy, Legal and Institutional Framework

The Government of Kenya has recognized the need for equal and full participation of women and men at all levels of national development. The Government is committed to empowering women, reducing gender inequality, and eliminating all forms of discrimination. The Vision 2030 highlights progress made on gender equality and the commitments for ensuring compliance with international and regional obligations on gender equality and women's empowerment. The National Gender Policy 2011 and the Strategic Plan of the National Gender and Equality Commission (2019-2024) were designed to provide a framework for improving gender equality. Policies such as the Article 27 (8) of the 2010 constitution of the Republic of Kenya requires progressive realisation of the enforcement of the two-thirds gender rule and requires this to be implemented during general elections. In addition, Article 27 (6) provides for strong affirmative action, aimed at ensuring full realisation of gender equality.

The Government of Kenya has responded to the challenge of climate change by enacting and assenting to various policy, legislation, and strategies at the national and international levels. The link between gender and climate was first recognized at the global level in 2001 with the United Nations Framework Convention on Climate Change (UNFCCC) formally addressing representation and participation of women. At continental level, the African Union (AU) is

committed to addressing climate change issues with a gender perspective. Kenya has sought to operationalize AU commitments by integrating a gendered lens in their respective climate change policies, strategies and action plans as well as green growth and low carbon strategies.

Kenya's Climate Change Act of 2016 establishes institutions at national, sectoral and subnational levels to address climate change challenges. These include the National Climate Change Council, the Climate Change Directorate and the Climate Change Fund. Kenya has had a gender policy since the year 2000. It provides a framework for stakeholders to address the gender inequalities in a cross-cutting manner. The State Department for Gender in the Ministry of Public Service, Youth and Gender Affairs develops relevant policies and coordinates the mainstreaming of gender and women's empowerment in government departments and agencies at national and subnational levels. Additionally, the National Gender and Equality Commission (NGEC) coordinates, audits, monitors, facilitates and advises both state and non-state actors on gender equality and freedom from discrimination. NGEC hosts the secretariat of the African Working Group on Gender and Climate Change. The sectoral institutions include the Climate Change Units and Gender Focal Points in all Ministries, Departments and Agencies.

Kenya has done well to ensure robust climate change policies and legislative frameworks; however, gender disparities persist largely due to the slow implementation processes. This can be attributed to gaps in the laws, delayed enactment of gender-related legislation, ignorance of existing legislations, gaps in some of the policies and laws, inadequate accountability by citizens and duty bearers.

The Rationale for Gender Activities in BREFOL Kenya Project

The project will cover the six counties that were part of DRSLP-1 and Garissa county. The Gender Report stemming out of DRSLP-1 indicates that means of productions remain dominated by men in all six counties covered in the first phase, a situation dictated by patriarchal traditions. Labour, land and equipment, were controlled by adult and elderly males. The latter also determined how resources were allocated and used at household and communal levels.

The communities are traditionally pastoralist with livestock being the main source of livelihood. Livestock production activities (herding, watering, securing livestock, treatment, slaughtering, and sale) were dominated by men. In Samburu and Marsabit counties did women and girls participate in herding. The seasonal migration of men in search of pasture and water for livestock activities and to attend market days meant their reduced availability for farming activities during these periods. Women had full control in milking and selling milk as well as in chicken rearing, sale of eggs and sale of chicken. Key challenges to livestock production as a livelihood were: lack of water; diseases; low levels of value addition; decreasing pasture space; insecurity; limited access to credit; infiltration of livestock trade by brokers; and price fluctuations. In Samburu, a man could assign some livestock to his wives to sustain their households. However, the wives had no authority to dispose of the stock and must consult their husbands before doing so. In Baringo County, there were cases of single women and widows owning dairy cattle and married women owning goats and sheep but the latter needed to consult their husbands if they wished to dispose of the livestock. Although ownership and control of livestock mainly lay with men, all family members benefited from and accessed livestock products. However, income from livestock was under men's management.

Though crop farming is new to these communities, it is now the second main source of livelihoods. Communities do not have the capacity yet to adopt advanced crop farming technologies. In crop farming, land preparation is dominated by men while the sorting of the produce is exclusively done by women. Planting, weeding, harvesting, storage and transport of produce to the market was done by both men and women and the distribution of labour varied by county and by crop. There was a shared control on crops and related income but women largely controlled benefits related to subsistence farming activities. In crop farming, key challenges were lack of water, poor access to extension services, and low levels of value addition.

Women bore the burden of reproductive activities (fetching water, cooking, fetching firewood, cleaning houses, laundry; and care for children, sick and elderly). These activities were conducted daily and manually. Most of these activities were done in the domestic sphere and, as a result, limited women's exposure. Men, on the other hand, were in charge of activities viewed as technical.

A majority of respondents (85%) across the six counties reported being married. Household data confirmed that polygamy was common and in Samburu County practiced almost as much as monogamy. In all six counties, the man was the head of the household and the main decision maker on family assets and community matters. The wife was the daily manager of food and other family resources. In the absence of the man, the first son took over decision making if he had attained maturity age. In the case of polygamous relationships, the first wife shared responsibilities with her co-wives and alternated in herding, child care and other duties. Her first son was the senior son regardless of his chronological age relative to sons of other wives.

In all six counties, there was male dominance in community management activities such as gathering intelligence about community security and welfare; dispute resolution (land, domestic feuds, resources, theft etc); disciplining young men; carrying out religious activities and ceremonies, organising initiation ceremonies; and blessing warriors. Contemporary leadership was male-dominated and allowed women representation solely to abide by the constitutional gender rule. Only in Samburu and Marsabit counties were things different with female chiefs who were given respect due to their positions.

Rites of passage such as circumcision for boys and female genital mutilation (FGM) for girls were still practiced in some counties. Circumcision across the other counties was carried out during periodic ceremonies associated with good harvests. FGM continues to have disastrous effect on girls' and women's health. Early marriage was responsible for the high dropout for girls in most of the communities.

The issue of land ownership and acquisition remains key to any intervention aimed at building resilience. Study findings indicated that individual ownership of land across the six counties was common, exception in Samburu County where land was owned through group ranches. Where land titling was available, men are the ones registered as land owners. Though land was accessed and used by all the community members, women were unable to use such land as collateral for development loan. Pasture could be accessed by all but control was exercised by men in the community. Similar for water resources which could be accessed by everyone in the community,

but Water Resources Users Associations (WRUAs) were dominated by men. The considerable workload of women in both productive and reproductive activities restricted their availability to participate and conduct WRUA activities.

Female-headed households faced more constraints than male-headed households. Such constraints included: lack of traditional physical protection provided by men; higher levels of poverty; lower incomes due to reliance on one breadwinner; inadequate resources due to patriarchal ownership and inheritance patterns; and high workload. Fewer children from such households went to school and quite a number became labourers to earn a livelihood for the family. One advantage of female-headed households was cited as more economical use of resources.

Various institutions operate in the counties covered by the DRSLP. They include co-operative societies, government institutions, NGOs, faith-based organisations, and community-based organisations and self-help groups. Information from the cooperatives indicated low levels of attention to gender differentials. This was largely because they were voluntary organisations by nature and furthered the interests of their members without necessarily encouraging any particular sex to join. Membership is dictated by the ability to fulfil membership conditions such as payment of registration fees, regular contribution of shares, involvement in a specific productive activity, production capacity and ability to repay loans. Other information indicated that women were constrained by lack of capital and ignorance regarding sources of credit due to lower levels of education and exposure.

Recommendations

- 10) DSRLP project activities that depend on men's inputs need to be synchronised with seasons when they are not migrating. In parallel, the capacity of women to engage in activities dominated by men should be built.
- 11) Deliberate strategies should be developed to include women in all committees related to the project, including community management activities on decision making, conflict management and land management. Strong reference should be made to the constitutional edict on inclusion through the two-thirds gender rule. Traditional and contemporary leaders should be mobilised to facilitate such dialogues and lead the desired changes. Where necessary, the project should rely on separate spaces for women to ensure that women articulate their views on community management issues.
- 12) Livestock investment should be spent on value chains benefitting both men and women equitably. Key challenges should be addressed, including: water, pasture, insecurity, diseases and marketing. For water, adequacy during the dry season and accessibility through shorter distance travelled should be tackled as priorities.
- 13) With the growing interest in crop farming as an alternative livelihood, the project should support simple crop production technologies that would enhance food production and security. More complex technologies should be introduced at a later stage when acceptance and capacity of beneficiaries have been secured. Crop farming activities have shown great potential in the gender balancing of responsibilities.
- 14) Invest in value chains that benefit women. Poultry rearing is the domain of women. Commercial poultry rearing should be prioritized as an income generating activity for women with safeguards in place to ensure that the trade is not hijacked by men once it

becomes lucrative. Processing of dairy products can also be envisaged as an investment that will benefit women.

- 15) Move away from the assumption that heads of households are men. With polygamy commonly practiced, assuming so would be counter-productive and omit women who are *de facto* heads of their individual households. This would also ensure that households headed by widows and single mothers are reached.
- 16) Support advocacy work to recognise girls' and women's right to inherit family property and the succession rights of widows.
- 17) Run awareness campaigns on the elimination of FGM, early marriages, discriminatory dietary practices that negatively impact the development of girls and initiation practices that negatively impact boys' education.
- 18) Maximise the use of all structures created under the project as entry points for advocating gender equality and mobilising the community for gender training. For instance, using a facility for the distribution of educational and behaviour change materials on gender; to campaign against domestic violence and girls' education; and to dialogue and promote women's and girls' rights.
- 19) The WRUAs should adhere to the two-thirds gender rule in their leadership and include women in positions of authority, not just as treasurers. Ideally, equal representation of men and women is preferred to ensure that both perspectives are considered, including the inputs of women when setting user fees. Ensure the WRUAs are not dominated by men by discontinuing the practice of registering men as the household heads and instead registering the adult that manages water within the household.
- 20) Train field staff on gender mainstreaming in the project cycle. In addition, farmers and elders need to receive targeted training on gender issues, the roles they play in preserving the status quo, on how to create change as well as on the key constitutional and legal provisions on gender equality.

BREFOL KENYA GENDER ACTION PLAN/ GENDER ACTIVITIES FOR IMPLEMENTATION IN KENYA

PROJECT TITLE	Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL) <i>Objective: To contribute to the improvement of living conditions and food and nutrition security in the Horn of Africa</i>					
CATEGORY (GMS)	GMS II					
COMPONENT	ACTIVITIES	BUDGET IN USD	TARGET	INDICATOR	RESPONSIBLE (S)	TIME LINE
Component 1: Strengthening Pastoral and Agropastoral Production Systems' Resilience to Climate Change	Ensure a minimum of 1/3 representation from each gender in all committees (community land management, conservation, etc.) and community consultations	Project (100,000)	30% women, 50% young people. (no tbc @ baseline study).	# of women # of young people who were consulted	PIU/ MoALF&C	Year 1
	Leadership training for women including public speaking training to boost participation in Climate Change Adaptation Plans and other processes	10,000	70% of women beneficiaries of the project	# of women trained in leadership and speaking (no tbc @ baseline study)	PIU/ MoALF&C	Year 1-2
	Facilitation of gender transformative forums for men and women - Gender Action Learning Systems (GALS) Approach and Methodology targeting women/men/traditional leaders (5 per county) across targeted counties (these trainings to be facilitated by GALS champions trained by IGAD)	50,000	35 Local leaders in all 7 counties trained in GALS	#of local leaders trained per county, of which women	IGAD	Year 2-4
	Documentation of best practices on Gender responsive Land Use	IGAD (part of Gender expert TORs)	1 best practice on gender responsive land use produced	Existence of documentation (video or print medium)	IGAD /Ministry of Public Service and Gender/NGEC	Year 3-5
Component 2: Supporting the Development of Agribusinesses	Support Agri-business Officers in providing training on business development, financial inclusion and digital literacy (+ functional literacy for women)	10,000	Training of trainer(s) (14) or refresher on climate-smart business development, financial inclusion & digital literacy 30% trainees are women	14 Ministry of Agriculture officers trained to be trainers or refreshed in agribusiness	PIU/ MoALF&C/ Gender Officer	Year 2
	Policy dialogues on land access (leasing and ownership for sustainability after the project; strengthening local districts' gender	20,000	1 county level	# of policy dialogues held	PIU/ MoALF&C/ Gender Officer +	Years 2-4

	unit and negotiating/lobbying traditional leaders depending on context)		policy dialogue		State Department of Gender + NGE	
	Study to improve gender responsiveness of poultry and dairy value chains where women are heavily involved (e.g. mechanization; information; negotiation skills; business plan; how to commercialise)	10000	1 study on improvement of gender responsiveness of value chains	Existence of a study on gender responsiveness of value chains	PIU/ MoALF&C/ Gender Officer + State Department of Gender	Years 2-3
	Partnerships with agronomy, engineers, business schools to implement best practices.	10,000	At least 3 partnerships established	No. of functional partnerships	PIU/ MoALF&C/ Gender Officer+ Min of Education	Years 2-6
	Eligibility criteria for agri-business start-ups acknowledge differences in young men and women's profiles	Project Budget (done by the PMU)	Inclusive eligibility criteria developed and applied	Existence of inclusive criteria; No. of youth and women owned businesses benefiting	PIU/ MoALF&C/ Gender Officer + NGE	Years 2-3
Component 3: Strengthening Agropastoral Communities' Capacity to Adapt to Climate Change	ToT for climate-smart agriculture practices and nutrition by/for women (30 participants)	10000	30 trainees trained	# of women and men trained	PIU/ MoALF&C/ Gender Officer	Year 2
	Climate-smart agriculture training to women groups of farmers throughout the project	Project (done by the national gender expert)	30 % women among the beneficiaries of the project (# tbc @ baseline)	# of women groups trained	PIU/ MoALF&C/ Gender Officer	Year 2-3
	Capacity building of beneficiaries on local natural resource management and sensitization on GBV	40,000	4 Community user committees (land, water, forest) per target county	# of persons (M/F) and stakeholders trained	PIU/ MoALF&C/ MoI	Year 2-4
Component 4: Project Coordination and Management	Gender Expert in PIU for an effective and efficient coordination of the GAP and gender mainstreaming in the project	420,000	1 gender expert (local)	1 gender expert recruited and working	PIU/MoALF	6 Months
	Gender-sensitive procurement plan	Project (done by the national gender expert)	Project guidelines on gender sensitive procurement	Existence and application of gender sensitive project procurement guidelines and	Gender Expert/PIU/ MoALF	Year 1

	Capacity building of staff and beneficiaries on gender mainstreaming and cross-cutting issues	10,000	100% of PIU 70% of beneficiaries	# of PIU staff, No. of beneficiaries' capacities built	PIU/ MoALF&C	Year 1-2
	Gender assessment of the project to obtain baseline data and inform project implementation	50,000	1 assessment	Gender Assessment report; Updated GAP	PIU/Gender Expert/MoALF	6 Months after inception
	Capacity building on Gender mainstreaming of key stakeholders including on GBV/SEAH prevention	20,000	At least 1 training every quarter	# of stakeholder trained	Contractor/Gender Expert/PIU/MoALF	Year 2- 6
	Learning and sharing good practices and lessons among the countries/in the continent/internationally)	10,000	2 virtual exchange workshops on gender communication	# of exchange workshops	Gender Expert/Communications Lead/PIU/MoALF	Years 3 and 6
Total		770,000				

SOMALIA

I. Introduction

The project is assigned a Category II on the Bank's Gender Marker System as the project will improve women's economic empowerment through specific targeting within the project activities. Interventions to address GBV will also ensure that prevention measures are in place to support community women from any risk associated with project activities. Indicators related to the outputs are sex-disaggregated or gender-relevant.

II. Gender Analysis

According to the 2019 country gender profile for Somalia by AfDB and UN Women, the Federal Government of Somalia has made commitments and efforts in realizing gender rights and inclusion as well as empowerment of women through participation in decision making highlighting the roles for peace building and rebuilding of Somalia. Information gathered from previous key informant interviews with gender and human rights advisor, office of the president in the FGS and reported in literature indicates that: three key country frameworks have been drafted that provide strong platforms for promoting gender equality and development, with support from international partners. At the state level, the Provisional Constitution (2012) has included important new gender provisions and an 'ambitious' articulation and protection of citizen rights. Notably, in 2013, a Draft Constitution Technical Review Committee was established, with two lawyers, including one woman. Meanwhile, the international community's Compact for Somalia under the 'New Deal Framework' (2013) and the New Partnership for Somalia (NPS) have mainstreamed and prioritized gender commitments, with a specific focus on addressing gender disparities across sectors. Finally, the National Development Plan (NDP) (2017-2019) articulated a national framework for country development, with a notable alignment to the Sustainable Development Goals (SDGs) and goals to promote gender equality, in particular SDG 5: Gender Equality as a 'platform for all other sustainable development goals.

Additionally, government ministries and focal points in gender have also been established and appointed respectively, for example, the Ministry for Women and Human Rights Development (MWHRD) was established to better support gender and rights development in Somalia. Based on this institutional and policy frameworks and leadership from the focal points, ministries have now developed (or are developing) their own gender policies and gender budgeting guidelines. The development of relevant monitoring frameworks and the tracking of gender budgeting across ministries remain key outstanding areas of action for the MWHRD .

National Gender Policies

The draft National Gender Policy of 2015 provides an overall assessment and guidance on gender sensitive and inclusive programming in the areas of economic empowerment, education, health, governance and political participation with various cited ‘indicators’, or gender achievements outlined to facilitate the monitoring and evaluation of this policy (MWHRD, 2015). As key national reference and guidance platforms for the promotion of gender equality and inclusion, National Gender Policies (NGPs) have been formulated in each of the three regions (Somaliland, Puntland, and Somalia), most recently at FGS level. These have been well received by civil society and women’s groups. Unfortunately, apart from the support of gender activists, public impacts of these gender policies are limited, with little knowledge or awareness of their existence among many social groups. The implications of such limited public awareness and involvement by the FGS, is that gender rights and equality have thus remained critical issues in the uncertain context of Somalia, with low human development indicators, instability, and climate change.

Gender Gaps in Somalia

The Federal Government of Somalia (FGS) is committed to advance gender equality and women’s empowerment. The Provisional Federal Constitution stresses women’s equality with men and emphasizes women’s effective participation in all spheres of life, including public offices and in decision-making. In 2016, the FGS adopted a National Gender Policy with the aim to further gender mainstreaming in its peace and state-building processes.

Women in Somalia experience higher unemployment rates than men: 74% for women and 61% for men. Despite the Somali private sector being dominated by micro, small and medium enterprises of which women are the main drivers, women have limited access to credit, technology, and information. Access to services such as health care and education, by women and girls is disproportionately low. Somalia’s maternal mortality rate is the highest in the world, at 1,600 per 100,000 live births. Literacy levels are low; only 26% of women can read and write compared to 36% for men. Wo

Somalia remains one of the most unequal countries with the Gender Index standing at 0.776(1 – complete inequality) placing Somalia at the fourth-highest position globally. Climate Change has exacerbated gender inequality in Somalia, with women and children affected more by climate change effects due to lack of access and control of critical resources, biased labour division and absence from decision making processes from household to the National level. On the other hand, women and youth can act as agents of change and play a key role in supporting the transition towards low emissions and climate resilient development in Somalia. There are critical gender issues across sectors and hence the promotion of gender equality is critical for effective climate adaptation and mitigation in Somalia.

Barriers to Achievement of Women’s Empowerment and Gender Equality in Somalia

The Somali culture is considered socially conservative for women, with women's lives historically restricted to the private sphere, leading to their constrained role in decision-making, and limited access to assets and productive resources in their communities. Yet, women have contributed significantly to peace building efforts in recent years, forging new public roles. New trends are also now visible in the political, social, and economic domains.

Women are 'absent in leadership', particularly in Somaliland, and a gap does exist in 'mentors and role models' in (formal) institutions and in society that discourages women to participate in public leadership spheres. In all regions, National Gender Policies have not been publicly discussed or disseminated. With limited government capacities and allocated budgets, there is also a lack of practical action in working towards gender targets with low public awareness and support available for women (and minority groups) as there are few national campaigns and limited government outreach.

Social cultural gender norms, traditions and religious interpretation are still highlighted as the major barriers in working towards gender equality and rights development. Particularly, religious, and traditional clan leaders may challenge the smooth implementation of gender laws and policies in fostering conservative narratives. Furthermore, parliament and decision-making institutions within the FGS remain male-dominated with gender/women's oversights in progressive laws and policies.

Violence against women and girls cuts across all aspects of women's lives and remains deeply rooted in the local culture and traditions according to a report by UNDP 2014. In the transition from a predominantly nomadic and pastoralist society, traditional structures and customary laws have been strained, with varying effects on women and girls. Gender relations and roles have evolved and shifted, particularly with war and displacement over the past three decades resulting in women's increasing household responsibilities as family breadwinners, and large numbers of female-headed households living as IDPs (with men killed in conflict, absent, or working as migrants). As highlighted by a Rift Valley Institute discussion forum, while Somali women have often suffered disproportionately from the civil war and gender-based violence, the conflict has also ushered in new opportunities for them to assert themselves socially and politically.

Traditional Gender Roles and Responsibilities in Somalia

Somali society is strongly patriarchal, with clearly delineated roles and responsibilities of men, women, boys, and girls. The traditional gender roles and responsibilities of Somali men, women, girls, and boys highlight the impact of war and poverty (particularly because of increasing rural livelihood pressures). Men are traditionally the protectors of family security, primary breadwinners, and the central decision makers in both public and private settings. Men also control household expenditures, and their place is outside the home.

On the other side, women are homemakers and are responsible for bearing and raising children, food production and all domestic tasks. Women face land pressures – land alienation, degraded resources, and vulnerability to drought. Conditions are even worse for women and girls from female headed households who experience heightened challenges due to limited or no access to productive resources and assets compared to women in male headed households (Muse and Gardner, 2013). The bias manifests itself in terms of family investment in girls' skills (e.g., education), with a preference for boys to be educated; and women's low access to formal employment, with positions and status reserved for men (EU, 2013).

III. Project Intervention Regions and Constraints resulting from Gender Relations in Agro Pastoral Communities

The project will be implemented in priority locations in six Regional States in Somalia: Somaliland, Puntland, Galmudug, Hirshabelle, Southwest, and Jubbaland. The selection of intervention locations was based on the following principles: (i) Focusing on agro-pastoral investments, (ii) Promoting community-managed infrastructures, (iii) Prioritizing cross border clusters (iv) Continuity of intervention districts to facilitate project implementation and supervision (v) Ensuring synergies with ongoing interventions in each Region and alignment with the HOAI economic corridors and IGAD Cluster Approach, (vi) Alignment with country policies and strategies, and (v) Building on the progress made with DRSLP II implementation and leveraging lessons learnt. The project is targeting a total of 500,000 people out of which 50% of the beneficiaries should be women from male headed and female headed households.

Gender relations among agro pastoral communities are varied. Review of gender relevant literature for Somalia and consultation with the relevant implementing Ministry of Livestock, Forestry and Range highlights the important role played by women as small-scale business owners in agro pastoral communities and in livelihood diversification as a strategy to cope with ongoing climate change risks and shocks. While men migrate with livestock in search of scarce resources and others leave for the bigger cities in search of paid employment, women are left home to attend to the families and households, ensure that their household are food and nutrition secure and generate income for varied household needs. It is therefore inevitable for women not to engage in diversified income earning activities to support their families. With these multiple roles at hand, women and girls are constrained by high labour intensity burdens on their activity time budgets resulting in time poverty constraints, childcare barrier due to the amount of time spent raising babies and small children, harmful social cultural practices like gender-based violence, female genital mutilation, and early marriages. In addition, women are also faced with bigger challenges hindering their participation in the economic sphere due to limited or no access to and control over productive resources like financial services, land, information on entrepreneurial activities, climate smart resilient agricultural practices and technology to cope with the recurring climate change risks due to drought.

IV. Program's Contribution to Gender Equality

The Program is categorized GEN II according to the Bank's Gender Marker System where the gender assessment has identified four key entry points to enhance livelihood diversification thereby resulting in socio-economic benefits for women (in male and female headed households) and youth through: (i) design and implementation of gender sensitive infrastructure and services for example installation of boreholes and use of solar systems and biodigesters to reduce labour intensity burden on women, child care barrier and time poverty constraints; (ii) development of agribusiness value chains as strategies for livelihood diversification targeting both women and youth and access to productive resources like financial services, entrepreneurship training, markets and market linkages and climate smart agricultural practices through their cooperatives and groups; (iii) transformation of socio cultural gender norms through GALS Approach and Methodology to support women's role in decision making and leadership in their agro pastoral communities; and (iv) and recruitment of a Gender Expert in the PIU to ensure gender mainstreaming of the GAP activities along the program components and strengthening the capacity of key stakeholders in the implementing FGS ministries to address the gender issues during the project implementation, monitoring and evaluation periods. Additionally, the gender expert will also build the capacity of PIU and regional implementing staff and focal points on

issues around gender mainstreaming to close the gender gaps identified resulting in empowerment of women and achievement of gender equality. By the end of the project, there should be targeted achievement of 30% - 50% participation by women depending on the country's commitment and number of women reached in this project with increased income for women and economic empowerment due to increase in per capita income among women compared to men. There will be collection of sex disaggregated data to monitor progress and the logframe has clear gender mainstreamed indicators of change related to the above activities to achieve the desired outcomes. The Gender Action Plan activities and budget below are designed to address the gender inequalities and accelerate opportunities for women and youth in agro pastoral communities in Somalia.

Identification of specific gender-based possible risks and mitigation measures in Somalia during the implementation of the program

- Project activities that are dominated by men, for example in design of infrastructure like boreholes, roads and rehabilitation of such structures, can tend to limit the participation of women due to socio norms. This will be minimized through training of women in leadership and ensuring that 30% of representation in these forums are women with again equal representation in leadership positions
- Efforts should be made to make sure that women are not underrepresented in all forums across the project components (water committees, sustainable management of agro pastoral lands, climate smart villages with demonstration sites, access to digital technology on climate resilience practices and services etc) with at least 30% representation. Engaging with norms will involve transforming the mindsets of both men and women and their traditional and customary leaders through gender transformative activities like community dialogues. To this effect, the GALS Approach and methodology should be implemented in all regions covered by the project. If it is not possible, then organized tours to the region where such methodology has been implemented should be arranged for other community members to learn. In cases where norms persist and hinder women's participation in the presence of men, then women cooperatives should be encouraged so that women can share their views on their needs, concerns, and interests with respect to implementation and management of the project activities so that they are not disadvantaged.
- Markets for agro pastoral value chains especially livestock tend to be dominated by men and thus reduce the bargaining power of women with respect to prices of livestock and other agro pastoral value chain products. These challenges can be addressed through capacity building in leadership for women, having digital platforms packaged and simplified to meet the market needs of women especially providing weekly market price targets on agro pastoral value chain products across the project regions.
- Other challenges related to this project could arise due to time poverty constraints based on women's heavy workload. Care should be taken to make sure alternative technologies like renewable energy – solar and biodigester systems are available to women to reduce their time poverty constraints imposed on them by their productive and reproductive roles. Boreholes should also be located within short distances of homesteads and villages so that women spend less time travelling to fetch water. For water, adequacy during the dry season and accessibility through shorter distance travelled should be tackled as priorities. Women should have quotas (at least 30% participation) in reseedling programs and in land rehabilitation programmes.

- Women have low literacy levels and therefore could face challenges in understanding livestock and crop production technologies in agro pastoral value chains. The project should support simplified and packaged crop production technologies and extension service training manuals and modules targeting low literacy levels of women to enhance food production and nutrition. Such modules and technology manuals should be packaged in the local languages to meet the needs of women. Training should be conducted near villages and homesteads where women are comfortable to avoid the fear of traveling for long distances away from their homesteads based on socio cultural gender norms which limit women's mobility to faraway places without permission from the male heads of households and due to their heavy workload. Training periods of time and dates of training should be agreed upon based on consultations between women and education extension service providers.
- Based on intrahousehold control of specific value chains by men and control of income from such value chains due to social cultural gender norms based on patrilineal systems in the agro pastoral communities, it is advisable to invest in value chains targeting women to make sure that women can earn income from value chains that have no value to men based on gender norms and traditions. Such value chains include poultry, goat and sheep fattening commonly termed as small stock value chains. Studies have shown that when a certain value chain becomes profitable, there is the possibility of control by men. Therefore, the project implementers at the community level should put in place safeguards to ensure that markets, and market prices and products from women only value chains are not taken over by men. Such safeguards can include women organising into cooperatives for poultry, sheep and goat keeping and marketing their products as a group and thereafter sharing the income. Processing plants for produce from women only value chains are more secure from male dominance if they are operated as women only cooperatives for economic empowerment of women.
- There could be gender biased implementation of project components by field staff if they are not trained on gender mainstreaming into the project cycle activities. The gender expert in the PIU should build the capacity of field staff and gender focal points at the local level so that gender is mainstreamed into the project activities to achieve empowerment for women and girls and to reduce or eliminate the challenges identified in agro pastoral communities in relation to food, health and nutrition insecurity.

GENDER ACTION PLAN (SPECIFIC ACTIVITIES) FOR SOMALIA

PROJECT TITLE	Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL) <i>Objective: To contribute to the improvement of living conditions and food and nutrition security in the Horn of Africa</i>					
CATEGORY (GMS)	GMS II					
COMPONENT	ACTIVITIES	BUDGET IN USD	TARGET	INDICATOR	RESPONSIBLE (S)	TIME LINE
Component 1: Strengthening Pastoral and Agropastoral Production Systems' Resilience to Climate Change	Community consultations including women/men groups during design and site identification for infrastructure development	Project (done by the national gender expert)	30% women, 50% young people. (no tbc @ baseline study).	# of women # of young people who were consulted	PIU	Year 1
	Participation of youth and women groups in the sustainable land management practices in agro pastoral lands' (reseeding programs, at least 30% women participation in water committees and in leadership positions)	10,000	70% of women beneficiaries of the project	# of women trained in leadership and speaking	PIU	Year 1-2
Component 2: Supporting the Development of Agribusinesses	Livelihood diversification strategies for women and youth using alternative energy systems – solar systems and biogas to reduce labour intensity burden	10,000	Training of trainer(s) (14) or refresher on climate-smart business development, financial inclusion & digital literacy 30% trainees are women	14 Ministry of Agriculture officers trained to be trainers or refreshed in agribusiness	PIU/ Ministry/Gender Officer	Year 2
	Packaging of training modules (on business development services, digital and financial literacy) in Somali language to meet the low literacy levels of women in male and female-headed households	20,000	1 county level policy dialogue	# of policy dialogues held	PIU/ Ministry/Gender Officer + State Department of Gender + NGE	Years 2-4
	Support enabling environment and TA to accelerate women's access to finance identifying local savings and credit organizations among the community members (women and youth groups) and linking them to MFIs.	20,000	1 study on improvement of gender responsiveness of value chains	Existence of a study on gender responsiveness of value chains	PIU/ Ministry/Gender Officer	Years 2-3

	Gender responsive Digital advisory services packaged to meet the needs of women low literacy	10000	At least 3 partnerships established	No. of functional partnerships	PIU/ Ministry/ Gender Officer	Years 2-6
Component 3: Strengthening Agropastoral Communities' Capacity to Adapt to Climate Change	Training to target women group farmers and women cooperatives on climate-smart agriculture and nutrition	10000	30 trainees trained	# of women and men trained	PIU/ Ministry/ Gender Officer	Year 2
	Capacity building on CSA practices and agricultural value chains that can withstand adverse climate change risks with special seed varieties (maize, millet and sorghum): fodder and food to women	Project (40,000)	30 % women among the beneficiaries of the project (# tbc @ baseline)	# of women groups trained	PIU/ Ministry/ Gender Officer	Year 2-3
	Network of climate vulnerability assessors equipped with early-warning systems with simplified manuals to target women in male and female headed households with low literacy levels	40,000	4 Community user committees (land, water, forest) per target county	# of persons (M/F) and stakeholders trained	PIU/ Ministry/ Gender Officer	Year 2-4
	Facilitation of gender transformative forums for men and women: Sub-national Gender Action Learning Systems (GALS) Approach and methodology targeting women/men/traditional leaders across all the target locations (these trainings to be facilitated by GALS champions trained by IGAD) – For each region and woreda, we could have 2 groups at minimum, comprised of 30 people (15 couples – husbands and wives)	50,000	35 Local leaders in all 7 counties trained in GALS	# of local leaders trained per county, of which women	PIU/ Ministry/ Gender Officer/IGAD	Year 2-4
Component 4: Project Coordination and Management	Gender assessment of the project to obtain baseline data and inform project implementation	50,000	1 assessment	Gender Assessment report; Updated GAP	PIU/ Ministry/ Gender Officer	6 Months after inception
	Gender Expert in PIU to implement and coordinate the GAP and gender mainstreaming in the project	420,000	1 gender expert (local)	1 gender expert recruited and working	PIU	6 Months
	Capacity building of staff and beneficiaries on gender mainstreaming and cross-cutting issues	Project	100% of PIU 70% of beneficiaries	# of PIU staff, No. of beneficiaries' capacities built	PIU	Year 1-2
	Capacity building on Gender mainstreaming of key stakeholders including on GBV/SEAH prevention	20,000	At least 1 training every quarter	# of stakeholder trained	Gender Expert/PIU	Year 2- 6

	Learning and sharing good practices and lessons among the countries/in the continent/internationally)	10,000	2 virtual exchange workshops on gender communication	# of exchange workshops	Gender Expert/PIU	Years 3 and 6
Total		710,000				

SOUTH SUDAN

PART I: GENDER ANALYSIS/ASSESSMENT

I. Gender Analysis

South Sudan is committed to supporting gender equality within its development priorities at the national and international levels, through its policies such as the National Gender Policy (NGP) (2012), South Sudan's National Women's Strategy 2016 and the National Development Policy. The GoSS through the Ministry of Gender, Child and Social welfare has developed a National Gender Policy with the objective to design efficient policy frameworks and mechanisms for women economic empowerment and gender responsive agriculture ecosystem. The National Gender Policy has identified the following strategies:

- Conduct research into gender dimensions of food security and the food-production sector including agriculture, livestock, fisheries, water transport and trade;
- Institutionalize mechanisms to address key barriers that reduce women's ability to produce and preserve sufficient food; and
- Facilitate adequate resource allocation for small-scale agriculture and better access to water, energy transport and markets.

Agriculture: Agriculture is the backbone of South Sudan's economy. Women in South Sudan turn predominantly to agriculture as a necessity to support their families rather than as a business opportunity. Evidence shows that the agriculture sector provides livelihoods for over 80% of the population living in rural areas. However, much of the rural sector activity is currently focused on low-input and low-output subsistence agriculture, instead of farming as business and production for markets and export. The vast majority of South Sudanese, especially women and youth, are non-waged employees who depend on natural resources, particularly animal husbandry or crop farming to survive. Female-headed households are a significant proportion of the rural and urban poor and face even greater challenges in securing livelihood. According to Oxfam South Sudan gender analysis report (2016)⁴², women suffer greater food insecurity due to their cultural and social roles as caregivers. This care economy/unpaid workload hinders them from involvement in productive activities.

Labour: The gendered division of labour in South Sudan is particularly marked, with women having responsibilities for searching for food, firewood and clean water; cooking; domestic chores and childcare; as well being responsible for much of the agricultural activities at the household and community levels. Similar to other African countries, their productivity is lowered by unequal access to land, access to markets, productivity inputs, limited technical and business capacity, and the need to balance between business and family/societal responsibilities. This has important consequences for women's opportunities for economic empowerment, the types and nature of employment available to them, and their vulnerability to poverty.

The regions covered by the project have different gender specificities that are analyzed within the framework of the project in order to inform adequate measures and ensure equitable participation of women and men. In the Eastern Equatoria, there were more issues of displacement, challenges

of women's land rights, cited incidences of challenges of widows and women headed households, issues of refugee and returnee rights and sexual and gender-based violence.

Land tenure: Despite the South-Sudanese Land Act (2009) and the Constitution stating women and men's equal right to property, social and customary norms, however, prioritize male land ownership and inheritance. Land owned by a married couple is often registered in the name of the male spouse (Deng, 2016). Evidence show significant gender imbalances in land ownership for the targeted regions. According to FAO's gender assessment, land ownership in Central Equatoria is 4.8% for women (compared to 90.5% for men), in Eastern Equatoria is 4.6% for women (compared to 95.4% for men). Although women constitute 80% of the agricultural workforce, ownership and management of larger scale farm tend to be male dominated. Women are under-represented in the commercial farming and overrepresented at the lower end of the informal market system. The project's targeted crops, maize, sorghum and sesame, demonstrate high potential for women empowerment with women being more represented from planting to processing and man being in charge of transportation to wholesale. These crops are also drought-resilient with high production capacity in selected states. Access to and control of lands has compounded effects on poverty alleviation through agricultural production, food security as well as maintaining sustainable peace.

Climate change: South Sudanese women and vulnerable groups such as persons with disabilities are also at greater climatic risks given that they are both more likely to be dependent on natural resources and have less access to education, decision-making spaces, resources, information, and are less likely to own land and face sharper conditions when facing climate-induced displacement. In addition, women are responsible for biomass and water collection for their homes and deforestation and droughts will increase their time burden and risks of gender-based violence when collecting from further water and forest points. Despite this, women remain considerable knowledge bearers in agriculture and natural resource management and involving them in climate-adaptive and climate-smart practices will be key in enhancing community resilience and mitigation during climatic events. The UNFCCC in fact, recognizes both climate-based gender vulnerabilities and the important role of women in climate-dependent sectors and the unique strengths they bring to climate change adaptation and mitigation. South Sudan designed a policy, the National Adaptation Programme of Actions (NAPA) to Climate Change with gender equality and eliminating food insecurity as part of guiding principles for projects.

The project aims to respond to some of the gender issues in alignment with the GoSS' continuous efforts towards addressing gender issues and improving gender equity. The project is classified as Category 2 under the Bank's Gender Marker System (GMS). The project's actions aim at fostering livelihood diversification for women and youth through agro pastoral value chains including support for alternative renewable energy systems (solar and biodigesters) to reduce their labour intensity burden and time related constraints through (i) accelerated access to gender-sensitive infrastructure. Community infrastructure will be designed in consultation with women and include gender-sensitive constructions (ii) gender-responsive land management with specific leadership training for women to participate in land management committees (iii) enhanced access to skills for agribusiness development/growth and sustainable income generation and (iv) strengthened capacity to adapt to climate change through access to climate-smart agriculture trainings, practices and tools.

II. Identification of specific gender-based risks and mitigation measures to address identified gender gaps, inequalities, or discriminatory practices in the regions targeted for program implementation in South Sudan

1. Project activities that are dominated by men for example in design of infrastructure like boreholes, roads and rehabilitation of such structures can tend to limit the participation of women due to socio norms. This will be minimized through training of women in leadership and making sure that 30% of representation in these forums are women with again equal representation in leadership positions
2. Efforts should be made to make sure that women are not underrepresented in all forums across the project components (water committees, sustainable management of agro pastoral lands, climate smart villages with demonstration sites, access to digital technology on climate resilience practices and services etc) with at least 30% representation. Also engaging with norms will involve transforming the mindsets of both men and women and their traditional and customary leaders especially through gender transformative activities like community dialogues. In addition, we encourage the GALS Approach and methodology to be implemented in all regions covered by the project. If it is not possible, then organized tours to the region where such methodology has been implemented should be arranged for other community members to learn. In cases where norms persist and hinder women's participation in the presence of men, then women only groups should be encouraged so that women can share their views on their needs, concerns, and interests with respect to implementation and management of the project activities so that they are not disadvantaged.
3. Other challenges related to this project could arise due to time poverty constraints based on women's heavy workload. Care should be taken to make sure alternative technologies like renewable energy – solar and biodigester systems are available to women to reduce their time poverty constraints imposed on them by their productive and reproductive roles. Boreholes should also be located within short distances of homesteads and villages so that women spend less time travelling to fetch water. For water, adequacy during the dry season and accessibility through shorter distance travelled should be tackled as priorities. Women should have quotas (at least 30% participation) in reseedling programs and in land rehabilitation programmes.
4. Women have low literacy levels and therefore could face challenges in understanding livestock and crop production technologies in agro pastoral value chains. The project should support simplified and packaged crop production technologies and extension service training manuals and modules targeting low literacy levels of women to enhance food production and nutrition Such modules and technology manuals should be packaged in the local languages to meet the needs of women. Training should be conducted near villages and homesteads where women are comfortable to avoid the fear of traveling for

long distances away from their homesteads based on socio cultural gender norms which limit women's mobility to far away places without permission from the male heads of households and due to their heavy workload. Training periods of time and dates of training should be agreed upon based on consultations between women and education extension service providers.

5. Based on intrahousehold control of specific value chains by men and control of income from such value chains due to social cultural gender norms based on patrilineal systems in the agro pastoral communities, it is advisable to invest in women only value chains targeting women to make sure that women have the opportunity to earn income from value chains that have no value to men based on gender norms and traditions. Such value chains include poultry, goat and sheep fattening commonly termed as small stock value chains. Studies have shown that when a certain value chain becomes profitable, there is the possibility of control by men. Therefore, the project implementers at the community level should put in place safeguards to ensure that markets, and market prices and products from women only value chains are not taken over by men. Such safeguards can include women organising into cooperatives for poultry, sheep and goat keeping and marketing their products as a group and thereafter sharing the income. Processing plants for produce from women only value chains are more secure from male dominance if they are operated as women only cooperatives for economic empowerment of women.
6. Women are not homogeneous, and care must be taken not to assume that the interests, needs and concerns of women are the same. Such an assumption could result in biases in identification and participation of women beneficiaries in the project activities. There are women from male headed households and female headed households and younger women. Women from female headed households like widows and single mothers should be involved in all project activities targeting women using a quota system of representation to make sure that all groups and categories of women benefit equally from the project activities through representation at all planned activities across the project components.
7. Harmful traditional and customary practices can result in low participation of women in the project. Community mobilization to create awareness against such practices should be encouraged through gender transformative action plans for example through community wide campaigns against female genital mutilation and early marriages, reproductive health awareness and use of capacity building materials simplified and packaged to target different local community groups on behaviour change and use of gender sensitive and diversified behaviour change communication strategies to promote women and girls empowerment.
8. There could be gender biased implementation of project components by field staff if they are not trained on gender mainstreaming into the project cycle activities. The gender expert in the PIU gender expert and coordinator should build the capacity of field staff and gender focal points at the local level so that gender is mainstreamed into the project activities to

achieve empowerment for women and close in the gender gaps identified in agro-pastoral communities.

PROJECT TITLE	Building Climate Resilience for Food and Livelihoods in the Horn of Africa (BREFOL) <i>Objective: To contribute to the improvement of living conditions and food and nutrition security in the Horn of Africa</i>					
CATEGORY (GMS)	GMS II					
COMPONENT	ACTIVITIES	BUDGET USD	TARGET	INDICATOR	RESPONSIBLE (S)	TIMELINE
Component 1: Strengthening Pastoral and Agropastoral Production Systems' Resilience to Climate Change	Leadership training for women to increase participation and decision making in Climate Adaptation Planning	15,000	At least 50% of women beneficiaries of the project	#of women trained in leadership and public speaking	PIU/MoA	Year 1-2
	Active participation of women (at least 30%) in community consultations in identifying lands to rehabilitate constructions in gender-sensitive manner (ie. Adequate WASH, street lighting, road safety measures etc.)	Project (to be done by national gender expert)	# in reference to baseline of Payams selected (TBC)	# of women # of young people who were consulted	PIU/MoA	Year 1-6
	Constructions include gender-sensitive infrastructure (boreholes), feeder roads connecting markets, and irrigation infrastructures.	Project (to be done by national gender expert)	100% gender-sensitive infrastructure	Proportion of gender-sensitive infrastructure	PIU	Year 2-6
	Training of 30 persons per targeted Payam (15 couples and traditional leaders) / twice a year on the Gender Action Learning Systems (GALS) methodology	50,000	# in reference to baseline of Payams selected (TBC)	# of persons trained on the GALS methodology per Payam # of traditional leaders trained per Payam	PIU/MoA/IGAD	Year 1-4
	Ensure a minimum of 1/3 representation from each gender in all committees (community land management, conservation, etc.) and community consultations	Project (to be managed by the national gender expert)	# in reference to baseline of Payams	#of M/F in all committees	PIU/MoA	Year 1-6

			selected (TBC)	#of M/F in community consultations		
	Documentation of best practices on Gender responsive Land access and use	10,000	1 best practice on gender responsive land use produced	Existence of documentation (video or print medium)	IGAD	Year 3-5
	Policy dialogues with traditional leaders and local government on land access towards development of bylaws to extend land ownership of women (towards leasing, ownership for sustainability after the project)	20,000	Three policy dialogues held and one bylaw developed.	#dialogues held	PIU/ IGAD	Year 2-4
	Training of communities for social cohesion	10,000	Biannual trainings held for social cohesion	#trainings held	PIU	Years 1-3
Component 2: Supporting the Development of Agribusinesses	Adapted business trainings (local language and low-literacy levels in business development, processing, cross-border trade) for female-headed households and women.	40,000	Quarterly trainings held	#trainings held	PIU	Years 2-3
	Gender-sensitive value chain assessment	10,000		Gender assessment report	PIU	Year 2-3
	Facilitating the organisation of women farmers into cooperatives and supporting formalisation to benefit from financing and linkages with MFIs	Project (to be managed by the national gender expert)	30 % women among the beneficiaries of the project (# tbc @ baseline)	# of women groups formalized	PIU/ MoA/MoF	Year 2-3
Component 3: Strengthening Agropastoral Communities' Capacity to Adapt to Climate Change	Climate-smart agriculture training to women groups of farmers throughout the project	Project (to be managed by the national gender expert)	70% of women among the	# of women groups trained	PIU/ MoA	Year 1-4

			beneficiaries of the project			
	Capacity building of beneficiaries on local natural resource management and sensitization on GBV/SEAH	25,000	4 Community user committees (land, water, forest) per target county	# of persons (M/F) and stakeholders trained	PIU/ MoA	Year 2-4
	ToT for climate-smart agriculture practices and nutrition by/for women (30 participants)	10,000	30 trainees trained	# of women and men trained	PIU/ MoA	Year 2
	Capacity building on CSA practices and agricultural value chains that can withstand adverse climate change risks with special seed varieties	Project (to be managed by the national gender expert)	70% of women among the beneficiaries of the project	# of women and men trained	PIU/ MoA	Year 1-4
Component 4: Project Coordination and Management	Gender assessment of the project to obtain baseline data and inform project implementation and gender action plan	50,000	1 assessment	Gender Assessment report; Updated GAP	PIU/Gender Expert/MoA	6 Months after inception
	Recruitment of gender and community development expert for an effective and efficient coordination of the GAP and gender mainstreaming in the project	420,000	1 gender expert (local)	1 gender expert recruited and working	PIU/MoA	6 Months
	Capacity building of staff and beneficiaries on gender mainstreaming and cross-cutting issues	10,000	100% of PIU 70% of beneficiaries	# of PIU staff, No. of beneficiaries' capacities built	PIU/ MoA	Year 1-2
	Capacity building on Gender mainstreaming of key stakeholders including on GBV/SEAH prevention	20,000	At least 1 training every quarter	# of stakeholder trained	Contractor/Gender Expert/PIU/MoA	Year 2- 6
	Collection of gender disaggregated data for Monitoring, Evaluation and Learning Communication on the project gender outcomes (Success/impact stories, sharing good practices and lessons learned)	Project	Gender disaggregated data in every IPR / Yearly communication on project	#M/F beneficiaries in every program component	PIU	Year 1-6

			gender outcomes	#gender programme reports		
TOTAL		690,000				

REGIONAL

Cross Cutting Gender Issues of Concern in the 6 HoA Countries (Sudan, South Sudan, Ethiopia, Kenya, Somalia, and Djibouti)

1. Although in country differences may arise among the agro sylvo pastoral communities based on ethnicity, religious orientation and other intra country and inter-ethnic specific norms and traditions, the key gender inequalities faced by women and girls in agro sylvo pastoral communities are generally similar. Limited access to, control over, ownership and decision making on productive resources and in community forums are common limiting factors to women's role in productive activities, economic empowerment, and achievement of gender equality. Other socio-cultural expectations on women place an extra burden on their activity time budgets for example caregiving and childcare, gathering of fuel wood, fodder, grass for house construction and overall reduced mobility outside the home into the community sphere. Other socio economic and political factors include low literacy levels leading low skills hence poor salaries and low levels of employment in formal jobs, poor maternal and child health care, reduced representation in political, legal and policy making institutions. This in turn affects the way women participate in livelihood diversification activities across the member states in the HoA. These inequalities have their root causes in socio-cultural gender norms, traditional practices and kinship rules and patrilineal rules that limit women's access to, control over and decision making on key productive resources and participation in socio economic and political leadership forums in the community, subnational and national institutions.

2. Access to and Control Over Resources in Agro Sylvo Pastoral Communities:

- Women's roles within livestock production system and the distribution of ownership of livestock is strongly related to social and cultural factors.
- Often, when the rearing of small animals becomes a more important source of family income, ownership management and control are often turned over to the man.
- Women entrepreneurs, involved in local and border trade have the challenge of limited access to marketing networks, credit, and other financial services e.g., bank loans.
- Women in cross border trade, may have challenges in quickly accessing travel documents or gender insensitive trade agreements.
- Men often own larger animals such as cattle and camels while small animals such as sheep, goats, pigs, and chicken that are reared near the house are kept by women.
- Women in livestock economy are often responsible for livestock management, processing and marketing, caregiving, feed gatherers, and milk production, but often lack decision making power and capabilities that is important for their social and economic empowerment.
- Despite this involvement, women's role in livestock production has often been underestimated.
- Women entrepreneurs, involved in local and boarder trade, also have the challenge of limited access to marketing networks, credit due to lack of collateral and technical knowledge.

- Women in cross border trade, may have challenges quickly accessing travel documents. Other factors that have negatively impacted on women cross border trade have been gender insensitive trade agreements, such the World Trade organization (WTO).

3. While women are faced with such challenges, it is worth noting that access to, control over and management of resources including hides, meat, grazing areas and feed resources empower women and lead to an overall welfare of the whole household. Furthermore, market access plays a significant role in the wellbeing of a household as it allows people to buy and sell goods and services, including household essentials; and it contributes to the diversification of livelihoods in agro sylvo pastoral communities. Market inaccessibility is, therefore, an important cause of poverty, inequality and in determining where public services are inadequately provided. The target regions in each of the 6 countries within the framework of this program are highly dependent on pastoral and agro-pastoral economy, and livestock and related products and the major markets products.

Gender Roles and Responsibilities

4. In these communities, women are responsible for milking, processing, and selling milk products, providing feed/fodder and water, caring for new-born lambs/kids and sick animals. Young girls are also involved in the grazing of goats and sheep, whereas married and young women are responsible for household activities. This means that often, women have access to these products and able to improve on the welfare of their families.

5. Typically, male tasks include herding, cutting branches for home feeding and administering modern medicines. Wool- shearing, giving traditional vaccines against sheep pox, and castration are exclusively the domain of men for socio-religious reasons. Men migrate over long distances with the livestock in search of pasture and water, women and girls remain at home with meager resources at their disposal and are expected to sustain their families, hence the need for livelihood diversification by women and youth in these communities.

Livelihoods Support and the role of Formal and Informal Institutions in Perpetuating Gender Inequalities

6. Studies show that among most of the communities living in the HoA region, pastoralism and agro pastoralism are the main sources of livelihood. A fundamental shift in strategizing pastoralist livelihoods, and especially that of women, is to recognize innovation and entrepreneurship, and not limiting activities to those of coping. Although gender relations in the pastoral communities vary widely among tribes, regions, and countries participating in the program, there are some shared realities. Not only are women time-poor with long daily household routine duties, from childrearing, preparation of the meals, collection of firewood and water for the household to livestock management, but opportunities to economic empowerment and contribution to household are not maximised due to low capacity, lack of inputs, tools and credit and financial illiteracy. However, gender inequalities are produced and reproduced through intra and extra household gender dynamics in policies, institutions (socio, political, economic, traditional, religious, and customary), and processes, both formal and informal, to reinforce these gender inequalities that are observed today in agri sylvo pastoral communities across the 6

countries. Among most communities living in the HoA region in the identified clusters to be covered in this program, pastoralism and agro pastoralism are the main sources of livelihood. Some of the key gender issues include:

- Gender inequalities in relation to livelihood support make women highly vulnerable to livelihood insecurity in times of disaster which impacts negatively on the economic and social development of the region due to underutilization of available human resources.
- Men often own and control the assets of production, including the land and the livestock. The men, therefore, often determine women's access to land which forms the basis of their livelihood such as rearing small livestock.
- Limited access to land means little collateral for obtaining credit.
- Livelihoods programmes in emergency situations can provide important opportunities to promote higher standards of gender equality in economic life, by affirming women's role as economic change agents and by promoting equal access to productive resources and to decision-making mechanisms.
- Differential access to and control of livelihood assets also means differential impact of livelihood vulnerabilities between men and women as well as on male-headed households (MHHs) versus female-headed households (FHHs).
- Men often own and control the assets of production, including land and livestock. The men, therefore, often determine women's access to land which form the basis of their source of livelihood such as rearing small livestock, and limited access to land means little collateral for obtaining credit.
- Ensuring food and feed resources, and livelihood security for poor smallholder livestock producers and processors continues to be a challenge.

7. The livelihood diversification approach can thus be used as a key tool in promoting early recovery as well as greater gender equality in the post crisis social context.

The effects of Disasters: Climate Change Risks, Droughts, Floods, War and Conflicts on Gender Dynamics and Women Empowerment

8. Previous studies in the HoA have looked at the effects of disasters on men's and women's lives in agro sylvo pastoral communities and found that:

- Women are more likely to be disproportionately affected and exposed to risks, increased loss of livelihoods, gender-based violence and even lives during and in the aftermath of disasters
- Despite the differences, the potential women have and their knowledge and experiences is overlooked and women's leadership in building resilience in communities is overlooked.
- Pre-existing gender inequalities constrain the influence and control of women and girls over decisions governing their lives as well as their access to resources of production such as finances, agricultural inputs, land, and property as well as technologies.
- Women are therefore more likely to be disproportionately affected and exposed to risks, increased loss of livelihoods, gender-based violence (GBV) and even loss of life during and in the aftermath of disasters.
- Despite the differences, the potential and leadership skills women have in building resilience in communities have been overlooked.

- Poverty and inequality shape women's and men's vulnerability to disaster and the impacts of climate change, and their capacity to cope and recover in the post-disaster period.
- Since women and men are affected differently by disaster and climate change, their different vulnerabilities and capacities must be analysed, and their gender-specific concerns and priorities addressed.
- These gender inequalities make women highly vulnerable to livelihood insecurity in times of disaster and have negative impacts on the economic and social development of the region due to underutilization of available human resources.

9. The program will cover six countries of the HoA namely Djibouti, Ethiopia, Somalia, South Sudan, Sudan, and Kenya targeting climate change affected, drought-prone, food and nutrition insecure areas and communities where pastoralism and agro-pastoralism are the main sources of livelihood. Out of the targeted beneficiaries, at least 50% should be women in both male and female headed households as well as youth. Other direct beneficiaries include the Governments of the States of the Region whose capacities will be strengthened to coordinate a regional approach to enhance enhancing drought resilience development and regional integration. The beneficiaries of the Program are the populations of the six (6) IGAD Member States found in agro sylvo pastoral cluster regions and areas.

10. The programme is expected to increase the income of agro-pastoralists through the improvement of the delivery of climate smart services ensuring food, nutrition, and health security through identified interventions along the program components. More importantly, diversification of sources of livelihood and the rehabilitation, construction, and mobilization/conservation of water resources will reduce the exposure of the population to the climate change shocks and related conflicts over natural resources like water and pastureland. Women and youth will particularly benefit from the project through targeted interventions (including income generation through livelihood diversification activities in establishment, growth, and maintenance of agro pastoral value chains, establishment of renewable energy sources like solar systems and biodigesters, climate smart agricultural practices, and simplified digital technology designed to target women to be able to respond to climate change related shocks on time) to enhance their economic and social integration/empowerment. More detailed information is provided in member country gender analysis and GAP activities.

11. The progress toward achieving this objective will be measured through i) the number of water and animal health and marketing related infrastructures developed, ii) number of women and men and livestock accessing water within a certain distance; iii) the number and adoption rate of alternative livelihoods sources with data stratified and disaggregated by gender, iv) the incremental productivity/production of livestock and main crops; v) the area of land rehabilitated and, vi) the change in the beneficiary's incomes (men and women), vii) number of personnel and beneficiaries (males and females) benefiting from capacity building programs including the gender transformative capacity building of national champions at the regional level through the Gender Action Learning Systems (GALS) Approach and Methodology. This approach and methodology as an intervention targets transformation of socio-cultural gender norms, and harmful traditional and customary practices with women as main agents of change in their communities and men as their allies to reduce gender-based inequalities in the 6 countries.

12. This program will address gender inequalities in the IGAD region in support of the ‘political will’, manifested by IGAD country ministers responsible for gender issues, who occasionally meet to advocate the engagement of women in all development initiatives in the region. The creation of an IGAD Gender and Resilience Working Group and the Gender Unit will work towards planning, coordination, monitoring and evaluation of the regional gender component and follow up supervisory roles, monitoring and evaluation in gender and socio norms research, documentation of best practices and collection of sex disaggregated data to ensure that gender mainstreaming is part of the program design, implementation, monitoring and evaluation in the 6 member countries. The main goal is to achieve women economic empowerment and gender equality and equity within the context of the program components for enhanced climate change resilience, food, nutrition, and health security among agro sylvo pastoral communities in the 6 member countries.

13. The economic and financial resources and opportunities provided by the project will be measured by sex-disaggregated indicators. In the Bank Gender Marker System (GMS), the project is then considered as a GEN II. By the end of the project period, it is expected that there will positive outcomes for example increased income for women from agro sylvo pastoral value chains with an output indicator in increased annual per capita income among women (in male and female headed households) and creation of additional jobs for women and youth. Furthermore, women should have increased access to social services and infrastructure to reduce their labour and time poverty constraints, reduce child and care giving barriers and be able to participate more in livelihood diversification activities through agribusiness development in agro sylvo pastoral value chains.

14. Gender analysis for each country, including action plan, has been presented in the respective Country Components/Reports. The Regional Gender Action Plan with the corresponding budget for the regional gender activities is provided below.

Gender Action Plan Regional Activities and Budget (*Regional Gender Activities for Implementation by IGAD and its Institutions*)

NOTE: Activities Presented here cut across the 4 components of the Project

Gender Activities	Estimated Cost USD
Regional Capacity Building for ToT (including Gender and Resilience Working Group members) on the Gender Action Learning Systems (GALS) Approach and Methodology adopted from OXFAM- Uganda Using GALS Champions from Uganda. <i>ToTs (National ToTs) to spearhead implementation of GALS activities in their respective countries</i>	200,000
Regional Gender Expert supporting national gender experts; leading regional gender action plan and producing knowledge products to capture lessons learnt.	420,000
Translation of GALS Tools into local Languages (Amharic, Arabic, French and Somali) for regional capacity building	20,000

Gender Activities	Estimated Cost USD
Regional mentorship support for National Member Country Gender Action Learning Systems ToTs	48,000
Regional Capacity Building on Gender Planning and Budgeting (Using GALS) for Member country institutions across the project components	45,000
Regional Coordination and Supervision of Gender and Social Cultural Norms Research in the Agro pastoral communities across the IGAD Region to identify barriers to women empowerment and devise appropriate initiatives	70,000
Documentation of Gender transformative best practices and knowledge management in agro pastoralists communities	52,000
Collection of gender disaggregated data with gender mainstreamed indicators of change in agro pastoral communities	60,000
Regional Cross Country Gender Learning Exchanges to countries where GALS methodology has been implemented in the agro pastoral communities	35,000
Regional monitoring and evaluation mechanism to ascertain gender compliance of the standards and tools	60,000
Cross country supervisory role by the Gender Unit including contribution to staff salary	80,000
Establishment of a coordination/strengthening of a coordination mechanism (Gender and Resilience Working Group)	30,000
Grand Total	1,120,000